

NATO STANDARD

AEP-4671

**UNMANNED AIRCRAFT SYSTEMS
AIRWORTHINESS REQUIREMENTS
(USAR)**

EDITION B VERSION 1

APRIL 2019



NORTH ATLANTIC TREATY ORGANIZATION

ALLIED ENGINEERING PUBLICATION

Published by the
NATO STANDARDIZATION OFFICE (NSO)
© NATO/OTAN

INTENTIONALLY BLANK

NORTH ATLANTIC TREATY ORGANIZATION (NATO)

NATO STANDARDIZATION OFFICE (NSO)

NATO LETTER OF PROMULGATION

2 April 2019

1. The enclosed Allied Engineering Publication AEP-4671, Edition B, Version 1, UNMANNED AIRCRAFT SYSTEMS AIRWORTHINESS REQUIREMENTS, which has been approved by the nations in the NATO NAVAL ARMAMENTS GROUP, is promulgated herewith. The agreement of nations to use this publication is recorded in STANAG 4671.
2. AEP-4671, Edition B, Version 1, is effective upon receipt and supersedes AEP-4671, Edition A, Version 1, which shall be destroyed in accordance with the local procedure for the destruction of documents.
3. No part of this publication may be reproduced, stored in a retrieval system, used commercially, adapted, or transmitted in any form or by any means, electronic, mechanical, photo-copying, recording or otherwise, without the prior permission of the publisher. With the exception of commercial sales, this does not apply to member or partner nations, or NATO commands and bodies.
4. This publication shall be handled in accordance with C-M(2002)60.



Zoltán GULYÁS
Brigadier General, HUNAF
Director, NATO Standardization Office

INTENTIONALLY BLANK

RESERVED FOR NATIONAL LETTER OF PROMULGATION

INTENTIONALLY BLANK

RECORD OF RESERVATIONS

[illegible]

INTENTIONALLY BLANK

RECORD OF SPECIFIC RESERVATIONS

[nation]	[detail of reservation]
DEU	Reservation to USAR.1309 and associated AMC.1309(b) By applying the original table (Figure 1, page 2-F-6 (238 of 258) to page 2-F-7 (239 of 258)) in combination with the recommended development standard ARP 4754A, an applicant would be allowed to substitute a FDAL B with two FDAL C. This would cause a safety gap, because the main difference between FDAL B and C is, that the vast majority of FDAL B objectives must be fulfilled with independence, whilst for FDAL C objectives this is not required. Original text of Figure 1, page 2-F-6 Classification of Failure No Safety Effect Minor Major Hazardous Catastrophic FDAL E D C C B This is not acceptable for functions that may cause Catastrophic or Hazardous effects. Therefore, Germany expands the failure classification with the following additional notes and modifications, in order to clarify the handling of FDAL substitution: Classification of Failure No Safety Effect Minor Major Hazardous Catastrophic FDAL E D Note 1 C Note 2 C Note 3 B Note 4 Note 1 Functional Failure sets with a single Member: FDAL D Functional failure sets with multiple members: FDAL D for one Member, additional Member(s) contributing to the top-level Failure Condition at the level associated with the most severe individual effects of an error in their development process for all applicable top-level Failure Conditions. Note 2 Functional Failure sets with a single Member: FDAL C Functional failure sets with multiple members: FDAL D for two of the Members leading to top-level Failure Condition. The other Members at the level associated with the most severe individual effects of an error in their development process for all applicable top-level Failure Conditions. Note 3 Functional Failure sets with a single Member: FDAL C Functional failure sets with multiple members: FDAL C for one Member, additional Member(s) contributing to the top-level Failure Condition at the level associated with the most severe individual effects of an error in their development process for all applicable top-level Failure Conditions (but no lower than level D for the additional Members). Note 4 Functional Failure sets with a single Member: FDAL B Functional failure sets with multiple members: FDAL B for one Member, additional Member(s) contributing to the top-level Failure Condition at the level associated with the most severe individual effects of an error in their development process for all applicable top-level Failure Conditions (but no lower than level C for the additional Members). Germany ratifies AEP-4671, Ed. B, Ver. 1, Ratification Draft 1, nevertheless it reserves the right to apply the proposed amendments immediately.

DEU	Reservation to USAR.1459 and associated AMC.1459) For USAR.1459 (page 1-F-21 (147/258)) GMAA postulates a national reservation and expands the requirement as follows. In addition to the data recording in the UCS, it must be mandatory to record all necessary flight data onboard the UA. A flight data recorder ("Crash Recorder") on board the UA is mandatory in order to reconstruct the whole flight in case of an accident or incident. With respect to the flight time of data to be recorded, at least the following apply: In case of a total link loss, the flight data recorder must store all relevant flight data at least from the point of time at which the total link loss occurred until the link is reestablished or the UA has landed or crashed. It should be the aim to record all relevant flight data throughout the whole flight time. Each flight data recorder must survive at least the maximum kinetic energy of a crash in order to allow the readout and processing of the stored data. Unintended deletion or overwrite of the stored data, in particular after an incident or accident, must be avoided. Germany ratifies AEP-4671, Ed. B, Ver. 1, Ratification Draft 1, nevertheless it reserves the right to apply the proposed amendments immediately.
-----	--

FRA	1. USAR.590 (c) : The following sentence: “The Arresting System shall include means to maintain UAS azimuth on landing strip after touchdown and arresting hook engagement also in a situation in which one or both arresting devices (e.g. drums) fails to operate correctly”. Will be replaced in Book 1 with: “The Arresting System shall include means to maintain UAS azimuth on landing strip after touchdown and arresting hook engagement. The Arresting System architecture must prevent failure of one or both arresting devices (e.g. drums) from resulting in a Hazardous or Catastrophic event. ” An addition will be made in Book 2: AMC.590: During system deployment, risk assessment should be conducted with regard to runway surroundings in order to take into account Arresting Device failure. 2. USAR.631 (a): France will not apply this requirement. 3. AMC.881 : France will not apply this AMC, but will use edition 2 AMC 4. USAR.1041: The current text : “(a) The powerplant and auxiliary power unit cooling provisions must maintain the temperatures of: (1) powerplant components (2) engine fluids (3) auxiliary power unit components and fluids within the limits established for those components and fluids” Will be replaced with : “The powerplant and auxiliary power unit cooling provisions must maintain the temperatures of powerplant components and engine fluids and auxiliary power unit components and fluids within the limits established for those components and fluids under the most adverse ground, water and flight operations to the maximum altitude and maximum ambient atmospheric temperature conditions for which approval is requested, and after normal engine and auxiliary power unit shutdown.” 5. USAR.1304 : The current text : “The navigation architecture must prevent navigation system failures external to the UAS from resulting in a Hazardous or more serious event.” Will be replaced with: “The navigation architecture must prevent navigation system failures external to the UAS from resulting in a Catastrophic event.”
-----	---

FRA	6. AMC.1308 : The current text: “(b) Where the operational environment may be more severe, STANAG 3614 or MIL STD 464 may be used.” Will be replaced with: “(b) Where the operational environment may be more severe, STANAG 4370 (AECTP 250) or MIL STD 464 may be used.” 7. AMC.1309 : France will apply Class I requirements for UAV up to 20 T of MTOW. 8. USAR.1316 (b) : France will not apply this requirement.
-----	---

USA	U.S. Reservations: NATO Standard AEP-4671 Edition B, Unmanned Aircraft Systems Airworthiness Requirements 1. USAR 1.2.1 Scope, Derivation and Structure of USAR: The U.S. will implement this paragraph with the following: "The airworthiness requirements in this code do not address engine or propeller certification. The Certifying Authority is responsible for certification of the UA engine and propeller." Rationale: STANAG 4671 does not address Engine or Propeller Certification. The Military Certifying Authority is responsible for engine and certification. This clarification is essential to bound the scope of the STANAG. 2. USAR 1.2.1 Scope, Derivation and Structure of USAR: The U.S. will implement this paragraph with the following: "United States (U.S.) Military Certifying Authorities will use MIL-HDBK-516 to establish the airworthiness certification basis for military fixed-wing UAS intended for unrestricted flight operations over all population densities. A MIL-HDBK-516-based certification basis will establish the airworthiness certification criteria, standards, and methods of compliance necessary to provide an equivalent level of safety to STANAG 4671 using MIL-STD-882 as the system safety framework. STANAG 4671 will be used as a partial airworthiness basis to establish standards and methods of compliance for military fixed-wing UAS intended for unrestricted flight operations over all population densities. This STANAG does not identify all of the airworthiness criteria, standards, or methods of compliance necessary to certify fixed-wing UAS by a U.S. Military Certifying Authority. Additional criteria, standards and certifications (or the data from these certifications) may be required by U.S. Military Certifying Authorities to address areas not adequately covered by the STANAG, such as those areas identified as "not covered" in Section 1.2.1 USAR Scope". 3. Annex A Glossary – Hazardous, AMC.1309 (b) and (c) (5) Severity Reference System - Hazardous (ii), AMC 1309 Fig 1 Severity Definition - Hazardous: The U.S. will implement the definition of 'Hazardous' in the standard as follows: "Failure conditions for which it can be reasonably expected that a fatality to UAS crew, ground staff, or third parties will not occur." with "A large reduction in safety margins or functional capabilities; or, Failure conditions which may result in serious injury to UAS crew, ground staff or third parties." Rationale: Including large reductions in safety margins or functional capabilities and serious injury to UAS crew, ground staff or third parties aligns with CS-23 definition of Hazardous failure conditions, MIL-STD-882, and provides discriminators between Major and Hazardous with respect to injury to personnel. 4. USAR.307(a) Proof of Structure: The U.S. will implement this requirement as follows: "Structural analysis may be used only if the structure conforms to those for which experience has shown this method to be reliable" with "Structural analysis (static or fatigue) may be used only if approved by the Certifying Authority, and the structure conforms to those structures for which experience has shown this method to be reliable. In other cases, substantiating load tests must be made up to a level considered to be sufficient and agreed with the Certifying Authority." Rationale: The requirement allows proof of structure by analysis without agreement by the Certifying Authority. The U.S. requires agreement by the Certifying Authority for proof of structure by analysis. 5. USAR.1353 Storage battery or emergency power supply design & installation: The U.S. will implement this requirement with the following sub-paragraph: "(j) Storage Batteries or Emergency Power Supplies utilizing Lithium chemistry are subject to special conditions approved by the Certifying Authority." Rationale: The STANAG does not address certification requirements for storage batteries or emergency power supplies utilizing Lithium chemistry batteries. These battery types present unique safety concerns and certification requirements not covered by the STANAG.
-----	---

USA	6. USAR.1435(a)(4) Hydraulic Systems. The U.S. will implement this requirement as follows: "Withstand the proof pressure without permanent deformation that would prevent it from performing its intended function, and the ultimate pressure without rupture. The proof and ultimate pressures are defined in terms of the design operating pressure (DOP) as follows: 1. Tubes and Fittings: Proof 1.5 x DOP; Ultimate 3.0 x DOP 2. Accumulators and other high pressure vessels containing high pressure gas: Proof 3.0 x DOP; Ultimate 4.0 x DOP. 3. Reservoirs and other low pressure vessels: Poof 1.5 x DOP; Ultimate 3.0 x DOP.4. Hoses: Proof 2.0 x DOP; Ultimate 4.0 x DOP. 5. All other elements: Proof 1.5 x DOP; Ultimate 2.0 x DOP." Rationale: CS-23 requirements do not reflect current best aerospace practice for hydraulic system design, nor tailor proof and ultimate loads to system components. The above requirements are derived from CS-25.1435 and align with military hydraulic system design practice. 7. USAR.1438 Pressurization and pneumatic systems: The U.S. will implement this requirement as follows: "Pneumatic systems (ducting and components) served by bleed air, such as engine bleed air, air conditioning, pressurization, engine starting and hot air ice-protection systems, which are essential for the safe operation of the UA or whose failure may adversely affect any essential or critical part of the UA must be so designed and installed as to comply the USAR.1309. In particular account must be taken of bursting or excessive leakage. See AMC.1438". Rationale: New text derived from CS-25 and focuses on meeting USAR.1309. The CS-23 burst and proof pressure test limits are less than typically used for aerospace best practice (see for example CS-25). The original text is proposed to be moved to an AMC, with (a) and (b) burst and proof limits identified as 'not less than' limits, and to be agreed by the certifying authority. 8. USAR.U1735 Motion and representation of controls: The U.S. will implement this requirement without the following: "The representation of these controls must be similar to conventional flight controls that exist in manned aircraft." Rationale: The requirement states that the representation of UCS controls must be designed so that they, or their representations are similar to conventional flight controls that exist in manned aircraft. The requirement is overly prescriptive. 9. USAR.U1785 Warning, Caution and advisory information color code: The U.S. will implement (a) through (c) as follows:(a) Red, for Warning information (information indicating the existence of a hazardous condition requiring immediate action to prevent loss of life, equipment damage, or abortion of the mission.(b) Amber, for Caution information (information indicating the existence of a condition requiring immediate attention but no immediate action." (c) Green for safe operation information (information indicating a safe or normal configuration, condition of performance or operation of essential equipment, or attracts attention and imparts information for routine action purposes)" Rationale: Color coding of warning caution and advisory information is not consistent with U.S. military design standards. The U.S. will use the definitions from MIL-STD-411 for Red, Amber and Green color codes respectively.
Note: The reservations listed on this page include only those that were recorded at time of promulgation and may not be complete. Refer to the NATO Standardization Document Database for the complete list of existing reservations.	

TABLE OF CONTENTS

1 – Introduction.....	1
1.1 – General.....	1
1.2 – Scope, Derivation and Structure of USAR.....	2
1.3 – Type Certification (or Equivalent) Process.....	5
1.4 – Special Conditions and Special Military Airworthiness Requirements.....	6
1.5 – Special Military Modes of Operation Resulting in Reduction of Level of Safety.....	6
Annex A to USAR Introduction: Glossary.....	A-1
Annex B to USAR Introduction: Abbreviations.....	B-1
Annex C to USAR Introduction.....	C-1
Book 1 – Airworthiness Code.....	1-A-1
Subpart A – General.....	1-A-1
Subpart B – UA Flight.....	1-B-1
Subpart C – UA Structure.....	1-C-1
Subpart D – UA Design and Construction.....	1-D-1
Subpart E – UA Powerplant.....	1-E-1
Subpart F – Equipment.....	1-F-1
Subpart G – Operating Limitations and Information.....	1-G-1
Subpart H – Command and Control Data Link.....	1-H-1
Subpart I – UA Control Station.....	1-I-1
Appendix C – Basic landing conditions.....	1-App C-1
Appendix D – Wheel Spin up Loads.....	1-App D-1
Appendix F – Test Procedures for Self-Extinguishing Materials in Accordance With USAR.1359.....	1-App F-1
Appendix G – Instructions for Continued Airworthiness....	1-App G-1
Appendix K – Other Aspects of Transportation and Factors to Consider.....	1-App K-1
Appendix L – HIRF Environments and Equipment HIRF Test Levels.....	1-App L-1
Book 2 – Acceptable Means of Compliance (AMC).....	2-A-1
Subpart A – General.....	2-A-1
Subpart B – UA Flight.....	2-B-1
Subpart C – UA Structure.....	2-C-1
Subpart D – UA Design and Construction.....	2-D-1
Subpart E – UA Powerplant.....	2-E-1
Subpart F – Equipment.....	2-F-1
Subpart G – Operating Limitations and Information.....	2-G-1
Subpart H – Command and Control Data Link.....	2-H-1
Subpart I – UA Control Station.....	2-I-1

INTENTIONALLY BLANK

1.0 Introduction

1.1 GENERAL

This document contains a set of technical airworthiness requirements intended primarily for the airworthiness certification of fixed-wing military Unmanned Aircraft Systems (UAS) with a maximum take-off weight (MTOW) between 150 and 20,000 kg that intend to regularly operate in non-segregated airspace. Certifying Authorities may apply these certification requirements outside these weight limits where appropriate.

These requirements represent the minimum applicable requirements to meet the safety objectives defined by paragraph 1309 and its associated AMC. It may be augmented by additional Special Conditions (i.e. additional airworthiness requirements) required by individual Certifying Authorities. Unmanned Aircraft Systems Airworthiness Requirements (USAR) is intended for application by Certifying Authorities within each country's relevant national regulatory framework.

These requirements may not be sufficient for the certification of UAS with unconventional, novel or extremely complex features. Additionally, USAR may be insufficient for UAS with a design usage spectrum significantly different from that of General Aviation. Nevertheless, the USAR may have significant value for assessing all or parts of such systems and Certifying Authorities are encouraged where appropriate to use these requirements as a basis for certification of such systems within their national regulatory frameworks.

A glossary is at Annex A and a listing of abbreviations used within the document is at Annex B to this introduction.

Throughout this document, the term 'Type Certificate' refers to any document issued by a National Military Certifying Authority that within the regulatory framework of that Nation certifies compliance (as determined by the National Military Certifying Authority) with a USAR-derived Type Certification Basis. Where the Certifying Authority issues an alternative document to a Type Certificate (such as a Release To Service, Military Aircraft Type Qualification Certificate or Flight Permit, which may include items outside the scope of USAR) it is expected that the degree of compliance with USAR will be clearly stated therein. The probability reference system used in this document assigns quantitative values to the terms "Frequent", "Probable", "Remote", "Extremely Remote", and "Extremely Improbable" (see AMC.1309 paragraph 5). These terms may also be used in a general, qualitative context in the USAR. When the quantitative meanings of these terms are intended, these terms are capitalized (e.g. Probable). These terms are not capitalized when used in a general sense (e.g. probable). Additionally, the terms "Catastrophic", "Hazardous", "Major", and "Minor" appear capitalized and in italics when used in the sense of AMC.1309.

The most recent version of any referenced document in effect shall apply unless otherwise specified or agreed with the certifying authority. In case of conflicting requirements between a referenced document and the STANAG 4671, STANAG 4671 shall take precedence. In case of a conflict between a referenced document and this document, it should be reported to the custodian of this document.

1.2 SCOPE, DERIVATION AND STRUCTURE OF USAR

1.2.1 USAR SCOPE

Whilst military aircraft are not specifically required to comply with civil airworthiness regulations as agreed in the ICAO¹ Convention on International Civil Aviation (Article 3), it is the intention of this document to correspond as closely as practicable to a comparable minimum level of airworthiness for fixed-wing aircraft as embodied in documents such as EASA² CS-23 (from which USAR is derived) and 14 CFR³ Title 14 Part 23 whilst recognizing from military experiences that there are certain unique features of UA that require modification to those civil airworthiness requirements and/or the addition of new individual requirements or new subparts. Civil codes were utilized as the foundation for USAR because they represent the minimum airworthiness requirements for flight in non-segregated airspace with minimal or no restrictions. Additionally, many military fixed-wing UA have flight profiles similar to that of General Aviation, making the selection of civil airworthiness codes a practical starting point.

In line with the JAA-Eurocontrol Taskforce recommendations, the following areas are not covered by this airworthiness code:

- Control station security,
- Security of the command and control data link from wilful interference,
- Airspace integration and segregation of aircraft (including “sense and avoid”),
- The competence, training and licensing of UAS crew, maintenance and other staff,
- Approval of operating, maintenance and design organizations,
- The type of operation,
- Vehicle Management and Navigation requirements,
- Frequency spectrum allocation,
- Noise, emission, and other environmental certification,
- Launch/landing equipment that is not safety critical and which does not form part of the Type Certification Basis,
- Operation of the payload (other than its potential to hazard the aircraft),
- Carriage and release of weapons, pyrotechnics and other functioning or non-functioning stores designed for release during normal operations,
- Non-deterministic flight, in the sense that UA flight profiles are not pre-determined or UA actions are not predictable to the UA crew,
- Sea-basing,
- Piloting from an external or internal control box,
- Supersonic flight.

It is expected that these areas will be subject to other forms of approval by Certifying Authority in order to ensure a total aviation safety approach. Where such approval requires technical assessment, the Certifying Authority may supplement these requirements with suitable additional conditions as appropriate.

¹ International Civil Aviation Organization

² European Aviation Safety Agency

³ Code of Federal Regulations

It is recognized that 'sense and avoid' is a key enabling issue for UA operations. The derivation and definition of 'sense and avoid' requirements is primarily an operational issue and hence outside the scope of USAR. However, once these requirements have been clarified, any system designed and installed to achieve these objectives is an item of installed equipment within a UA System and hence falls under the airworthiness requirements of USAR.

Cybersecurity is outside of the scope of the USAR. It is recognized that Cybersecurity may introduce unique safety risks to the UAS and it should therefore be considered.

The USAR does not identify all of the airspace access requirements. Access to all classes of airspace may require additional equipment and/or navigational accuracies not addressed in the USAR.

1.2.2 DERIVATION OF THE USAR

This document is an airworthiness code derived from EASA CS-23 (ex JAR 23) requirements supplemented by elements from the following UAS airworthiness and safety documents:

Title	Date
JAA Eurocontrol UAS Task Force – Final Report	05/2004
Airworthiness standard for Unmanned aerial vehicles, RAI-UAS - Ente Nazionale Aviazione Civile – (Italy)	1999
Design standards UAS - Civil Aviation Safety Authority (Australia)	05/2000
Design and airworthiness requirements for RPAS systems – DEF STAN 00-970 Part 9 (UK MOD)	05/2002
USICO (Unmanned Safety Issues for Civil Operations)– WP 2400 – Certification review item (CRI) “stall demonstration”	01/2004
AC23.1309-1C – Equipment, Systems, and Installations in Part 23 Airplanes – FAA. (USA)	03/1999
TSO C23d – Minimum Performance Standards for Parachute assemblies and Components, Personnel (USA)	07/1992
Special Conditions; Ballistic Recovery Systems Cirrus SR-20 Installation – 14 CFR Part 23 – FAA (USA)	10/1997

1.2.3 USAR STRUCTURE

These requirements consist of 9 interrelated sub parts, covering the following areas:

		UA System				
		UA	Command and control data link	Communication system	UA control station	Other ancillary elements
A	General	X	X	X	X	X
B	UA Flight	X				
C	UA Structure	X				X
D	UA Design and Construction	X				X
E	UA Powerplant	X				
F*	Equipment	X				
G	Operating limitations and information	X	X	X	X	X
H	Command and control data link Communication system		X	X		
I	UA control station				X	

*Note: Paragraph 1309 (in subpart F) and its AMC applies to the entire UAS and not only to the UA.

Subparts A-G are derived directly from CS-23. While subparts H and I follow the format of CS-23, they are unique to USAR.

Paragraph numbers throughout subparts A-G correspond directly to CS-23. Where an entire paragraph is not applicable to a UAS, it is deleted and annotated 'not applicable'. If a subparagraph is not applicable, it is deleted and annotated 'not applicable'. Where a paragraph is applicable or partly applicable but its location within sections A-G of CS-23 is inconsistent with UAS (e.g. the logical location in the context of UAS is in section H or I), a cross-reference is included at the original location with the annotation 'not applicable in this subpart'. Where a paragraph is unique to USAR, it is identified as such and its paragraph numbering is marked with the prefix 'U'.

A cross-reference table between USAR and CS-23 is available at Annex C. This cross-reference table states all USAR paragraphs that have been inspired or adapted from CS-23 but that, due to the structure of USAR could not be maintained in their original CS-23 numbering/position. Their new numbering/position in USAR is mentioned in the cross-reference table.

These requirements also include Book 2, consisting of material describing acceptable means of compliance. This feature is similar to FAA or EASA advisory material and allows a full set of UAS certification documentation to be referenced in a single volume.

1.3 TYPE CERTIFICATION (OR EQUIVALENT) PROCESS

USAR has been created to mirror as closely as possible the structure and content of CS-23. Safety assurance assumes that the requirements are used in a process using the same or broadly equivalent steps to Type Certification of 14 CFR Part 23/EASA CS-23 aircraft. Where the procedures used by a Certifying Authority differ substantially from this approach, the Authority is expected to determine that the process used ensures that an equivalent level of safety is achieved.

Therefore, it is expected that USAR will normally be used to define the UAS Type Certification Basis (or equivalent national document) using the applicable paragraphs of the USAR Airworthiness Code (Book 1), completed by the related USAR Acceptable Means of Compliance (Book 2). AMCs are non-exclusive means of demonstrating compliance with USAR, and Certifying Authorities, in parallel with civil regulatory systems, may define or approve alternate equivalent means where appropriate. Where the Certifying Authority has imposed additional conditions, it is expected that the Certifying Authority will approve the acceptable means of compliance.

It is further assumed that the Certifying Authority will issue a Military Type Certificate or equivalent national document, if applicable with a Type Certification Data Sheet, or equivalent national document(s), containing as a minimum:

- System Identification
- System configuration details
- Requested operating frequencies
- Statement of compliance with USAR (including if applicable additional conditions, exemptions and deviations)
- List of approved publications – Operating and maintenance
- Issuing Agency
- Date of Issue

1.4 SPECIAL CONDITIONS AND SPECIAL MILITARY AIRWORTHINESS REQUIREMENTS

In addition to the requirements of USAR, it is expected that national certifying authorities may impose extra airworthiness requirements (for instance cold soak). Where these are of a nature similar to Special Conditions usually imposed by civilian certifying authorities and potentially applicable to both civil and military applications, these will be known as Special Conditions.

It is also expected that UAS certified or assessed using USAR will be employed in a variety of military roles and/or modes, all or some of which may involve manoeuvres or use of special equipment or payloads that fall outside the scope of these requirements. It is expected that national authorities will address these modes by the use of special military airworthiness requirements. Special military airworthiness requirements recognize the unique nature of military operations, and are analogous to Special Conditions or similar terminology used by a civilian Certifying Authority. They are to be applied and assessed in a similar manner to Special Conditions.

It is expected that any special military airworthiness requirements that result in an actual or potential hazard condition that reduces the margin of safety below the levels required by paragraph 1309 and its associated AMC, whether temporary or permanent, will be addressed by suitable operational restrictions. Where this is not possible, the condition is to be clearly identified in the Type Certificate as resulting in the system operating at a level of safety below that required by USAR. The resulting technical and/or operational risk associated with the special military airworthiness requirements are expected to be assessed and accepted using relevant national procedures.

1.5 SPECIAL MILITARY MODES OF OPERATION RESULTING IN REDUCTION OF LEVEL OF SAFETY

Where for military reasons a UAS contains special military modes whose operation would result in a level of safety below that required by USAR, type certification to USAR may still be carried out providing that a sufficiently robust segregation is achieved between the special modes when inactive and the basic UAS. An example of a special military mode may be weapons or stores arming and release or operation of electromagnetic spectrum emitters.

Annex A to USAR Introduction: Glossary

Airfield	An area prepared for the accommodation (including any buildings, installations and equipment), landing and take-off of aircraft.
Antenna Margin (or link margin)	The amount (usually expressed in dB) by which a received signal lies above a predetermined lower limit for desired message quality.
Automatic	The execution of a predefined process or event that requires UAS crew initiation.
Availability (data link)	The long-term ratio of the data link actual RF channel operation time to scheduled RF channel operation time.
Catastrophic (failure condition)	Failure conditions that are expected to result in at least uncontrolled flight (including flight outside of pre-planned or contingency flight profiles/areas) and/or uncontrolled crash. Or Failure conditions may result in a fatality to UA crew, ground staff, or third parties.
Command and Control Link	A data transmission used for control of the UA that transmits UA crew commands from the UCS to the UA (uplink) and UA status data from the UA to the UCS (downlink).
Communication system	A means that allows ATC communication between the UA crew in the remote control station and the air traffic control service.
Critical Engine	The engine whose failure would most adversely affect the performance or flight characteristics of the UA.
Data Link (UAS)	A wireless communication channel between one or more UCS and one or more UA, or between multiple UA. UAS data link data exchange may include but is not limited to exchange of command and control or payload data between UA and UCS. A UAS data link may consist of: (1) Uplink – Transmittal of UA crew commands from the UCS to the UA. (2) Downlink – Transmittal of UA status data from the UA to the UCS.
Decision Point	The height below which a go around may not be safely performed (i.e., there will be ground contact that may damage the UA).
Degraded Mode	State of the UAS that reflects a loss of accuracy, capability, or performance in response to a failure of a component or system.
Design Usage Spectrum	Information that forms the basis for assumptions underpinning fatigue and damage tolerance and associated individual UA tracking.
Dormant Fault	An internal, inactive, fault that upon activation produces an error; if and when the erroneous behaviour affects the delivered service, a failure occurs (Laprie, J-C and Kanounk K (1996). Software Reliability and System Reliability. HDBK, Software Reliability (Lyu, M.R., Ed). IEEE Computer Society Press).
Effective Maximum Range (data link)	Measure of data link coverage over a horizontal distance that is a function of frequency, availability, bit error rate, climate area and altitude.
Electromagnetic Compatibility (EMC)	The ability of equipment or a system to function in its electromagnetic environment without causing intolerable electromagnetic disturbances to anything in that environment.
Electromagnetic Environment (EME)	EME is the totality of electromagnetic phenomena existing at a given location.
Electromagnetic Interference (EMI)	Any electromagnetic disturbance, whether intentional or not, which interrupts, obstructs, or otherwise degrades or limits the effective performance of electronic or electrical equipment.

Electromagnetic Vulnerability (EMV) The characteristics of a system that cause it to suffer degradation in performance of, or inability to perform, its specified task as a result of electro-magnetic interference.
Emergency Recovery Capability Procedure that is implemented through UA crew command or through design means in order to mitigate the effects of critical failures with the intent of minimising the risk to third parties. This may include automatic pre-programmed course of action to reach a predefined and unpopulated forced landing or recovery area.
Essential Load On a power supply generation or distribution system, a load which is required for the continued safe flight and landing, but not necessarily a system whose functioning is required for certification'.
Failure Conditions A condition having an effect on the UAS, UAS crew, ground staff or third parties, either direct or consequential, which is caused or contributed to by one or more failures or errors considering flight phase and relevant adverse operational or environmental conditions or external events.
Fireproof With respect to materials, components and equipment, the capability to withstand the application of heat by a specified flame, for a period of 15 minutes without any failure that would create a hazard to the UA. The flame will have the following characteristics: Temperature $1100^{\circ}\text{C} \pm 80^{\circ}\text{C}$ Heat Flux Density $116 \text{ KW/m}^2 \pm 10 \text{ KW/m}^2$ For materials this is considered to be equivalent to the capability of withstanding a fire at least as well as steel or titanium in dimensions appropriate for the purposes for which they are used.
Fire-resistant With respect to materials, components and equipment, the capability to withstand the application of heat by a specified flame, as defined for 'Fireproof', for a period of 5 minutes without any failure that would create a hazard to the UA. For materials this may be considered to be equivalent to the capability of withstanding a fire at least as well as aluminium alloy in dimensions appropriate for the purposes for which they are used.
Flight Control System The flight control system comprises sensors, actuators, computers and all those elements of the UAS, necessary to control the attitude, speed and flightpath of the UA. The flight control system can be divided into 2 parts: Flight control computer – A programmable electronic system that operates the flight controls in order to carry out the intended inputs. Flight controls – sensors, actuators and all those elements of the UAS (except the flight control computer), necessary to control the attitude, speed and flightpath of the UA. Flight controls can further be defined as: Primary flight control – Primary flight controls are those used in the UA by the flight control system for the immediate control of pitch, roll, yaw and speed. Secondary flight control - Secondary controls are those controls other than primary flight controls, such as wheel brakes, spoilers and tab controls.
Flight Load Factor The ratio of the aerodynamic force component (acting normal to the assumed longitudinal axis of the UA) to the weight of the UA. A positive flight load factor is one in which the aerodynamic force acts upward, with respect to the UA.
Flight Envelope Protection A system that prevents the UA from exceeding its designed operating limits.
Flight Termination System A system to immediately terminate UA flight.
Forced Landing

A condition resulting from one or a combination of failure conditions that prevents the UA from normal landing on its planned main landing site although the flight control system is still able to maintain the UA controllable and manoeuvrable.
Full-time (data display context) Display data required to be displayed continuously during all phases of flight.
Functional Hazard Assessment (FHA) A systematic, comprehensive examination of UA and system functions to identify potential Minor, Major, Hazardous and Catastrophic failure conditions that may arise as a result of a malfunction or failure to function.
Ground Staff Qualified personnel necessary for ground operations (such as supplying the UA with fuel and maintenance) as stated in the UA System Flight Manual or in the UA Maintenance Manual.
Handover The operation that consists in performing a UA command and control transfer from one UCS to another one or from one workstation to another one in the same UCS.
Hazardous Failure conditions that either by themselves or in conjunction with increased crew workload, are expected to result in a controlled-trajectory termination or forced landing potentially leading to the loss of the UA where it can be reasonably expected that a fatality will not occur. Or Failure conditions for which it can be reasonably expected that a fatality to UAS crew, ground staff, or third parties will not occur.
Hidden Failures (Software) All software errors not identified during requirement-based (functional) testing. (Certification Authorities Software Team (CAST) Position Paper CAST-9).
Horizontal Surface Balancing Load A load necessary to maintain equilibrium in any specified flight condition with no pitching acceleration.
Infrastructure (UCS) The basic facilities, services and installations needed for the functioning of the UCS which may include power supply, shelter, communication systems etc.
Intervisibility (data link) The performance of a LOS (line of sight) data link signal, taking into consideration the interposed land mass between the UA antennas and the UCS antennas.
Landing The phase of a UA system mission that involves the return of a UA to the ground or sea surface. This also includes the return of the UA to the surface via parachute or other capture/recovery device.
Latency (data transfer) Delay in time between the sending of a unit of data at one end of a connection, until the receipt of that unit at the destination.
Latent Failures An existing fault that has not yet been recognized by the UA crew or ground staff(BSI, BS 4778-3.2 1991; IEC 50-191, 1990).
Launch Catapult and/or rocket assisted take-off.
Launch Safety Trace The area, associated with a UA launch, in which there may be a hazard which could result in a risk to personnel, equipment or property.
Line of Sight A visually unobstructed straight line through space between the transmitter and receiver. In communications, a direct propagation path that does not go below the radio horizon.
Link Budget The allocation of the maximal acceptable total loss of a radio link between attenuations, gains and operating margins. Link Budget calculation considers the gain and loss factors associated with the antennas, transmitters, transmission lines and propagation environment used to determine the maximum distance at which a transmitter and receiver can successfully operate.
Major

Failure conditions that either by themselves or in conjunction with increased crew workload, are expected to result in an emergency landing of the UA on a predefined site where it can be reasonably expected that a serious injury will not occur. Or Failure conditions which may result in injury to UAS crew, ground staff, or third parties.
Masking (data link) Blockage of data link due to fuselage blockage or unfavourable UA attitude.
Minor Failure conditions that do not significantly reduce UAS safety and involve UAS crew actions that are well within their capabilities. These conditions may include a slight reduction in safety margins or functional capabilities, and a slight increase in UAS crew workload.
Minimum Demonstration Speed ($V_{\min \text{ DEMO}}$) The minimum speed demonstrated by the Applicant by flight test, while possibly adjusting or inhibiting flight control protection features, using the procedure and meeting the flight characteristics specified in USAR.201
Minimum Engine Performance The lowest level of acceptable engine performance. Minimum engine performance is due to deterioration and variation within a family of engines due to manufacturing and control tolerances. This represents a predetermined variation below the specification performance of a family of engines. This performance level is predetermined for an aircraft and usually expressed as a percentage of Specification. 95% is often, but not always, the minimum level used. Engines falling below this level of performance would be removed from service.
Must Used to indicate a mandatory requirement (see also “shall”).
Parachute Landing Safety Trace. The area, associated with a UA parachute landing, in which there may be a hazard which could result in a risk to UA crew, ground staff, third parties, equipment or property.
Part-time (data display context) Display data only required to be displayed during certain phases of flight or upon UA crew request.
Payload Device or equipment carried by the UA, which performs the mission assigned. The useful payload comprises all elements of the aircraft that are not necessary for flight but are carried for the purpose of fulfilling specific mission objectives.
Refusal Speed (V_{Rf}) The speed above which a take-off may not be safely aborted. V_{Rf} is equivalent to V_1 as used for manned aircraft.
Pcumcat The cumulative probability resulting from the probabilities per flight hour of all Individual Catastrophic Failure Conditions caused by all system
Safety Critical Control A control requiring immediate action to ensure continued safe flight or safely terminate the flight.
Shall Used to indicate a mandatory requirement (see also “must”).
Should Used to indicate a preferred, but not mandatory, method of accomplishment.
Switchover The operation that consists of performing the transfer of the UA command and control from one data link channel to another channel within the same UCS.
Take-off The process by which a UA leaves the surface and attains controlled flight (includes launch via catapult or rocket assistance).
Type Certificate Any document issued by a National Military Certifying Authority that within the regulatory framework of that Nation certifies compliance as determined by the National Military Certifying Authority with a USAR-derived Type Certification Basis.
Type Certification Basis

The document elaborated by the Applicant with the Certifying Authority based on the airworthiness code and is specifically applicable to the design of the UAS to be certified. It may also include Special Conditions as detailed in the Introduction of USAR.
UCS Flight Control Flight controls used by the UA crew in the UCS to operate the UA in the semi-automatic mode of control as defined in USAR.U1330
Unmanned Aircraft (UA) An aircraft that does not carry a human operator and is operated remotely using varying levels of automated functions. Moreover a UA: • Is capable of sustained flight by aerodynamic means, • Is remotely piloted or automatically flies a pre-programmed flight profile, • Is reusable, • Is not classified as a guided weapon or similar one shot device designed for the delivery of munitions.
UA Control Station (UCS) A facility or device from which the UA is controlled and/or monitored for all phases of flight considering USAR.U2 (a).
UA Crew One or more qualified people responsible for monitoring and controlling the flightpath, flight status, and functions of one or more UA. Includes the UA Operator and also all support crewmembers responsible for operating on-board systems (e.g. payload).
UA Operator The UA crew member in the UA Control Station tasked with overall responsibility for operation and safety of the UAS. See STANAG 4670 for UA Operator training guidance
Unmanned Aircraft System (UAS) A system whose components include the Unmanned Aircraft (UA), the UA control station and any other UA System elements necessary to enable flight such as a command and control data link, communication system and take-off and landing element. There may be multiple UA, UCS, or take-off and landing elements within a UAS.
Uncontrolled Crash A condition resulting from one or a combination of failure conditions that prevents the flight control system from maintaining the UA controllable and manoeuvrable until impact on the ground.
Uncontrolled Flight A condition resulting from one or a combination of failure conditions that result in loss of UA control and/or manoeuvrability. Uncontrolled flight includes flight outside of pre-planned or contingency flight profiles/areas
Workload The amount of work assigned to or expected from a person in a specified time.
Workstation A computer interface between an individual UA crew member and the UA to perform the functions of mission planning, flight control and monitoring, and/or for display and evaluation of the downloaded image and data.

INTENTIONALLY BLANK

Annex B to USAR Introduction: Abbreviations

°C	Degree Celsius
°F	Degree Fahrenheit
AC	Advisory Circular
AIT	Auto-Ignition Temperature
AMC	Acceptable Means of Compliance
ANC	Army Navy Civil Committee
AOA	Angle of Attack
APU	Auxiliary Power Unit
ARP	Aerospace Recommended Practices
ATOL	Automatic Take-Off and Landing
BLOS	Beyond Line of Sight
BVID	Barely Visible Impact Damage
C.G. or c.g.	Centre of Gravity
CAS	Calibrated Air Speed
cc	Cubic centimetre
CCA	Common Cause Analysis
CEP	Circular Error Probability
CFR	Code of Federal Regulations
C _L	Lift coefficient of aircraft
cm	Centimetre
CNA	Aerodynamic normal force coefficient
CRT	Cathode Ray Tube
CS	Certification Specification
CS-23	Certification Specification for Normal, Utility, Aerobatic and Commuter category aeroplanes
CS-25	Certification Specification for Large aeroplanes
CS-VLA	Certification Specification for Very Light aeroplanes
DAL	Development Assurance Level
Def Stan	Defence Standard
DUS	Design Usage Spectrum
EAS	Equivalent Air Speed
EASA	European Aviation Safety Agency
ECS	Environmental Control System
EFIS	Electronic Flight Information System
EMC	Electromagnetic Compatibility
EME	Electromagnetic Environment
EMI	Electromagnetic Interference
EMV	Electromagnetic Vulnerability
FAA	Federal Aviation Administration
ft	Feet
FCS	Flight Control System
FDAL	Functional Design Assurance Level
FEM	Finite Elements Model
FHA	Functional Hazard Assessment
FMEA	Failure Mode Effects Analysis
FOD	Foreign Object Damage
fps	feet per second
FTR	Fatigue Type Record
g	Acceleration due to gravity
GPS	Global Positioning System
h	Hour
HDBK	Handbook
IAS	Indicated Air Speed
in	Inch
JAA	Joint Aviation Authorities

KCAS	Knots Calibrated Air Speed
kg	Kilogram
km	Kilometre
kPa	Kilo Pascal
kt	Knot
l	Litre
lb	Pound
LOS	Line of Sight
m	Metre
mm	Millimetre
MIL-HDBK	Military Handbook
MIL-STD	Military Standard
MoD	Ministry of Defence
N	Newton
n	Load Factor
NATO	North Atlantic Treaty Organization
NDT/I	Non Destructive Techniques and Inspection
psig	Pounds per square inch gauge
PSSA	Preliminary System Safety Assessment
RAI	Registro Aeronautico Italiano
RF	Radiofrequency
RPM	Revolutions per Minute
s	Second
SPC	Sortie Profiles Codes
SSA	System Safety Assessment
STANAG	Standard Agreement (NATO)
STR	Static Type Record
TSO	Technical Standard Order
UA	Unmanned Aircraft
UCS	UA Control Station
UK	United Kingdom
US	United States of America
USAR	UAS Airworthiness Requirements
USICO	UA Safety Issues for Civil Aviation
UV	Ultra Violet
$V_{min\ DEMO}$	Minimum Demonstrated Airspeed
V_A	Design manoeuvring speed
V_C / M_C	Design Cruising Speed / Mach
V_D / M_D	Design Dive Speed /Mach
V_{EF}	Speed at which the Critical Engine is assumed to fail
V_F	Speed at which Flaps are Fully Extended
V_{FE}	Flap Extended Speed
V_G	Negative Manoeuvring Load Factor Speed
VLA	Very Light Aircraft
V_{MO} / M_{MO}	Maximum Operating Limit Speed.
V_{MC}	Control Speed with Critical Engine Inoperative
V_{MCG}	Minimum Control Speed on the Ground
V_{NE}	Never Exceed Speed
V_{NO}	maximum structural cruising speed
V_R	Rotation Speed
V_{REF}	Reference Landing Approach Speed
V_{Rf}	Refusal Speed
V_S	Stalling Speed
V_{S0}	Stalling Speed or Minimum Steady Flight Speed in Landing Configuration
V_{S1}	Stalling Speed or Minimum Steady Flight Speed in Take-off Configuration
V_{SF}	Computed Stalling Speed with Flaps fully extended at the Design Weight

V_{SSE}
W
WP

Minimum safe speed with one Engine inoperative
Watts
Work Package

INTENTIONALLY BLANK

Annex C to USAR Introduction

Cross-reference table with EASA CS-23 for paragraphs transferred to subpart I

CS §	USAR § in subpart I	CS Subject
207 (a), (d)	U1789 (a)	Stall warning
207 (b)	U1789 (b)	Stall warning
671 (b)	U1731 (a)	Controls arrangement
679 (a)	U1825	Control system lock warning
699	U1791	Wing flap position indicator
729 part of (e)	U1793 (a)	Landing Gear position indicator
729 (f)	U1793 (b)	Landing Gear warning
771	U1703	Pilot compartment
777	U1731	Cockpit controls
779	U1735	Motion and effect of cockpit controls
781 (a) and (b)	U1733 (c) and (d)	Cockpit control and knob shape
841 (b) (5), (6) and (7)	U1795	Pressurised cabins
863 (c)	U1817	Flammable fluid fire protection
991 (c)	U1797	Fuel pumps warning
995 (a) (b) and (g)	U1743 (a) (b) and (c)	Fuel controls
1001 (f) and (g)	U1745 (a) and (b)	Fuel jettisoning system
1091 (b)(4)	U1747	Air induction system
1091 (b)(5)	U1799	Air induction system
1141 (g)	U1803	Powerplant controls: general
1142	U1751	APU controls
1143 (a) through (f)	U1751	Engine controls
1145	U1753	Ignition switches
1147 (a)	U1755	Mixture controls
1149	U1757	Propeller and pitch controls
1153	U1759	Propeller feathering control
1155	U1761	Turbine engine reverse thrust and propeller pitch settings below the flight regime
1157	U1763	Carburettor air temperature controls
1165 (d)	U1801	Engine ignition warning
1189 (c)	U1805	Shut-off controls
1189 (a) (6)	U1765 (a)	Shut-off means
1303	U1723	Flight and navigation instruments
1305	U1725	Powerplant instruments
1311	U1727	Electronic display instrument system
1321	U1721	Arrangement and visibility
1322	U1785	Warning, caution and advisory lights
1326	U1819	Pitot heat indication system
1329 (h)	U1790	AP mode of operation
1331 (a) 1 st sentence	U1821	Instruments using a power source – power indicator
1331 (a) 2 nd sentence	~U1721	Instruments using a power source – location
1335	U1790	Flight director systems
1337 (b) 1 st paragraph	U1729 (a)	Fuel quantity indicator
1337 (b)(5), (6)	U1729 (a)(1) and (2)	Fuel quantity indicator
1337 (d)(2)	U1729 (b)	Oil quantity indicator
1351(b)(1)(i) and (iii)	U1712 (a)(1) and (2)	Risk of electrical shock to the crew
1351 (c)(4)	U1809 (a)	Warning and indications (electrical systems)
1351 (d)	U1809 (b)	Warning and indications (electrical systems)
1381	U1705	Instrument lights
1416 (c)	U1811	Pneumatic de-icer boot system indication means

1431 (c), (d) and (e) and AMC (e)	U1707	Electronic equipment
1435 (a)(2)	U1813	Hydraulic systems indicators
1457	U1709	Cockpit voice recorders
1523	U1704	Minimum flight crew
1543	U1733 (b)	Instrument markings: general
1545	U1835	Airspeed data
1547	U1837	Magnetic direction indicator
1549	U1839	Powerplant and auxiliary power unit instruments
1551	U1841	Oil quantity indicator
1553	U1843	Fuel quantity indicator
1555	U1845	Control markings
1559	U1849	Operating limitations indications
1563	U1835 (e)	Airspeed data

When “~” is used it means “approximates”.

BOOK 1 – AIRWORTHINESS CODE

Subpart A - GENERAL

1	USAR.1 Applicability (a) This airworthiness code is primarily applicable to fixed wing UAS of maximum take-off weight of more than 150 kg and less than 20,000 kg. It may also be applied to UAS of any other maximum take-off weight where considered applicable by the Certifying Authority. (b) A UAS where an operator manually controls pitch, roll, yaw and/or thrust for the conduct of normal flight operations is not covered by USAR. Additional criteria are necessary to address the flight control system for manual piloting operations of this nature.
U2	USAR.U2 Assumptions (a) A UAS comprises individual UA System elements consisting of the UA, the UA control station (UCS) and any other UA System elements necessary to enable flight such as a command and control data link, communication system and take-off and landing element. There may be multiple UA, UCS, or take-off and landing elements within a UAS. (b) It is assumed in USAR that the UAS is designed in order to provide to the UAS crew the capability to command and control the UA for all phases of flight in normal, degraded and emergency operation, except for some specific points mentioned in USAR paragraphs (e.g. pre-programmed or pre-defined flight in case of data link loss). (c) Special Conditions can be prescribed by the Certifying Authority if the airworthiness requirements of this code do not contain adequate or appropriate safety standards, because: (1) The UAS has novel or unusual design features relative to the design practices on which the applicable USAR is based; or (2) The intended use of the UAS is unconventional; or (3) Experience from other similar UAS in service or UAS having similar design features, has shown that unsafe conditions may develop. (d) Every Special Condition must be defined at the beginning of the certification process on the request of the Applicant in order to define the Type Certification Basis applicable to the type of UAS to be certificated. (e) USAR requirements are mostly based upon CS-23 requirements as a reference code tailored to UAS. When establishing the Type Certification Basis for a particular UAS, the Applicant is entitled to propose the replacement of specific paragraphs by alternative criteria, based upon the use and/or the tailoring of other recognised airworthiness code requirements (such as CS-VLA, CS-25,...), pending on the UAS under consideration and presentation of appropriate rationale. Such alternative approach shall be subject to the Certifying Authority acceptance. (f) As USAR is based on CS-23 airworthiness code, it is assumed that twin or multi-engine UA are designed in such a manner that no single engine failure might affect the safety of the flight. As a consequence, USAR paragraphs request that engine installation and associated systems be independent and that no single failure might affect the safe operation of more than an engine. Nevertheless it might be possible to consider and certify (under a Special Condition "Multi engine, single propulsion system") any UAS as a single propulsion UA whatever the number of engines installed. Whatever the failure, UA safety objectives will have to be satisfied, and it shall be

	demonstrated that following any failure of the propulsion component the UA will have a behaviour similar or better than a single engine UA. (g) Where a UA is designed with more than 2 engines, a Special Condition shall be established with the Certifying Authority to accommodate conditions where more than one engine is inoperative.
	3 Aeroplane categories Not Applicable.
U15	USAR.U15 UAS ancillary elements Where a UAS includes any ancillary elements necessary to enable safe flight (such as, for instance, launch and landing elements), Special Conditions in addition to USAR.1581 (a)(2) must be established and agreed with the Certifying Authority to ensure safe operations.
U17	USAR.U17 Design Usage Spectrum See AMC.U17 (a) The Applicant must present to the Certifying Authority the Design Usage Spectrum (DUS) for the UAS for which certification is requested. The DUS shall include details of the usage in the air, on the ground and during transportation that the UA is designed. This statement shall form part of the UAS document set. (b) The certification of the UA is tied to a specific Design Usage Spectrum. Modification and/or addition of missions may require the UA to be re-certified for these missions.
U19	USAR.U19 Special military modes of operation See AMC.U19 The special military modes of operation, when inactive, must be shown not to reduce the UA System level of safety below that required by USAR.

Subpart B – UA FLIGHT

GENERAL	
21	USAR.21 Proof of Compliance See AMC.21 Each requirement of this subpart must be met at each appropriate combination of weight and centre of gravity within the range of loading conditions for which certification is requested. This must be shown: (a) By tests upon a UA of the type for which certification is requested, or by calculations and, where appropriate, simulations, based on, and equal in accuracy to, the results of testing; and (b) By systematic investigation of each probable combination of weight and centre of gravity, if compliance cannot be reasonably inferred from combinations investigated.
23	USAR.23 Load Distribution Limits (a) Ranges of weight and centres of gravity for each payload configuration within which the UA may be safely operated must be established. They must include the range for lateral centres of gravity if possible loading conditions can result in significant variation of their positions that could ultimately result in flight characteristics changes. (b) The load distribution shall consider the following: (1) Any load configuration (considering partial or complete installation) specified by the Applicant and agreed with the Certifying Authority; (2) Expenditure of any expendable useful load items (e.g. fuel, payload); and (3) The extremes of the above plus the most critical combination of special or alternate load items. (c) The load distribution must not exceed: (1) The limits selected by the Applicant; (2) The limits at which the structure is proven; or (3) The limits at which compliance with each applicable flight requirement of this subpart is shown.
25	USAR.25 Weight Limits (a) Maximum weight. The maximum weight is the highest weight at which compliance with each applicable requirement of USAR (other than those complied with at the design landing weight) is shown. The maximum weight must be established so that it is: (1) Not more than the least of: (i) The highest weight selected by the Applicant; or (ii) The design maximum weight, which is the highest weight at which compliance with each applicable structural loading condition of USAR (other than those complied with at the design landing weight) is shown; or (iii) The highest weight at which compliance with each applicable flight requirement is shown, and

	(2) Not less than the weight with: (i) maximum payload in terms of weight, oil at full tank capacity, and at least enough fuel for one half-hour of maximum continuous power operation; or (ii) minimum payload in terms of weight, and oil and fuel at full tank capacity. (b) Minimum weight. The minimum weight (the lowest weight at which compliance with each applicable requirement of USAR is shown) must be established so that it is not more than the sum of: (1) The basic weight determined under USAR.29; (2) Not applicable. (3) The weight of: (i) For turbojet powered UA, 5% of the total fuel capacity of that particular fuel tank arrangement under investigation; and (ii) For other UA, the fuel necessary for one-half hour of operation at maximum continuous power.
29	USAR.29 Basic Weight and Corresponding Centre of Gravity (a) The basic weight and corresponding centre of gravity must be determined by weighing the UA with: (1) Fixed ballast (if applicable); (2) Unusable fuel determined under USAR.959; and (3) Full operating fluids, including: (i) Oil; (ii) Hydraulic fluid; and (iii) Other fluids required for normal operation of UA. (4) The payload or load configuration specified by the Applicant and agreed with the Certifying Authority, or without payload if such a configuration is to be approved. (b) The condition of the UA at the time of determining basic weight must be one that is well defined and can be easily repeated.
31	USAR.31 Removable Ballast Removable ballast may be used in showing compliance with the flight requirements of this subpart, if: (a) The place(s) for carrying ballast is properly designed and installed; and (b) Instructions are included in the UAS Flight Manual, approved manual material, or markings and placards, for the proper placement of the removable ballast under each loading condition for which removable ballast is necessary.
33	USAR.33 Propeller Speed and Pitch Limits (a) General. The propeller speed and pitch must be limited to values that will assure safe operation under normal operating conditions. (b) Propellers not controllable in flight. For each propeller whose pitch cannot be controlled in flight:

	(1) During take-off and initial climb at the all-engine(s)-operating climb speed specified in USAR.65, the propeller must limit the engine rpm, at full throttle or at maximum allowable take-off manifold pressure, to a speed not greater than the maximum allowable take-off rpm; and (2) During a closed throttle glide at maximum achievable speed maintained by the flight control system, the propeller may not cause an engine speed above 110% of maximum continuous speed. (c) Controllable pitch propellers without constant speed controls. Each propeller that can be controlled in flight, but that does not have constant speed controls, must have a means to limit the pitch range so that: (1) The lowest possible pitch allows compliance with sub-paragraph (b) (1) of this paragraph; and (2) The highest possible pitch allows compliance with sub-paragraph (b) (2) of this paragraph. (d) Controllable pitch propellers with constant speed controls. Each controllable pitch propeller with constant speed controls must have: (1) With the governor in operation, a means at the governor to limit the maximum engine speed to the maximum allowable take-off rpm; and (2) With the governor inoperative, a means to limit the maximum engine speed to 103% of the maximum allowable take-off rpm with the propeller blades at the lowest possible pitch and with take-off power setting, the UA stationary, and no wind.
--	---

PERFORMANCE

45	USAR.45 General (a) Unless otherwise prescribed, the performance requirements of this subpart must be met for: (1) Still air and standard atmosphere at sea level, and (2) Ambient atmospheric conditions, and (3) Minimum engine performance. (b) Performance data must be determined over not less than the following ranges of conditions: (1) Airfield or launch site altitude from sea-level to maximum take-off altitude at which certification is requested; and (2) temperatures from standard to 30°C above standard; or (3) the maximum ambient atmospheric temperature at which compliance with the cooling provisions of USAR.1041 to USAR.1047 is shown, if lower. (c) Performance data must be determined with the cowl flaps or other means for controlling the engine cooling air supply in the position used in the cooling tests required by USAR.1041 to USAR.1047. (d) The available propulsive thrust must correspond to engine power or thrust, not exceeding the approved power or thrust, less: (1) Installation losses; and (2) The power absorbed by the accessories and services appropriate to the particular ambient atmospheric conditions and the particular flight condition.
----	--

	(e) The performance as affected by engine power or thrust must be based on a relative humidity of: (1) 80% at and below standard temperature; and (2) 34% at and above standard temperature plus 28°C (plus 50°F). Between the two temperatures listed in sub-paragraphs (e) (1) and (e) (2) of this paragraph the relative humidity must vary linearly. (f) Unless otherwise prescribed in determining the take-off and landing distances, changes in the UA's configuration, speed and power or thrust must be made in accordance with procedures established by the Applicant for operation in service. These procedures must be able to be executed consistently by UA crew of average skill in atmospheric conditions reasonably expected to be encountered in service. (g) The following, as applicable, must be determined on a smooth, dry, hard-surfaced runway and zero headwind: (1) Take-off distance of USAR.53 (b); (2) Accelerate-stop distance or critical field length of USAR.55; (3) Not applicable. (4) Landing distance of USAR.75. The effect on these distances of operation on other types of surface (e.g. grass, gravel) when dry, may be determined or derived and these distances listed in accordance with USAR.1583 (p). (h) Not applicable.
49	USAR.49 Stalling Speed (a) V_{S0} and V_{S1} are the stalling speeds or the minimum steady flight speed, in knots (CAS), at which the UA is controllable with: (1) For reciprocating engine-powered UA, engine(s) idling, the throttle(s) closed or at not more than the power necessary for zero thrust at a speed not more than 110% of the stalling speed; and (2) For turbine engine-powered UA, the propulsive thrust may not be greater than zero at the stalling speed, or, if the resultant thrust has no appreciable effect on the stalling speed, with engine(s) idling and throttle(s) closed; (3) Propeller(s) in the take-off position; (4) The UA in the condition existing in the test or calculation in which V_{S0} and V_{S1} are being used; (5) Centre of gravity in the position which results in the highest value of V_{S0} and V_{S1}; and (6) Weight used when V_{S0} or V_{S1} are being used as a factor to determine compliance with a required performance standard. (b) V_{S0} and V_{S1} must be determined by: (1) analysis based on a calculation method agreed with the Certifying Authority, or (2) by flight tests using the procedure and meeting the flight characteristics specified in USAR.201 and USAR.203

	(c) Not applicable.
U50	USAR.U50 Minimum demonstration speed If the stalling speed is not demonstrated by flight tests, a “minimum demonstration speed” must be demonstrated. (a) The minimum demonstration speed $V_{\min \text{ DEMO}}$ is the minimum speed demonstrated by the Applicant by flight test, while possibly adjusting or inhibiting flight control protection features, using the procedure and meeting the flight characteristics specified in USAR.201, USAR.203, USAR.U334 and AMC.203 (b) The minimum demonstration speed $V_{\min \text{ DEMO}}$ must be less than r times the minimum steady flight speed (except take-off and landing) allowed by the flight envelope protection maintained by the flight control system as defined in USAR.U334 and AMC.203. The r ratio shall not be above 0.95 and shall be agreed with the Certifying Authority.
51	USAR.51 Take-off Speeds Except for catapult assisted or rocket assisted take-off UA, the following applies: (a) The rotation speed V_R (if applicable), is the speed at which the UA crew or the flight control system makes a control input with the intention of lifting the UA out of contact with the runway. (1) For multi-engine UA, V_R must not be less than the greater of: (i) $1.05 V_{MC}$, and (ii) $1.10 V_{S1}$ (2) For single engine UA, V_R must not be less than V_{S1} (3) Not applicable. (b) The speed at 15 m (50 ft) must not be less than: (1) For multi-engine UA, the greater of: (i) A speed that is shown to be safe for continued flight (or land-back, if applicable) under all reasonably expected conditions, including turbulence and complete failure of the critical engine and compliant with the requirement established in USAR.63; and (ii) $1.10 V_{MC}$; and (iii) $1.20 V_{S1}$ (2) For single-engine UA, the greater of: (i) A speed that is shown to be safe under all reasonably expected conditions, including turbulence and complete engine failure and compliant with the requirement established in USAR.63; and (ii) $1.20 V_{S1}$ (c) Not applicable.
53	USAR.53 Take-off Performance Except for catapult assisted or rocket assisted take-off UA, the following applies: (a) The take-off distance must be determined in accordance with sub-paragraph (b), using speeds determined in accordance with USAR.51 (a) and (b).

	(b) The distance required to take-off and climb to a height of 15 m (50 ft) above the take-off surface must be determined for each weight, altitude and temperature within the operational limits established for take-off with: (1) Take-off power or thrust on each engine; (2) Wing flaps in the take-off position(s); and (3) Landing gear extended in the take-off position. (c) Take-off performance as required by USAR.53 (a) and USAR.55 must be determined with the operating engines within approved operating limitations. (d) Maximum rotation rate (if applicable) is to be determined such that resulting dynamic effects do not lead to unsafe conditions or reduction in loading or manoeuvres safety margins.
55	USAR.55 Accelerate-stop Distance or Critical Field Length See AMC.55 Except for catapult assisted or rocket assisted take-off UA, the critical field length must be determined as follows: (a) For multi engine UA, the critical field length is the sum of the distances necessary to: (1) Accelerate the UA from a standing start to V_{EF} with all engines operating; (2) Within the same distance either accelerate the UA from V_{EF} to V_{Rf}, assuming the critical engine fails at V_{EF}; or come to a full stop from the point at which V_{EF} is reached. V_{EF} is the calibrated airspeed at which the critical engine is assumed to fail and the same distance is required to either continue the take-off or stop. The V_{EF} must be selected for the UA, but must not be less than V_{MCG} determined under USAR.149 (f). (b) For single engine UA, the accelerate-stop distance is the sum of the distances necessary to: (1) Accelerate the UA from a standing start to V_{Rf} with engine operating; (2) Come to a full stop from the point at which V_{Rf} is reached. (c) Means other than wheel-brakes may be used to determine the critical field length if that means: (1) Is safe and reliable; and (2) Is used so that consistent results can be expected under normal operating conditions; and (3) Provides that all wheels remain on the ground during braking. (d) The following shall be included in the ground roll calculation: (1) Engine spool down characteristics. (2) System and UA crew reaction time to sense a failure and make the appropriate response to the failure. (3) If applicable, the time for UA configuration changes (e.g. flap retractions, chute deployment, etc.)

	57 Take-off path A rejected take-off at VR shall not lead to a Hazardous or more serious failure condition.
	59 Take-off distance and take-off run Not applicable.
	61 Take-off flight path Not applicable.
63	USAR.63 Climb: General (a) Compliance with the requirements of USAR.65, USAR.66, USAR.67, USAR.69 and USAR.77 must be shown: (1) Out of ground effect; and (2) At speeds which are not less than those at which compliance with the powerplant cooling requirements of USAR.1041 to USAR.1047 have been demonstrated. (3) Unless otherwise specified, with one engine inoperative, at a bank angle not exceeding 5 degrees. (4) For catapult assisted or rocket assisted take-off UA when the UA leaves the flight safety area associated to the launch safety trace required in USAR.U283 (b) Compliance must be shown with USAR.65, USAR.67, where appropriate and USAR.77 at maximum take-off or landing weight, as appropriate in a standard atmosphere, or (c) At weights, as a function of airfield or launch site altitude and ambient temperature, within the operational limits established for take-off and landing respectively. (d) Not applicable.
65	USAR.65 Climb: All Engines Operating (a) Each UA must have a steady gradient of climb at sea level of at least 5 % with: (1) Not more than maximum continuous power on each engine; (2) The landing gear retracted (if such a configuration is designed); (3) The wing flaps in the take-off position(s); and (4) A climb speed not less than the greater of $1.1 V_{MC}$ and $1.2 V_{S1}$ for multi-engine UA and not less than $1.2 V_{S1}$ for single-engine UA. (b) For configurations with retractable landing gear, each UA must have a steady gradient of climb at sea level of at least 2.5 % with: (1) Not more than maximum continuous power on each engine; (2) The landing gear extended; (3) The wing flaps in the take-off position(s); and (4) A climb speed not less than the greater of $1.1 V_{MC}$ and $1.2 V_{S1}$ for multi-engine UA and not less than $1.2 V_{S1}$ for single-engine UA.

66	USAR.66 Take-off Climb: One-engine-inoperative for multi-engine UA The steady gradient of climb or descent must be determined at each weight, altitude and ambient temperature within the operational limits established by the Applicant with: (a) The critical engine inoperative and its propeller in the position it rapidly and automatically assumes; (b) The remaining engine at take-off power or thrust; (c) The landing gear extended in the take-off position except that, if the landing gear can be retracted in not more than 7 seconds and the gear retraction sequence does not have a higher drag profile than the gear deployed configuration, it may be assumed to be retracted; (d) The wing flaps in the take-off position(s); (e) The wings level; and (f) A climb speed equal to that achieved at 15 m (50 ft) in the demonstration of USAR.53
67	USAR.67 Climb: One-engine-inoperative for multi-engine UA (a) The steady gradient of climb must not be less than 2% at an altitude of 122 m (400 ft) above the take-off surface with: (1) The critical engine in-operative and its propeller in the minimum drag position; (2) The remaining engine at take-off power or thrust; (3) The landing gear retracted (if applicable); (4) The wing flaps in the take-off position(s); and (5) A climb speed equal to that achieved at 15 m (50 ft) in the demonstration of USAR.53 (b) For configurations with retractable landing gear, the steady gradient of climb must not be less than 0.5% at an altitude of 122 m (400 ft) above the take-off surface, as appropriate with: (1) The critical engine in-operative and its propeller in the minimum drag position; (2) The remaining engine at take-off power or thrust; (3) The landing gear extended; (4) The wing flaps in the take-off position(s); and (5) A climb speed equal to that achieved at 15 m (50 ft) in the demonstration of USAR.53 (c) The steady gradient of climb must not be less than 0.8 % at an altitude of 422 m (1,500 ft) above the take-off or landing surface, as appropriate with: (1) The critical engine in-operative and its propeller in the minimum drag position; (2) The remaining engine at not more than maximum continuous power or thrust; (3) The landing gear retracted; (4) The wing flaps retracted; and (5) A climb speed not less than $1.2 V_{S1}$
69	USAR.69 En-route Climb/Descent

	(a) All engines operating. The steady gradient and rate of climb must be determined at each weight, altitude and ambient temperature within the operational limits established by the Applicant with: (1) Not more than maximum continuous power or thrust on each engine; (2) The landing gear retracted; (3) The wing flaps retracted; and (4) A climb speed not less than $1.3 V_{S1}$ (b) One-engine-inoperative. The steady gradient and rate of climb/descent must be determined at each weight, altitude and ambient temperature within the operational limits established by the Applicant with: (1) The critical engine inoperative and its propeller in the minimum drag position; (2) The remaining engine at not more than maximum continuous power or thrust; (3) The landing gear retracted; (4) The wing flaps retracted; and (5) A climb speed not less than $1.2 V_{S1}$
71	USAR.71 Glide See AMC.71 The maximum horizontal distance travelled in still air, in nautical miles per 1,000 ft of altitude lost in a glide, and the speed necessary to achieve this, must be determined with the engine inoperative and its propeller in the minimum drag position, landing gear and wing flaps in the most favourable available position.
73	USAR.73 Reference Landing Approach Speed Except where a UA is designed to be recovered by parachute, the reference landing approach speed, V_{REF}, must not be less than the greater of: (a) V_{MC}, determined under USAR.149 (c), and (b) $1.3 V_{SO}$, except if it is demonstrated that a lower speed does not affect safe landing due to UA performance whatever the combination of environmental conditions.
75	USAR.75 Landing Distance Except where a UA is designed to be recovered by parachute, the horizontal distance necessary to land and come to a complete stop from a point 15 m (50 ft) above the landing surface must be determined, for standard temperatures at each weight and altitude within the operational limits established for landing, as follows: (a) A steady approach at not less than V_{REF}, determined in accordance with USAR.73 as appropriate, must be maintained down to 15 m (50 ft) height and the steady approach must be at a gradient of descent selected by the Applicant, called “standard slope”, down to the 15 m (50 ft) height. (b) A constant configuration must be maintained throughout the manoeuvre;

	(c) The landing must be made without excessive (limiting factor should be brakes, structure, landing gear, structural fatigue) vertical acceleration or tendency to bounce, nose-over, ground loop or porpoise. (d) It must be shown that a safe transition to the balked landing conditions of USAR.77 can be made from the conditions that exist at the 15 m (50 ft) height, at maximum landing weight or the maximum landing weight for altitude and temperature of USAR.63 (c). (e) The brakes must not be used so as to cause excessive wear of brakes or tyres. (f) Retardation means other than wheelbrakes may be used if that means: (1) Is safe and reliable; (2) Is used so that consistent results can be expected in service; and (g) If any device is used that depends on the operation of any engine, and the landing distance would be increased when a landing is made with that engine inoperative, the landing distance must be determined with that engine inoperative unless the use of other compensating means will result in a landing distance not more than that with each engine operating.
77	USAR.77 Balked Landing (a) Except where a UA is designed to be recovered by parachute, the steady gradient of climb must not be less than 2.5% with: (1) Take-off power on each engine; (2) The landing gear extended; (3) The wing flaps in the landing position; except that if the flaps may be safely retracted without loss of altitude and without sudden changes of angle of attack, they may be retracted; and (4) A climb speed equal to V_{REF}, as defined in USAR.73 (b) Not applicable. (c) Not applicable. (d) Minimum balked landing height shall be determined. This is defined as the minimum height above the ground where a successful balked landing could be performed safely.

FLIGHT CHARACTERISTICS

141	USAR.141 General See AMC.141 (a) The UA must meet the requirements of USAR.143 to USAR.253 When operated in the automatic control mode the UA should be shown to have acceptable controllability, manoeuvrability and stability characteristics throughout the flight envelope protection (see USAR.U334 and USAR.U1330) maintained by the flight control system, without requiring exceptional skill or alertness from the UA crew. (b) The UA must demonstrate the fulfilment of the “flight characteristics” requirements at all practical loading conditions and all operating altitudes, not exceeding the maximum operating altitude established under USAR.1527, for which certification has been requested. (c) The UA must meet the “flight characteristics” requirements at all environmental conditions (including turbulence and gust) for which certification has been requested.
-----	---

CONTROLLABILITY AND MANOEUVRABILITY

143	USAR.143 General (a) The UA must be safely controllable and manoeuvrable during all flight phases including: (1) Take-off (except for rocket assisted or catapult assisted take-off UA, see USAR.U282); (2) Climb; (3) Level flight, including mission relevant special manoeuvres; (4) Descent; (5) Go-around (except where UA is designed to be recovered by parachute, see USAR.U290); and (6) Landing (power on and power off) with the wing flaps extended and retracted (except where UA is designed to be recovered by parachute, see USAR.U290). (7) Ground taxi (see USAR.231 to USAR.235). (b) It must be possible to make a smooth transition from one flight phase and/or condition to another (including turns and slips) without danger of exceeding the limit load factor, under any probable operating condition, (including, for multi-engine UAs, those conditions normally encountered in the sudden failure of any engine). (c) Not applicable.
	145 Longitudinal control Not applicable. (see USAR.171)
	147 Directional and lateral control Not applicable. (see USAR.171)
149	USAR.149 Minimum Control Speed See AMC.149 (b) (a) V_{MC} is the calibrated airspeed at which, when the critical engine is suddenly made inoperative, it is possible to maintain control of the UA, with that engine still inoperative, and thereafter maintain straight flight at the same speed with an angle of bank not more than 5°. The method used to simulate critical engine failure must represent the most critical mode of powerplant failure with respect to controllability expected in service. (b) V_{MC} for take-off must not exceed $1.2 V_{S1}$, (where V_{S1} is determined at the maximum take-off weight) and must be determined with the most unfavourable weight and centre of gravity position and with the UA airborne and the ground effect negligible, for the take-off configuration(s) with: (1) Maximum available take-off power or thrust initially on each engine; (2) The UA trimmed for take-off; (3) Flaps in the take-off position(s); (4) Landing gear retracted; and (5) All propeller controls in the recommended take-off position throughout. Note: For UAs with sudden engine failure control characteristics, the dynamic VMC may be several knots higher than the static VMC. Dynamic VMC should be assessed with the landing gear extended

	and retracted. (c) Except where UA is designed to be recovered by parachute, the requirements of sub-paragraph (a) must also be met for the landing configuration with: (1) Maximum available take-off power or thrust initially on each engine; (2) The UA trimmed for an approach with all engines operating at V_{REF} at an approach gradient equal to the steepest used in the landing distance demonstration of USAR.75; (3) Flaps in the landing position; (4) Landing gear extended; and (5) All propeller controls throughout in the position recommended for approach with all engines operating. (d) A minimum speed to intentionally render the critical engine inoperative must be established and designated as the safe, intentional, one-engine inoperative speed, V_{SSE} (e) At V_{MC}, it must not be necessary to reduce power or thrust of the operative engine. During the manoeuvre the UA must not assume any dangerous attitude and it must be possible to prevent a heading change of more than 20°. (f) Except for rocket assisted or catapult assisted take-off UA, V_{MCG}, the minimum control speed on the ground, is the calibrated airspeed during the take-off run, at which, when the critical engine is suddenly made inoperative and with its propeller, if applicable, in the position it automatically achieves, it is possible for the flight control system to maintain control of the UA to enable the take-off to be safely continued. In the determination of V_{MCG}, assuming that the path of the UA accelerating with all engines operating is along the centreline of the runway, its path from the point at which the critical engine is made inoperative to the point at which recovery to a direction parallel to the centreline is completed, may not deviate more than 9.1m (30ft) laterally from the centreline at any point. V_{MCG} must be established, with: (1) The UA in each take-off configuration or, at the option of the Applicant, in the most critical take-off configuration; (2) Maximum available take-off power or thrust on the operating engines; (3) The most unfavourable centre of gravity; (4) The UA trimmed for take-off; and (5) The most unfavourable weight in the range of take-off weights.
	151 Aerobatic manoeuvres Not applicable.
	153 Control during landings Not applicable.
	155 Elevator control force in manoeuvres Not applicable.
	157 Rate of roll Not applicable.

161	USAR.161 Trim The Flight Control System (FCS) must trim the UA in such a manner that a maximum of control remains and that dynamic characteristics and safety margins are not compromised.
-----	--

STABILITY

171	USAR.171 General See AMC.171 (a) The UA, augmented by the FCS including all degraded modes, and including the effects of sensor and computational errors and delays must be longitudinally, directionally and laterally stable in any condition normally encountered in service, at any combination of weight and centre of gravity for which certification is requested. (b) Transient response in all axes during transition between different flight conditions and flight modes must be smooth, convergent, and exhibit damping characteristics with minimal overshoot of the intended flight path. (c) In addition to data obtained by computation or modelling, stability analysis must be supported by the results of relevant flight tests.
	173 Static longitudinal stability Not applicable.
	175 Demonstration of static longitudinal stability Not applicable.
	177 Static directional and lateral stability Not applicable.
	181 Dynamic stability Not applicable.

STALLS

201	USAR.201 Wings Level Stall See AMC.203 (a) Flight tests shall be conducted in straight flight for each relevant UA flaps configuration, with the engine at idle position and for the most appropriate combination of weight and centre of gravity while reducing the speed at a decelerating rate of approximately 1kt/s: (1) up to the time the UA stalls, or (2) until $V_{min\ DEMO}$, if the stalling speed is not to be demonstrated in compliance with USAR.U50, and (i) no stall tendency shall occur down to $V_{min\ DEMO}$, (ii) $V_{min\ DEMO}$ shall be lower by the margin established under USAR.U50 than the minimum steady flight speed (except take-off and landing) allowed by the flight envelope protection maintained by the flight control system. (b) These flight tests may be conducted, while possibly adjusting or inhibiting flight control protection features.
203	USAR.203 Stall protection in wing level and turning flight See AMC.203

	(a) Flight tests shall be conducted in straight flight and in the maximum bank angle allowed by the flight control protection features for each relevant UA flaps configuration for the most unfavourable combination of weight, centre of gravity and engine setting while abruptly reducing speed command as per relevant flight control mode. (b) During these tests, it should be shown that: (1) the steady speed achieved should remain greater than or equal to the minimum steady flight speed (except take-off and landing) allowed by the flight envelope protection maintained by the flight control system. (2) No unsafe characteristics occur.
	207 Stall warning Not applicable in this subpart. (see USAR.U1789 Low speed warning)

SPINNING

221	USAR.221 Spinning and tumbling The UA must be designed characteristically incapable of intentional spinning/tumbling (all spin and tumbling modes) due to the flight envelope protection maintained by the flight control system or other means to be substantiated by the Applicant unless in defined circumstances agreed with the Certifying Authority (i.e. used as a flight termination system in compliance with USAR.U1412).
-----	---

GROUND HANDLING CHARACTERISTICS

231	USAR.231 Longitudinal Stability and Control A UA may have no uncontrollable tendency to nose over in any reasonably expected operating condition, including rebound during landing (except for parachute operations) or take-off. Wheel brakes (where fitted) must operate smoothly and may not induce any undue tendency to nose over.
233	USAR.233 Directional Stability and Control (a) A 90° cross-component of wind velocity, demonstrated to be safe for taxiing (except for UA not designed for taxiing), take-off and landing must be established and must be not less than 0.2 V_{S0} (b) Except where UA is designed to be recovered by parachute only, it must be satisfactorily controllable in power-off landings at normal landing speed, without using brakes or engine power to maintain a straight path until the speed has decreased to less than 50% of the speed at touchdown. (c) Except for UA not designed for taxiing it must have adequate directional control during taxiing. (d) Not applicable.
235	USAR.235 Operation On Unpaved Surfaces See AMC.235 The UA must be demonstrated to have satisfactory characteristics and the shock-absorbing mechanism must not damage the UA when the UA is taxied on the roughest ground that may reasonably be expected in normal operation (except for UA not designed for taxiing) and when take-off and landings are performed on unpaved runways having the roughest surface that may reasonably be expected in normal operation. UAS Flight Manual restrict UA operation to paved taxiways and runways only: (a) the requirements for taxi, take-off and landing on unpaved runway are not applicable, and (b) the UAS Flight Manual shall give all operational indications to face emergency landing on unpaved runways.
	237 Operation on water Not applicable.

	239 Spray characteristics Not applicable.
U240	USAR.U240 Wet Runway Operations See AMC.U240 Aircraft shall be designed to withstand runway spray if intended to operate under wet runway conditions.
U249	USAR.U249 Transportation and storage See Appendix K (a) Where a UA System or part of the System is designed to be transportable by any means during normal operations or System use, it shall be demonstrated that no environmental factors associated with the means of transportation shall adversely affect any requirement of these standards. (b) It shall be demonstrated that any special equipment used for transportation during normal operations (for instance special containers, cradles etc.) provides the appropriate level of environmental protection for the method of transport used. (c) Where a UA System or part of the System is reconfigured for transportation, it shall be shown that the expected number of reconfigurations in any System life cycle will not adversely affect any requirement of USAR. (d) Where a UA System or part of the System is designed to be placed in storage as part of the normal usage pattern, it shall be demonstrated that no environmental factors associated with preparation for storage, storage, or recovery from storage shall adversely affect any factor/requirement of these standards. (e) In this part environmental factors relating to transportation or storage shall include all shock, vibration, water and moisture, particulate matter, electromagnetic, thermal, and other foreseeable conditions or effects likely to be encountered during transportation or storage that would adversely affect any requirement of these standards.

MISCELLANEOUS FLIGHT REQUIREMENTS

251	USAR.251 Vibration and Buffeting There must be no vibration or buffeting severe enough to result in structural damage and each part of the UA must be free from excessive vibration, under any appropriate speed and power or thrust conditions up to at least the minimum value of V_D allowed in USAR.335. In addition there must be no vibration or buffeting in any normal flight condition severe enough to interfere with the satisfactory control of the UA.
253	USAR.253 High Speed Characteristics If a maximum operating speed V_{MO}/M_{MO} is established under USAR.1505 (c), the following speed increase and recovery characteristics must be met: (a) Operating conditions and characteristics likely to cause inadvertent speed increases (including upsets in pitch and roll) must be simulated with the UA trimmed at any likely speed up to V_{MO}/M_{MO}. These conditions and characteristics include gust upsets, levelling off from climb and descent from Mach to airspeed limit altitude. (b) Allowing for UA crew or flight control system reaction time after occurrence of effective inherent or artificial speed warning specified in USAR.U1723, it must be shown that the UA can be recovered to a normal attitude and its speed reduced to V_{MO}/M_{MO} without: (1) Exceeding V_D/M_D, the maximum speed shown under USAR.251, or the structural limitations; or

	(2) Buffeting that would impair the UA ability for recovery. (c) There may be no control reversal about any axis at any speed up to the maximum speed shown under USAR.251
--	--

CATAPULT ASSISTED AND ROCKET ASSISTED TAKE-OFF UA

U280	USAR.U280 Launch performance (a) The UA must achieve sufficient energy and controllability at the end of the launch phase to ensure safe and controllable fly-away under the most adverse combination of environmental and operating conditions: (1) at a minimum of $1.15 V_{S1}$ or $1.15 V_{MC}$, whichever is the higher; (2) in order to comply with requirements of USAR.51 (b), USAR.65 and USAR.67 The launch phase ends when the UA leaves the flight safety area associated to the launch safety trace required in USAR.U283 (b) The launch performance (launch parameters settings, launch speed) must be determined for each weight, altitude, temperature and wind condition within the operational limits established for take-off. (c) It must be shown by tests that the acceleration sustained by the UA during the launch phase do not lower UA engine performance in a manner that could be inadequate for safe operation. (d) A manual abort function must be easily accessible to the UA crew in order to cancel the UA launch at any time before the irreversible catapult or rocket ignition phase.
U281	USAR.U281 Transition to normal flight attitude (a) The transition to normal flight attitude or normal in-flight UA configuration must be such that no possibility of conflict exists between the UA and its launch platform or any other object under any combination of environmental conditions. (b) The UA must remain in a predictable flight condition that does not exhibit any tendencies to depart from controlled flight throughout the launch phase.
U282	USAR.U282 UA active control In case of launch without active control by the flight control system of the aircraft attitude or direction, the UA must not diverge beyond its recoverable limit and the active control must be reinstated before the UA reaches the boundary of its launch safety trace.
U283	USAR.U283 Launch safety trace See AMC.U283 The limits of the launch safety trace around the launch platform must be determined for each weight, altitude, wind conditions, and temperature within the operational limits established for take-off.

PARACHUTE LANDING SYSTEM

U290	USAR.U290 UA performance before parachute landing (a) The UA flight performance and control characteristics must be adequate for all intended parachute landing procedure under all specified operational conditions. (b) Two modes of landing by parachute can be foreseen: (1) a normal landing mode where a parachute is used in a regular way after every flight, and (2) an emergency landing mode where a parachute is used in case of emergency.
------	---

	(c) It must be possible to abort the normal landing procedure at any point prior to the initiation of the final deployment sequence and it must be shown that a safe transition to a normal flight mode or go around conditions can be made. (d) The normal and emergency parachute landing sequence must be precisely defined in the UAS Flight Manual including for normal landing the approach phase and the go around procedure.
U291	USAR.U291 Parachute landing characteristics (a) The normal landing under parachute must be made without excessive vertical acceleration or tendency to bounce, nose over, ground loop or porpoise. (b) The minimum parachute safety height must ensure a correct parachute deployment sequence and must ensure that the UA descent under a fully inflated parachute is stabilised whatever the combination of environmental conditions. (c) The parachute must be deployed at a height greater or equal to the minimum parachute safety height above ground, which depends on the timing of the parachute sequence. (d) The minimum parachute safety height must be stated in the UAS Flight Manual. (e) Special attention must be given to the static and dynamic stability of the UA during the parachute landing phase: movements of the UA and of the parachute must not lead to unsafe characteristics.
U292	USAR.U292 Parachute landing performance (a) The normal parachute landing must be designed to allow a precise landing on the ground surface with a CEP to be stated in the UAS Flight Manual under and calculated under any combination of environmental conditions. (b) It must be shown that the parachute landing sequence is a reliable, repeatable and predictably safe operation: (1) at every combination of weight and balance of the UA for which certification is requested, (2) in the most adverse weather conditions (wind, rain, icing, ...) for which approval is requested, (3) throughout the life cycle of the UA. (c) The features of the terrain over which the parachute landing can be performed in normal condition must be stated in the UAS Flight Manual, in particular its acceptable slope.
U293	USAR.U293 Parachute landing safety trace See AMC.U293 The limits of the parachute landing safety trace must be determined for each weight, altitude and temperature within the operational limits established for landing.

Subpart C – UA STRUCTURE

GENERAL	
301	USAR.301 Loads See AMC.301 (a) Strength requirements are specified in terms of limit loads (the maximum loads to be expected in service) and ultimate loads (limit loads multiplied by prescribed factors of safety). Unless otherwise provided, prescribed loads are limit loads. (b) Unless otherwise provided, the air and ground loads must be placed in equilibrium with inertia forces, considering each item of mass in the UA. These loads must be distributed to conservatively approximate or closely represent actual conditions. Methods used to determine load intensities and distribution on canard and tandem wing configurations must be validated by flight test measurement unless the methods used for determining those loading conditions are shown to be reliable or conservative on the configuration under consideration. (c) If deflections under load would significantly change the distribution of external or internal loads, this redistribution must be taken into account. (d) Simplified structural design criteria may be used when agreed with the Certifying Authority if they result in design loads not less than those prescribed in USAR.331 to USAR.511 (e) The requirements of subpart C must be assessed for each payload configuration.
302	USAR.302 Canard or Tandem Wing Configurations The forward structure of a canard or tandem wing configuration must: (a) Meet all requirements of subpart C and subpart D of USAR applicable to a wing; and (b) Meet all requirements applicable to the function performed by these surfaces.
303	USAR.303 Factor of Safety The factor of safety shall not be lower than 1.5 for structure whose failure would lead to a Hazardous or more serious failure condition. For other structure, the factor of safety shall not be lower than 1.25. For a factor of safety less than 1.5, the Applicant must provide justification to be agreed with the Certifying Authority.
U304	USAR.U304 Structural Performance See AMC.U304 For aeroplanes equipped with systems that affect structural performance, either directly or as a result of a failure or malfunction, the influence of these systems and their failure conditions must be taken into account when showing compliance with the requirements of Subparts C and D.
305	USAR.305 Strength and Deformation (a) The structure must be able to support limit loads without detrimental, permanent deformation at any load up to limit loads, the deformation may not interfere with safe operation. (b) The structure must be able to support ultimate loads without failure for at least three seconds, except local failures or structural instabilities between limit and ultimate load are acceptable only if the structure can sustain the required ultimate load for at least three seconds. However, when proof of strength is shown by dynamic tests simulating actual load conditions, the three second limit does not apply.
307	USAR.307 Proof of Structure See AMC.307

	(a) Compliance with the strength and deformation requirements of USAR.305 must be shown for each critical load condition; including fire conditions (see USAR.865). Structural analysis may be used only if the structure conforms to those for which experience has shown this method to be reliable. In other cases, substantiating load tests must be made up to a level considered to be sufficient and agreed with the Certifying Authority. Dynamic tests, including structural flight tests, are acceptable if the design load conditions have been simulated. (b) Certain parts of the structure must be tested as specified in Subpart D of USAR. (c) When analytical methods are used to show compliance with the ultimate load strength, it must be shown that: (1) The effects of deformation are not significant; or (2) The deformations involved are fully accounted for in the analysis; or the methods and assumptions used are sufficient to cover the effects of these deformations.
--	---

FLIGHT LOADS

321	USAR.321 General See AMC.321 (c) (a) Flight load factors represent the ratio of the aerodynamic force component (acting normal to the assumed longitudinal axis of the UA) to the weight of the UA. A positive flight load factor is one in which the aerodynamic force acts upward, with respect to the UA. (b) Compliance with the flight load requirements of this subpart must be shown at each critical combination of: (1) altitude within the range in which the UA may be expected to operate; (2) weight from the design minimum weight to the design maximum weight; (3) centre of gravity between the allowable center of gravity limits; (4) altitude, weight and centre of gravity, for any practicable distribution of disposable load within the operating limitations specified in USAR.1583 to USAR.1589 (c) When significant the effects of compressibility must be taken into account. (d) The significant forces acting on the UA must be placed in equilibrium in a rational or conservative manner. The linear inertia forces must be considered in equilibrium with power or thrust and all aerodynamic loads, while the angular (pitching) inertia forces must be considered in equilibrium with power or thrust and all aerodynamic moments. Critical power or thrust values in the range from zero to maximum continuous power or thrust must be considered. (e) The manoeuvres which need to be considered in the load establishment are those resulting from any combination of possible (taking into account the UA System design) control surface deflections and power or thrust settings. The resulting load conditions must be established in a rational or conservative manner and must consider: (1) the UAS nominal modes of control, (2) the UAS failure modes where probability of occurrence is higher than extremely remote. Such conservative manoeuvre conditions may be, if convenient, based on conventional type manoeuvre (symmetrical manoeuvres, roll manoeuvres, yaw manoeuvres) such as defined in USAR.331, USAR.349 and USAR.351
-----	--

	(f) The UAS design shall be such that the loads possibly encountered during the defined usage spectrum, considering approved procedures may not exceed the loads to which the UAS is certified.
331	USAR.331 Symmetrical Flight Conditions (a) The appropriate balancing horizontal tail load must be accounted for in a rational or conservative manner when determining the wing loads and linear inertia loads corresponding to any of the symmetrical flight conditions specified in USAR.331 to USAR.341 (b) The incremental horizontal tail loads due to manoeuvring and gusts must be reacted by the angular inertia of the UA in a rational or conservative manner. (c) Mutual influence of the aerodynamic surfaces must be taken into account when determining flight loads.
333	USAR.333 Flight Envelope (a) General. Compliance with the strength requirements of this subpart must be shown at any combination of airspeed and load factor on and within the boundaries of a flight envelope (similar to the one in sub-paragraph (d) of this paragraph) that represents the envelope of the flight loading conditions specified by the manoeuvring and gust criteria of sub-paragraphs (b) and (c) of this paragraph respectively. (b) Manoeuvring envelope. Except where limited by maximum (static) lift coefficients, the UA is assumed to be subjected to symmetrical manoeuvres resulting in the following limit load factors: (1) The positive manoeuvring load factor specified in USAR.337 at speeds up to V_D; (2) The negative manoeuvring load factor specified in USAR.337 at V_C; and (3) Factors varying linearly with speed from the specified value at V_C to 0.0 at V_D

	(c) Gust envelope (1) The UA is assumed to be subjected to symmetrical vertical and lateral gusts in level flight. The resulting limit load factors must correspond to the conditions determined as follows: positive (up) and negative (down) gust values at V_C and V_D should be determined by rational analysis of the intended use of the UAS, considering the time spent at low altitude levels and the cruise speed (consistent with the design usage spectrum defined in USAR.U17), however as a minimum: (i) Positive (up) and negative (down) gusts of 15.2 m/s (50 fps) at V_C must be considered at altitudes between sea level and 6096 m (20,000 ft). The gust velocity may be reduced linearly from 15.2 m/s (50 fps) at 6096 m (20,000 ft) to 7.6 m/s (25 fps) at 15,240 m (50,000 ft); and (ii) Positive and negative gusts of 7.6 m/s (25 fps) at V_D must be considered at altitudes between sea level and 6096 m (20,000 ft). The gust velocity may be reduced linearly from 7.6 m/s (25 fps) at 6096 m (20,000) ft to 3.8 m/s (12.5 fps) at 15,240 m (50,000 ft). (2) The following assumptions must be made: (i) The Shape of the gust is: $U = \frac{U_{de}}{2} \left(1 - \cos \frac{2\pi s}{25\bar{C}} \right)$ where: s = Distance penetrated into gust (ft.) $\bar{C}$ = Mean geometric chord of wing (ft.) U_{de} = Derived gust velocity referred to in sub-paragraph (1) linearly with speed between V_C and V_D (ii) Gust load factors vary linearly with speed between V_C and V_D. (d) Flight envelope: see figure above. where: V_A: design manoeuvring speed V_C: design cruising speed V_D: design diving speed V_G: negative manoeuvring load factor speed V_S: stalling speed
U334	USAR.U334 Flight envelope protection (a) Flight envelope protection shall be implemented in the flight control system as follows: (1) Characteristics of each envelope protection feature must be smooth, appropriate to the phase of flight and type of manoeuvre. (2) Limit values of protected flight parameters must be compatible with: (i) UA structural limits, (ii) required safe and controllable manoeuvring of the UA, (iii) margin to Hazardous or more serious failure conditions.

	(iv) The environmental conditions (including turbulence and gust) for which certification has been requested. (3) The minimum speed allowed by the flight control system must be compatible with the margin specified in USAR.U50 (4) The UA must respond to intentional dynamic manoeuvring within a suitable range of parameter limit. (5) Dynamic characteristics such as damping and overshoot must also be appropriate for the manoeuvre and limit parameter concerned. (6) Characteristics of the flight control system must not result in residual oscillations in commanded output due to combinations of flight envelope protection limits and any other flight control internal limit. (b) When simultaneous envelope protection limits are engaged, adverse coupling or adverse priority must not result. (c) The Applicant must define clearly the borders and prioritization within the control system of the flight envelope protection maintained by the flight control system.
--	---

335	USAR.335 Design Airspeeds The selected airspeeds are equivalent airspeeds (EAS). (a) Design cruising speed, V_C. For V_C, the following apply: (1) V_C shall be defined according to UAS operating requirements. (2) Not applicable. (3) Not applicable. (4) At altitudes where an M_D is established, a cruising speed M_C limited by compressibility may be selected. (b) Design dive speed, V_D. For V_D the following apply: (1) V_D/M_D may not be less than $1.25 V_C/M_C$; however where credit may be claimed from existing experience and / or high speed flight envelope protection, lesser margin values may be considered. (2) Not applicable. (3) Not applicable. (4) Compliance with sub-paragraph (1) of this paragraph need not be shown if V_D/M_D is selected so that the minimum speed margin between V_C/M_C and V_D/M_D is the greater of the following: (i) The speed increase resulting when, from the initial condition of stabilised flight at V_C/M_C, the UA is assumed to be upset, flown for 20 seconds along a flight path 7.5° below the initial path and then pulled up with a load factor of 1.5 (0.5g acceleration increment). At least 75% maximum continuous power for reciprocating engines and maximum cruising thrust for turbines, or, if less, the thrust required for V_C/M_C for both kinds of engines, must be assumed until the pull-up is initiated, at which point power reduction devices may be used; and (ii) Mach 0.05 (at altitudes where M_D is established). (c) Design manoeuvring speed V_A. For V_A, the following applies: (1) V_A may not be less than $V_S \cdot n^{1/2}$ where: (i) V_S is a computed stalling speed with flaps retracted at the design weight, normally based on the maximum UA normal force coefficients, C_{NA}; and (ii) n is the limit manoeuvring load factor used in design and specified at USAR.337 (2) The value of V_A need not exceed the value of V_C used in design. (d) Not applicable.
337	USAR.337 Limit Manoeuvring Load Factors (a) The minimum positive limit manoeuvring load factor n is the minimum of $2.1 + 10900/(W+4536)$ (where W = design maximum take-off weight in kg) or 3.8; (b) The negative limit manoeuvring load factor may not be less than 0.4 times the positive load factor; (c) Manoeuvring load factors lower than those specified in this section may be used if the UAS has design features that make it impossible to intentionally exceed these values in flight.

341	USAR.341 Gust Load Factors See AMC.341 (b) (a) Each UA must be designed to withstand loads on each lifting surface resulting from gusts specified in USAR.333(c). (b) The gust load for a canard or tandem wing configuration must be computed using a rational analysis, or may be computed in accordance with sub-paragraph (c) provided that the resulting net loads are shown to be conservative with respect to gust criteria of USAR.333(c). (c) In the absence of a more rational analysis the gust load factors must be computed as follows: $N = 1 \pm \frac{k g \rho_0 U_{de} V a}{2(W/S)}$ Where: $k g = \frac{0.88 \mu g}{5.3 + \mu g} = \text{gust alleviation factor}$ $\mu g = \frac{2(W/S)}{\rho \bar{C} a g} = \text{UA mass ratio}$ U_{de} = Derived gust velocities referred to in USAR.333 (c) (m/s) ρ_0 = Density of air at sea-level (kg/m³) $\bar{\rho}$ = Density of air (kg/m³) at the altitude considered W/S = Wing loading due to the applicable weight of the aeroplane in the particular load case (N/m²) $\bar{C}$ = Mean geometric chord (m) g = Acceleration due to gravity (m/sec²) V = Aeroplane equivalent speed (m/s) a = Slope of the aeroplane normal force coefficient curve C_{NA} per radian if the gust loads are applied to the wings and horizontal tail surfaces simultaneously by a rational method. The wing lift curve slope C_L per radian may be used when the gust load is applied to the wings only and the horizontal tail gust loads are treated as a separate condition.
343	USAR.343 Design Fuel Loads See AMC.343 (b) (a) The disposable load combinations must include each fuel load in the range from zero fuel to the selected maximum fuel load. The zero fuel load does not include the unusable fuel remaining. (b) If fuel is carried in the wings, the maximum allowable weight of the UA without any fuel in the wing tank(s) must be established as "maximum zero wing fuel weight" if it is less than the maximum weight. (c) Not applicable.

345	USAR.345 High Lift Devices See AMC.345 (d) (a) If flaps or similar high lift devices are to be used for take-off, approach or landing, the UA, with the flaps fully extended at V_F, is assumed to be subjected to symmetrical manoeuvres and gusts within the range determined by: (1) Manoeuvring, to a positive limit load factor of 2.0 or according to those values defined by USAR.337 (c) application if lower; and (2) Positive and negative gust of 7.6 m/s (25 fps) acting normal to the flight path in level flight. (b) V_F must be assumed to be not less than $1.4 V_S$ or $1.8 V_{SF}$, whichever is greater, where V_S is the computed stalling speed with flaps retracted at the design weight; and V_{SF} is the computed stalling speed with flaps fully extended at the design weight. However, if an automatic flap load limiting device is used, the UA may be designed for the critical combinations of airspeed and flap position allowed by that device. (c) In determining external loads on the UA as a whole, thrust, slipstream and pitching acceleration may be assumed to be zero. (d) The flaps, their operating mechanism and their supporting structures, must be designed for the conditions prescribed in sub-paragraph (a) of this paragraph. In addition, with the flaps fully extended at speed V_F the following conditions, taken separately, must be accounted for: (1) A head-on gust having a velocity of 7.6 m/s (25 fps) (EAS), combined with propeller slipstream corresponding to 75% of maximum continuous power or thrust; and (2) The effects of propeller slipstream corresponding to maximum take-off power or thrust.
347	USAR.347 Asymmetrical Flight Conditions The UA is assumed to be subjected to the asymmetrical flight conditions of USAR.349 and USAR.351. Unbalanced aerodynamic moments about the centre of gravity must be reacted in a rational or conservative manner, considering the principal masses furnishing the reacting inertia forces.
349	USAR.349 Rolling Conditions The wing and wing bracing must be designed for the following loading conditions: (a) Asymmetrical wing loads. Unless the following values result in unrealistic loads, the rolling accelerations may be obtained by modifying the symmetrical flight conditions in USAR.333 (d) as follows: in condition A, assume that 100% of the semi-span wing air load acts on one side of the UA and 75% of this load acts on the other side. (b) The loads resulting from the aileron deflections and speeds specified in USAR.455, in combination with a UA load factor of at least two thirds of the positive manoeuvring load factor used for design. Unless the following values result in unrealistic loads, the effect of aileron displacement on wing torsion may be accounted for by adding the following increment to the basic airfoil moment coefficient over the aileron portion of the span in the critical condition determined in USAR.333 (d). $\Delta C_m = -0.01\delta \text{ where --}$ ΔC_m is the moment coefficient increment; and δ is the down aileron deflection in degrees in the critical condition.
351	USAR.351 Yawing Conditions

	The UA must be designed for yawing loads on the vertical surfaces resulting from the loads specified in USAR.441 to USAR.445
361	USAR.361 Engine Torque (a) The mounting arrangement for each engine and its supporting structure must be designed for the effects of: (1) A limit engine torque corresponding to take-off power or thrust and propeller speed acting simultaneously with 75% of the limit loads from flight condition A of USAR.333 (d); (2) A limit engine torque corresponding to maximum continuous power or thrust and propeller speed acting simultaneously with the limit loads from flight condition A of USAR.333 (d); and (3) For turbo-propeller installations, in addition to the conditions specified in sub-paragraphs (a) (1) and (a) (2) of this paragraph, a limit engine torque corresponding to take-off power or thrust and propeller speed, multiplied by a factor accounting for propeller control system malfunction, including quick feathering, acting simultaneously with 1g level flight loads. In the absence of a rational analysis, a factor of 1.6 must be used. (b) For turbine-engine installations, the mounting arrangement for each engine and supporting structure must be designed to withstand each of the following: (1) A limit engine torque load imposed by sudden engine stoppage due to malfunction or structural failure (such as compressor jamming); and (2) A limit engine torque load imposed by the maximum acceleration of the engine. (c) The limit engine torque to be considered under sub-paragraph (a) of this paragraph must be obtained by multiplying the mean torque by a factor of: (1) 1.25 for turbo-propeller installations; (2) 1.33 for engines with five or more cylinders; and (3) Two, three, or four, for engines with four, three or two cylinders, respectively.
363	USAR.363 Sideload On Engine Mount (a) The mounting arrangement for each engine and its supporting structure must be designed for a limit load factor in a lateral direction, for the sideload on the engine mount, of not less than: (1) 1.33; or (2) One-third of the limit load factor for flight condition A. (b) The sideload prescribed in sub-paragraph (a) of this paragraph may be assumed to be independent of other flight conditions.
365	USAR.365 Pressurised Compartment Loads For each pressurised compartment, the following apply: (a) The UA structure must be strong enough to withstand the flight loads combined with pressure differential loads from zero up to the maximum relief valve setting. (b) The external pressure distribution in flight and any stress concentrations, must be accounted for. (c) If landings may be made, with the compartment pressurised, landing loads must be combined with pressure differential loads from zero up to the maximum allowed during landing.

	(d) The UA structure must be strong enough to withstand the pressure differential loads corresponding to the maximum relief valve setting multiplied by a factor of 1.33 omitting other loads. (e) If a pressurised compartment has two or more compartments, separated by bulkheads or a floor, the primary structure must be designed for the effects of sudden release of pressure in any compartment with external opening. This condition must be investigated for the effects of failure of the largest opening in the compartment. The effects of intercompartmental venting may be considered.
367	USAR.367 Asymmetrical Loads Due to Engine Failure (a) The UA must be designed for the asymmetrical loads resulting from the failure of the critical engine. Turbopropeller UA must be designed for the asymmetrical loads resulting from the failure of the critical engine including the following conditions in combination with a single malfunction of the propeller drag limiting system: (1) At speeds between V_{MC} and V_D, the loads resulting from power failure because of fuel flow interruption are considered to be limit loads; (2) At speeds between V_{MC} and V_C, the loads resulting from the disconnection of the engine compressor from the turbine or from loss of the turbine blades are considered to be ultimate loads; (3) The time history of the thrust decay and drag build-up occurring as a result of the prescribed engine failures must be substantiated by test or other data applicable to the particular engine-propeller combination; and (4) Not applicable. (see USAR.367 (c)) (b) Not applicable. (c) The timing and magnitude of the corrective action performed automatically by the flight control system must be conservatively estimated, considering the characteristics of the particular engine-propeller-UA combination.
369	USAR.369 Rear Lift Truss Loads (a) If a rear lift truss is used, it must be designed for conditions of reversed airflow at a design speed of; $V = 8.7 \sqrt{\frac{W}{S}} + 8.7 \text{ (knots)}$ where W/S = wing loading at design maximum take-off weight (lbs/ft²). (b) Either aerodynamic data for the particular wing section used, or a value of C_L equalling 0.8 with a chordwise distribution that is triangular between a peak at the trailing edge and zero at the leading edge, must be used.

371	USAR.371 Gyroscopic and Aerodynamic Loads See AMC.371 The mounting arrangement for each engine and its supporting structure must be designed for the gyroscopic, inertia and aerodynamic loads that result, with the engine(s) and propeller(s), if applicable at maximum continuous rpm, under either: (a) The conditions prescribed in USAR.351 and USAR.423; or (b) All possible combinations of the following in the limits allowed by the flight control system: (1) a yaw velocity of 150% of the maximum predicted yaw rotational velocity within the flight envelope maintained by the flight control system; (2) a pitch velocity of 150% of the maximum predicted pitch rotational velocity within the flight envelope maintained by the flight control system; (3) a normal load factor of 150% of the maximum predicted load factor within the flight envelope maintained by the flight control system; and (4) Maximum continuous thrust.
373	USAR.373 Speed Control Devices If speed control devices (such as spoilers and drag flaps) are incorporated for use in en-route conditions: (a) The UAS must be designed for the symmetrical manoeuvres and gusts prescribed in USAR.333, USAR.337 and USAR.341 and the yawing manoeuvres and lateral gusts in USAR.441 and USAR.443, with the device extended at speeds up to the maximum operational speed for that device; and (b) If the device has automatic operating or load limiting features, the UA must be designed for the manoeuvre and gust conditions prescribed in sub-paragraph (a) of this paragraph at the speeds and corresponding device positions that the mechanism allows.

CONTROL SURFACE AND SYSTEM LOADS

391	USAR.391 Control Surface Loads The control surface loads specified in USAR.397 to USAR.459 are assumed to occur in the conditions described in USAR.331 to USAR.351
393	USAR.393 Loads Parallel to Hinge Line See AMC.393 (a) and AMC.393 (b) (a) Control surfaces and supporting hinge brackets must be designed to withstand inertia loads acting parallel to the hinge line. (b) In the absence of more rational data, the inertia loads may be assumed to be equal to KW, where: (1) K = 24 for vertical surfaces; (2) K = 12 for horizontal surfaces; and (3) W = weight of the movable surfaces.
395	USAR.395 Control System Loads (a) Each flight control system and its supporting structure must be designed for loads corresponding to at least 125% of the computed hinge moments of the movable control surface in the conditions prescribed in USAR.391 to USAR.459. In addition, the following apply:

	(1) The system limit loads need not exceed the higher of the loads that can be sustained by the servocontrols or actuators. (2) The design must, in any case, provide a rugged system for service use, considering jamming, ground gusts, taxiing downwind (if the UA is designed to taxi), control inertia and friction. (b) A 125% factor on computed hinge movements must be used to design elevator, aileron and rudder systems. However, a factor as low as 1.0 may be used if hinge moments are based on accurate flight test data, the exact reduction depending upon the accuracy and reliability of the data. (c) Forces occurring from the actuating system are assumed to act at the appropriate attachments of the control system to the control surface horns.
397	USAR.397 Limit Control Forces and Torques (a) In the control surface flight loading condition, the air loads on movable surfaces and the corresponding deflections need not exceed those that would result in flight from the application of any force occurring from the actuating system within the range specified in USAR.397 (b). (b) The control system must be able to bear the maximum loads and torques generated by the actuating system.
	399 Dual control system Not applicable.
405	USAR.405 Secondary Flight Control Secondary controls, (i.e. all flight controls other than primary flight controls (see definition of primary flight controls in USAR.673(a)), such as wheel brakes, spoilers and tab controls, must be designed for the greater of limit load or the maximum forces that the actuating system is likely to apply to those controls.
407	USAR.407 Trim Tab Effects The effects of trim tabs on the control surface design conditions must be accounted for only where the surface loads are limited by maximum effort of the actuating system. In these cases, the tabs are considered to be deflected in the direction that would assist the system. These deflections must correspond to the maximum degree of "out of trim" expected at the speed for the condition under consideration.
409	USAR.409 Tabs Control surface tabs must be designed for the most severe combination of airspeed and tab deflection likely to be obtained within the flight envelope protection for any usable loading condition.
415	USAR.415 Ground Gust Conditions (a) The control system must be investigated as follows for control surface loads due to ground gusts and taxiing downwind (if the UA is designed to taxi): (1) If an investigation of the control system for ground gust loads is not required by subparagraph (2) of this paragraph, but the Applicant elects to design a part of the control system for these loads, these loads need only be carried from control surface horns through the nearest stops or gust locks and their supporting structures. (2) The effects of surface loads due to ground gusts and taxiing downwind must be investigated for the entire control system according to the formula: $H = K c S q$ where H = limit hinge moment (Nm);

c = mean chord of the control surface aft of the hinge line (m);

S = area of control surface aft of the Hinge line (m²);

q = dynamic pressure (psf) based on a design speed not less than

$14.6 \sqrt{\frac{W}{S}} + 14.6 (fps)$ (where W/S =wing loading at design maximum weight (lbs/ft²))

except that the design speed need not exceed 26.8 m/s (88 fps); and

K = limit hinge moment factor for ground gusts derived in subparagraph (b) of this paragraph. (For ailerons and elevators, a positive value of K indicates a moment tending to depress the surface and a negative value of K indicates a moment tending to raise the surface.

(b) The limit hinge moment factor K for ground gusts must be derived as follows:

Surface	K	Position of controls
(a) Aileron	0.75	Mid-position locked
(b) Aileron	± 0.50	Ailerons at full throw: + moment on one aileron, - moment on the other
(c) } Elevator	± 0.75	(c) Elevator full up (-)
(d) }		(d) Elevator full down (+)
(e) } Rudder	± 0.75	(e) Rudder in neutral
(f) }		(f) Rudder at full throw

(c) At all weights between the Empty Weight and the maximum weight declared for tie-down stated in the appropriate Manual, any declared tie-down points and surrounding structure, control system, surfaces and associated gust locks must be designed to withstand the limit load conditions arising when tied-down, resulting from wind speeds of up to 120 km/h (65 knots) horizontally from any direction.

HORIZONTAL TAIL SURFACES

421 USAR.421 Balancing Loads

(a) A horizontal surface balancing load is a load necessary to maintain equilibrium in any specified flight condition with no pitching acceleration.

(b) Horizontal balancing surfaces must be designed for the balancing loads occurring at any point on the limit manoeuvring envelope and in the flap conditions specified in USAR.345

423 USAR.423 Manoeuvring Loads

Each horizontal surface and its supporting structure, and the main wing of a canard or tandem wing configuration, if that surface has pitch control, must be designed for manoeuvring loads imposed by the following conditions:

(a) A sudden movement of the pitching control, at the speed V_A , as limited by the control stops, the maximum rate allowed by the FCS, or, maximum force of the servocontrols or actuators, whichever is critical, to:

- (1) the maximum aft movement, and
- (2) the maximum forward movement.

(b) A sudden aft movement of the pitching control at speeds above V_A , followed by a forward movement of the pitching control, as limited by the control stops, the maximum rate allowed by the FCS, or, maximum force of the servocontrols or actuators, whichever is critical, resulting in the following combinations of normal and angular acceleration:

	Condition Normal acceleration (n) Angular acceleration (radian/sec²) Nose-up pitching 1.0 $+\frac{72.2}{V_{/km/h}} n_m (n_m - 1.5) \left(+\frac{39}{V_{/kt}} n_m (n_m - 1.5) \right)$ Nose-down pitching n_m $-\frac{72.2}{V_{/km/h}} n_m (n_m - 1.5) \left(-\frac{39}{V_{/kt}} n_m (n_m - 1.5) \right)$ Where: n_m = positive limit manoeuvring load factor used in the design of the UA; and $V_{/km/h}$ = initial speed in km/h. The conditions in this paragraph involve loads corresponding to the loads that may occur in a "checked manoeuvre" (a manoeuvre in which the pitching control is suddenly displaced in one direction and then suddenly moved in the opposite direction). The deflections and timing of the "checked manoeuvre" must avoid exceeding the limit manoeuvring load factor. The total horizontal surface load for both nose-up and nose-down pitching conditions should be based on the actual control surface movement in response to the pitch demand and is the sum of the balancing loads at V and the specified value of the normal load factor n, plus the manoeuvring load increment due to the specified value of the angular acceleration. (c) A movement of the pitching control to cause a transition from steady level flight at a speed within the flight envelope protection maintained by the flight control system or on the boundary of the flight envelope as specified in USAR.333 to the appropriate extreme steady normal acceleration condition.
425	USAR.425 Gust Loads (a) Each horizontal surface other than a main wing, must be designed for loads resulting from: (1) Gust velocities specified in USAR.333 (c) with flaps retracted; and (2) Positive and negative gusts of 7.6 m/s (25 fps) nominal intensity at V_F corresponding to the flight conditions specified in USAR.345(a)(2). (b) Reserved. (c) When determining the total load on the horizontal surfaces for the conditions specified in subparagraph (a) of this paragraph, the initial balancing loads for steady unaccelerated flight at the pertinent design speeds, V_F, V_C and V_D must first be determined. The incremental load resulting from the gusts must be added to the initial balancing load to obtain the total load. (d) In the absence of a more rational analysis, the incremental load due to the gust must be computed as follows only on UA configurations with aft-mounted, horizontal surfaces, unless its use elsewhere is shown to be conservative: $\Delta L_{ht} = \frac{\rho_0 K_g U_{de} V a_{ht} S_{ht}}{2} \left[1 - \frac{d_\varepsilon}{d_\alpha} \right]$ Where: ΔL_{ht} = Incremental horizontal tail load (N); ρ_0 - Density of air at sea level (kg/m³);

	U_{de} = Derived gust velocity (m/s); V = Aeroplane equivalent speed (m/s); a_{ht} = Slope of aft horizontal tail lift curve (per radian); S_{ht} = Area of aft horizontal tail (m²); $\left[1 - \frac{d\varepsilon}{d\alpha}\right]$ = Downwash factor; and K_g = Gust alleviation factor as defined by: $K_g = \frac{0.88\mu g}{5.3 + \mu g} \text{ and}$ $\mu g = \frac{2(W/S)}{\rho \bar{C} a g}$ where: ρ = Density of air (kg/m³) at the altitude considered. W/S = Wing loading due to the applicable weight of the aeroplane in the particular load case (N/m²); $\bar{C}$ = Mean geometric chord (m); g = Acceleration due to gravity (m/sec²); a = Slope of the aeroplane normal force coefficient curve CNA per radian if the gust loads are applied to the wings and horizontal tail surfaces simultaneously by a rational method. The wing lift curve slope CL per radian may be used when the gust load is applied to the wings only and the horizontal tail gust loads are treated as a separate condition. μg = Aeroplane mass ratio.
427	USAR.427 Asymmetric Loads (a) Horizontal surfaces other than main wing and their supporting structure must be designed for asymmetric loads arising from yawing and slipstream effects, in combination with the loads prescribed for the flight conditions set forth in USAR.421 to USAR.425 (b) In the absence of more rational data for UA that are conventional in regard to location of engines, wings, horizontal surfaces other than main wing, and fuselage shape: (1) 100% of the maximum loading from the symmetrical flight conditions may be assumed on the surface on one side of the plane of symmetry; and (2) The following percentage of that loading must be applied to the opposite side: % = 100-10(n-1), where n is the specified positive manoeuvring load factor, but this value may not be more than 80%. (c) For UAS that are not conventional (such as UA with horizontal surfaces other than main wing having appreciable dihedral or supported by the vertical tail surfaces) the surfaces and supporting structures must be designed for combined vertical and horizontal surface loads resulting from each prescribed flight condition taken separately.
VERTICAL SURFACES	
441	USAR.441 Manoeuvring loads

	See AMC.441 (a) At speeds up to V_A the vertical surfaces must be designed to withstand the following conditions. In computing the loads, the yawing velocity may be assumed to be zero: (1) With the UA in unaccelerated flight at zero yaw, it is assumed that the rudder control is suddenly displaced to the maximum deflection, as limited by the control stops or by the maximum effort provided by the servocontrols or actuators. (2) With the rudder deflected as specified in sub-paragraph (1), it is assumed that the UA yaws to the overswing sideslip angle. In lieu of a rational analysis, an overswing angle equal to 1.5 times the static sideslip angle of sub-paragraph (3) may be assumed. (3) A yaw angle of 15° with the rudder control maintained in the neutral position (except as limited by the maximum effort sustained by the servocontrols or actuators). (b) The loads imposed by the following additional manoeuvre must be substantiated at speeds from V_A to V_D/M_D. When computing the vertical surface loads: (1) The UA must be yawed to the largest attainable steady state sideslip angle with the rudder at maximum deflection caused by any one of the following: (i) Control surface stops; (ii) Maximum available effort sustained by the servocontrols or actuators; (iii) Not applicable. (2) The rudder must be suddenly displaced from the maximum deflection to the neutral position. (c) The yaw angles specified in sub-paragraph (a) (3) may be reduced if the yaw angle chosen for a particular speed cannot be exceeded in: (1) Steady slip conditions; (2) Uncoordinated rolls from steep banks; or (3) Sudden failure of the critical engine with delayed corrective action.
443	USAR.443 Gust Loads See AMC.443 (a) Vertical surfaces must be designed to withstand, in unaccelerated flight at speed V_C, lateral gusts of the values prescribed for V_C in USAR.333 (c). (b) Not applicable. (c) In the absence of a more rational analysis, the gust load must be computed as follows: $\Delta L_{vt} = \frac{\rho \phi K_{gt} U_{de} V_{avn} S_{vt}}{2}$ where:

	ΔL_{vt} = Vertical surface loads (N); $K_{gt} = \frac{0.88\mu gt}{5.3 + \mu gt}$ = gust alleviation factor; $\mu gt = \frac{2W}{\rho \bar{C}_t g a_{vt} S_{vt}} \left(\frac{K}{l_{vt}} \right)^2$ = lateral mass ratio; ρ_o = Density of air at sea – level (kg/m^3); U_{de} = Derived gust velocity (m/s); ρ = Air density (Kg/m^3); W = the applicable weight of the aeroplane in the particular load case (N); S_{vt} = Area of vertical surface (m^2); $\bar{C}_t$ = Mean geometric chord of vertical surface (m); a_{vt} = Lift curve slope of vertical surface (per radian); K = Radius of gyration in yaw (m); l_{vt} = Distance from aeroplane c.g. to lift centre of vertical surface (m); g = Acceleration due to gravity (m/sec^2); and V = Aeroplane equivalent speed (m/s);
445	USAR.445 Outboard fins or winglets (a) If outboard fins or winglets are included on the horizontal surfaces or wings, the horizontal surfaces or wings must be designed for their maximum load in combination with loads induced by the fins or winglets and moment or forces exerted on horizontal surfaces or wings by the fins or winglets. (b) If outboard fins or winglets extend above and below the horizontal surface, the critical vertical surface loading (the load per unit area as determined under USAR.441 and USAR.443) must be applied to: (1) The part of the vertical surfaces above the horizontal surface with 80% of that loading applied to the part below the horizontal surface; and (2) The part of the vertical surfaces below the horizontal surface with 80% of that loading applied to the part above the horizontal surface; (c) The endplate effects of outboard fins or winglets must be taken into account in applying the yawing conditions of USAR.441 and USAR.443 to the vertical surfaces in sub-paragraph (b). (d) When rational methods are used for computing loads, the manoeuvring loads of USAR.441 on the vertical surfaces and the one-g horizontal surface load, including induced loads on the horizontal surface and moments or forces exerted on the horizontal surfaces by the vertical surfaces, must be applied simultaneously for the structural loading condition.
AILERONS AND SPECIAL DEVICES	
455	USAR.455 Ailerons The ailerons must be designed for the loads to which they are subjected, allowance may be made for the capability of the flight control system (such as rate of movement or limitations on deflection). (a) In the neutral position during symmetrical flight conditions; and (b) By the following deflections, except as limited by the maximum force provided by the servocontrols or actuators, during asymmetrical flight conditions: (1) Sudden maximum displacement of the aileron control at V_A. Suitable allowance may be made for control system deflections.

	(2) Sufficient deflection at V_C, where V_C is more than V_A, to produce a rate of roll not less than obtained in sub-paragraph (b)(i). (3) Sufficient deflection at V_D to produce a rate of roll not less than one third of that obtained in sub-paragraph (b)(i).
459	USAR.459 Special devices The loading for special devices using aerodynamic surfaces (such as slats and spoilers) must be determined from test data or using proven or acceptable design procedures.

GROUND LOADS

471	USAR.471 General See AMC.471 (a) The limit ground loads specified in this subpart are considered to be external loads and inertia forces that act upon a UA structure. In each specified ground load condition, the external reactions must be placed in equilibrium with the linear and angular inertia forces in a rational or conservative manner. (b) The requirements described in USAR.473 to USAR.499 are not applicable to UAS designed to be recovered by parachute.
473	USAR.473 Ground load conditions and assumptions (a) The ground load requirements of this subpart must be complied with at the design maximum weight except that USAR.479, USAR.481 and USAR.483 may be complied with at a design landing weight (the highest weight for landing conditions at the maximum descent velocity) allowed under sub-paragraphs (b) and (c). (b) The design landing weight may be as low as: (1) 95% of the maximum weight if the minimum fuel capacity is enough for at least one-half hour of operation at maximum continuous power or thrust plus a capacity equal to a fuel weight which is the difference between the design maximum weight and the design landing weight; or (2) The design maximum weight less the weight of 25% of the total fuel capacity. (c) The design landing weight of a multi-engine UA may be less than that allowed under sub-paragraph (b) if: (1) The UA meets the one engine-inoperative climb requirements of USAR.67; and (2) Compliance is shown with the fuel jettisoning system requirements of USAR.1001. (d) The selected limit vertical inertia load factor at the centre of gravity of the UA for the ground load conditions prescribed in this subpart may not be less than that which would be obtained when landing with a descent velocity (V), in meters per second (feet per second), equal to $4.4 (W/S)^{1/4}$ (with W in lbs and S in ft^2), except that this velocity need not be more than 3.0 m (10 ft) per second and may not be less than 2.1 m (7 ft) per second. These prescribed descent velocities may be modified if it is shown that the UAS has design features that make it impossible to develop these velocities or it can be shown that alternate safe values may be chosen based upon previous experience. (e) Wing lift not exceeding two-thirds of the weight of the UA may be assumed to exist throughout the landing impact and to act through the centre of gravity. The ground reaction load factor may be equal to the inertia load factor minus the ratio of the above assumed wing lift to the UA weight. (f) If energy absorption tests are made to determine the limit load factor corresponding to the required limit descent velocities, these tests must be made under USAR.723 (a).

	(g) No inertia load factor used for design purposes may be less than 2.67, nor may the limit ground reaction load factor be less than 2.0 at design maximum weight, unless these lower values will not be exceeded in taxiing at speeds up to take-off speed over terrain as rough as that expected in service, as specified in USAR.235.
477	USAR.477 Landing gear arrangement USAR.479 to USAR.483, or the conditions in Appendix C, apply to UA with conventional wheeled / skids arrangements of main and nose gear, or main and tail gear.
479	USAR.479 Level landing conditions (a) For a level landing, the UA is assumed to be in the following attitudes: (1) For UA with tail wheels/skids, a normal level flight attitude; (2) For UA with nose wheels, attitudes in which: (i) The nose and main wheels contact the ground simultaneously; and (ii) The main wheels contact the ground and the nose wheel is just clear of the ground. The attitude used in subdivision (i) of this sub-paragraph may be used in the analysis required under subdivision (ii) of this sub-paragraph. (b) When investigating landing conditions, the drag components simulating the forces required to accelerate the tyres and wheels up to the landing speed (spin-up) must be properly combined with the corresponding instantaneous vertical ground reactions, and the forward-acting horizontal loads resulting from rapid reduction of the spin-up drag loads (spring-back) must be combined with vertical ground reactions at the instant of the peak forward load, assuming wing lift and a tyre sliding coefficient of friction of 0.8. However, the drag loads may not be less than 25% of the maximum vertical ground reaction (neglecting wing lift). (c) In the absence of specific tests or a more rational analysis for determining the wheel spinup and spring-back loads for landing conditions, the method set forth in Appendix D must be used. If Appendix D is used, the drag components used for design must not be less than those given by Appendix C. (d) For UA with tip tanks or large overhung masses (such as turbo-propeller or jet engines) supported by the wing, the tip tanks and the structure supporting the tanks or overhung masses must be designed for the effects of dynamic responses under the level landing conditions of either sub-paragraph (a) (1) or (a) (2) (ii). In evaluating the effects of dynamic response, a UA lift equal to the weight of the UA may be assumed. The most unfavourable configuration has to be retained for the application of this paragraph.
481	USAR.481 Tail down landing conditions See AMC.481 (a) For a tail down landing, the UA is assumed to be in the following attitudes: (1) For UA with tail wheels, an attitude in which the main and tail wheels contact the ground simultaneously. (2) For UA with nose wheels, a stalling attitude (or maximum allowed by the flight envelope protection system maintained by the flight control system), or the maximum angle allowing ground clearance by each part of the UA, whichever is less. (b) For UA with either tail or nose wheels / skids, ground reactions are assumed to be vertical, with the wheels up to speed before the maximum vertical load is attained. (c) For design of tail skid and affected structure and empennage including balancing weight attachment, the tail skid load in a tail down landing (main landing gear free from ground) must be calculated as follows:

	$P = 4mg \frac{i_y^2}{i_y^2 + L^2}$ where: P = tail skid load (N) m = mass of the UA (kg) g = acceleration of gravity (m/s²) i_y = radius of gyration of the UA (m) L = distance between tail skid and UA c.g. (m)
483	USAR.483 One-wheel landing conditions For the one-wheel landing condition, the UA is assumed to be in the level attitude and to contact the ground on one side of the main landing gear. In this attitude, the ground reactions must be the same as those obtained on that side under USAR.479.
485	USAR.485 Sideload conditions (a) For the sideload condition, the UA is assumed to be in a level attitude with only the main wheels contacting the ground and with the shock absorbers and tyres in their static positions. (b) The limit vertical load factor must be 1.33, with the vertical ground reaction divided equally between the main wheels. This value can be reduced to 1.2 if UA operations are restricted to paved taxiways and runways only (subject to statement in the UAS Flight Manual). (c) The limit side inertia factor must be 0.83, with the side ground reaction divided between the main wheels so that: (1) 0.5 (W) is acting inboard on one side; and (2) 0.33 (W) is acting outboard on the other side. (d) The side loads prescribed in sub-paragraph (c) are assumed to be applied at the ground contact point and the drag loads may be assumed to be zero.
493	USAR.493 Braked roll conditions Under braked roll conditions, with the shock absorbers and tyres in their static positions, the following applies: (a) The limit vertical load factor must be 1.33 (b) The attitudes and ground contacts must be those described USAR.479 for level landings. (c) A drag reaction equal to the vertical reaction at the wheel multiplied by a coefficient of friction of 0.8 must be applied at the ground contact point of each wheel with brakes, except that the drag reaction need not exceed the maximum value based on limiting brake torque.
497	USAR.497 Supplementary conditions for tail wheels In determining the ground loads on the tail wheel and affected supporting structures, the following applies: (a) For the obstruction load, the limit ground reaction obtained in the tail down landing condition is assumed to act up and aft through the axle at 45°. The shock absorber and tyre may be assumed to be in their static positions.

	(b) For the sideload, a limit vertical ground reaction equal to the static load on the tail wheel, in combination with a side component of equal magnitude, is assumed. In addition: (1) If a swivel is used, the tail wheel is assumed to be swivelled 90° to the UA longitudinal axis with the resultant ground load passing through the axle; (2) If a lock, steering device, or shimmy damper is used, the tail wheel is also assumed to be in the trailing position with the sideload acting at the ground contact point; and (3) The shock absorber and tyre are assumed to be in their static positions. (c) If a tail wheel, bumper, or an energy absorption device is provided to show compliance with USAR.925 (b), the following applies: (1) Suitable design loads must be established for the tail wheel, bumper, or energy absorption device; and (2) The supporting structure of the tail wheel, bumper, or energy absorption device must be designed to withstand the loads established in sub-paragraph (c) (1). (d) In determining the ground loads on the tail skid and affected supporting structures, the following applies: (1) Except as provided in (d)(2), if the c.g. of the unloaded UA – in side view – is situated behind the ground contact area of the main landing gear, the rear portion of the fuselage, the tail skid and the empennage must be designed to withstand the loads arising when the tail landing skid is raised to its highest possible position, consistent with the main wheel remaining on the ground, and is then released and allowed to fall freely. (2) If the c.g. in all loading conditions is situated behind the ground contact area of the main landing gear (d)(1) need not be applied.
499	USAR.499 Supplementary conditions for nose wheels In determining the ground loads on nose wheels and affected supporting structures and assuming that the shock absorbers and tyres are in their static positions, the following conditions must be met: (a) For aft loads, the limit force components at the axle must be: (1) A vertical component of 2.25 times the static load on the wheel; and (2) A drag component of 0.8 times the vertical load. (b) For forward loads, the limit force components at the axle must be: (1) A vertical component of 2.25 times the static load on the wheel; and (2) A forward component of 0.4 times the vertical load. (c) For sideloads, the limit force components at ground contact must be: (1) A vertical component of 2.25 times the static load on the wheel; and (2) A side component of 0.7 times the vertical load. (d) For UA with a steerable nose wheel which is controlled by hydraulic or other power, at maximum weight with the nose wheel in any steerable position the application of 1.33 times the full steering torque combined with a vertical reaction equal to 1.33 times the maximum static reaction on the nose gear must be assumed. However, if a torque limiting device is installed, the steering torque can be reduced to the maximum value allowed by that device.

	(e) Not applicable.													
	505 Supplementary conditions for ski-planes Not applicable.													
507	USAR.507 Jacking and lifting loads (a) The UA must be designed for the loads developed when it is supported on jacks lifting equipment or by appropriate number of authorized ground staff at the design maximum weight assuming the following load factors for landing gear jacking points or lifting points at a three-point attitude and for primary flight structure jacking or lifting points in the level attitude. (1) Vertical load factor of 1.35 times the static reactions. (2) Fore, aft and lateral load factors of 0.4 times the vertical static reactions. (b) The horizontal loads at the jack or lift points must be reacted by inertia forces so as to result in no change in the direction of the resultant loads at the jack or lift points. (c) The horizontal loads must be considered in all combinations with the vertical load.													
509	USAR.509 Towing loads For UA designed to be towed, the towing loads must be applied to the design of tow fittings and their immediate attaching structure. (a) The towing loads specified in sub-paragraph (d) must be considered separately. These loads must be applied at the towing fittings and must act parallel to the ground. In addition: (1) A vertical load factor equal to 1.0 must be considered acting at the centre of gravity; and (2) The shock struts and tyres must be in their static positions. (b) For towing points not on the landing gear but near the plane of symmetry of the UA, the drag and side tow load components specified for the auxiliary gear apply. For towing points located outboard of the main gear, the drag and side tow load components specified for the main gear apply. Where the specified angle of swivel cannot be reached, the maximum obtainable angle must be used. (c) The towing loads specified in sub-paragraph (d) must be reacted as follows: (1) The side component of the towing load at the main gear must be reacted by a side force at the static ground line of the wheel to which the load is applied. (2) The towing loads at the auxiliary gear and the drag components of the towing loads at the main gear must be reacted as follows: (i) A reaction with a maximum value equal to the vertical reaction must be applied at the axle of the wheel to which the load is applied. Enough UA inertia to achieve equilibrium must be applied. (ii) The loads must be reacted by UA inertia. (d) The prescribed towing loads are as follows, where W is the design maximum weight: <table><tr><th rowspan="2">Tow point</th><th rowspan="2">Position</th><th colspan="3">Load</th></tr><tr><th>Magnitude</th><th>No.</th><th>Direction</th></tr><tr><td>Main Gear</td><td></td><td>0.225 W per main gear unit</td><td>1</td><td>Forward, parallel to drag axis</td></tr></table>	Tow point	Position	Load			Magnitude	No.	Direction	Main Gear		0.225 W per main gear unit	1	Forward, parallel to drag axis
Tow point	Position			Load										
		Magnitude	No.	Direction										
Main Gear		0.225 W per main gear unit	1	Forward, parallel to drag axis										

					2	Forward, at 30° to drag axis	
					3	Aft, parallel to drag axis	
					4	Aft, at 30° to drag axis	
		Auxiliary Gear	Swivelled forward	0.30W	5	Forward	
					6	Aft	
			Swivelled aft	0.30W	7	Forward	
					8	Aft	
			Swivelled 45° from forward	0.15W	9	Forward, in plane of wheel	
					10	Aft, in plane of wheel	
			Swivelled 45° from aft	0.15W	11	Forward, in plane of wheel	
					12	Aft, in plane of wheel	
511	USAR.511 Ground load; asymmetrical loads on multiple-wheel units (a) Pivoting loads. The UA is assumed to pivot about one side of the main gear with: (1) The brakes on the pivoting unit locked; and (2) Loads corresponding to a limit vertical load factor of 1 and coefficient of friction of 0.8, applied to the main gear and its supporting structure. (b) Unequal tyre loads. The loads established under USAR.471 to USAR.483 must be applied in turn, in a 60/40% distribution, to the dual wheels and tyres in each dual wheel landing gear unit. (c) Deflated tyre loads. For the deflated tyre condition: (1) 60% of the loads established under USAR.471 to USAR.483 must be applied in turn to each wheel in a landing gear unit; and (2) 60% of the limit drag and sideloads and 100% of the limit vertical load established under USAR.485 and USAR.493 or lesser vertical load obtained under sub-paragraph (1), must be applied in turn to each wheel in the dual wheel landing gear unit.						
	521 Water load conditions Not applicable.						
	523 Design weights and centre of gravity positions Not applicable.						
	525 Application of loads Not applicable.						
	527 Hull and main float load factors Not applicable.						
	529 Hull and main float landing conditions Not applicable.						
	531 Hull and main float landing conditions Not applicable.						
	533 Hull and main float bottom pressures Not applicable.						
	535 Auxiliary float loads						

	Not applicable.
537 Seawing loads	Not applicable.
561 Emergency landing conditions – General	Not applicable.
562 Emergency landing dynamic conditions	Not applicable.

FATIGUE EVALUATION

U570	USAR.U570 General See AMC.U570, AMC.603 (a) and AMC 20-29 (a) Fatigue and damage tolerance demonstration shall depend on the predicted life time, usage spectrum (see USAR.U17) and maintenance philosophy of the UAS for which certification is requested. (b) For UAS for which certification is requested for very short life time (less than 5 cycles) (1) static strength demonstration using ultimate loads, see USAR.305(b), is required, and (2) USAR.572 to USAR.575 may not be applicable however this must be agreed with the Certifying Authority. (c) For other UAS, fatigue and damage tolerance analysis must be carried out. (1) The demonstration level and exhaustivity shall be adapted to predicted life time using USAR.572 to USAR.575 (2) Consideration shall be given to fatigue monitoring, in agreement with the Certifying Authority. (d) The requirements of USAR.U570 are applicable for either metallic or composite structures.
571	Metallic pressurised cabin Structures Not applicable.
572	USAR.572 Metallic fuselage, wing, empennage and associated structures See AMC.572 (a) The strength, detail design, and fabrication of those parts of the airframe structure whose failure would be <i>Catastrophic</i> must be evaluated using a rational analysis of the intended use, construction method and maintenance philosophy of the aircraft under one or more of the following unless it is shown that the structure, operating stress level, materials and expected uses are comparable, from a fatigue standpoint, to a similar design that has had extensive satisfactory service experience. Neither Safe Life nor Fail Safe strength investigation is sufficient by itself as a criteria to assure structural integrity under repeated loads unless approved by the Certifying Authority. (1) A safe life approach supported by a fatigue strength investigation in which the structure is shown by tests, or by analysis supported by test evidence, to be able to withstand the repeated loads of variable magnitude expected in service; or (2) A fail-safe strength investigation in which it is shown by analysis, tests, or both, that Catastrophic failure of the structure is not probable after fatigue failure, or obvious partial failure, of a principal structural element, and that the remaining structure is able to withstand a static ultimate load factor of 75% of the critical limit load factor at Vc. These loads must be multiplied

	by a factor of 1.15 unless the dynamic effects of failure under static load are otherwise considered. (3) The damage tolerance evaluation of USAR.573(b). (b) Each evaluation required by this paragraph must: (1) Include typical loading spectra (e.g. taxi if the UA is designed to, ground-air-ground cycles, manoeuvre, gust); (2) Account for any significant effects due to the mutual influence of aerodynamic surfaces; and (3) Consider any significant effects from propeller slipstream loading, and buffet from vortex impingements.
573	USAR.573 Damage tolerance and fatigue evaluation of structure See AMC.573 (a) (1) & (3), and AMC.573 (b) (a) Composite airframe structure. Composite airframe structure must be evaluated under this paragraph instead of USAR.572 The composite airframe structure, the failure of which would result in Catastrophic loss of the UA, in each wing (including canards, tandem wings, and winglets), empennage, their carrythrough and attaching structure, moveable control surfaces and their attaching structure, fuselage, and pressurised compartments must be evaluated using the damage-tolerance criteria prescribed in sub-paragraphs (a)(1) through (a)(4) unless shown to be impractical. If the Applicant establishes that damage-tolerance criteria is impractical for a particular structure, the structure must be evaluated in accordance with sub-paragraphs (a)(1) and (a)(6) for safe-life designed UA. Where bonded joints are used, the structure must also be evaluated in accordance with sub-paragraph (a)(5). The effects of material variability and environmental conditions on the strength and durability properties of the composite materials must be accounted for in the evaluations required by this paragraph. (1) It must be demonstrated by tests, or by analysis supported by tests, that the structure is capable of carrying ultimate load with damage up to the threshold of detectability considering the inspection procedures employed. (2) The growth rate or no-growth of damage that may occur from fatigue, corrosion, manufacturing flaws or impact damage, under repeated loads expected in service, must be established by tests or analysis supported by tests. (3) The structure must be shown by residual strength tests, or analysis supported by residual strength tests, to be able to withstand critical limit flight loads, considered as ultimate loads, with the extent of detectable damage consistent with the results of the damage tolerance evaluations. For pressurised compartment, the following loads must be withstood: (i) Critical limit flight loads with the combined effects of normal operating pressure and expected external aerodynamic pressures. (ii) The expected external aerodynamic pressures in 1g flight combined with a compartment differential pressure equal to 1.1 times the normal operating differential pressure without any other load. (4) The damage growth, between initial detectability and the value selected for residual strength demonstrations, factored to obtain inspection intervals, must allow development of an inspection program suitable for application by operation and maintenance personnel. (5) For any bonded joint, the failure of which would result in <i>Catastrophic</i> loss of the UA, the limit load capacity must be substantiated by one of the following methods:

	(i) The maximum disbonds of each bonded joint consistent with the capability to withstand the loads in sub-paragraph (a)(3) must be determined by analysis, test, or both. Disbonds of each bonded joint greater than this must be prevented by design features; or (ii) Proof testing must be conducted on each production article that will apply the critical limit design load to each critical bonded joint; or (iii) Repeatable and reliable non-destructive inspection techniques must be established that ensure the strength of each joint. (6) Structural components for which the damage tolerance method is shown to be impractical must be shown by component fatigue tests, or analysis supported by tests, to be able to withstand the repeated loads of variable magnitude expected in service. Sufficient component, subcomponent, element, or coupon tests must be done to establish the fatigue scatter factor and the environmental effects. Damage up to the threshold of detectability and ultimate load residual strength capability must be considered in the demonstration. (b) Metallic airframe structure. If the Applicant elects to use USAR.572(a)(3), then the damage tolerance evaluation must include a determination of the probable locations and modes of damage due to fatigue, corrosion, or accidental damage. The determination must be by analysis supported by test evidence and, if available, service experience. Damage at multiple sites due to fatigue must be included where the design is such that this type of damage can be expected to occur. The evaluation must incorporate repeated load and static analyses supported by test evidence. The extent of damage for residual strength evaluation at any time within the operational life of the UA must be consistent with the initial detectability and subsequent growth under repeated loads. The residual strength evaluation must show that the remaining structure is able to withstand critical limit flight loads, considered as ultimate, with the extent of detectable damage consistent with the results of the damage tolerance evaluations. Fatigue test article tear-down and storage requirements must be approved by the Certifying Authority. For pressurised compartment, the following load must be withstood: (1) The normal operating differential pressure combined with the expected external aerodynamic pressures applied simultaneously with the flight loading conditions specified in this subpart, and (2) The expected external aerodynamic pressures in 1g flight combined with a compartment differential pressure equal to 1.1 times the normal operating differential pressure without any other load.
	574 Metallic damage tolerance and fatigue evaluation of commuter category aeroplanes Not applicable.
575	USAR.575 Inspections and other procedures Each inspection or other procedure, based on an evaluation required by USAR.572 and USAR.573, must be established to prevent <i>Catastrophic</i> failure and must be included in the limitations section of the instructions for continued airworthiness required by USAR.1529 Cracks meeting the inspection-based requirements in terms of crack growth and residual strength shall not result in the rupture of critical parts of the structure. See also USAR.603

CATAPULT ASSISTED AND ROCKET ASSISTED TAKE-OFF UAs

U585	USAR.U585 Launch load factor The maximum forces applied by the launch system to the relevant components of the UA must be consistent with the structural acceleration limitation of the UA; therefore: (a) The relevant components of the UA must be designed to sustain a longitudinal load compliant with the maximum continuous load factor applied by the launch system combined with the factor prescribed in USAR.303 This requirement must be complied with at the maximum and minimum take-off weight. (b) The design of the launcher must ensure that the acceleration and the rate of change of acceleration (jerk) are controlled such that the UA does not sustain damage during launch.
------	--

	(c) The fatigue effects of repeated launches shall be considered for the launch system and, in accordance with USAR.U570, for the aircraft including payloads.
U586	USAR.U586 Use of trolley or shuttle (a) Any structure such as trolley or shuttle which is used to transfer the accelerating forces to the aircraft or to stabilise the aircraft during launch, must be capable of withstanding the acceleration and subsequent decelerations without causing any damage to the aircraft. (b) The interface of the trolley or shuttle and the UA must be such as to ensure controlled, reliable and safe separation at lift off. (c) The moving trolley or shuttle must exhibit adequate control and stability, both during acceleration and subsequent retardation following separation.
U587	USAR.U587 Rocket assisted take-off The use of rocket must not cause damage to the airframe of the UA arising from heat, blast or aerodynamic interference from the rocket efflux. The UA integrity must be maintained following the release of rocket(s) in flight.

ARRESTED LANDING SYSTEMS

U590	USAR.U590 Arresting system See AMC.U590 (a) The design of the Arresting System shall ensure that it retards the AV without imparting forces and accelerations in excess of those for which the airframe and its internal equipment are designed. (b) The Arresting System shall accommodate deviations from the intended recovery trajectory resulting from local environmental conditions. (Wind, gust). (c) The Arresting System shall include means to maintain UA azimuth on landing strip after touchdown and arresting hook engagement also in a situation in which one or both arresting devices (e.g. drums) fails to operate correctly. (d) Arrested landings shall be performed to demonstrate that the aircraft can consistently engage the arresting gear and that aircraft motions during the arrested run-out are conducive to safe arrestment. The following where applicable shall be demonstrated: (1) That there is no cable impact damage to the aircraft during arrested landings for all landing conditions and landing configurations. (2) Satisfactory swivelling of the nose gear for roll back in the arresting gear. (3) Satisfactory anti-hunting and shimmy characteristics of the nose gear. (4) Abrupt application of brakes during the post arresting roll-back.
------	--

PARACHUTE LANDING SYSTEM

U595	USAR.U595 Limit load factor See AMC.U595 (a) (a) The structural design of the UA and the integrity of onboard equipment must be consistent with the maximum deceleration due to: (1) the deployment of the parachute. (2) the impact of the UA on the ground taking into account any shock absorber system, for the worst case conditions, agreed to with the Certifying Authority.
------	---

	(b) The maximum deceleration forces must take into account all the conditions under which the deployment of the parachute is possible within the flight envelope protection as prescribed in USAR.U334, especially if the parachute is to be used under emergency conditions.
U596	USAR.U596 UA dragging on the ground The parachute system must include means for preventing the dragging of the UA by the wind after impact with the landing surface.
U597	USAR.U597 Sacrificial elements Where sacrificial elements of the structure are used to deploy the parachute or to activate the shock absorber system: (a) the number of such elements must be minimised. (b) the structural integrity of the UA airframe must not be reduced after the release or operation of these sacrificial elements.
U598	USAR.U598 Extracting devices See AMC.U598 The use of extracting device (mechanical drogue, etc.,) must not cause any damage to the UA air frame or to the landing system assets outside of that necessary to perform its intended function.
U599	USAR.U599 Installation of the parachute in the airframe Consideration shall be given to the following aspects of the parachute installation: (a) The packing of the parachute, installation in the airframe and attachments to the airframe should take account of all potential operational conditions and in particular, launch accelerations, which may affect its safety. (b) All components of the parachute landing system should be made for ventilation and drainage of the parachute canister and associated structure to ensure the sound condition of the system. (c) Adequate means should be provided to permit the close examination of the parachute and other system components to ensure proper functioning, alignment, lubrication, and adjustment during the required inspection of the system. (d) Each fitting point of the parachute straps and the surrounding structures should be designed for the maximum expected opening loads with a safety factor not less than 1.5 (e) The installation of the parachute in the airframe should be designed in order to mitigate the effects of possible oscillation of the UA once the parachute is deployed in the air and at the impact on the ground if such oscillations present a hazard to third parties, ground staff, or UA survivability.

Subpart D – UA DESIGN AND CONSTRUCTION

GENERAL	
601	USAR.601 General See AMC.601 The suitability of each questionable design detail and part having an important bearing on safety in operations, as determined by the Certifying Authority, must be established by tests.
603	USAR.603 Materials and workmanship For metallic materials, see AMC.603 (a); for composite materials, see AMC.603 (b) and AMC 20-29 (a) The suitability and durability of materials used for parts, the failure of which could adversely affect safety, must: (1) Be established by experience or tests; (2) Meet approved specifications that ensure they have the strength and other properties to meet the design requirements; and (3) Take into account the effects of environmental conditions, such as temperature, humidity, UV, chemical, solvent, fuel, electromagnetic radiation, and airborne particulates expected in service. (b) Workmanship must be of a high standard. (c) The UA materials shall obtain environmental qualification, as applicable, at the component and system level for both natural and induced environments. (d) All hazardous materials shall be identified and their safe use will be accounted for.
605	USAR.605 Fabrication methods (a) The methods of fabrication used must produce consistently sound structures. If a fabrication process (such as gluing, spot welding, or heat-treating) requires close control to reach this objective, the process must be performed under an approved process specification. (b) Each new UAS fabrication method must be substantiated by a test programme.
607	USAR.607 Fasteners See AMC.607 (b) (a) Each removable fastener must incorporate two retaining devices if the loss of such fastener would preclude continued safe flight and landing. (b) Fasteners and their locking devices must not be adversely affected by the environmental conditions associated with the particular installation. (c) No self-locking nut may be used on any bolt subject to rotation in operation unless a nonfriction locking device is used in addition to the self-locking device.
609	USAR.609 Protection of structure See AMC.609 Each part of the structure must: (a) Be suitably protected against deterioration or loss of strength in service due to any cause, including:

	(1) Weathering; (2) Corrosion; and (3) Abrasion; and (b) Have adequate provisions for ventilation and drainage. Unless it is detrimental to the function, there must be adequate provisions for ventilation and drainage.
611	USAR.611 Accessibility provisions See AMC.611 For each part that requires maintenance, inspection, or other servicing, appropriate means must be incorporated into the UAS design to allow such servicing to be accomplished. The inspection means for each item must be appropriate to the inspection interval for the item.
613	USAR.613 Material strength properties and design values See AMC.613 (a) Material strength properties must be based on enough tests of material meeting specifications to establish design values on a statistical basis. (b) The design values must be chosen to minimise the probability of structural failure due to material variability. Except as provided in sub-paragraph (e), compliance with this paragraph must be shown by selecting design values that assure material strength with the following probability: (1) Where applied loads are eventually distributed through a single member within an assembly, the failure of which would result in loss of structural integrity of the component; 99% probability with 95% confidence. (2) For redundant structure, in which the failure of individual elements would result in applied loads being safely distributed to other load carrying members; 90% probability with 95% confidence. (c) The effects of temperature on allowable stresses used for design in an essential component or structure must be considered where thermal effects are significant under normal operating conditions. Also, thermal degradation from an in-flight fire condition shall be considered (see USAR.865). (d) The design of structure must minimise the probability of <i>Catastrophic</i> fatigue failure particularly at points of stress concentration. (e) Design values greater than the guaranteed minimum's required by this paragraph may be used where only guaranteed minimum values are normally allowed if a "premium selection" of the material is made in which a specimen of each individual item is tested before use to determine that the actual strength properties of the particular item will equal or exceed those used in design. (f) Appropriate Non-Destructive Techniques Testing and Inspection (NDT/I) processes and technology must be implemented throughout the UA life cycle (from design through operational support) to assure that the UA is produced with integrity, properly evaluated, tested and maintained to meet design quality levels and performance requirements. The Designer must ensure the required inspections are established throughout the UA life(e.g. damage tolerance based inspections, calendar based inspections, event triggered inspections). The inspection regime must be promulgated in accordance with USAR.1529 (g) Alloys prone to insidious failure modes (e.g. hydrogen embrittlement) must not be used in the design unless specifically approved by the Certifying Authority.
619	USAR.619 Special factors The factor of safety prescribed in USAR.303 must be multiplied by the highest pertinent special factors of safety prescribed in USAR.621 to USAR.625 for each part of the structure whose strength is:

	(a) Uncertain; (b) Likely to deteriorate in service before normal replacement; or (c) Subject to appreciable variability because of uncertainties in manufacturing processes or inspection methods.
621	USAR.621 Casting factors The approved national standards regarding the casting factors shall be considered and agreed with the Certifying Authority.
623	USAR.623 Bearing factors (a) Each part that has clearance (free fit) and that is subject to pounding or vibration, must have a bearing factor large enough to provide for the effects of normal relative motion. (b) For control surface hinges and control system joints, compliance with the factors prescribed in USAR.657 and USAR.693 respectively, meets paragraph (a).
625	USAR.625 Fitting factors For each fitting (a part or terminal used to join one structural member to another), the following applies: (a) For each fitting whose strength is not proven by limit and ultimate load tests in which actual stress conditions are simulated in the fitting and surrounding structures, a fitting factor of at least 1.15 must be applied to each part of: (1) The fitting; (2) The means of attachment; and (3) The bearing on the joined members. (b) No fitting factor need be used for joint designs based on comprehensive test data (such as continuous joints in metal plating, welded joints and scarf joints in wood). (c) For each integral fitting, the part must be treated as a fitting up to the point at which the section properties become typical of the member. (d) Not applicable.
627	USAR.627 Fatigue strength The structure must be designed, as far as practicable, to avoid points of stress concentration where variable stresses above the fatigue limit are likely to occur in normal service.
629	USAR.629 Flutter See AMC.629 and AMC.629 (j) (a) It must be shown by the methods of (b) and (c), that the UA is free from flutter, control reversal and divergence for any condition of operation within the limit V-n envelope defined in USAR.333 In addition: (1) Adequate tolerances must be established for quantities which affect flutter; including speed, damping, mass balance and control system stiffness; and (2) The natural frequencies of main structural components must be determined by vibration tests or other approved methods. (b) Flight flutter tests are only required if rational flutter analysis based on the results of ground vibration tests show any predicted flutter is less than a 20% speed margin over V_D/M_D or any critical flutter mechanism has a total damping (aerodynamic and structural) of less than 2% at V_D/M_D. In case such flight tests are required, they must be made to show that the UA is free from flutter, control reversal and divergence and to show by these tests that:

	(1) Proper and adequate attempts to induce flutter have been made within the speed range of the flight envelope protection maintained by the flight control system. (2) The vibratory response of the structure during the test indicates freedom from flutter; (3) A proper margin of damping exists at the maximum speed allowed by the flight envelope protection; and (4) There is no large and rapid reduction in damping as the maximum speed allowed by the flight envelope protection is approached. (c) Any rational analysis used to predict freedom from flutter, control reversal and divergence for normal conditions without failures, malfunctions or adverse conditions shall cover all combinations of altitudes and speed encompassed by the V_D/M_D versus altitude envelope enlarged at all points by an increase of 15% in equivalent airspeed at constant Mach number and constant altitude. In addition a proper margin of stability must exist at all speeds up to V_D/M_D and there must be no large and rapid reduction in stability as V_D/M_D is approached. (d) Not applicable. (e) For UA where the propeller is mounted on the lifting surfaces, the dynamic evaluation must include: (1) Whirl mode degree of freedom which takes into account the stability of the plane of rotation of the propeller and significant elastic, inertial and aerodynamic forces; and (2) Propeller, engine, engine mount and UA structure stiffness and damping variations appropriate to the particular configuration. (f) Freedom from flutter, control reversal and divergence must be shown at speed up to V_D after failure, malfunction, or disconnection of any single element in the primary flight control system, any tab control surface, or any flutter damper if the failure is shown not to be extremely remote. (g) For UA showing compliance with the fail-safe criteria of USAR.572, the UA must be shown by analysis to be free from flutter up to V_D/M_D after fatigue failure, or obvious partial failure of a principal structural element. (h) For UA showing compliance with the damage-tolerance criteria of USAR.573, the UA must be shown by analysis to be free from flutter up to V_D/M_D with the extent of damage for which residual strength is demonstrated. (i) For modifications to the type design which could affect the flutter characteristics compliance with sub-paragraph (a) must be shown, except that analysis alone, which is based on previously approved data, may be used to show freedom from flutter, control reversal and divergence. (j) Control surface backlash should be kept to a minimum.
U631	USAR.U631 Bird strike SeeAMC.U631 (a) The UA must be designed to assure that impact with a 2 lb (0.91 Kg) bird shall not result in an uncontrolled flight and/or uncontrolled crash, when the velocity of the UA at impact (relative to the bird along the UA's flight path) is equal to V_c at sea level, selected under USAR.335 Compliance may be shown by analysis only when based on tests carried out on sufficiently representative structures of similar design. (b) The degree of protection of the UA against bird strike shall be determined by rational analysis of the intended size and use of the UA and shall be agreed with the Certifying Authority.
U635	USAR.U635 Ground equipment affecting flight safety

	If a UAS contains ground equipment affecting flight safety not defined in this code, such equipment shall be designed using the applicable standards and shown not to adversely affect any requirement of this code.
--	--

WINGS

641	USAR.641 Proof of strength The strength of wings must be proven by load tests or by combined structural analysis and load tests.
-----	---

CONTROL SURFACES

651	USAR.651 Proof of strength (a) Rational and conservative analysis or limit load tests or combined structural analysis and load tests of control surfaces are required. These tests must include the horn or fitting to which the control system is attached. (b) In structural analyses, rigging loads due to wire bracing must be accounted for in a rational or conservative manner.
655	USAR.655 Installation (a) Movable surfaces must be installed so that there is no interference between any surfaces, their bracing or adjacent fixed structure, when one surface is held in its most critical clearance positions and the others are operated through their full movement. (b) If an adjustable stabiliser is used, it must have stops that will limit its range of travel to that allowing safe flight and landing.
657	USAR.657 Hinges (a) Control surface hinges, except ball and roller bearing hinges, must have a factor of safety of not less than 6.67 with respect to the ultimate bearing strength of the softest material used as a bearing. (b) For ball or roller bearing hinges, the approved rating of the bearing may not be exceeded.
659	USAR.659 Mass balance The supporting structure and the attachment of concentrated mass balance weights used on control surfaces must be designed for: (a) 24g normal to the plane of the control surface; (b) 12g fore and aft; and (c) 12g parallel to the hinge line.

CONTROL SYSTEMS

671	USAR.671 General See AMC.671 (a) Each control must operate easily, smoothly and positively enough to allow proper performance of its functions. (b) Not applicable in this subpart. (see USAR.U1731 Controls - General)
	672 Stability augmentation and automatic power operating systems Not applicable.
673	USAR.673 Primary and secondary flight controls (a) Primary flight controls are those used in the UA by the flight control system for the immediate control of pitch, roll and yaw. (b) Secondary flight controls are defined in USAR.405
675	USAR.675 Stops

	(a) Each flight control must have stops that positively limit the range of motion of each movable aerodynamic surface controlled by the system. (b) Each stop must be located so that wear, slackness, or take-up adjustments will not adversely affect the control characteristics of the UA because of a change in the range of surface travel. (c) Each stop must be able to withstand any loads corresponding to the design conditions for the control system.
677	USAR.677 Trim systems (a) Proper precautions must be taken to prevent inadvertent, improper, or abrupt auto-trim tab operation by the flight control system. (b) Trimming devices must be designed so that, when any one connecting or transmitting element of the primary flight controls fails, adequate control for safe flight and landing is available with: (1) For single-engine UA, the longitudinal trimming devices; or (2) For multi-engine UA, the longitudinal and directional trimming devices. (c) Tab controls must be irreversible unless the tab is properly balanced and has no unsafe flutter characteristics. Irreversible tab systems must have adequate rigidity and reliability in the portion of the system from the tab to the attachment of the irreversible unit to the UA structure. (d) It must be demonstrated that the UA is safely controllable by the flight control system and that the UAS crew can perform all the manoeuvres and operations necessary to effect a safe landing following any powered trim system runaway that reasonably might be expected in service. The demonstration must be conducted at the critical UA weights and centre of gravity positions.
679	USAR.679 Primary or secondary flight controls lock If there is a device to lock the flight controls: (a) Not applicable in this subpart. (see USAR.U1825 Flight control system lock warning) (b) There must be a means to: (1) Warn the ground staff when the device is engaged; (2) Not applicable. (c) The device must have a means to preclude the possibility of it becoming inadvertently engaged in flight.
681	USAR.681 Limit load static tests (a) Compliance with the limit load requirements of USAR must be shown by rational and conservative analysis or by tests in which: (1) The direction of the test loads produces the most severe loading in the primary flight controls; and (2) Each fitting, pulley and bracket used in attaching the system to the main structure is included. (b) Compliance must be shown (by analyses or individual load tests) with the special factor requirements for control system joints subject to angular motion.
683	USAR.683 Operation tests See AMC.683 (a) It must be shown by operation tests that, when the controls are operated with the system loaded as prescribed in sub-paragraph (b), the system is free from:

	(1) Jamming; (2) Excessive friction; (3) Excessive deflection; (4) Excessive freeplay. (b) The prescribed test loads are: (1) For the entire system, loads corresponding to the limit air loads on the appropriate surface, or the maximum loads and torques generated by the actuating system, whichever are less; and (2) For secondary controls, loads not less than those corresponding to the maximum servocontrols or actuators force.
685	USAR.685 Control system details (a) Each detail of each control system must be designed and installed to prevent jamming, chafing and interference from payloads, loose objects, or the freezing of moisture. (b) There must be means to prevent the entry of foreign objects into places where they would jam the system. (c) There must be means to prevent the slapping of cables or tubes against other parts. (d) Each element of the flight control system must have design features, or must be distinctively and permanently marked, to minimise the possibility of incorrect assembly that could result in malfunctioning of the control system. (e) UAS control systems need to be resistant to external and internal sources of electromagnetic interference in accordance with USAR.1431
687	USAR.687 Spring devices The reliability of any spring device used in the control system must be established by tests simulating service conditions unless failure of the spring will not cause flutter or unsafe flight characteristics.
689	USAR.689 Cable systems (a) Each cable, cable fitting, turn-buckle, splice and pulley used must meet approved specifications. In addition: (1) Primary flight control cables shall be designed and sized such that limit loads do not exceed 67% of the rated breaking strength of the cable. (2) Each cable system must be designed so that there will be no hazardous change in cable tension throughout the range of travel under operating conditions and temperature variations; and (3) There must be means for visual inspection at each fairlead, pulley, terminal and turnbuckle. (b) Each kind and size of pulley must correspond to the cable with which it is used. Each pulley must have closely fitted guards to prevent the cables from being misplaced or fouled, even when slack. Each pulley must lie in the plane passing through the cable so that the cable does not rub against the pulley flange. (c) Fairleads must be installed so that they do not cause a change in cable direction of more than 3°. (d) Clevis pins subject to load or motion and retained only by cotter pins may not be used in the control system.

	(e) Turnbuckles must be attached to parts having angular motion in a manner that will positively prevent binding throughout the range of travel. (f) Tab control cables may be less than 3.2mm (1/8 inch) diameter in UA that are safely controllable with the tabs in the most adverse positions.
693	USAR.693 Joints See AMC.693 (a) Control system joints (in push-pull systems) that are subject to angular motion, except those in ball and roller bearing systems, must have a special factor of safety of not less than 3.33 with respect to the ultimate bearing strength of the softest material used as a bearing. This factor may be reduced to 2.0 for joints in cable control systems. For ball or roller bearings, the approved ratings may not be exceeded. (b) All elastomeric bearings shall be: (1) Suitable for the expected loads and the environment that the bearing will experience. (2) Selected from a Qualified Products List.
697	USAR.697 Wing flap controls (a) Each wing flap control must be designed so that, when the flap has been placed in any position upon which compliance with the performance requirements of USAR is based, the flap will not move from that position unless the control is adjusted or is moved by the automatic operation of a flap load limiting device. (b) The rate of movement of the flaps in response to the operation of the UAS crew's control or flight control system must give satisfactory flight and performance characteristics under steady or changing conditions of airspeed, engine power or thrust and attitude.
	699 Wing flap position indicator Not applicable in this subpart. (see USAR.U1791 Wing flap position indicator)
701	USAR.701 Flap interconnection See AMC.701 (a) The main wing flaps and related movable surfaces as a system must: (1) Be synchronised by means independent of the flap drive system; or (2) Be designed so that the occurrence of any failure of the flap system that would result in an unsafe flight characteristic of the UA is Extremely Improbable; or (b) The UA must be shown to have safe flight characteristics with any combination of extreme positions of individual movable surfaces (mechanically interconnected surfaces are to be considered as a single surface). (c) If an interconnection is used in multi-engine UA, it must be designed to account for the asymmetrical loads resulting from flight with the engine on one side of the plane of symmetry inoperative and the remaining engine at take-off power or thrust. For single-engine UA and multi-engine UA with no slipstream effects on the flaps, it may be assumed that 100% of the critical air load acts on one side and 70% on the other.
703	USAR.703 Take-off protection If the UA is in an unsafe take-off configuration, either: (a) the UAS crew and ground staff (where applicable) must be notified; or (b) the initiation of take-off must be automatically prevented.

LANDING GEAR

	721 General
	Not applicable.
U722	USAR.U722 Landing gear - General
	The following section is for conventional landing gear arrangements. If novel designs are proposed the acceptance methods shall be agreed with the Certifying Authority.
723	USAR.723 Shock absorption tests
	(a) It must be shown that the limit load factors selected for design in accordance with USAR.473 for take-off and landing weights, respectively, will not be exceeded. This must be shown by energy absorption tests except that analysis based on tests conducted on a landing gear system with identical energy absorption characteristics may be used for increases in previously approved take-off and landing weights. (b) The landing gear may not fail, but may yield, in a test showing its reserve energy absorption capacity, simulating a descent velocity of 1.2 times the limit descent velocity, assuming wing lift equal to the weight of the UA.
725	USAR.725 Limit drop tests
	(a) If compliance with USAR.723 (a) is shown by free drop tests, these tests must be made on the complete UA, or on units consisting of wheel, tyre and shock absorber, in their proper relation, from free drop heights not less than those determined by the following formula: $h \text{ (m)} = 0.0132 \text{ (Mg/S)}^{1/2}$ However, the free drop height may not be less than 0.234 m (9.2 inches) and need not be more than 0.475 m (18.7 inches). (b) If the effect of wing lift is provided for in free drop tests, the landing gear must be dropped with an effective weight equal to: $M_e = M (h + [1-L]d) / (h + d)$ Where: M_e = the effective weight to be used in the drop test (kg); h = Specified free drop height (m); d = deflection under impact of the tyre (at the approved inflation pressure) plus the vertical component of the axle travel relative to the drop mass (m); M = MM for main gear units (kg), equal to the static weight on that unit with the UA in the level attitude (with the nose wheel clear in the case of the nose wheel type UA); M = MT for tail gear units (kg), equal to the static weight on the tail unit with the UA in the tail-down attitude; M = MN for nose wheel units (kg), equal to the vertical component of the static reaction that would exist at the nose wheel, assuming that the mass of the UA acts at the centre of gravity and exerts a force of 1.0g downward and 0.33g forward; and L = the ratio of the assumed wing lift to the UA weight, but not more than 0.667 g = The acceleration due to gravity (m/s²) (c) The limit inertia load factor must be determined in a rational or conservative manner, during the drop test, using a landing gear unit attitude and applied drag loads, that represent the landing conditions. (d) The value of d used in the computation of M_e in sub-paragraph (b) may not exceed the value actually obtained in the drop test.

	(e) The limit inertia load factor must be determined from the drop test in sub-paragraph (b) according to the following formula: $n = n_j (M_e / M) + L$ where: n_j = the load factor developed in the drop test (that is, the acceleration (dv/dt) in g's recorded in the drop test) plus 1.0; and M_e, M and L are the same as in the drop test computation. (f) The value of n determined in accordance with sub-paragraph (e) may not be more than the limit inertia load factor used in the landing conditions in USAR.473
726	USAR.726 Ground load dynamic tests (a) If compliance with the ground load requirements of USAR.479 to USAR.483 is shown dynamically by drop test, one drop test must be conducted that meets USAR.725 except that the drop height must be: (1) 2.25 times the drop height prescribed in USAR.725 (a); or (2) Sufficient to develop 1.5 times the limit load factor. (b) The critical landing condition for each of the design conditions specified in USAR.479 to USAR.483 must be used for proof of strength.
727	USAR.727 Reserve energy absorption drop tests (a) If compliance with the reserve energy absorption requirements in USAR.723 (b) is shown by free drop tests, the drop height may not be less than 1.44 times that specified in USAR.725 (b) If the effect of wing lift is provided for, the units must be dropped with an effective mass equal to $M_e = M (h / [h + d])$ when the symbols and other details are the same as in USAR.725
729	USAR.729 Landing gear extension and retraction system See AMC.729 (g) (a) General. For UA with retractable landing gear, the following apply: (1) Each landing gear retracting mechanism and its supporting structure must be designed for maximum flight load factors with the gear retracted and must be designed for the combination of friction, inertia, brake torque and air loads, occurring during retraction at any airspeed up to $1.6 V_{S1}$ with flaps retracted and for any load factor up to those specified in USAR.345 for the flaps-extended condition. (2) The landing gear and retracting mechanism, including the wheel well doors, must withstand flight loads, including loads resulting from all yawing conditions specified in USAR.351, with the landing gear extended at any speed up to at least $1.6 V_{S1}$ with the flaps retracted. (b) Landing gear lock. There must be positive means (other than the use of hydraulic pressure) to keep the landing gear extended. (c) Emergency operation. For a UAs having retractable landing gear, there must be means to extend the landing gear in the event of either: (1) Any reasonably probable failure in the normal landing gear operation system; or (2) Any reasonably probable failure in a power source that would prevent the operation of the normal landing gear operation system. (d) Operation test. The proper functioning of the retracting mechanism must be shown by operation tests.

	(e) If a retractable landing gear is used, there must be landing gear position sensor and switches to actuate the indicator in the UCS (see USAR.U1793) to inform the UAS crew that each gear is secured in the extended (or retracted) position. If switches are used, they must be located and coupled to the landing gear mechanical system in a manner that prevents an erroneous indication of either “down and locked” if each gear is not in the fully extended position, or of “up and locked” if each landing gear is not in the fully retracted position. (f) Not applicable in this subpart. (see USAR.U1793 Landing gear position and warning) (g) Equipment located in the landing gear bay. If the landing gear bay is used as the location for equipment other than the landing gear, that equipment must be designed and installed to minimise damage.
731	USAR.731 Wheels (a) The maximum static load rating of each wheel may not be less than the corresponding static ground reaction with: (1) Design maximum weight; and (2) Critical centre of gravity. (b) The maximum limit load rating of each wheel must equal or exceed the maximum radial limit load determined under the applicable ground load requirements of USAR.
733	USAR.733 Tyres (a) Each landing gear wheel must have a tyre whose approved tyre ratings (static and dynamic) are not exceeded: (1) By a load on each main wheel tyre (to be compared to the static rating approved for such tyres) equal to the corresponding static ground reaction under the design maximum weight and critical centre of gravity; and (2) By a load on nose wheel tyres (to be compared with the dynamic rating approved for such tyres) equal to the reaction obtained at the nose wheel, assuming the mass of the UA to be concentrated at the most critical centre of gravity and exerting a force of $1.0 M_g$ downward and $0.31 M_g$ forward (where M_g is the design maximum weight), with the reactions distributed to the nose and main wheels by the principles of statics and with the drag reaction at the ground applied only at wheels with brakes. (b) If specially constructed tyres are used, the wheels must be plainly and conspicuously marked to that effect. The markings must include the make, size, number of plies and identification marking of the proper tyre. (c) Each tyre installed on a retractable landing gear system must, at the maximum size of the tyre type expected in service, have a clearance to surrounding structure and systems that is adequate to prevent contact between the tyre and any part of the structure or systems.
735	USAR.735 Brakes See AMC.735 (c) Arresting systems not covered by USAR.U590 are out of scope.

	(a) The landing brake kinetic energy capacity rating of each main wheel brake assembly must not be less than the kinetic energy absorption requirements determined under either of the following methods: (1) The brake kinetic energy absorption requirements must be based on a conservative rational analysis of the sequence of events expected during landing at the design landing weight. (2) Instead of a rational analysis, the kinetic energy absorption requirements for each main wheel brake assembly may be derived from the following formula: $KE = \frac{1}{2}MV^2/N$ Where: KE = Kinetic energy per wheel (Joules); M = Mass at design landing weight (kg); V = UA speed in m/s. V must be not less than V_{SO}, the power off stalling speed of the UA at sea level, at the design landing weight, and in the landing configuration; and N = Number of main wheels with brakes. (3) Arresting cable, barrier net or other unconventional braking/arresting devices are out of scope regarding brake energy absorption on wheels and must belong to Special Condition category. The design of any arrestor system shall ensure that it retards the aircraft without imparting forces and accelerations in excess of those for which its airframe and internal equipment are designed. (b) Brakes must be able to prevent the wheels from rolling on a paved runway with take-off power or thrust in the critical engine, but need not prevent movement of the UA with wheels locked. (c) During the landing distance determination required by USAR.75, the pressure in the wheel braking system must not exceed the pressure specified by the brake manufacturer. (d) If anti-skid devices are installed, the devices and associated systems must be designed so that no single probable malfunction or failure will result in a Hazardous loss of braking ability or directional control of the UA. (e) The rejected take-off brake kinetic energy capacity rating of each mainwheel brake assembly must not be less than the kinetic energy absorption requirements determined under either of the following methods: (1) The brake kinetic energy absorption requirements must be based on a conservative rational analysis of the sequence of events expected during a rejected take-off at maximum weight. (2) Instead of a rational analysis, the kinetic energy absorption requirements for each mainwheel brake assembly may be derived from the following formula: $KE = \frac{1}{2} MV^2/N$ Where: KE = Kinetic energy per wheel (Joules); M = Mass at maximum weight (kg); V = Ground speed in m/s associated with the maximum value of VRf selected in accordance with USAR.51 N = Number of main wheels with brakes.
	737 Skis
	Not applicable.
745	USAR.745 Nose/tail-wheel steering

	(a) If nose/tail-wheel steering is installed, it must be demonstrated that it operates correctly during take-off and landing, in cross-winds and in the event of an engine failure or its use must be limited to low speed manoeuvring. (b) Movement of the steering control must not interfere with correct retraction or extension of the landing gear.
	USAR.751 Main float buoyancy Not applicable.
	USAR.753 Main float design Not applicable.
	USAR.755 Hull Not applicable.
	USAR.757 Auxiliary floats Not applicable.

PAYLOAD AND EQUIPMENT COMPARTMENTS

	USAR.771 Pilot compartment Not applicable in this subpart. (see USAR.U1703 UAS crew workplace)
	USAR.773 Pilot compartment view Not applicable.
775	USAR.775 Payload transparencies The design of payload transparencies and radomes in pressurised compartments must be based on factors specific to high altitude operation, including: (a) The effects of continuous and cyclic pressurisation loading; (b) The inherent characteristics of the material used; (c) The effects of temperatures and temperature gradients; (d) The effects on the structural integrity of the UA in the occurrence of wall pressurisation fracture, either by flaw or by explosion; and (e) Viewing areas of camera and sensor windows whose obstruction could result in an adverse safety effect shall be maintained free of fog, frost and other obstructions for all steady state and transient ground and flight operating conditions within the specified UAS environmental envelope. They shall be designed to withstand foreign object damage (FOD) from birds, hail, runway, taxiway, and ramp debris.
	USAR.777 Cockpit controls Not applicable in this subpart. (see USAR.U1731 General)
	USAR.779 Motion and effect of cockpit controls Not applicable in this subpart. (see USAR.U1735 Motion and representation of controls)
	USAR.781 Cockpit control knob shape Not applicable in this subpart. (see USAR.U1733 Conventional controls and indicators)
783	USAR.783 Doors, covers and hatches (a) Not applicable. (b) Not applicable.

	(c) Not applicable. (d) Not applicable. (e) Each external door and hatch must comply with the following requirements: (1) There must be a means to lock and safeguard each external door and hatch, including payload and service type doors, against inadvertent opening in flight, as a result of mechanical failure or failure of a single structural element, either during or after closure. (2) There must be a provision for direct visual inspection of the locking mechanism to determine if the external door or hatch, for which the initial opening movement is not inward, is fully closed and locked. The provisions must be discernible, under operating lighting conditions, by inspection and maintenance staff using a flashlight or an equivalent lighting source.
	USAR.785 Seats, berths, litters, safety belts and shoulder harnesses Not applicable.
787	USAR.787 Payload compartments Each payload compartment must: (a) Be designed for the maximum weight and distribution of contents and for the critical load distributions at the appropriate maximum load factors corresponding to the flight and ground load conditions of USAR. (b) Have means to prevent the contents of any compartment from becoming a hazard by shifting, and to protect any controls, wiring, lines, equipment, or accessories whose damage or failure would affect safe operations.
	USAR.791 Passenger information signs Not applicable.
	USAR.803 Emergency evacuation Not applicable.
	USAR.805 Flight crew emergency exits Not applicable.
	USAR.807 Emergency exits Not applicable.
	USAR.811 Emergency exit marking Not applicable.
	USAR.812 Emergency lighting Not applicable.
	USAR.813 Emergency exit access Not applicable.
	USAR.815 Width of aisle Not applicable.
	USAR.831 Ventilation Not applicable.
PRESSURISATION	
841	USAR.841 Pressurised compartments (a) Not applicable.

	(b) If necessary for structural protection, pressurised compartments should have the following valves (or their equivalent) (1) A pressure relief valve (or its equivalent) to automatically limit the positive pressure differential to a predetermined value at the maximum rate of flow delivered by the pressure source. The pressure differential is positive when the internal pressure is greater than the external. (2) A reverse pressure differential relief valve (or its equivalent) to automatically prevent a negative pressure differential that would damage the structure. (3) Not applicable. (4) Not applicable. (5) Not applicable in this subpart. (see USAR.U1795 Pressurised compartment indicator) (6) Not applicable. (7) Not applicable in this subpart. (see USAR.U1795 Pressurised compartment indicator) (8) Not applicable.
843	USAR.843 Pressurisation tests (a) Strength test. The complete pressurised compartment, including doors, windows, canopy and valves, must be tested as a pressure vessel for the pressure differential specified in USAR.365 (d). (b) Functional tests. The following functional tests must be performed: (1) Tests of the functioning and capacity of the positive and negative pressure differential valve. (2) Tests of the pressurisation system to show proper functioning under each possible condition of pressure, temperature and moisture, up to the maximum altitude for which certification is requested.

FIRE PROTECTION

U850	USAR.U850 Fire protection – General See AMC.U850 Specific UAS fire protection requirements presented in USAR aim at minimizing the risk of fire that may lead to uncontrolled UA flight and crash and potential damages to third parties.
	USAR.851 Fire extinguishers Not applicable.
	USAR.853 Passenger and crew compartment interiors Not applicable.
	USAR.855 Cargo and baggage compartment fire protection Not applicable.
	USAR.859 Combustion heater fire protection Not applicable.

863	USAR.863 Flammable fluid fire protection See AMC.863 (a) In each area where flammable fluids or vapours might escape by leakage of a fluid system, there must be means to minimise the probability of ignition of the fluids and vapours and the resultant hazard if ignition does occur. (b) Compliance with sub-paragraph (a) must be shown by analysis or tests and the following factors must be considered: (1) Possible sources and paths of fluid leakage and means of detecting leakage. (2) Flammability characteristics of fluids, including effects of any combustible or absorbing materials. (3) Possible ignition sources, including electrical faults, over-heating of equipment, static electricity, lightning and malfunctioning of protective devices. (4) Means available for controlling or extinguishing a fire, such as stopping flow of fluids, shutting down equipment, fireproof containment, or use of extinguishing agents. (5) Ability of UA components that are critical to safety of flight to withstand fire and heat. (c) Not applicable in this subpart. (see USAR.U1817 Fire protection warning) (d) Each area where flammable fluids or vapours might escape by leakage of a fluid system must be identified and defined.
865	USAR.865 Fire protection of flight control system components, engine mounts and other flight structure See AMC.865 Flight control system components, engine mounts, and other flight structure located in designated fire zones, or in adjacent areas that would be subjected to the effects of fire in the designated fire zones, must be constructed of fireproof material or be shielded so that they are capable of withstanding the effects of a fire. Engine vibration isolators must incorporate suitable features to ensure that the engine is retained if the non-fireproof portions of the isolators deteriorate from the effects of a fire.
ELECTRICAL BONDING AND LIGHTNING PROTECTION	
867	USAR.867 Electrical bonding and protection against lightning and static electricity See AMC.867 (a) (a) The UA must be protected against <i>Catastrophic</i> effects from lightning and static electricity. A lightning analysis assessment has to be carried out and agreed with the Certifying Authority. (b) For metallic components, compliance with sub-paragraph (a) may be shown by: (1) Bonding the components and grounding them properly to the airframe; or (2) Designing the components so that a strike will not result in a <i>Catastrophic</i> event. (c) For non-metallic components, compliance with sub-paragraph (a) may be shown by: (1) Designing the components to minimise the effect of a strike; or (2) Incorporating acceptable means of diverting the resulting electrical current so as not to result in a <i>Catastrophic</i> event.
871	Levelling means Not applicable.

PARACHUTE LANDING SYSTEM

U881	USAR.U881 Parachute design See AMC.U881 (a) Where a UA is designed to be recovered by parachute: (a) materials and workmanship shall be of a quality which documented experience or tests have conclusively demonstrated to be suitable for the manufacture of parachute assemblies and components. (b) All materials shall remain functional for storage and use from -40°C to +93.3°C, and from 0 to 100% relative humidity. (c) All plated ferrous parts shall be treated to minimise hydrogen embrittlement. (d) Stitching shall be of a type that will not unravel when broken. (e) Information concerning parachute assemblies and components must be furnished in the UAS documentation, including: (1) part number. (2) manufacturer's name and address. (3) maximum operating limits. (4) instruction for packing method and inspection at approved intervals. (5) instruction for continued airworthiness. (f) Where parachute assemblies are designed for re-use, parachute attachments shall have a fatigue evaluation determined in accordance with USAR.U570, unless otherwise agreed with the Certifying Authority.
------	---

Subpart E – UA POWERPLANT

GENERAL	
901	USAR.901 Installation See AMC.901 (a) For the purpose of USAR, the UA powerplant installation includes each component that: (1) Is necessary for propulsion; and (2) Affects the safety of the major propulsive units. (b) Each powerplant installation must be constructed and arranged to: (1) Ensure safe operation over the full envelope of operating conditions (in particular, the maximum operating altitude) for which approval is requested. (2) Be accessible for necessary inspections and maintenance, (3) Account for all static, dynamic and oscillatory characteristics of the UA. (c) Engine covers, cowls and nacelles must be easily removable or openable by the inspection and maintenance staff to provide adequate access to and exposure of the engine compartment for the required pre-flight checks. (d) Each turbine engine installation must be constructed and arranged to: (1) Result in carcass vibration characteristics that do not exceed those established during the type certification of the engine. (2) Provide continued safe operation without a Hazardous loss of power or thrust while being operated in rain for at least 3 minutes with the rate of water ingestion being not less than 4% by weight, of the engine induction airflow rate at the maximum installed power or thrust approved for take-off and at flight idle. (e) The powerplant installation must comply with: (1) The installation instructions provided under: (i) The engine Type Certificate or any qualification accepted by the Certifying Authority, and (ii) The propeller Type Certificate or any qualification accepted by the Certifying Authority. (2) The applicable provisions of this subpart, (3) USAR.903 (a)(2) requirement. (f) Each auxiliary power unit installation must meet the applicable portions of USAR.

903	USAR.903 Engines and auxiliary power units See AMC.903(a) and AMC.903(f) (a) Engine Type Certificate (1) Each engine must have a Type Certificate or any qualification accepted by the Certifying Authority. (2) In addition, each turbine engine must either: (i) Comply with: - ingestion of hail, rain and foreign objects risk qualification, and - bird strike and ingestion risk qualification agreed with the Certifying Authority, or (ii) Be shown to have a foreign object ingestion service history in similar installation locations which has not resulted in any unsafe condition. (b) Turbine engine installations. For turbine engine installations (1) Design precautions must be taken to minimise the hazards to the UA in the event of an engine rotor failure or of a fire originating inside the engine which burns through the engine case. (2) The powerplant systems associated with engine control devices, systems and instrumentation must be designed to give reasonable assurance that those operating limitations that adversely affect turbine rotor structural integrity will not be exceeded in service. (c) Engine isolation. In multiple-engine applications, the powerplants must be arranged and isolated from each other to allow operation, in at least one configuration, so that the failure or malfunction of any engine, or the failure or malfunction (including destruction by fire in the engine compartment) of any system that can affect an engine will not: (1) Prevent the continued safe operation of the remaining engines; or (2) Require immediate action by any UAS crew for continued safe operation of the remaining engine. (d) Starting and stopping (1) Any techniques and associated limitations for engine starting and stopping must be established and included in the UAS Flight Manual and in the UAS Maintenance Manual. (2) For safety purpose, there must be a means to prevent inadvertent engine starting on the ground. (3) The design of the installation must be such that risk of fire or mechanical damage to the engine or UA, as a result of starting the engine in any conditions in which starting is to be permitted, is reduced to a minimum. (e) Restart capability (1) Engine restart capability and demonstration should be weighed with the risk of engine loss and corresponding emergency procedures as defined in USAR.U1413 and related operational limitations. (2) Where restart capability is required:
-----	--

	(i) Any techniques and associated limitations must be established and included in the UAS Flight Manual, or applicable operating placards must be included in the UCS. (ii) It must be demonstrated in flight that when restarting engines following a false start, all fuel or vapour is discharged in such a way that it does not constitute a fire hazard. (iii) Restart envelope. An altitude and airspeed envelope must be established for the UA for in-flight engine restarting. (iv) For turbine engine-powered UA, if the minimum windmilling speed of the engines, following the in-flight shutdown of all engines, is insufficient to provide the necessary electrical power for engine ignition, a power source independent of the engine-driven electrical power generating system must be provided to permit in-flight engine ignition for restarting. (f) Auxiliary power units. Each APU must comply with APU qualification agreed with the Certifying Authority.
	904 Automatic power reserve system Not applicable.
905	USAR.905 Propellers See AMC.905 (d), AMC.905 (e) and AMC.905 (g) (a) Each propeller must have a Type Certificate or any qualification accepted by the Certifying Authority. (b) Engine power and propeller shaft rotational speed may not exceed the limits for which the propeller is certificated. (c) Each featherable propeller must have a means to unfeather it in flight. (d) Propeller systems that allow variation of blade pitch angle during flight shall be designed such that failure of the variable pitch mechanism or components or of its control system, causes the blades to revert to a condition that enables the UA to continue in safe, controlled flight and landing. (e) If certified for icing conditions, all areas of the UA that are likely to accumulate and shed ice into the propeller disc during any operating condition must be suitably protected to prevent ice formation, or it must be shown that any ice shed into the propeller disc will not create a Hazardous condition. (f) Each propeller must be marked so that the disc is conspicuous under normal daylight ground conditions. (g) If the engine exhaust gases are discharged into the pusher propeller disc, it must be shown by tests, or analysis supported by tests, that the propeller is capable of continuous safe operation. (h) All engine cowlings, access doors, and other removable items must be designed to ensure that they will not separate from the UA and contact the pusher propeller. (i) If certified for icing conditions, it must be shown that any ice shed from or by the propeller disc will not create an unsafe condition.
907	USAR.907 Propeller vibration See AMC.907 (a) (a) Each propeller other than a conventional fixed pitch wooden propeller must be shown to have vibration stresses, in normal operating conditions, that do not exceed values that have been shown by the propeller manufacturer to be safe for continuous operation. This must be shown by:

	(1) Measurement of stresses through direct testing of the propeller; (2) Comparison with similar installations for which these measurements have been made; or (3) Any other acceptable test method or service experience that proves the safety of the installation. (b) Proof of safe vibration characteristics for any type of propeller, except for conventional, fixed-pitch, wood propellers must be shown where necessary.
909	USAR.909 Turbo charger systems See AMC.909 (a) and AMC.909 (d) (1) (a) Each turbo charger must be approved under the engine Type Certificate or it must be shown that the turbo charger system, while in its normal engine installation and operating in the engine environment: (1) Can withstand, without defect, an endurance test of a time specified by the Certifying Authority, and (2) Will have no adverse effect upon the engine. (b) Control system malfunctions, vibrations and abnormal speeds and temperatures expected in service may not damage the turbo charger compressor or turbine. (c) Each turbo charger case must be able to contain fragments of a compressor or turbine that fails at the highest speed that is obtainable with normal speed control devices in-operative. (d) Each intercooler installation, where provided, must comply with the following: (1) The mounting provisions of the intercooler must be designed to withstand the loads imposed on the system; (2) It must be shown that, under the installed vibration environment, the intercooler will not fail in a manner allowing portions of the intercooler to be ingested by the engine, and (3) Airflow through the intercooler must not discharge directly on any UA component unless such discharge is shown to cause no hazard to the UA under all operating conditions. (e) Engine power, cooling characteristics, operating limits, and procedures affected by the turbocharger system installations must be evaluated. Turbocharger operating procedures and limitations, where appropriate, must be included in the UAS Flight Manual in accordance with USAR.1581
925	USAR.925 Propeller clearance Propeller clearances with the UA at the most adverse combination of weight and centre of gravity and with the propeller in the most adverse pitch position, may not be less than the following: (a) Ground clearance. (1) For UAs designed for conventional take-off or landing on a runway, there must be a clearance of at least 18 cm (7 in) (for each UA with nose wheel landing gear) or 23 cm (9 in) (for each UA with tail wheel landing gear) between each propeller and the ground with the landing gear statically deflected and in the level, normal take-off, or taxiing attitude, whichever is the most critical. In addition, for each UA with conventional landing gear struts using fluid or mechanical means for absorbing landing shocks, there must be positive clearance between the propeller and the ground in the level take-off attitude with the critical tyre completely deflated and the corresponding landing gear strut bottomed. Positive clearance for UA using leaf spring

	struts is shown with a deflection corresponding to 1.5g. Exemptions from clearance limits described above have to be agreed with the Certifying Authority. (2) For rocket assisted take-off or catapult assisted take-off UAs, there must be a clearance of at least 10 cm (3.9 in) between the propeller and the launch system (catapult, launching ramp, ...). This clearance must be kept during all the launching phase. (3) For UAs designed to be recovered by parachute and except for UA using wood propeller, the propeller must be subject to appropriate inspection after each recovery except if it has been shown by analysis, simulation or test that the integrity of the propeller after ground impact is never affected whatever the combination of weight, balance and environmental conditions. Inspection procedures must be documented. (b) Aft mounted propellers. Except for catapult assisted and rocket assisted take-off UA, in addition to the clearance specified in sub-paragraph (a) a UA with an aft mounted propeller must be designed such that the propeller will not contact the runway surface when the UA is in the maximum pitch attitude attainable during normal take-off and landings. (c) Not applicable. (d) Structural clearance. In the most adverse conditions, there must be: (1) At least 25 mm (1 in) radial clearance between the blade tips and the UA structure, plus any additional radial clearance necessary to prevent harmful vibration; (2) At least 12.7 mm (½ in) longitudinal clearance between the propeller blades or cuffs and stationary parts of the UA; and (3) Positive clearance between other rotating parts of the propeller or spinner and stationary parts of the UA.
929	USAR.929 Engine installation ice protection Propellers and other components of complete engine installations must be protected against the accumulation of ice as necessary to enable satisfactory functioning without appreciable loss of thrust when operated in the icing conditions for which certification is requested.
933	USAR.933 Thrust reversing systems (a) For turbojet and turbofan thrust reversing systems: (1) Each system intended for ground operation only must be designed so that during any reversal in flight the engine will produce no more than flight idle thrust. In addition, it must be shown by analysis or test, or both, that: (i) Each operable reverser can be restored to the forward thrust position; or (ii) The UA is capable of continued safe flight and landing under any possible position of the thrust reverser. (2) Each system intended for in-flight use must be designed so that no unsafe condition will result during normal operation of the system. Failure conditions of each system must be assessed in accordance with USAR.1309 under all operating conditions including ground operation. Failure of structural elements need not be considered if the probability of this kind of failure is Extremely Remote. (3) Each system must have means to prevent the engine from producing more than idle thrust when the reversing system malfunctions, except that it may produce any greater thrust that is shown to allow directional control to be maintained, with aerodynamic means alone, under the most critical reversing condition expected in operation.

	(b) For propeller thrust reversing systems: (1) Each system must be designed so that no single failure (or reasonably likely combination of failures) or malfunction of the system will result in unwanted reverse thrust under any expected operating condition. Failure of structural elements need not be considered if this kind of failure is extremely remote. (2) Compliance with sub-paragraph (b) (1) may be shown by failure analysis or testing, or both, for propeller systems that allow propeller blades to move from the flight low pitch position to a position that is substantially less than that at the normal flight low-pitch position.
934	USAR.934 Turbojet and turbofan engine thrust reverser system tests See AMC.934 Thrust reverser systems of turbojet or turbofan engines must meet the appropriate requirements agreed with the Certifying Authority.
937	USAR.937 Turbopropeller-drag limiting systems (a) Turbopropeller-powered UA propeller-drag limiting systems must be designed so that no single failure or malfunction of any of the systems during normal or emergency operation results in propeller drag in excess of that for which the UA was designed under the structural requirements of USAR. Failure of structural elements of the drag limiting systems need not be considered if the probability of this kind of failure is extremely remote. (b) As used in this paragraph, drag limiting systems include manual or automatic devices that, when actuated after engine power loss can move the propeller blades toward the feather position to reduce windmilling drag to a safe level.
939	USAR.939 Powerplant operating characteristics See AMC.939 (c) (a) Turbine engine powerplant operating characteristics must be investigated in flight to determine that no adverse characteristics (such as stall, surge, or flameout) are present, to a <i>Hazardous</i> degree, during normal and emergency operations within the range of operating limitations of the UA and of the engine. (b) Turbocharged reciprocating engine operating characteristics must be investigated in flight to an appropriate degree to assure that no adverse characteristics, such as a result of an inadvertent overboost, surge, flooding, or vapour lock, are present during normal or emergency operation of the engine(s) throughout the range of operating limitations of both UA and engine. (c) For turbine engines, the air inlet system must not, as a result of airflow distortion during normal operation, cause vibration harmful to the engine or instability greater than that allowed by the engine specification.
943	USAR.943 Negative acceleration No Hazardous malfunction of an engine, an auxiliary power unit approved for use in flight, or any component or system associated with the powerplant or auxiliary power unit may occur when the UA is operated at the negative accelerations within the flight envelope protection maintained by the flight control system as detailed in USAR.U334 This must be shown for the greatest value and duration of the acceleration expected in service.

FUEL SYSTEM

951	USAR.951 General (a) Each fuel system must be constructed and arranged to ensure fuel flow at a rate and pressure established for proper engine and auxiliary power unit functioning under each likely operating condition, including any manoeuvre for which certification is requested and during which the engine or auxiliary power unit is permitted to be in operation.
-----	---

	(b) Each fuel system must be arranged so that: (1) No fuel pump can draw fuel from more than one tank at a time; or (2) There are means to prevent introducing air into the system. (c) Each fuel system for a turbine engine must be capable of sustained operation throughout its flow and pressure range with fuel initially saturated with water at 27°C (80°F) and having 0.75cc of free water per 3.8 litres (0.83 Imperial gallon/1 US gallon) added and cooled to the most critical condition for icing likely to be encountered in operation.
953	USAR.953 Fuel system independence (a) Each fuel system for a multi-engine UA must be arranged so that, in at least one system configuration, the failure of any one component will not result in the loss of power of more than one engine or require immediate action by the UAS crew to prevent the loss of power of more than one engine. Nevertheless, automatic actions are compliant with USAR.953 (b) If a single fuel tank (or series of fuel tanks interconnected to function as a single fuel tank) is used on a multi-engine UA, the following must be provided: (1) Independent tank outlets for each engine, each incorporating a shutoff valve at the tank. This shutoff valve may also serve as the firewall shutoff valve required if the line between the valve and the engine compartment does not contain more than one litre (one US quart) of fuel (or any greater amount shown to be safe) that can escape into the engine compartment. (2) At least two vents arranged to minimize the probability of both vents becoming obstructed simultaneously. (3) Filler caps designed to minimize the probability of incorrect installation or inflight loss. (4) A fuel system in which those parts of the system from each tank outlet to any engine are independent of each part of the system supplying fuel to any other engine.
954	USAR.954 Fuel system lightning protection The fuel system must be designed and arranged to prevent the ignition of fuel vapour within the system in consistency with the lightning protection requirements defined in USAR.867 by: (a) Direct lightning strikes to areas having a high probability of stroke attachment; (b) Swept lightning strokes on areas where swept strokes are highly probable; and (c) Corona or streamering at fuel vent outlets.
955	USAR.955 Fuel flow (a) General. The ability of the fuel system to provide fuel at the rates specified in this paragraph and at a pressure sufficient for proper engine operation must be shown in the attitude that is most critical with respect to fuel feed and quantity of unusable fuel. These conditions may be simulated in a suitable mock-up. In addition: (1) The quantity of fuel in the tank may not exceed the amount established as the unusable fuel supply for that tank under USAR.959 (a) plus that necessary to show compliance with this paragraph; (2) If there is a fuel flowmeter, it must be blocked during the flow test and the fuel must flow through the test-meter or its by-pass. (3) If there is a flowmeter without a bypass, it must not have any failure mode that would restrict fuel flow below the level required in this fuel flow demonstration;

	(4) The fuel flow must include that flow needed for vapour return flow, jet pump drive flow and for all other purposes for which fuel is used. (b) Gravity systems. The fuel flow rate for gravity systems (main and reserve supply) must be 150% of the fuel consumption of the engine at the maximum take-off power or thrust approved under USAR. (c) Pump systems. The fuel flow rate for each pump system (main and reserve supply) for each reciprocating engine, must be 125% of the fuel flow required by the engine at the maximum take-off power or thrust approved under USAR. (1) This flow rate is required for each main pump and each emergency pump, and must be available when the pump is operating as it would during take-off; (2) Not applicable. (3) The fuel pressure, with main and emergency pumps operating simultaneously, must not exceed the fuel inlet pressure limits of the engine, unless it can be shown that no adverse effect occurs. (d) Auxiliary fuel systems and fuel transfer systems. Sub-paragraphs (b), (c) and (f) apply to each auxiliary and transfer system, except that: (1) The required fuel flow rate must be established upon the basis of maximum continuous power or thrust and engine rotational speed, instead of take-off power and fuel consumption; and (2) Where appropriate, if there are operating instructions related to fuel management in the UCS, a lesser flow rate may be used for transferring fuel from any auxiliary tank into a larger main tank. This lesser flow rate must be adequate to maintain maximum continuous power or thrust but the flow rate must not overfill the main tank at lower engine power or thrust. (e) Multiple fuel tanks. For reciprocating engines that are supplied with fuel from more than one tank, if engine power loss becomes apparent due to fuel depletion from the tank selected, it must be possible after switching to any full tank, in level flight, to obtain 75% maximum continuous power or thrust on that engine in not more than: (1) 10 seconds for naturally aspirated single-engine UA; (2) 20 seconds for turbocharged single-engine UA, provided that 75% maximum continuous naturally aspirated power is regained within 10 seconds; or (3) 20 seconds for multi-engine UA. (f) Turbine engine fuel systems. Each turbine engine fuel system must provide at least 100% of the fuel flow required by the engine under each intended operation condition and manoeuvre. The conditions may be simulated in a suitable mock-up. This flow must: (1) Be shown with the UA in the most adverse fuel feed condition (with respect to altitudes, attitudes and other conditions) that is expected in operation; and (2) For multi-engine UA, notwithstanding the lower flow rate allowed by sub-paragraph (d), be automatically uninterrupted with respect to any engine until all the fuel scheduled for use by that engine has been consumed. In addition: (i) For the purposes of this paragraph, "fuel scheduled for the use by that engine" means all fuel in any tank intended for use by a specific engine. (ii) The fuel system design must clearly indicate the engine for which fuel in any tank is scheduled.
--	--

	(iii) Compliance with this paragraph must require no UAS crew action after completion of the engine starting phase of operations. (3) For single engine UA, require no UAS crew action after completion of the engine starting phase of operations.
957	USAR.957 Flow between interconnected tanks (a) It must be impossible, in a gravity feed system with interconnected tank outlets, for enough fuel to flow between the tanks to cause an overflow of fuel from any tank vent under the conditions in USAR.959, except that full tanks must be used. (b) If fuel can be pumped from one tank to another in flight, the fuel tank vents and the fuel transfer system must be designed so that no structural damage to any UA component can occur because of overfilling of any tank.
959	USAR.959 Unusable fuel supply See AMC.959 (a) (a) The unusable fuel supply for each tank must be established as not less than that quantity at which the first evidence of malfunctioning occurs under the most adverse fuel feed condition occurring under each intended operation and flight manoeuvre involving that tank. Fuel system component failures need not be considered. (b) In addition, the effect on the unusable fuel quantity as a result of a failure of any pump must be determined.
961	USAR.961 Fuel system hot weather operation See AMC.961 (a) Each fuel system must be free from vapour lock when using fuel at its critical temperature, with respect to vapour formation, when operating the UA in all critical operating and environmental conditions (including ambient pressure) for which approval is requested. (b) For turbine fuel, the initial temperature must be 43°C, -0°C, + 2.7°C (110°F, -0°F, +5°F) or the maximum outside air temperature for which approval is requested, whichever is more critical.
963	USAR.963 Fuel tanks: general (a) Each fuel tank must be able to withstand, without failure, the vibration, inertia, fluid and structural loads that it may be subjected to in operation. (b) Each flexible fuel tank liner must be shown to be suitable for the particular application. (c) Each integral fuel tank must have adequate facilities for interior inspection and repair. (d) The total usable capacity of the fuel tanks must be enough for at least ½ hour of operation at maximum continuous power or thrust. (e) Each fuel quantity indicator must be adjusted, as specified USAR.U1729 (b), to account for the unusable fuel supply determined under USAR.959 (a).
965	USAR.965 Fuel tank tests (a) Each fuel tank must be able to withstand the following pressures without failure or leakage: (1) For each conventional metal tank and non-metallic tank with walls not supported by the UA structure, a pressure of 24 kPa (3.5 psig), or that pressure developed during maximum ultimate acceleration with a full tank, whichever is greater.

	(2) For each integral tank, the pressure developed during the maximum limit acceleration of the UA with a full tank, with simultaneous application of the critical limit structural loads. (3) For each non-metallic tank with walls supported by the UA structure and constructed in an acceptable manner using acceptable basic tank material and with actual or simulated support conditions, a pressure of 14 kPa (2 psig) for the first tank of a specific design. The supporting structure must be designed for the critical loads occurring in the flight or landing strength conditions combined with the fuel pressure loads resulting from the corresponding accelerations. (b) Each fuel tank with large, unsupported, or unstiffened flat surfaces, whose failure or deformation could cause fuel leakage, must be able to withstand the following test without leakage, failure or excessive deformation of the tank walls: (1) Each complete tank assembly and its support must be vibration tested while mounted to simulate the actual installation. (2) Except as specified in sub-paragraph (b) (4), the tank assembly must be vibrated for 25 hours (or the maximum UA endurance for which certification is requested if more than 25 hours) at a total displacement of not less than 0.8 mm (1/32 in) (unless another displacement is substantiated) while 2/3 filled with water or other suitable test fluid. (3) The test frequency of vibration must be as follows: (i) If no frequency of vibration resulting from any rpm within the normal operating range of engine or propeller speeds is critical, the test frequency of vibration is the number of cycles per minute obtained by multiplying the maximum continuous propeller speed in rpm by 0.9 for propeller-driven UA, except that for non-propeller driven UA, the test frequency of vibration is 2 000 cycles per minute. (ii) If only one frequency of vibration resulting from any rpm within the normal operating range of engine or propeller speeds is critical, that frequency must be the test frequency. (iii) If more than one frequency of vibration resulting from any rpm within the normal operating range of engine or propeller speeds is critical, the most critical of these frequencies must be the test frequency. (4) Under sub-paragraph (3) (ii) and (iii), the time of test must be adjusted to accomplish the same number of vibration cycles that would be accomplished in 25 hours (or the maximum UA endurance for which certification is requested if more than 25 hours) at the frequency specified in sub-paragraph (b) (3) (i). (5) During the test, the tank assembly must be rocked at a rate of 16 to 20 complete cycles per minute, through an angle of 15° on either side of the horizontal (30° total), about an axis parallel to the axis of the fuselage, for 25 hours (or the maximum UA endurance for which certification is requested if more than 25 hours). (c) Each integral tank using methods of construction and sealing not previously proven to be adequate by test data or service experience must be able to withstand the vibration test specified in sub-paragraphs (1) to (4) of paragraph (b). (d) Each tank with a non-metallic liner must be subjected to the sloshing test outlined in sub-paragraph (5) of paragraph (b), with the fuel at room temperature. In addition, a specimen liner of the same basic construction as that to be used in the UA must, when installed in a suitable test tank, withstand the sloshing test with fuel at a temperature of 43°C (110°F).
967	USAR.967 Fuel tank installation See AMC.967

- (a) Each fuel tank must be supported so that tank loads are not concentrated. In addition:
- (1) There must be pads, if necessary, to prevent chafing between each tank and its supports;
 - (2) Padding must be non-absorbent or treated to prevent the absorption of fuel;
 - (3) If a flexible tank liner is used, it must be supported so that it is not required to withstand fluid loads;
 - (4) Interior surfaces adjacent to the liner must be smooth and free from projections that could cause wear, unless:
 - (i) Provisions are made for protection of the liner at those points; or
 - (ii) The construction of the liner itself provides such protection.
 - (5) A positive pressure must be maintained within the vapour space of each bladder cell under all conditions of operation except for a particular condition for which it is shown that a zero or negative pressure will not cause the bladder cell to collapse; and
 - (6) Siphoning of fuel (other than minor spillage) or collapse of bladder fuel cells may not result from improper securing or loss of the fuel filler cap.
- (b) Each tank compartment must be able to be ventilated and drained to prevent the accumulation of flammable fluids or vapours. Each compartment adjacent to a tank that is an integral part of the UA structure must also be ventilated and drained.
- (c) No fuel tank may be on the engine side of the firewall. There must be at least 13 mm (½ in) of clearance between the fuel tank and the firewall. No part of the engine nacelle skin that lies immediately behind a major air opening from the engine compartment may act as the wall of an integral tank.
- (d) Not applicable.
- (e) Fuel tanks must be designed, located and installed so as to retain fuel:
- (1) In consistency with safety objectives as stated in AMC.1309 regarding forced landing conditions, and
 - (2) Except where a UA is designed to be recovered by parachute under conditions likely to occur when the UA lands on a paved runway at a normal landing speed under each of the following conditions:
 - (i) The UA in a normal landing attitude and its landing gear retracted.
 - (ii) The most critical landing gear leg collapsed and the other landing gear legs extended. In showing compliance with sub-paragraph (e) (2), the tearing away of an engine mount must be considered unless all the engines are installed above the wing or on the tail or fuselage of the UA.
 - (3) Where a UA is designed to be recovered by parachute, so as to retain fuel under conditions likely to occur at the impact on the ground under following conditions:
 - (i) The UA in a normal landing attitude and its shock absorber cushions not inflated,
 - (ii) The most critical shock absorber cushions configuration taking into account inflated and non- inflated cushions.

	(4) Where a UA is designed for rocket assisted take-off or catapult assisted take-off, so as to retain fuel under conditions likely to occur on the launch ramp and during the initial take-off phase.
969	USAR.969 Fuel tank expansion space Each fuel tank must have an expansion space of not less than 2% of the tank capacity, unless the tank vent discharges clear of the UA (in which case no expansion space is required). It must be impossible to fill the expansion space inadvertently with the UA in the normal ground attitude.
971	USAR.971 Fuel tank sump (a) Where appropriate, each fuel tank must have a drainable sump with an effective capacity, in the normal ground and flight attitudes, of 0.25% of the tank capacity, or 0.24 litres (0.05 Imperial gallon/ 0.625 US-gallon), whichever is greater. (b) Each fuel tank must allow drainage of any Hazardous quantity of water from any part of the tank to its sump with the UA in the normal ground attitude. (c) Where required by the Certifying Authority, each reciprocating engine fuel system must have a sediment bowl or chamber that is accessible for drainage; has a capacity of 30 cm³ (1 fl oz) for every 75.7 litres (16.7 Imperial gallon/20 US-gallon) of fuel tank capacity; and each fuel tank outlet is located so that, in the normal flight attitude, water will drain from all parts of the tank except the sump to the sediment bowl or chamber. (d) Each sump, sediment bowl and sediment chamber drain required by sub-paragraphs (a), (b) and (c) must comply with the drain provisions of USAR.999 (b) (1) and (2).
973	USAR.973 Fuel tank filler connection See AMC.973 (a) Each fuel tank filler connection must be marked as prescribed in USAR.1557 (c). (b) Spilled fuel must be prevented from entering the fuel tank compartment or any part of the UA other than the tank itself. (c) Each filler cap must provide a fuel-tight seal for the main filler opening. However, there may be small openings in the fuel tank cap for venting purposes or for the purpose of allowing passage of a fuel gauge through the cap provided such openings comply with the requirements of USAR.975 (a). (d) There must be a means to insure that there is no electrical potential difference between the fuel filling points (except the pressure fuelling connection points) and the ground fuel equipment. (e) For UA with engines requiring gasoline as the only permissible fuel, the inside diameter of the fuel filler opening must be no larger than 60 mm (2.36 in). (f) For UA with turbine engines, the inside diameter of the fuel filler opening must be no smaller than 75 mm (2.95 in).
975	USAR.975 Fuel tank vents and carburettor vapour vents See AMC.975 (a) Each fuel tank must be vented from the top part of the expansion space. In addition: (1) Each vent outlet must be located and constructed in a manner that minimises the possibility of it being obstructed by ice or other foreign matter; (2) Each vent must be constructed to prevent siphoning of fuel during normal operation; (3) The venting capacity must allow the rapid relief of excessive differences of pressure between the interior and exterior of the tank;

	(4) Airspaces of tanks with interconnected outlets must be inter-connected; (5) There may be no points in any vent line where moisture can accumulate with the UA in either the ground or level flight attitudes unless drainage is provided. (6) No vent may terminate at a point where the discharge of fuel from the vent outlet will constitute a fire hazard; and (7) Vents must be arranged to prevent the loss of fuel, except fuel discharged because of thermal expansion, when the UA is parked in any direction on a ramp having a 1% slope and when the UA is on the launching ramp or catapult for the UAS designed for rocket assisted or catapult assisted take-off. (8) Vents must be arranged to prevent re-ingestion of vented fuel or fuel vapours back into the UA. (b) Each carburettor with vapour elimination connections and each fuel injection engine employing vapour return provisions must have a separate vent line to lead vapours back to the top of one of the fuel tanks. If there is more than one tank and it is necessary to use these tanks in a definite sequence for any reason, the vapour vent line must lead back to the fuel tank to be used first, unless the relative capacities of the tanks are such that return to another tank is preferable. (c) Not applicable.
977	USAR.977 Fuel tank outlet (a) There must be a fuel strainer for the fuel tank outlet or for the booster pump. This strainer must: (1) For reciprocating engine-powered UA, have appropriate mesh density; and (2) For turbine engine-powered UA, prevent the passage of any object that could restrict fuel flow, damage any fuel system component or result in loss of power. (b) The clear area of each fuel tank outlet strainer must be at least five times the area of the outlet line. (c) The diameter of each strainer must be at least that of the fuel tank outlet. (d) Each strainer must be accessible for inspection and cleaning.
979	USAR.979 Pressure fuelling systems For pressure fuelling systems, the following applies: (a) Each pressure fuelling system fuel manifold connection must have means to prevent the escape of hazardous quantities of fuel from the system if the fuel entry valve fails. (b) An automatic shut-off means must be provided to prevent the quantity of fuel in each tank from exceeding the maximum quantity approved for that tank. This means must allow checking for proper shut-off operation before each fuelling of the tank; (c) A means must be provided to prevent damage to the fuel system in the event of failure of the automatic shut-off means prescribed in sub-paragraph (b). (d) All parts of the fuel system up to the tank which are subjected to fuelling pressures must have a proof pressure of 1.33 times and an ultimate pressure of at least 2.0 times, the surge pressure likely to occur during fuelling.
FUEL SYSTEM COMPONENTS	
991	USAR.991 Fuel pumps

	(a) Main pumps. For main pumps, the following apply: (1) For reciprocating engine installations having fuel pumps to supply fuel to the engine, at least one pump for each engine must meet USAR.955. This pump is a main pump. (2) For turbine engine installations, each fuel pump required for proper engine operation, or required to meet the fuel system requirements of this subpart (other than those in sub-paragraph (b)), is a main pump. In addition: (i) There must be at least one main pump for each turbine engine; (ii) The power supply for the main pump for each engine must be independent of the power supply for each main pump for any other engine; and (iii) For each main pump, provision must be made to allow the by-pass of each positive displacement fuel pump other than a fuel injection pump approved as part of the engine. (b) Emergency pumps. There must be an emergency pump immediately available to supply fuel to the engine if any main pump (other than a fuel injection pump approved as part of an engine) fails. The power supply for each emergency pump must be independent of the power supply for each corresponding main pump. (c) Not applicable in this subpart. (see USAR.U1797 Fuel pumps warning) (d) Operation of any fuel pump may not affect engine operation so as to create a hazard, regardless of the engine power or thrust setting or the functional status of any other fuel pump, except for jet engines fitted with an afterburning system, where an emergency pump is limited to engine dry operation.
993	USAR.993 Fuel system lines and fittings See AMC.993 (a) Each fuel line must be installed and supported to prevent excessive vibration and to withstand loads due to fuel pressure and accelerated flight conditions. (b) Each fuel line connected to components of the UA between which relative motion could exist must have provisions for flexibility. (c) Each flexible connection in fuel lines that may be under pressure and subjected to axial loading must use flexible hose assemblies. (d) Each flexible hose must be shown to be suitable for the particular application. (e) No flexible hose that is shown to be adversely affected by exposure to high temperatures may be used where excessive temperatures will exist during operation or after shut-down of an engine or auxiliary power unit. (f) Fuel system fittings shall be designed to prevent degradation caused by electrical potentials developed by dissimilar metals.
994	USAR.994 Fuel system components Fuel system components in an engine nacelle or in the fuselage shall be protected from damage which could result in spillage of enough fuel to constitute a fire hazard as a result of a wheels-up landing on a paved runway in accordance with safety objectives as stated in AMC.1309 regarding forced landing conditions.
995	USAR.995 Fuel valves and controls See AMC.995

	(see also USAR.U1743 Fuel controls) (a) There must be a means on board, commanded from the UCS, to allow UAS crew to rapidly shut off, in flight, the fuel to each engine individually. (b) No shut-off valve, if present, may be on the engine side of any firewall. (c) Each valve and fuel system control must be supported so that loads resulting from its operation or from accelerated flight conditions are not transmitted to the lines connected to the valve. (d) Each valve and fuel system control must be installed so that gravity and vibration will not affect the selected position. (e) Each fuel valve handle and its connections to the valve mechanism must have design features that minimise the possibility of incorrect installation. (f) Each valve must be constructed, or otherwise incorporate provisions, to preclude incorrect assembly or connection of the valve. (g) If fuel tank valves are installed on the UA, these fuel tank selector valves must: (1) Require a separate and distinct action to place the selector in the “OFF” position; and (2) Have the tank selector positions located in such a manner that it is impossible for the selector to pass through the “OFF” position when changing from one tank to another.
997	USAR.997 Fuel strainer or filter See AMC.997 There must be a fuel strainer or filter between the fuel tank outlet and the inlet of either the fuel metering device or an engine driven positive displacement pump, whichever is nearer the fuel tank outlet. This fuel strainer or filter must: (a) Be accessible for draining and cleaning and must incorporate a screen or element which is easily removable; (b) Have a sediment trap and drain except that it need not have a drain if the strainer or filter is easily removable for drain purposes; (c) Be mounted so that its weight is not supported by the connecting lines or by the inlet or outlet connections of the strainer or filter itself, unless adequate strength margins under all loading conditions are provided in the lines and connections; and (d) Have the capacity (with respect to operating limitations established for the engine) to ensure that engine fuel system functioning is not impaired, with the fuel contaminated to a degree (with respect to particle size and density) that is greater than that established for the engine during its type certification. (e) Not applicable.
999	USAR.999 Fuel system drains See AMC.999 (a) There must be at least one drain to allow safe drainage of the entire fuel system with the UA in its normal ground attitude. (b) Each drain required by sub-paragraph (a) and USAR.971 must: (1) Discharge clear of all parts of the UA;

	(2) Have a drain valve: (i) That has manual or automatic means for positive locking in the closed position; (ii) That is readily accessible; (iii) That can be easily opened and closed; (iv) That allows the fuel to be caught for examination; (v) That can be observed for proper closing; and (vi) That is either located or protected to prevent fuel spillage in the event of a landing with landing gear retracted, for UA incorporating a retractable landing gear.
1001	USAR.1001 Fuel jettisoning system See AMC.1001 (a) If the design landing weight is less than that permitted under the requirements of USAR.473 (b), the UA must have a fuel jettisoning system installed that is able to jettison enough fuel to bring the maximum weight down to the design landing weight. The average rate of fuel jettisoning must be at least 1% of the maximum weight per minute, except that the time required to jettison the fuel need not be less than 10 minutes. (b) Fuel jettisoning must be demonstrated at maximum weight with flaps and landing gear up and in: (1) A power-off glide at $1.4 V_{S1}$; and (2) A climb, at the speed at which one engine in-operative en-route climb data has been established in accordance with USAR.69 (b), with the critical engine inoperative and the remaining engine at maximum continuous power or thrust; and (3) Level flight at $1.4 V_{S1}$, if the results of the tests in the conditions specified in sub-paragraphs (1) and (2) show that this condition could be critical. (c) During the flight tests prescribed in sub-paragraph (b), it must be shown that: (1) The fuel jettisoning system and its operation are free from fire hazard; (2) The fuel discharges clear of any part of the UA; (3) Fuel or fumes do not enter any parts of the UA; and (4) The jettisoning operation does not adversely affect the controllability of the UA. (d) For reciprocating engine powered UA, the jettisoning system must be designed so that it is not possible to jettison the fuel in the tanks used for take-off and landing below the level allowing 45 minutes flight at 75% maximum continuous power or thrust. However, if there is an auxiliary control independent of the main jettisoning control, the system may be designed to jettison all the fuel. (e) For turbine engine-powered UA, the jettisoning system must be designed so that it is not possible to jettison fuel in the tanks used for take-off and landing below the level allowing climb from sea level to 3048 m (10,000 ft) and thereafter allowing 45 minutes cruise at a speed for maximum range. (f) Not applicable in this subpart. (see USAR.U1745 Fuel jettisoning control) (g) Not applicable in this subpart. (see USAR.U1745 Fuel jettisoning control)

	(h) The fuel jettisoning system must be designed so that any reasonably probable single malfunction in the system will not result in a <i>Hazardous</i> condition due to asymmetrical jettisoning of, or inability to jettison, fuel. (i) Fuel jettisoning must be performed by the UAS crew. Nevertheless, in case of total loss of data link, automatic procedures of fuel jettisoning must be assessed according to safety objectives stated in USAR.1309
--	--

OIL SYSTEM

1011	USAR.1011 General See AMC.1011 (b) (a) For oil systems and components that have been approved under the engine airworthiness requirements and where those requirements are equal to or more severe than the corresponding requirements of subpart E of USAR, that approval need not be duplicated. Where the requirements of subpart E of USAR are more severe, substantiation must be shown to the requirements of subpart E. (b) Each engine and auxiliary power unit must have an independent oil system that can supply it with an appropriate quantity of oil at a temperature not above that safe for continuous operation. (c) The usable oil tank capacity may not be less than the product of the endurance of the UA under critical operating conditions and the maximum oil consumption of the engine under the same conditions, plus a suitable margin to ensure adequate circulation and cooling. (d) For an oil system without an oil transfer system, only the usable oil tank capacity may be considered. The amount of oil in the engine oil lines, the oil radiator and the feathering reserve, may not be considered. (e) If an oil transfer system is used and the transfer pump can pump some of the oil in the transfer lines into the main engine oil tanks, the amount of oil in these lines that can be pumped by the transfer pump may be included in the oil capacity. (f) Where engine lubrication is based on a fuel/oil mixture, an appropriate and reliable means for providing it must be substantiated under any combination of environmental conditions.
1013	USAR.1013 Oil tanks (a) Installation. Each oil tank must be installed to: (1) Meet the requirements of USAR.967 (a) and (b); and (2) Withstand any vibration, inertia and fluid loads expected in operation. (b) Expansion space. Oil tank expansion space must be provided so that: (1) Each oil tank used with a reciprocating engine has an expansion space of not less than the greater of 10% of the tank capacity or 1.9 litres (0.42 Imperial gallon/0.5 US-gallon) and each oil tank used with a turbine engine has an expansion space of not less than 10% of the tank capacity; and (2) It is impossible to fill the expansion space inadvertently with the UA in the normal ground attitude. (c) Filler connection. Each oil tank filler connection must be marked as specified in USAR.1557 (c). Each recessed oil tank filler connection of an oil tank used with a turbine engine, that can retain any appreciable quantity of oil, must have provisions for fitting a drain. (d) Vent. Oil tanks must be vented as follows:

	(1) Each oil tank must be vented to the engine from the top part of the expansion space so that the vent connection is not covered by oil under any normal flight condition. (2) Oil tank vents must be arranged so that condensed water vapour that might freeze and obstruct the line cannot accumulate at any point. (3) Not applicable. (e) Outlet. No oil tank outlet may be enclosed by any screen or guard that would reduce the flow of oil below a safe value at any operating temperature. No oil tank outlet diameter may be less than the diameter of the engine oil pump inlet. Each oil tank used with a turbine engine must have means to prevent entrance into the tank itself, or into the tank outlet, of any object that might obstruct the flow of oil through the system. There must be a shut-off valve at the outlet of each oil tank used with a turbine engine, unless the external portion of the oil system (including oil tank supports) is fire-proof. (f) Flexible liners. Each flexible oil tank liner must be of an acceptable kind. (g) Each oil tank filler cap of an oil tank that is used with an engine must provide an oil tight seal.
1015	USAR.1015 Oil tank tests Each oil tank must be tested under USAR.965, except that: (a) The applied pressure must be 34 kPa (5 psig) for the tank construction instead of the pressures specified in USAR.965 (a). (b) For a tank with a non-metallic liner the test fluid must be oil rather than fuel as specified in USAR.965 (d) and the slosh test on a specimen liner must be conducted with the oil at 121°C (250°F); and (c) For pressurised tanks used with a turbine engine, the test pressure may not be less than 34 kPa (5 psig) plus the maximum operating pressure of the tank.
1017	USAR.1017 Oil lines and fittings (a) Oil lines. Oil lines must meet USAR.993 and must accommodate a flow of oil at a rate and pressure adequate for proper engine functioning under any normal operating conditions. (b) Breather lines. Breather lines must be arranged so that: (1) Condensed water vapour or oil that might freeze and obstruct the line cannot accumulate at any point; (2) The breather discharge will not constitute a fire hazard if foaming occurs, or cause emitted oil to pollute critical sensors in terms of flight safety. (3) The breather does not discharge into the engine air induction system; (4) Not applicable. (5) The breather outlet is protected against blockage by ice or foreign matter. (c) Oil system fittings shall be designed to prevent degradation caused by electrical potentials developed by dissimilar metals.
1019	USAR.1019 Oil strainer or filter (a) Each turbine engine installation must incorporate an oil strainer or filter through which all of the engine oil flows and which meets the following requirements:

	(1) Each oil strainer or filter that has a by-pass must be constructed and installed so that oil will flow at the normal rate through the rest of the system with the strainer or filter completely blocked. (2) The oil strainer or filter must have the capacity (with respect to operating limitations established for the engine) to ensure that engine oil system functioning is not impaired when the oil is contaminated to a degree (with respect to particle size and density) that is greater than that established for the engine for its type certification. (3) The oil strainer or filter, unless it is installed at an oil tank outlet, must incorporate a means to indicate contamination before it reaches the capacity established in accordance with subparagraph (2). (4) The by-pass of a strainer or filter must be constructed and installed so that the release of collected contaminants is minimised by appropriate location of the by-pass to ensure that collected contaminants are not in the bypass flow path. (5) An oil strainer or filter that has no by-pass, except one that is installed at an oil tank outlet, must have a means to connect it to the warning system required in USAR.U1725 (c). (b) Each oil strainer or filter in a powerplant installation using reciprocating engines must be constructed and installed so that oil will flow at the normal rate through the rest of the system with the strainer or filter element completely blocked.
1021	USAR.1021 Oil system drains A drain or drains must be provided to allow safe drainage of the oil system. Each drain must: (a) Be accessible; (b) Have drain valves, or other closures, employing manual or automatic shut-off means for positive locking in the closed position; and (c) Be located or protected to prevent inadvertent operation.
1023	USAR.1023 Oil radiators Each oil radiator and its supporting structures must be able to withstand the vibration, inertia and oil pressure loads to which it would be subjected in operation.
1027	USAR.1027 Propeller feathering system (a) If the propeller feathering system uses engine oil and that oil supply can become depleted due to failure of any part of the oil system, a means must be incorporated to reserve enough oil to operate the feathering system. (b) The amount of reserved oil must be enough to accomplish feathering and must be available only to the feathering pump. (c) The ability of the system to accomplish feathering with the reserved oil must be shown. (d) Provision must be made to prevent sludge or other foreign matter from affecting the safe operation of the propeller feathering system.

COOLING

1041	USAR.1041 General The powerplant and auxiliary power unit cooling provisions must maintain the temperatures of: (a) powerplant components. (b) engine fluids.
------	---

	(c) auxiliary power unit components and fluids within the limits established for those components, and fluids.
1043	USAR.1043 Cooling tests (a) General. Compliance with USAR.1041 must be shown on the basis of tests, for which the following apply: (1) If the tests are conducted under ambient atmospheric temperature conditions deviating from the maximum for which approval is requested, the recorded powerplant temperatures must be corrected under sub-paragraphs (c) and (d), unless a more rational correction method is applicable. (2) Corrected temperatures determined under sub-paragraph (a) (1) must not exceed established limits. (3) The fuel used during the cooling tests must be of the minimum grade approved for the engine(s). (4) For turbocharged engines, each turbocharger must be operated through that part of the climb profile for which operation with the turbocharger is requested. (5) For reciprocating engines with manual mixture control, the mixture settings must be the leanest recommended for climb. (b) Maximum ambient atmospheric temperature. A maximum ambient atmospheric temperature corresponding to sea-level conditions of at least the intended maximum operating ambient temperature plus 5°C (9°F). The Applicant may select a maximum ambient atmospheric temperature corresponding to sea level conditions of less than 38°C (100°F). (c) Correction factor (except cylinder barrels). Temperatures of engine fluids and powerplant components (except cylinder barrels) for which temperature limits are established, must be corrected by adding to them the difference between the maximum ambient atmospheric temperature for the relevant altitude for which approval has been requested and the temperature of the ambient air at the time of the first occurrence of the maximum fluid or component temperature recorded during the cooling test. (d) Correction factor for cylinder barrel temperatures. Cylinder barrel temperatures must be corrected by adding to them 0.7 times the difference between the maximum ambient atmospheric temperature for the relevant altitude for which approval has been requested and the temperature of the ambient air at the time of the first occurrence of the maximum cylinder barrel temperature recorded during the cooling test.
1045	USAR.1045 Cooling test procedures for turbine engine-powered UA See AMC.1045 and AMC.1045 (b) (a) Compliance with USAR.1041 must be shown for all phases of operation. The UA must be flown in the configurations, at the speeds and following the procedures recommended in the UAS Flight Manual for the relevant stage of flight, corresponding to the applicable performance requirements, which are critical relative to cooling. (b) Temperatures must be stabilised under the conditions from which entry is made into each stage of flight being investigated, unless the entry condition normally is not one during which component and engine fluid temperatures would stabilise (in which case, operation through the full entry condition must be conducted before entry into the stage of flight being investigated in order to allow temperatures to reach their natural levels at the time of entry). The take-off cooling test must be preceded by a period during which the powerplant component and engine fluid temperatures are stabilised with the engines at ground idle. (c) Cooling tests for each stage of flight must be continued until:

	(1) The component and engine fluid temperatures stabilise; or (2) The stage of flight is completed; or (3) An operating limitation is reached.
1047	USAR.1047 Cooling test procedures for reciprocating engine-powered UA See AMC.1047 Compliance with USAR.1041 must be shown for the climb (or descent, for multi-engine UA with negative one-engine-inoperative rates of climb) stage of flight. The UA must be flown in the configurations, at the speeds and following the procedures recommended in the UAS Flight Manual, corresponding to the applicable performance requirements, which are critical relative to cooling.

LIQUID COOLING

1061	USAR.1061 Installation (a) General. Each liquid-cooled engine must have an independent cooling system (including coolant tank) installed so that: (1) Each coolant tank is supported so that tank loads are distributed over a large part of the tank surface; (2) There are pads or other isolation means between the tank and its supports to prevent chafing; and (3) Pads or any other isolation means that is used must be non-absorbent or must be treated to prevent absorption of flammable fluids; and (4) No air or vapour can be trapped in any part of the system, except the coolant tank expansion space, during filling or during operation. (b) Coolant tank. The tank capacity must be at least 3.8 litres (0.83 Imperial gallon/1 US gallon), plus 10% of the cooling system capacity. The Applicant may apply for a lower tank capacity on the basis of a rational analysis taking into account the cooling provisions for the engine. In addition: (1) Each coolant tank must be able to withstand the vibration, inertia and fluid loads to which it may be subjected in operation; (2) Each coolant tank must have an expansion space of at least 10% of the total cooling system capacity; and (3) It must be impossible to fill the expansion space inadvertently with the UA in the normal ground attitude. (c) Filler connection. Each coolant tank filler connection must be marked as specified in USAR.1557 (c). In addition: (1) Spilled coolant must be prevented from entering the coolant tank compartment or any part of the UA other than the tank itself; and (2) Each recessed coolant filler connection must have a drain that discharges clear of the entire UA. (d) Lines and fittings. Each coolant system line and fitting must meet the requirements of USAR.993, except that the inside diameter of the engine coolant inlet and outlet lines may not be less than the diameter of the corresponding engine inlet and outlet connections.
------	--

	(e) Radiators. Each coolant radiator must be able to withstand any vibration, inertia and coolant pressure load to which it may normally be subjected. In addition: (1) Each radiator must be supported to allow expansion due to operating temperatures and prevent the transmittal of harmful vibration to the radiator; and (2) If flammable coolant is used, the air intake duct to the coolant radiator must be located so that (in case of fire) flames from the nacelle cannot strike the radiator. (f) Drains. There must be an accessible drain that: (1) Drains the entire cooling system (including the coolant tank, radiator and the engine) when the UA is in the normal ground attitude; (2) Discharges clear of the entire UA; and (3) Has means to positively lock it closed.
1063	USAR.1063 Coolant tank tests Each coolant tank must be tested under USAR.965, except that: (a) The test required by USAR.965 (a) (1) must be replaced with a similar test using the sum of the pressure developed during the maximum ultimate acceleration with a full tank or a pressure of 24 kPa (3.5 psig), whichever is greater, plus the maximum working pressure of the system; and (b) For a tank with a non-metallic liner the test fluid must be coolant rather than fuel as specified in USAR.965 (c) and the slosh test on a specimen liner must be conducted with the coolant at operating temperature.

INDUCTION SYSTEM

1091	USAR.1091 Air induction system See AMC.1091 (a) The air induction system for each engine and auxiliary power unit and their accessories must supply the air required by that engine and auxiliary power unit under the operating conditions for which certification is requested. (b) Each reciprocating engine installation must be shown to have adequate air intake sources and must meet the following: (1) Primary air intakes may open within the cowl if that part of the cowl is isolated from the engine accessory section by a fire-resistant diaphragm or if there are means to prevent the emergence of backfire flames. (2) Each alternate air intake (if applicable) must be located in a sheltered position and may not open within the cowl if the emergence of backfire flames will result in a hazard. (3) The supplying of air to the engine through the alternate air intake (if applicable) system may not result in an excessive loss of power or thrust in addition to the power loss due to the rise in air temperature. (4) Not applicable in this subpart. (see USAR.U1747 Air induction control) (5) Not applicable in this subpart. (see USAR.U1799 Air induction indicator) (c) For turbine engine-powered UA:
------	--

	(1) There must be means to prevent hazardous quantities of fuel leakage or overflow from drains, vents or other components of flammable fluid systems from entering the engine or auxiliary power unit and their accessories intake system; and (2) The UA must be designed to prevent water or slush on the runway, taxi way, or other airfield operating surfaces from being directed into the engine or auxiliary power unit air intake ducts in hazardous quantities, and the air intake ducts must be located or protected so as to minimise the ingestion of foreign matter during take-off, landing and taxiing. (3) The air inlet system must not, as a result of airflow distortion during normal operation, cause vibration harmful to the engine or instability that would result in surge, flameout, or instability greater than that allowed by the engine specification, in line with USAR.939 (c).
1093	USAR.1093 Induction system icing protection See AMC.1093 and AMC.1093 (b) (a) Reciprocating engines. Each reciprocating engine air induction system must have means to prevent and eliminate icing. Unless this is done by other means, it must be shown that, in air free of visible moisture at a temperature of -1°C (30°F) (1) Each UA with sea-level engines using conventional venturi carburetors has a preheater that can provide a heat rise of 50°C (90°F) with the engines at 75% of maximum continuous power or thrust; (2) Each UA with altitude engines using conventional venturi carburetors has a preheater that can provide a heat rise of 67°C (120°F) with the engines at 75% of maximum continuous power or thrust; (3) Each UA with altitude engines using carburetors tending to prevent icing has a preheater that, with the engines at 60% of maximum continuous power or thrust, can provide a heat rise of: (i) 56°C (100°F); or (ii) 22°C (40°F), if a fluid de-icing system meeting the requirements of USAR.1095 to USAR.1099 is installed; (4) Each UA with sea level engine(s) using fuel metering device tending to prevent icing has a sheltered alternate source of air with a preheat of not less than 16°C (60°F) with the engines at 75% of maximum continuous power; (5) Each UA with sea level or altitude engine(s) using fuel injection systems having metering components on which impact ice may accumulate has a preheater capable of providing a heat rise of 42°C (75°F) when the engine is operating at 75% of its maximum continuous power; (6) Each UA with sea level or altitude engine(s) using fuel injection systems not having fuel metering components projecting into the airstream on which ice may form, and introducing fuel into the air induction system downstream of any components or other obstruction on which ice produced by fuel evaporation may form, has a sheltered alternate source of air with a preheat of not less than 16°C (60°F) with the engines at 75% of its maximum continuous power or thrust. (b) Turbine engines (1) Each turbine engine and its air inlet system must operate throughout the flight power range of the engine (including idling), without the accumulation of ice on engine or inlet system components that would adversely affect engine operation or cause a serious loss of power or thrust: (i) Under the icing conditions specified by the Certifying Authority; and

	(ii) In snow, both falling and blowing, within the limitations established for the UA for such operation. (2) Each turbine engine must idle for 30 minutes on the ground, with the air bleed available for engine icing protection at its critical condition, without adverse effect, in an atmosphere that is at a temperature between -9°C and -1°C (between 15°F and 30°F) and has a liquid water content not less than 0.3 grams per cubic metre in the form of drops having a mean effective diameter not less than 20 microns, followed by momentary operation at take-off power or thrust. During the 30 minutes of idle operation, the engine may be run up periodically to a moderate power or thrust setting in a manner acceptable to the Certifying Authority. (c) Reciprocating engines with superchargers. For UA with reciprocating engines having superchargers to pressurise the air before it enters the carburettor, the heat rise in the air caused by that supercharging at any altitude may be utilised in determining compliance with sub-paragraph (a) if the heat rise utilised is that which will be available, automatically, for the applicable altitudes and operating condition because of supercharging.
1095	USAR.1095 Carburettor de-icing fluid flow rate See AMC.1095 (a) If a carburettor de-icing fluid system is used: (a) It must be able to simultaneously supply each engine with a rate of fluid flow as agreed with the Certifying Authority. (b) The fluid must be introduced into the air induction system: (1) Close to, and upstream of, the carburettor; and (2) So that it is equally distributed over the entire cross section of the induction system air passages.
1097	USAR.1097 Carburettor de-icing fluid system capacity (a) The capacity of each carburettor de-icing fluid system: (1) May not be less than the greater of: (i) That required to provide fluid at the rate specified in USAR.1095 for a time equal to 3% of the maximum endurance of the UA; or (ii) 20 minutes at that flow rate; and (2) Need not exceed that required for a duration of operations as agreed with the Certifying Authority. (b) If the available preheat increases the temperature of more than 28°C (50°F) but less than 56°C (100°F), the capacity of the system may be decreased in proportion to the heat rise available in excess of 28°C (50°F).
1099	USAR.1099 Carburettor de-icing fluid system detail design Each carburettor de-icing fluid system must meet the applicable requirements for the design of a fuel system, except as specified in USAR.1095 and USAR.1097
1101	USAR.1101 Induction air preheater design Each exhaust-heated, induction air preheater must be designed and constructed to: (a) Ensure ventilation of the preheater when the induction air preheater is not being used during engine operation.

	(b) Allow inspection of the exhaust manifold parts that it surrounds; and (c) Allow inspection of critical parts of the preheater itself.
1103	USAR.1103 Induction system ducts See AMC.1103 (a) Each induction system duct must drain to prevent the accumulation of fuel or moisture in the normal ground and flight attitudes. No drain may discharge where it will cause a fire hazard. (b) Each duct connected to components between which relative motion could exist must have means for flexibility. (c) Each flexible induction system duct must be capable of withstanding the effects of temperature extremes, fuel, oil, water, and solvents to which it is expected to be exposed in service and maintenance without hazardous deterioration or delamination. (d) For reciprocating engine installations, each induction system duct must be: (1) Strong enough to prevent induction system failures resulting from normal backfire conditions; and (2) Fire resistant in any compartment for which a fire extinguishing system is required. (e) Each inlet system duct for an auxiliary power unit must be: (1) Fireproof within the auxiliary power unit compartment; (2) Fireproof for a sufficient distance upstream of the auxiliary power unit compartment to prevent hot gas reverse flow from burning through the duct and entering any other compartment of the UA in which a hazard would be created by the entry of the hot gases; (3) Constructed of materials suitable to the environmental conditions expected in service, except in those areas requiring fireproof or fire resistant materials; and (4) Constructed of materials that will not absorb or trap Hazardous quantities of flammable fluids that could be ignited by a surge or reverse-flow condition. (f) Not applicable. (g) For turbine engine installations, air inlet system shall have adequate structural margin to withstand over-pressures generated by inlet/compressor anomalies.
1105	USAR.1105 Induction system screens If induction system screens are used on reciprocating engines: (a) Each screen must be upstream of the carburettor or fuel injection system; (b) No screen may be in any part of the induction system that is the only passage through which air can reach the engine, unless: (1) The available preheat increases the temperature of more than 56°C (100°F); and (2) The screen can be de-iced by heated air; (c) No screen may be de-iced by alcohol alone; and (d) It must be impossible for fuel to strike any screen.

1107	USAR.1107 Induction system filters On reciprocating-engine installations, if an air filter is used to protect the engine against foreign material particles in the induction air supply: (a) Each air filter must be capable of withstanding the effects of temperature extremes, rain, fuel, oil, and solvents to which it is expected to be exposed in service and maintenance; and (b) Each air filter must have a design feature to prevent material separated from the filter media from interfering with proper fuel metering operation.
1109	USAR.1109 Turbocharger bleed air system (a) The pressurized air system may not be subject to hazardous contamination following any probable failure of the turbocharger or its lubrication system. (b) The turbocharger supply air must be taken from a source where it cannot be contaminated by harmful or hazardous gases or vapours following any probable failure or malfunction of the engine exhaust, hydraulic, fuel, or oil system causing a fire hazard or safety issues for the UA.
1111	USAR.1111 Turbine engine bleed air system For turbine engine bleed air systems, the following applies: (a) No <i>Major</i> or more serious event may result if duct rupture or failure occurs anywhere between the engine port and the UA unit served by the bleed air. (b) Engine performance degradation shall be quantified as a function of bleed air usage. (c) Not applicable. (d) Adequate compartment ventilation must be provided for cooling hot duct surfaces where the Auto-Ignition Temperature (AIT) of combustible materials is exceeded. (e) Ducts whose surface temperature exceeds the AIT must be insulated such that the outside surface temperature is less than AIT. (f) The design must prohibit combustible fluids from direct impingement on duct surfaces.

EXHAUST SYSTEM

1121	USAR.1121 General For powerplant and auxiliary power unit installations, the following applies: (a) Each exhaust system must ensure safe disposal of exhaust gases without fire hazard. (b) Each exhaust system part with a surface hot enough to ignite flammable fluids or vapours must be located or shielded so that leakage from any system carrying flammable fluids or vapours will not result in a fire caused by impingement of the fluids or vapours on any part of the exhaust system including shields for the exhaust system. (c) Each exhaust system must be separated by fireproof shields from adjacent flammable parts of the UA that are outside of the engine and auxiliary power unit compartment. (d) No exhaust gases may discharge dangerously near any fuel or oil system drain. (e) No exhaust gases may be discharged where they will cause any hazard: (1) to critical sensors in terms of flight safety, and (2) to ground staff necessary for ground operations as stated in the UAS Flight Manual or in the UAS Maintenance Manual.
------	---

	(f) Each exhaust system component must be ventilated to prevent points of excessively high temperature. (g) If significant traps exist, each turbine engine and auxiliary power unit exhaust system must have drains discharging clear of the UA, in any normal ground and flight attitude, to prevent fuel accumulation after the failure of an attempted engine or auxiliary power unit start. (h) Each exhaust heat exchanger must incorporate means to prevent blockage of the exhaust port after any internal heat exchanger failure. (i) For the purposes of compliance with USAR.603 the failure of any part of the exhaust system will adversely affect safety.
1123	USAR.1123 Exhaust system For powerplant installations, the following apply: (a) Exhaust piping must be heat and corrosion resistant, and must have provisions to prevent failure due to expansion by operating temperatures. (b) Piping must be supported to withstand any vibration and inertia loads to which it would be subjected in operation; and (c) Piping connected to components between which relative motion could exist must have means for flexibility.
1125	USAR.1125 Exhaust heat exchangers For reciprocating engine-powered UA, each exhaust heat exchanger must be constructed and installed to withstand the vibration, inertia and other loads that it may be subjected to in normal operation. In addition: (a) Each exchanger must be suitable for continued operation at high temperatures and resistant to corrosion from exhaust gases; (b) There must be means for inspection of critical parts of each exchanger, if fitted. (c) Not applicable.

POWERPLANT CONTROLS AND ACCESSORIES

1141	USAR.1141 UA Powerplant controls: General See AMC.1141 (e) (a) Not applicable. (b) Each flexible control must be shown to be suitable for the particular application. (c) Each control must be able to maintain any necessary position without tendency to creep due to control loads or vibration. (d) Each control must be able to withstand operating loads without failure or excessive deflection. (e) For turbine engine-powered UA, no single failure or malfunction in any powerplant control system shall cause a <i>Hazardous</i> or more serious event. (f) The portion of each powerplant control located in the engine compartment that is required to be operated in the event of fire must be at least fire resistant. (g) Not applicable in this subpart. (see USAR.U1803 Powerplant power assisted valve indicator)
------	---

	h) All engine controls which are part of the engine installation must be implemented properly to ensure that: (1) Each engine control performs its intended function. (2) Each engine control rigging, electrical connector, etc. must be able to withstand operating loads and vibration without failure or excessive deflection.
	1142 Auxiliary power unit controls Not applicable in this subpart. (see USAR.U1751 Engine and APU controls)
1143	USAR.1143 Engine controls See AMC.1143 (a) Not applicable in this subpart. (see USAR.U1751 Engine controls) (b) Not applicable in this subpart. (see USAR.U1751 Engine controls) (c) Not applicable in this subpart. (see USAR.U1751 Engine controls) (d) Not applicable in this subpart. (see USAR.U1751 Engine controls) (e) Not applicable in this subpart. (see USAR.U1751 Engine controls) (f) Not applicable in this subpart. (see USAR.U1751 Engine controls) (g) For reciprocating single-engine UA, each power or thrust control must be designed so that if the control separates at the engine fuel metering device, the UA is capable of continuing safe flight.
	1145 Ignition switches Not applicable in this subpart. (see USAR.U1753 Ignition switches)
1147	USAR.1147 Mixture controls See AMC.1147 (a) Not applicable in this subpart. (see USAR.U1755 Mixture controls) (b) Each engine mixture control must be designed so that, if the control separates at the engine fuel metering device, the UA is capable of continuing safe flight.
	1149 Propeller speed and pitch controls Not applicable in this subpart. (see USAR.U1757 Propeller speed and pitch controls)
	1153 Propeller feathering controls Not applicable in this subpart. (see USAR.U1759 Propeller feathering controls)
	1155 Turbine engine reverse thrust and propeller pitch settings below the flight regime Not applicable in this subpart. (see USAR.U1761 Turbine engine reverse thrust and propeller pitch settings below the flight regime)
	1157 Carburettor air temperature controls Not applicable in this subpart. (see USAR.U1763 Carburettor air temperature controls)
1163	USAR.1163 Powerplant accessories (a) Each engine mounted accessory must: (1) Be approved for mounting on the engine involved and use the provisions on the engines for mounting; or

	(2) Have torque limiting means on all accessory drives in order to prevent the torque limits established for those drives from being exceeded; and (3) In addition to sub-paragraphs (a) (1) or (a) (2), be sealed to prevent contamination of the engine oil system and the accessory system. (b) Electrical equipment subject to arcing or sparking must be installed to minimise the probability of contact with any flammable fluids or vapours that might be present in a free state. (c) Each generator rated at or more than 6 kilowatts must be designed and installed to minimise the probability of a fire hazard in the event it malfunctions. (d) If the continued rotation of any accessory remotely driven by the engine is hazardous when malfunctioning occurs, a means to prevent rotation without interfering with the continued operation of the engine must be provided. (e) Each accessory driven by a gearbox that is not approved as part of the powerplant driving the gearbox must: (1) Have torque limiting means to prevent the torque limits established for the affected drive from being exceeded; (2) Use the provisions on the gearbox for mounting; and (3) Be sealed to prevent contamination of the gearbox oil system and the accessory system.
1165	USAR.1165 Engine ignition systems Where engine restart capability is required: (a) Each battery ignition system must be supplemented by a generator that is automatically available as an alternate source of electrical energy to allow continued engine operation if any battery becomes depleted. (b) The capacity of batteries and generators must be large enough to meet the simultaneous demands of the engine ignition system and the greatest demands of any electrical system components that draw from the same source. (c) The design of the engine ignition system must account for: (1) The condition of an inoperative generator; (2) The condition of a completely depleted battery with the generator running at its normal operating speed; and (3) The condition of a completely depleted battery with the generator operating at idling speed if there is only one battery. (d) Not applicable in this subpart. (see USAR.U1801 Battery discharge warning) (e) Each turbine engine ignition system must be independent of any electrical circuit that is not used for assisting, controlling or analysing the operation of that system. (f) Not applicable.
POWERPLANT FIRE PROTECTION	
1181	USAR.1181 Designated fire zones; regions included For powerplants, designated fire zones are:

	(a) For reciprocating engines. (1) The power section; (2) The accessory section; (3) Any complete powerplant compartment in which there is no isolation between the power section and the accessory section. (b) For turbine engines. (1) The compressor and accessory sections; (2) The combustor, turbine and tailpipe sections that contain lines or components carrying flammable fluids or gases. (3) Any complete powerplant compartment in which there is no isolation between compressor, accessory, combustor, turbine and tailpipe sections. (c) Any auxiliary power unit compartment; and (d) Any fuel burning heater and other combustion equipment installation.
	1182 Nacelle areas behind firewalls Not applicable.
1183	USAR.1183 Lines, fittings and components (a) Except as provided in sub-paragraph (b), each component, line and fitting carrying flammable fluids, gas or air in any area subject to engine fire conditions must be at least fire resistant, except that flammable fluid tanks and supports which are part of and attached to the engine must be fireproof or be enclosed by a fireproof shield unless damage by fire to any nonfireproof part will not cause leakage or spillage of flammable fluid. Components must be shielded or located so as to safeguard against the ignition of leaking flammable fluid. Flexible hose assemblies (hose and end fittings) must be shown to be suitable for the particular application. (b) Sub-paragraph (a) does not apply to: (1) Lines, fittings and components which are already approved as part of a Type Certificated engine; and (2) Vent and drain lines and their fittings, whose failure will not result in, or add to, a fire hazard.
1189	USAR.1189 Shut-off means See AMC.1189 (a) (5) (a) For each multi-engine UA the following apply: (1) Each engine installation must have means to shut off or otherwise prevent hazardous quantities of fuel, oil, de-icing fluid and other flammable liquids from flowing into, within, or through any engine compartment, except in lines, fittings and components forming an integral part of an engine. (2) The closing of the fuel shut-off valve for any engine may not make any fuel unavailable to the remaining engine that would be available to that engine with that valve open. (3) Operation of any shut-off means may not interfere with the later emergency operation of other equipment such as propeller feathering devices.

	(4) Each shut-off must be outside of the engine compartment unless an equal degree of safety is provided with the shut-off inside the compartment. (5) No Hazardous amount of flammable fluid may drain into the engine compartment after shut-off. (6) Not applicable in this subpart. (see USAR.U1765 Shut-off controls) (b) Turbine engine installations need not have an engine oil system shut-off if: (1) The oil tank is integral with, or mounted on, the engine; and (2) All oil system components external to the engine are fireproof or located in areas not subject to engine fire conditions. (c) Power-operated valves must be designed so that the valve will not move from the selected position under vibration conditions likely to exist at the valve location (see also USAR.U1805 Shut-off valve indicator).
1191	USAR.1191 Firewalls See AMC.1191 (e) (a) Each engine, auxiliary power unit, fuel burning heater and other combustion equipment must be isolated from the rest of the UA by firewalls, shrouds or equivalent means. (b) Each firewall or shroud must be constructed, so that no hazardous quantity of liquid, gas or flame can pass from that compartment to other parts of the UA. (c) Each opening in the firewall or shroud must be sealed with close fittings, fireproof grommets, bushings or firewall fittings offering at least as much thermal and flame protection as the firewall itself. (d) Reserved. (e) Each firewall, shroud and firewall fitting must be fireproof and protected against corrosion.
1192	USAR.1192 Engine accessory compartment diaphragm For air-cooled radial engines, the engine power section and all portions of the exhaust system must be isolated from the engine accessory compartment by a diaphragm that meets the firewall requirements of USAR.1191
1193	USAR.1193 Cowling and nacelle See AMC.1193 (a) Each cowling must be constructed and supported so that it can resist any vibration, inertia and air loads to which it may be subjected in operation. (b) There must be means for rapid and complete drainage of each part of the cowling in the normal ground and flight attitudes. No drain may discharge where it will cause a fire hazard. (c) Cowling must be fireproof. (d) Each part around an opening in the engine compartment cowling must be at least fire-resistant for a distance not less than 61 cm (24 in) from of the opening. (e) Not applicable. (f) Each nacelle of a multi-engine UA with turbocharged engines must be designed and constructed so that with the landing gear retracted, a fire in the engine compartment will not burn through a cowling or nacelle and enter a nacelle area other than the engine compartment.

1195	USAR.1195 Fire extinguishing systems In case the propulsion system fire hazard has a probability greater than the acceptability threshold for <i>Catastrophic</i> event, detailed requirements associated with the extinguishing systems must be agreed with the Certifying Authority.
	USAR.1197 Fire extinguishing agents See USAR.1195
	USAR.1199 Extinguishing agent containers See USAR.1195
	USAR.1201 Fire extinguishing system materials See USAR.1195
1203	USAR.1203 Fire detector system A fire detection system must be installed in each designated fire zone, as defined in USAR.1181, and comply with the following: (a) Each fire detector system must be constructed and installed to withstand the vibration, inertia and other loads and environmental stresses (salt fog, humidity, dirt, etc.) to which it may be subjected in operation. (b) No fire detector, wiring or components may be adversely affected by any fuel, oil, water, other fluids, or fumes that might be present. (c) There must be means to allow the UAS crew to check, in flight, the functioning of each fire detector electric circuit. (d) The detector, wiring and other components of each fire detection system in a designated fire zone must be at least fire-resistant. (e) The fire detection system must be designed to minimise false warnings or inappropriate warnings and if they occur shall not result in a <i>Hazardous</i> or more serious event.

Subpart F - EQUIPMENT

GENERAL	
1301	USAR.1301 Function and installation Each item of installed equipment must: (a) Be of a kind and design appropriate to its intended function; (b) Be labelled as to its identification, function or operating limitations, or any applicable combination of these factors; (c) Be installed according to limitations specified for that equipment; (d) Function properly when installed.
	1303 Flight and navigation instruments Not applicable in this subpart. (see USAR.U1723 Flight and navigation data)
U1304	USAR.U1304 Navigation Architecture The navigation architecture must prevent navigation system failures external to the UAS from resulting in a <i>Hazardous</i> or more serious event.
	USAR.1305 Powerplant instruments Not applicable in this subpart. (see USAR.U1725 Powerplant data)
U1307	USAR.U1307 Environmental control system (ECS) See AMC.U1307 Cooling must be provided for equipment as required for it to meet its intended function and reliability for the intended lifetime. (a) The ECS design shall incorporate the system safety requirements of the UAS. (b) The ECS shall meet all safety requirements when operating under installed conditions over the design envelope and maintain integration integrity to ensure the UAS safety-of-flight. (c) The UA shall incorporate an alternate means of cooling of safety-critical avionics when the primary ECS is non-operational. (d) The ECS design (including emergency equipment and/or auxiliary methods) shall provide an acceptable pressure environment for equipment affecting safety-of-flight. (e) Normal and emergency pressurization requirements and status shall be indicated at the UCS. (f) Safety-critical items such as flight controls, avionics and communications shall function long enough to safely land the aircraft if ECS function is lost and alternate methods are not available to ensure airworthy operations. (g) ECS normal and emergency procedures shall be included in the UAS Flight Manual. (h) Adequate controls and displays for the ECS shall be installed in the UCS or other appropriate locations to allow the ECS to function as intended. Sufficient cautions, warnings, and advisories shall be provided to alert the UAS crew to problems in time for corrective action to be taken from a safety of flight perspective.

	(i) No single ECS subsystem failure (including UCS functions that are critical to aircraft flight safety) shall result in loss of UA. (j) Bleed air or other compressed air duct system shall be monitored for leaks and structural integrity. Hot air leaking from damaged ducting shall not cause ignition of any flammable fluids or other materials or cause damage to safety-critical equipment. Shutdown capability, with an appropriate UCS alert, shall be provided when a potentially damaging or fire-producing leak occurs. The sensors for the leak detection system shall recover their required leak detection function following exposure to a leak. (k) The UAS thermal management system shall be stable for all flight conditions and environments. The mass flow and delivery temperature of cooling medium shall be sufficient for the aircraft heat loads and provide the necessary thermal stability to ensure safety-of-flight.
1308	USAR.1308 High-intensity Radiation Fields (HIRF) Protection See AMC.1308 and AMC 20-158 (a) Each electrical and electronic system that performs a function whose failure would prevent the continued safe flight and landing of the UA must be designed and installed so that: (1) The function is not adversely affected during and after the time the UA is exposed to HIRF environment I, as described in Appendix L. (2) The system automatically recovers normal operation of that function, in a timely manner, after the UA is exposed to HIRF environment I, as described in Appendix L, unless the system's recovery conflicts with other operational or functional requirements of that system that would prevent continued safe flight and landing of the UA, and (3) The system is not adversely affected during and after the time the UA is exposed to HIRF environment II, as described in Appendix L. (b) Each electrical and electronic system that performs a function whose failure would significantly reduce the capability of the UA or the ability of the flight crew to respond to an adverse operating condition must be designed and installed so the system is not adversely affected when the equipment providing the function is exposed to equipment HIRF test level 1 or 2, as described in Appendix L. (c) Each electrical and electronic system that performs a function whose failure would reduce the capability of the UA or the ability of the flight crew to respond to an adverse operating condition must be designed and installed so the system is not adversely affected when the equipment providing the function is exposed to equipment HIRF test level 3, as described in Appendix L.
1309	USAR.1309 Equipment, systems and installations See AMC.1309 (b) and AMC.1309 (c) (a) The UAS must be designed to reduce the risk to people including UAS crew, ground staff and third parties to a level acceptable to the Certifying Authority. It must also be designed to reduce the risk of material loss or damage to a level acceptable to the Certifying Authority. (1) Where any function of a UA System is essential to, or can prejudice, continued safe flight and landing of the UA, that function, and the equipment performing the function, (including equipment remote from the UAS), shall be considered as part of the UAS for the purposes of the validity of the certificate of airworthiness. It must comply with the applicable airworthiness requirements set out in this code and AMC.1309 should be used as guidance. (2) Each item of equipment, each system, and each installation: (i) When performing its intended function, may not adversely affect the response, operation, or accuracy of any: - Equipment essential to safe operation; or - Other equipment unless there is a means to inform the UAS crew of the effect. (ii) Must be designed to prevent hazards to the UAS in the event of a probable malfunction or failure.

(b) The design of each item of equipment, each system, and each installation must be examined separately and in relationship to other systems and installations to determine:

- (1) if the UAS is dependent upon its function for continued safe flight and landing; and
- (2) if failure of a system would significantly reduce the capability of the UAS or the ability of the UAS crew to cope with adverse operating conditions.

(c) Each item of identified equipment, system and installations categorised by (b)(1) or (2) must be designed to comply with the following additional requirements:

- (1) It must perform its intended function under any foreseeable operating condition.
- (2) When systems and associated components are considered separately and in relation to other Systems:
 - (i) The occurrence of any *Catastrophic* failure condition of the UAS must be extremely improbable; and
 - (ii) The occurrence of any Hazardous failure condition of the UAS must be extremely remote; and
 - (iii) The combination of all *Catastrophic* failure conditions must be extremely remote.
- (3) Warning and/or Caution information must be provided to alert the UAS crew to unsafe system operating conditions and to enable the UAS crew to take appropriate corrective action. Systems, controls, and associated monitoring and warning means must be designed to minimise UAS crew errors that could create additional hazards.
- (4) Compliance with the requirements of sub-paragraph (b) (2) may be shown by analysis and, where necessary, by appropriate ground, flight, or simulator test. The analysis must consider:
 - (i) Possible modes of failure, including malfunctions and damage from external sources;
 - (ii) The probability of multiple failures, and the probability of undetected faults;
 - (iii) The resulting effects on the UAS and third parties, considering the stage of flight and operating conditions; and
 - (iv) The UAS crew alerting cues, corrective action required, and the UAS crew's capability of determining faults.

(d) Each item of equipment, each system, and each installation whose functioning is required for certification and that requires a power supply and is required for continued safe flight and landing of the UA, is an "essential load" on the power supply. This power supply must not affect the safety of the UAS. The power sources and the system must be able to supply the following power loads in all foreseeable operating combinations and for probable maximum duration:

- (1) Loads connected to the power distribution system with the system functioning normally.
- (2) Essential loads after failure of:
 - (i) Any one engine on a multi-engine UA; or
 - (ii) Any power converter or energy storage device.

	(3) Essential loads for which an alternate source of power is required by the operating rules or to enable the desired or specified level of redundancy, after any failure or malfunction in any one power supply system, distribution system, or other utilization system. (4) Redundant systems shall utilize redundant power sources where possible in order to maintain the desired level of safety (or failure probability) realized by that level of redundancy. (e) In determining compliance with sub-paragraph (c) (2), the power loads may be assumed to be reduced under a monitoring procedure consistent with safety in the kinds of operations authorised. (f) In demonstrating compliance with this paragraph with regard to the electrical power system and to equipment design and installation, critical environmental and atmospheric conditions, including radio frequency energy and the effects (both direct and indirect) of lightning strikes, must be considered in accordance with USAR.867. For electrical generation, distribution, and utilisation equipment required by or used in complying with this subpart, the ability to provide continuous, safe service under foreseeable environmental conditions may be shown by environmental tests, design analysis, or reference to previous comparable service experience on other UAS. (g) As used in this paragraph, "systems" refers to all pneumatic systems, fluid systems, electrical systems, mechanical systems, powerplant systems and computer systems included in the UA design, UCS, command and control data link and communication system except for the following: (1) UA powerplant systems provided as part of the certificated engine, (2) The UA flight structure whose requirements are specific in subpart C and D of USAR.
	USAR.1311 Electronic display instrument systems Not applicable in this subpart. (see USAR.U1727 Electronic data display)
1316	USAR.1316 System Lightning Protection (a) For functions whose failure would contribute to or cause a condition that would prevent the continued safe flight and landing of the UA, each electrical and electronic system that performs these functions must be designed and installed to ensure that the operation and operational capabilities of the systems to perform these functions are not adversely affected when the UAS is exposed to lightning. (b) For functions whose failure would contribute to or cause a condition that would reduce the capability of the UAS to cope with adverse operating conditions, each electrical and electronic system that performs these functions must be designed and installed to ensure that these functions can be recovered in a timely manner after the UAS is exposed to lightning. (c) Compliance with the lightning protection criteria prescribed in sub-paragraphs (a) and (b) of this paragraph must be shown for exposure to a severe lightning environment. The UAS must be designed for and it must be verified that aircraft electrical/electronic systems are protected against the effects of lightning by: (1) Determining the lightning strike zones; (2) Establishing the external lightning environment for the zones; (3) Establishing the internal environment; (4) Identifying all the electrical and electronic systems that are subject to the requirements of this paragraph, and their locations on or within the UAS; (5) Establishing the susceptibility of the systems to the internal and external lightning environment; (6) Designing protection; and

	(7) Verifying that the protection is adequate.
	USAR.1321 Arrangement and visibility Not applicable in this subpart. (see USAR.U1721 Arrangement and visibility)
	USAR.1322 Warning caution and advisory lights Not applicable in this subpart. (see USAR.U1785 Warning caution and advisory information colour code)

MEASURING DEVICES INSTALLATION

1323	USAR.1323 Airspeed measuring device If a pitot tube is used then the following applies: (a) The design and installation of each airspeed measuring device must provide positive drainage of moisture from the pitot static plumbing. (b) Each airspeed measuring device must have a heated pitot tube or an equivalent means of preventing malfunction due to icing. For all airspeed measuring devices: (c) Each airspeed measurement sensor must be calibrated to measure true airspeed at sea-level with a standard atmosphere and with a minimum practicable measuring device calibration error when the corresponding pitot and static pressures are applied. The airspeed measuring device system must further be calibrated in flight to determine the system error exclusive of instrument error. The system error may not exceed 3% of the calibrated airspeed or 9.3 km/h (5 knots), whichever is greater, throughout the following speed ranges: (1) $1.3 V_{S1}$ to V_{MO}/M_{MO} or V_{NE}, whichever is appropriate with flaps retracted. (2) $1.3 V_{S1}$ to V_{FE} with flaps extended. (d) Where dual or greater airspeed measurements are required by system redundancy and flight safety requirements, the respective pitot tubes or other airspeed measuring devices must be far enough apart to avoid damage to both tubes in a collision with a bird.
1325	USAR.1325 Static pressure measuring device See AMC.1325(e) (a) Each measuring device provided with static pressure case connections must be so vented that the influence of UA speed, airflow variations, moisture, or other foreign matter will have minimum effect on the accuracy of the device except as noted in sub-paragraph (b) (3). (b) If a static pressure measuring device is necessary for the functioning of systems, or devices, it must comply with the provisions of sub-paragraphs (1) to (3). (1) The design and installation of a static pressure measuring device must be such that: (i) Positive drainage of moisture is provided; (ii) Chafing of the tubing and excessive distortion or restriction at bends in the tubing, is avoided; and (iii) The materials used are durable, suitable for the purpose intended and protected against corrosion. (2) A proof test must be conducted to demonstrate the integrity of the static pressure measuring device in the following manner:

	(i) Unpressurised UA. Evacuate the static pressure system to a pressure differential of approximately 3.4 kPa (1 inch of mercury) or to a reading on the altimeter, 305 m (1 000 ft) above the UA elevation at the time of the test. Without additional pumping for a period of 1 minute, the loss of indicated altitude must not exceed 30 m (100 ft) on the altimeter. (ii) Pressurised UA. Evacuate the static pressure system until a pressure differential equivalent to the maximum compartment pressure differential for which the UAS is Type Certificated is achieved. Without additional pumping for a period of 1 minute, the loss of indicated altitude must not exceed 2% of the equivalent altitude of the maximum compartment differential pressure or 30 m (100 ft), whichever is greater. (3) If a static pressure system is provided for any device, or system required by the operating rules, each static pressure port must be designed or located in such a manner that the correlation between air pressure in the static pressure system and true ambient atmospheric static pressure is not altered when the UA encounters icing conditions. An anti-icing means or an alternate source of static pressure may be used in showing compliance with this requirement. If the altitude data provided by the alternate static pressure system differs from the altitude data provided by the primary static system by more than 15m (50 ft), an automatic correction must be provided in the UCS and/or the automatic control system for the alternate static system. (c) Except as provided in sub-paragraph (d), if the static pressure measuring device incorporates both a primary and an alternate static pressure source, the means for selecting one or the other source must be designed so that: (1) When either source is selected, the other is blocked off; and (2) Both sources cannot be blocked off simultaneously. (d) For unpressurised UA, sub-paragraph (c) (1) does not apply if it can be demonstrated that the static pressure system calibration, when either static pressure source is selected, is not changed by the other static pressure source being open or blocked. (e) Each static pressure measuring device must be calibrated in flight to determine the system error. The system error, in indicated pressure altitude, at sea-level, with a standard atmosphere, excluding measuring device calibration error, may not exceed that determined with the Certifying Authority for an agreed appropriate configuration and speed range. (f) Reserved. (g) For UAS prohibited from flight under known icing conditions in accordance with USAR.1525, sub-paragraph (b) (3) does not apply.
	USAR.1326 Pitot heat indication systems Not applicable in this subpart. (see USAR.U1819 Pitot heat indicator)
1327	USAR.1327 Magnetic heading measuring device (a) Except as provided in sub-paragraph (b): (1) Each magnetic heading measuring device, if existing, must be installed so that its accuracy is minimally affected by the UA's vibration or magnetic fields; and (2) The compensated installation may not have a deviation, in level flight, greater than 10° on any heading. (b) A magnetic non-stabilised direction measuring device, if existing, may deviate more than 10° due to the operation of electrically powered systems if either a magnetic stabilised direction measuring

	device, which does not have a deviation in level flight greater than 10° on any heading, or a gyroscopic direction measuring device is installed. Deviations of a magnetic non-stabilised direction measuring device of more than 10° must be automatically compensated in accordance with USAR.U1837
	USAR.1329 Flight control system Not Applicable. See USAR.U1330

U1330	USAR.U1330 Flight control performance See AMC.U1330 (e), AMC.U1330 (i) and AMC.U1330 (j) The flight control system comprises sensors, actuators, computers and all those elements of the UAS, necessary to control the attitude, speed and trajectory of the UA. The flight control system must meet the following: (a) The modes of control of the UAS must be the following categories, which may be selected at any time in flight by the UAS crew: (1) automatic: in this mode the UA attitude, speed and flight path are fully controlled by the flight control system. No input from the UCS is needed other than to load or modify the required flight plan. (2) semi-automatic: with this type of control the UAS crew commands outer loop parameters such as altitude, heading and air speed. The flight control system operates the UA controls to achieve the commanded outer loop parameter value. (b) The flight control system must be designed so that a UAS crew of average skill can operate the UAS with acceptable workload, (c) The flight control system must apply limits to manoeuvres to keep the UA in the flight envelope protection as defined in USAR.U334 (d) The UAS crews must have the opportunity to intervene at any time during the flight to manage safe control of the UAS, except: (1) during emergency situations such as total loss of data link, (2) during launch phase before achieving the minimum safe flight parameters, (3) during landing phase after reaching the decision point as defined in USAR.U1490 and USAR.U1492, (4) for UA designed to be recovered by parachute, during the landing phase under parachute, (5) for rocket or catapult assisted take-off UAS, during the launch phase before reaching the limits specified in USAR.U282 (e) The flight control system must be designed and adjusted so that, within the range of adjustment (if any) available to UAS crew, it cannot produce an unsafe condition. (f) The flight control system must be designed so that a single malfunction will not produce a hardover signal. If the flight control system integrates signals from auxiliary controls or furnishes signals for operation of other equipment, positive interlocks and sequencing of engagement to prevent improper operation are required. (g) There must be protection against adverse interaction of integrated components, resulting from a malfunction. (h) Not applicable in this subpart. (see USAR.U1790 UAS mode of control indicator) (i) Use of active flight controls for load alleviation, stability augmentation, and/or flutter suppression must comply with the control system stability requirements as agreed with the Certifying Authority. (j) The flight control system must have a comprehensive self-test available and operating during all phases of flight, including preflight.
-------	--

	(k) Data exchanged between components of the flight control system or received from components external to the flight control system must be verified for the integrity of the information prior to use. Information received from external sources must be verified within appropriate rate of change and range boundaries for the appropriate phase of flight before using in the computations.
1331	USAR.1331 Measuring devices using a power source For each measuring device that is safety critical for continued safe operation that uses a power source, it must be ensured, that the power required is measured adequately and that there are means to indicate a malfunctioning measuring device to the UAS and the UAS crew. See also USAR.U1809 Furthermore the following apply: (a) The installation and power supply systems must be designed so that: (1) The failure of a measuring device will not interfere with the proper supply of energy to the remaining device; and (2) The failure of the energy supply from one source will not interfere with the proper supply of energy from any other source. (b) There must be at least two independent sources of power (not driven by the same engine on multi-engine UA), and a manual or an automatic means to select each power source.
	USAR.1335 Flight director systems Not applicable in this subpart. (see USAR.U1790 UAS mode of control indicator)
1337	USAR.1337 Powerplant installation measuring device (a) The following requirements apply: (1) Each powerplant and auxiliary power unit measuring device line, hose and pipe must meet the requirements of USAR.993. (2) Each line, hose or pipe carrying flammable fluids under pressure must: (i) Have restricting orifices or other safety devices at the source of pressure to prevent the escape of excessive fluid if the line, hose or pipe fails; and (ii) Be installed and located so that the escape of fluids would not create a hazard. (3) Each powerplant and auxiliary power unit measuring device that utilises flammable fluids must be installed and located so that the escape of fluid would not create a hazard. (b) See USAR.U1729. In addition, the following requirements apply: (1) Each fuel quantity measuring device must be calibrated to read “zero” during level flight when the quantity of fuel remaining in the tank is equal to the unusable fuel supply determined under USAR.959 (a); (2) Each exposed sight gauge used as a fuel quantity indicator must be protected against damage (leakage, etc.); (3) Each sight gauge that forms a trap in which water can collect and freeze must have means to allow drainage on the ground; (4) There must be a means to indicate the amount of usable fuel in each tank when the UA is on the ground.

	(5) Not applicable in this subpart. (see USAR.U1729 Fuel quantity and oil quantity data) (6) Not applicable in this subpart. (see USAR.U1729 Fuel quantity and oil quantity data) (c) Fuel flowmeter system. If a fuel flowmeter system is installed, each metering component must have a means to by-pass the fuel supply if malfunctioning of that component severely restricts fuel flow. (d) Oil quantity indicator. There must be a means to indicate the quantity of oil in each tank when the UA is on the ground. (see also USAR.U1729 Fuel quantity and oil quantity data)
--	---

ELECTRICAL SYSTEMS AND EQUIPMENT

1351	USAR.1351 General See AMC.1351 (a)(2) (a) Electrical system capacity. Each electrical system must be adequate for the intended use. In addition: (1) Electric power sources, their transmission cables, and their associated control and protective devices, must be able to furnish the required power at the proper voltage to each load circuit essential for safe operation; and (2) Compliance with sub-paragraph (1) must be shown by an electrical load analysis, or by electrical measurements, that account for the electrical loads applied to the electrical system in all probable combinations and for probable duration. Consideration shall be given to every payload configuration. (b) Functions. For each electrical system, the following apply: (1) Each system, when installed, must be: (i) Free from hazards in itself, in its method of operation, and in its effects on other parts of the UAS; (ii) Protected from fuel, oil, water, other detrimental substances and mechanical damage; (iii) Not applicable in this subpart. (see USAR.U1712 UCS electrical systems) (2) Electric power sources must function properly when connected in combination or independently. (3) No failure or malfunction of any electric power source may impair the ability of any remaining source to supply load circuits essential for safe operation. (4) Reserved. (c) UA Generating system. There must be at least one electrical power generating device if the electrical system supplies power to load circuits essential for safe operation. In addition: (1) Each electrical power generating device must be able to deliver its continuous rated power, or such power as is limited by its regulation system; (2) Electrical power generating device voltage control equipment must be able to dependably regulate the electrical power generating device output within rated limits; (3) Automatic means must be provided to prevent either damage to any electrical power generating device, or adverse effects on the UA electrical system, due to reverse current. A means must also be provided to disconnect each electrical power generating device from the battery and the other electrical power generating device(s).
------	---

	(4) Not applicable in this subpart. (see USAR.U1809 UAS electrical systems warning and indicator) (5) Each electrical power generating device must have an overvoltage protection, independent of voltage control, designed and installed to prevent damage to the electrical system, or to equipment supplied by the electrical system, that could result if that electrical power generating device were to develop an overvoltage condition. (d) Not applicable in this subpart. (see USAR.U1809 UAS electrical systems warning and indicator) (e) Fire resistance. Electrical equipment must be so designed and installed that in the event of a fire in the engine compartment, during which the surface of the firewall adjacent to the fire is heated to 1,093°C (2,000°F) for 5 minutes or to a lesser temperature substantiated for the UA, the equipment essential to continued safe operation and located behind the firewall will function satisfactorily and will not create an additional fire hazard. For UAS designed to be recovered by parachute, the complete landing sequence must be still possible; this may be shown by tests or by analysis. (f) External power. If provisions are made for connecting external power to the UAS and that external power can be electrically connected to equipment other than that used for engine starting, means must be provided to ensure that no incompatible external power supply (e.g. a reverse polarity, or a reverse phase sequence, can supply power to the UA's electrical system). The external power connection must be located so that its use will not result in a hazard to the UAS or ground crew. The location should be such that there is no detrimental effect to the UA structure. (g) Not applicable.
1353	USAR.1353 Storage battery or emergency power supply design and installation (a) Each storage battery or emergency power supply must be designed and installed as prescribed in this paragraph. (b) Safe cell temperatures and pressures must be maintained during any probable charging and discharging condition. No uncontrolled increase in cell temperature may result when the battery is recharged (after previous complete discharge) (1) At maximum regulated voltage or power; (2) During a flight of maximum duration; and (3) Under the most adverse cooling condition likely to occur in service. (c) Compliance with sub-paragraph (b) must be shown by tests unless experience with similar batteries and installations has shown that maintaining safe cell temperatures and pressures presents no problem. (d) No explosive gases emitted by any battery or emergency power supply in normal operation, or as the result of any probable malfunction in the charging system or battery or emergency power supply installation, may accumulate in hazardous quantities within the UA. (e) No corrosive fluids or gases that may escape from the battery or emergency power supply may damage surrounding structures or adjacent essential equipment. (f) Each battery or emergency power supply installation capable of being used to start an engine or auxiliary power unit must have provisions to prevent any hazardous effect on structure or essential systems that may be caused by the maximum amount of heat the battery or emergency power supply can generate during a short circuit of the battery or emergency power supply or of any individual cells. (g) Battery installations must have:

	(1) A system to control the charging rate of the battery automatically so as to prevent battery overheating; or (2) A battery temperature sensing and over temperature warning system with a means for automatically disconnecting the battery from its charging source in the event of an over temperature condition; or (3) A battery failure sensing and warning system with a means for disconnecting the battery from its charging source in the event of battery failure. (h) In the event of a complete loss of the primary electrical power generating system, any battery or emergency power supply must be capable of providing enough electrical power to those loads that are essential to perform emergency procedures as defined in USAR.U1413 during the associated time duration. This time duration includes the time needed for the UAS crew to recognise the loss of generated power and to take appropriate action. (i) Where applicable, battery installations must be compliant with the appropriate national regulations regarding battery safety.
1357	USAR.1357 Circuit protective devices See AMC.1357 (a) (a) Protective devices, such as fuses or circuit breakers, must be installed in all electrical circuits other than: (1) The main circuits of starter motors used during starting only; and (2) Circuits in which no hazard is presented by their omission. (b) A protective device for a circuit essential to flight safety may not be used to protect any other circuit. (c) Where installed, each remote resettable circuit protective device ("trip free" device in which the tripping mechanism cannot be over-ridden by the operating control) must be designed so that: (1) A remote operation to be done by the UAS crew is required to restore service after tripping; and (2) If an overload or circuit fault exists, the device will open the circuit regardless of the position of the operating control. (3) Where automatic resettable circuit protection devices are used they must be designed so they comply with (c) (2) and restore circuit integrity on removal of fault condition. (d) Not applicable. (e) Not applicable.
1359	USAR.1359 Electrical system fire protection (a) Components of the electrical system must meet the applicable fire protection requirements of USAR.U850 and USAR.865 (b) Electrical cables, terminals and equipment in designated fire zones, that are used during emergency procedures, must be at least fire-resistant. (c) Insulation on electrical wire and cable must be self-extinguishing when tested at an angle of 60° in accordance with the applicable portions of Appendix F of USAR or other approved equivalent methods. The average burn length must not exceed 76 mm (3 in) and the average flame time after

	removal of the flame source must not exceed 30 seconds. Drippings from the test specimen must not continue to flame for more than an average of 3 seconds after falling.
1361	USAR.1361 Master switch arrangement See AMC.1361 (a) There must be a master switch arrangement on the UAS to allow ready disconnection by ground staff of each electric power source from the power distribution systems when the UA is on the ground and except as provided in sub-paragraph (b). (b) Load circuits may be connected so that they remain energised when the master switch is open; if: (1) The circuits are isolated, or physically shielded, to prevent their igniting flammable fluids or vapours that might be liberated by the leakage or rupture of any flammable fluid systems; and (2) The circuits are required for continued operation of the engine; or (3) The circuits are protected by circuit protective devices with a rating of five amperes or less adjacent to the electric power source. In addition, two or more circuits installed in accordance with the requirements of sub-paragraph (b) (2) must not be used to supply a load of more than five amperes. (c) Not applicable in this subpart. (see USAR.U1732 Safety critical controls)
1365	USAR.1365 Electric cables and equipment (a) Each electric connecting cable must be sized by gauge and temperature rating equal to or above the full load amperage of its respective circuit. (b) Any equipment that is associated with any electrical cable installation and that would overheat in the event of a circuit overload or fault must be flame resistant. (c) Means of identification must be provided for electrical cables, connectors and terminals. (d) Electrical cables must be installed such that the risk of mechanical damage and/or damage caused by fluids, vapours or sources of heat, is minimised. (e) Main power cables (including generator cables) must be designed to allow a reasonable degree of deformation and stretching without failure and must: (1) Be separated from flammable fluid lines; or (2) Be shrouded by means of electrically insulated flexible conduit or equivalent, which is in addition to the normal cable insulations. (f) Where a cable cannot be protected by a circuit protection device or other overload protection it must not cause a fire hazard under fault conditions.
1367	USAR.1367 Switches Each switch must be: (a) Able to carry its rated current; (b) Constructed with enough distance or insulating material between current carrying parts and the housing so that vibration in flight will not cause shorting; (c) Accessible to appropriate maintenance staff; and (d) Labelled as to operation and the circuit controlled.

LIGHTS

	1381 Instrument lights
	Not applicable in this subpart. (see USAR.U1705 UAS crew workplace lights)
1383	USAR.1383 Taxi and landing lights If needed, each taxi and landing light must be designed and installed so that: (a) If a sensor is required for taxi, take-off or landing phase then there should be no dangerous glare from taxi and landing lights that would affect UAS operational safety. (b) Not applicable. (c) It provides enough light for all intended operations; and (d) It does not cause a fire hazard in any configuration.
1385	USAR.1385 Position light system installation (a) General. Each part of each position light system must meet the applicable requirements and each system as a whole must meet the requirements of USAR.1387 to USAR.1397 (b) Left and right position lights. Left and right position lights must consist of a red and a green light spaced laterally as far apart as practicable and installed on the UA such that, with the UA in the normal flying position, the red light is on the left side and the green light is on the right side. (c) Rear position light. The rear position light must be a white light mounted as far aft as practicable on the tail or on each wing tip. (d) Light covers and colour filters. Each light cover or colour filter must be at least flame resistant and may not change colour or shape or lose any appreciable light transmission during normal use. (e) The position lights must be able to be switched on and off from the UCS and while the UA is in flight.
1387	USAR.1387 Position light system dihedral angles (a) Except as provided in sub-paragraph (e), each position light must, as installed, show unbroken light within the dihedral angles described in this paragraph. (b) Dihedral angle L (left) is formed by two intersecting vertical planes, the first parallel to the longitudinal axis of the UA, and the other at 110° to the left of the first, as viewed when looking forward along the longitudinal axis. (c) Dihedral angle R (right) is formed by two intersecting vertical planes, the first parallel to the longitudinal axis of the UA, and the other at 110° to the right of the first, as viewed when looking forward along the longitudinal axis. (d) Dihedral angle A (aft) is formed by two intersecting vertical planes making angles of 70° to the right and to the left, respectively, to a vertical plane passing through the longitudinal axis, as viewed when looking aft along the longitudinal axis. (e) If the rear position light, when mounted as far aft as practicable in accordance with USAR.1385 (c), cannot show unbroken light within dihedral angle A (as defined in sub-paragraph (d)), a solid angle or angles of obstructed visibility totalling not more than 0.04 steradians is allowable within that dihedral angle, if such solid angle is within a cone whose apex is at the rear position light and whose elements make an angle of 30° with a vertical line passing through the rear position light.
1389	USAR.1389 Position light distribution and intensities (a) General. The intensities prescribed in this paragraph must be provided by new equipment with each light cover and colour filter in place. Intensities must be determined with the light source

	operating at a steady value equal to the average luminous output of the source at the normal operating voltage of the UAS. The light distribution and intensity of each position light must meet the requirements of sub-paragraph (b) (b) Position lights. The light distribution and intensities of position lights must be expressed in terms of minimum intensities in the horizontal plane, minimum intensities in any vertical plane and maximum intensities in over-lapping beams, within dihedral angles L, R and A, must meet the following requirements: (1) Intensities in the horizontal plane. Each intensity in the horizontal plane (the plane containing the longitudinal axis of the UA and perpendicular to the plane of symmetry of the UA) must equal or exceed the values in USAR.1391 (2) Intensities in any vertical plane. Each intensity in any vertical plane (the plane perpendicular to the horizontal plane) must equal or exceed the appropriate value in USAR.1393, where I is the minimum intensity prescribed in USAR.1391 for the corresponding angles in the horizontal plane. (3) Intensities in overlaps between adjacent signals. No intensity in any overlap between adjacent signals may exceed the values in USAR.1395, except that higher intensities in overlaps may be used with main beam intensities substantially greater than the minima specified in USAR.1391 and USAR.1393, if the overlap intensities in relation to the main beam intensities do not adversely affect signal clarity. When the peak intensity of the left and right position lights is more than 100 candelas, the maximum overlap intensities between them may exceed the values in USAR.1395 if the overlap intensity in Area A is not more than 10% of peak position light intensity and the overlap intensity in Area B is not more than 2.5% of peak position light intensity. (c) Rear position light installation. A single rear position light may be installed in a position displaced laterally from the plane of symmetry of a UA if: (1) The axis of the minimum cone of illumination is parallel to the flight path in level flight; and (2) There is no obstruction aft of the light and between planes 70° to the right and left of the axis of maximum illumination.															
1391	USAR.1391 Minimum intensities in the horizontal plane of position lights Each position light intensity must equal or exceed the applicable values in the following table: <table><tr><th>Dihedral angle (light included)</th><th>Angle from right or left of longitudinal axis measured from dead ahead</th><th>Intensity (candelas)</th></tr><tr><td>L and R</td><td>0° to 10°</td><td>40</td></tr><tr><td>(red and green)</td><td>10° to 20°</td><td>30</td></tr><tr><td></td><td>20° to 110°</td><td>5</td></tr><tr><td>A (rear white)</td><td>110° to 180°</td><td>20</td></tr></table>	Dihedral angle (light included)	Angle from right or left of longitudinal axis measured from dead ahead	Intensity (candelas)	L and R	0° to 10°	40	(red and green)	10° to 20°	30		20° to 110°	5	A (rear white)	110° to 180°	20
Dihedral angle (light included)	Angle from right or left of longitudinal axis measured from dead ahead	Intensity (candelas)														
L and R	0° to 10°	40														
(red and green)	10° to 20°	30														
	20° to 110°	5														
A (rear white)	110° to 180°	20														

1393	USAR.1393 Minimum intensities in any vertical plane of position lights Each position light intensity must equal or exceed the applicable values in the following table: <table><tr><th>Angle above or below the horizontal plane</th><th>Intensity</th></tr><tr><td>0</td><td>1.00 I.</td></tr><tr><td>0° to 5°</td><td>0.90 I.</td></tr><tr><td>5° to 10°</td><td>0.80 I.</td></tr><tr><td>10° to 15°</td><td>0.70 I.</td></tr><tr><td>15° to 20°</td><td>0.50 I.</td></tr><tr><td>20° to 30°</td><td>0.30 I.</td></tr><tr><td>30° to 40°</td><td>0.10 I.</td></tr><tr><td>40° to 90°</td><td>0.05 I.</td></tr></table>	Angle above or below the horizontal plane	Intensity	0	1.00 I.	0° to 5°	0.90 I.	5° to 10°	0.80 I.	10° to 15°	0.70 I.	15° to 20°	0.50 I.	20° to 30°	0.30 I.	30° to 40°	0.10 I.	40° to 90°	0.05 I.					
Angle above or below the horizontal plane	Intensity																							
0	1.00 I.																							
0° to 5°	0.90 I.																							
5° to 10°	0.80 I.																							
10° to 15°	0.70 I.																							
15° to 20°	0.50 I.																							
20° to 30°	0.30 I.																							
30° to 40°	0.10 I.																							
40° to 90°	0.05 I.																							
1395	USAR.1395 Maximum intensities in overlapping beams of position lights No position light intensity may exceed the applicable values in the following table, except as provided in USAR.1389 (b) (3): <table><tr><th rowspan="2">Overlaps</th><th colspan="2">Maximum Intensity</th></tr><tr><th>Area A (Candelas)</th><th>Area B (Candelas)</th></tr><tr><td>Green in dihedral angle L</td><td>10</td><td>1</td></tr><tr><td>Red in dihedral angle R</td><td>10</td><td>1</td></tr><tr><td>Green in dihedral angle A</td><td>5</td><td>1</td></tr><tr><td>Red in dihedral angle A</td><td>5</td><td>1</td></tr><tr><td>Rear White in dihedral angle L</td><td>5</td><td>1</td></tr><tr><td>Rear White in dihedral angle R</td><td>5</td><td>1</td></tr></table> Where: (a) Area A includes all directions in the adjacent dihedral angle that pass through the light source and intersect the common boundary plane at more than 10° but less than 20°; and (b) Area B includes all directions in the adjacent dihedral angle that pass through the light source and intersect the common boundary plane at more than 20°	Overlaps	Maximum Intensity		Area A (Candelas)	Area B (Candelas)	Green in dihedral angle L	10	1	Red in dihedral angle R	10	1	Green in dihedral angle A	5	1	Red in dihedral angle A	5	1	Rear White in dihedral angle L	5	1	Rear White in dihedral angle R	5	1
Overlaps	Maximum Intensity																							
	Area A (Candelas)	Area B (Candelas)																						
Green in dihedral angle L	10	1																						
Red in dihedral angle R	10	1																						
Green in dihedral angle A	5	1																						
Red in dihedral angle A	5	1																						
Rear White in dihedral angle L	5	1																						
Rear White in dihedral angle R	5	1																						
1397	USAR.1397 Colour specifications Each position light colour must have the applicable International Commission on Illumination chromaticity co-ordinates as follows: (a) Aviation red – “y” is not greater than 0.335; and “z” is not greater than 0.002. (b) Aviation green – “x” is not greater than 0.440 – 0.320y; “x” is not greater than y – 0.170; and “y” is not less than 0.390 – 0.170x. (c) Aviation white – “x” is not less than 0.300 and not greater than 0.540;																							

	"y" is not less than "x – 0.040" or "y° – 0.010", whichever is the smaller; and "y" is not greater than "x + 0.020" nor "0.636 – 0.400x"; Where "y°" is the "y" co-ordinate of the Planckian radiator for the value of "x" considered.		
	1399 Riding light Not applicable.		
1401	USAR.1401 Anti-collision light system (a) General. The UA must have an anti-collision light system that: (1) Consist of one or more approved anti-collision lights located so that their light will not detract from the conspicuity of the position lights; and (2) Meet the requirements of sub-paragraphs (b) to (f). (b) Field of coverage. The system must consist of enough lights to illuminate the vital areas around the UA, considering the physical configuration and flight characteristics of the UA. The field of coverage must extend in each direction within at least 75° above and 75° below the horizontal plane of the UA, except that there may be solid angles of obstructed visibility totalling not more than 0.5 steradians. (c) Flashing characteristics. The arrangement of the system, that is, the number of light sources, beam width, speed of rotation, and other characteristics, must give an effective flash frequency of not less than 40, nor more than 100, cycles per minute. The effective flash frequency is the frequency at which the UA's complete anti-collision light system is observed from a distance, and applies to each sector of light including any overlaps that exist when the system consists of more than one light source. In overlaps, flash frequencies may exceed 100, but not 180, cycles per minute. (d) Colour. Each anti-collision light must be either aviation red or aviation white and must meet the applicable requirements of USAR.1397 (e) Light intensity. The minimum light intensities in any vertical plane, measured with the red filter (if used) and expressed in terms of "effective" intensities, must meet the requirements of sub-paragraph (f). The following relation must be assumed: $I_e = \frac{\int_{t_1}^{t_2} I(t) dt}{0.2 + (t_2 - t_1)}$ Where: I_e = effective intensity (candelas). I(t) = instantaneous intensity as a function of time. (t₂ – t₁) = flash time interval (seconds). Normally, the maximum value of effective intensity is obtained when t₂ and t₁ are chosen so that the effective intensity is equal to the instantaneous intensity at t₂ and t₁. (f) Minimum effective intensities for anti-collision lights. Each anti-collision light effective intensity must equal or exceed the applicable values in the following table: <table border="1"> <thead> <tr> <th>Angle above or below the horizontal plane</th><th>Intensity (candelas)</th></tr> </thead> </table>	Angle above or below the horizontal plane	Intensity (candelas)
Angle above or below the horizontal plane	Intensity (candelas)		

		0° to 5°	400	
		5° to 10°	240	
		10° to 20°	80	
		20° to 30°	40	
		30° to 75°	20	

SAFETY EQUIPMENT AND EMERGENCY CAPABILITY

	1411 General
	Not applicable.
U1412	USAR.U1412 Emergency recovery capability See AMC.U1412 (a)(2) and AMC.U1412 (e) (a) The UAS must integrate an emergency recovery capability that consists of: (1) a flight termination system, procedure or function that aims to immediately end normal flight, or (2) an emergency recovery procedure that is implemented through UAS crew command or automatically without operator command by design means in order to mitigate the effects of critical failures with the intent of minimising the risk to third parties, or (3) any combination of USAR.U1412 (a) (1) and USAR.U1412 (a) (2). (b) The emergency recovery capability must function as desired over the whole flight envelope under the most adverse combination of environmental conditions. (c) The emergency recovery capability must be safeguarded from interference leading to inadvertent operation. (d) The emergency recovery capability must receive its electrical power, if needed, from the bus that provides the maximum reliability for operation. In case of complete loss of the primary electrical power generating system, it must automatically switch to the battery. (e) Use of explosives to perform in-flight total destruction of the aircraft is not an acceptable means of compliance with USAR.U1412

	(f) Where the emergency recovery capability includes a pre-programmed course of action to reach a predefined site where it can be reasonably expected that fatality will not occur, the dimensions of such areas must be stated in the UAS Flight Manual.
U1413	USAR.U1413 Engine shut down procedure In the event of an engine failure that causes shutdown, the following requirements apply: (a) the UAS must be designed to retain sufficient control and manoeuvrability until it has reached a forced landing area. (b) therefore, the emergency electrical power must be designed in such a way that its reliability and duration are compatible with USAR.U1413 (a). The time period needed to perform a glide from maximum certificated altitude to sea level and reach a forced landing area includes the time needed for the UAS crew to recognise the failure and to take appropriate action, if required. (c) the engine shut down procedure must be analysed considering the existence of the emergency recovery capability specified in USAR.U1412
	1415 Ditching equipment Not applicable.
1416	USAR.1416 De-icer system If certification with ice protection provisions is desired and a de-icer system is installed: (a) The system must meet the requirements specified in USAR.1419 (b) The system and its components must be designed to perform their intended function under any normal system operating temperature or pressure. (c) Not applicable in this subpart. (see USAR.U1811 Pneumatic de-icer boot system indicator)
1419	USAR.1419 Ice protection (See AMC.1419) If certification with ice protection provisions is desired, compliance with the following requirements must be shown: (a) The recommended procedures for the use of the ice protection equipment must be set forth in the UAS Flight Manual or in approved manual material. (b) An analysis must be performed to establish, on the basis of the UAS's operational needs, the adequacy of the ice protection system for the various components of the UAS. In addition, tests of the ice protection system must be conducted to demonstrate that the UAS is capable of operating safely in continuous maximum and intermittent maximum icing conditions. (c) Compliance with all or portions may be accomplished by reference, where applicable because of similarity of the designs to analysis and tests performed for the type certification of a Type Certificated UAS. (d) When monitoring of the external surfaces of the UA by the UAS crew is required for proper operation of the ice protection equipment, it must be ensured that the monitoring can be done in all operating and environmental conditions.
MISCELLANEOUS EQUIPMENT	
1431	USAR.1431 Electronic equipment See AMC.1431 (a) In showing compliance with USAR.1309 (c)(1) and (2) with respect to radio and electronic equipment and their installations, critical environmental conditions must be considered.

	(b) Radio and electronic equipment, controls, and wiring must be installed so that operation of any unit or system of units will not adversely affect the simultaneous operation of any other radio or electronic unit, or system of units. (c) Not applicable in this subpart. (see USAR.U1707 Communication system) (d) Not applicable in this subpart. (see USAR.U1707 Communication system) (e) Not applicable in this subpart. (see USAR.U1707 Communication system) (f) Electronic payload equipment and wiring must be installed so that operation will not adversely affect the simultaneous operation of any other radio or electronic unit, or system of units. (g) All sensitive and essential equipment as identified in (a) must be protected against internal and external sources of electromagnetic interference (EMI) and, if applicable, electromagnetic pulse (EMP).
1435	USAR.1435 Hydraulic systems (a) Design. Each hydraulic system must be designed as follows: (1) Each hydraulic system and its elements must withstand, without yielding, the structural loads expected in addition to hydraulic loads. (2) Not applicable in this subpart. (see USAR.U1813 Hydraulic systems indicator) (3) There must be means to ensure that the pressure, including transient (surge) pressure, in any part of the system will not exceed the safe limit above design operating pressure and to prevent excessive pressure resulting from fluid volumetric changes in all lines which are likely to remain closed long enough for such changes to occur. (4) The minimum design burst pressure must be 2.5 times the operating pressure. (5) There must be adequate means to protect hydraulic systems critical to continued safe flight resulting from fluid loss. (6) All materials in contact with the hydraulic fluid shall be compatible with the hydraulic fluid over the temperature range, functional, service and storage conditions the hydraulic system will experience. (b) Tests. Each system must be substantiated by proof pressure tests. When proof-tested, no part of any system may fail, malfunction, or experience a permanent set. The proof load of each system must be at least 1.5 times the maximum operating pressure of that system. (c) Accumulators. Hydraulic accumulators or reservoirs may be installed on the engine side of any firewall if: (1) It is an integral part of an engine or propeller system, or (2) The reservoir is non-pressurised and the total capacity of all such non-pressurised reservoirs is one litre (one US-quart) or less.
1437	USAR.1437 Accessories for multi-engine UA For multi-engine UA, engine-driven accessories essential to safe operation must be distributed among the engines so that the failure of any one engine will not impair safe operation through the malfunctioning of these accessories.
1438	USAR.1438 Pressurisation and pneumatic systems

	(a) Pressurisation system elements must be burst pressure tested to 2.0 times, and proof pressure tested to 1.5 times, the maximum normal operating pressure. (b) Pneumatic system elements must be burst pressure tested to 3.0 times, and proof pressure tested to 1.5 times, the maximum normal operating pressure. (c) An analysis, or a combination of analysis and test, may be substituted for any test required by subparagraph (a) or (b) if the Certifying Authority finds it equivalent to the required test.
	USAR.1441 Oxygen equipment and supply Not applicable.
	USAR.1443 Minimum mass flow of supplemental oxygen Not applicable.
	USAR.1445 Oxygen distribution system Not applicable.
	USAR.1447 Equipment standards for oxygen dispensing units Not applicable.
	USAR.1449 Means for determining use of oxygen Not applicable.
	USAR.1450 Chemical oxygen generators Not applicable.
	USAR.1451 Fire protection for oxygen equipment Not applicable.
	USAR.1453 Protection for oxygen equipment from rupture Not applicable.
	USAR.1457 Cockpit voice recorder Not applicable in this subpart. (see USAR.U1709 Voice recorders)
1459	USAR.1459 UA onboard flight recorders See AMC.1459 (a) If required, each flight recorder must be installed so that: (1) It is supplied with airspeed, altitude, and directional data obtained from sources that meet the accuracy requirements of USAR.1323, USAR.1325 and USAR.1327, as appropriate; (2) The vertical acceleration sensor is rigidly attached; (3) It receives its electrical power from the bus that provides the maximum reliability for operation of the flight recorder without jeopardising service to essential or emergency loads; (4) There is an aural or visual means for pre-flight checking of the recorder for proper recording of data in the storage medium. (5) Except for recorders powered solely by the engine-driven electrical generator system, there is an automatic means to simultaneously stop a recorder that has a data erasure feature and prevent each erasure feature from functioning, within 10 minutes after crash impact; (6) A universal time reference signal is recorded. For instance, the GPS time signal can be used for this purpose.

	(b) Each non-ejectable record container must be located and mounted so as to minimise the probability of container rupture resulting from crash impact and subsequent damage to the record from fire. (c) A correlation must be established between the flight recorder readings of airspeed, altitude, and heading and the corresponding readings (taking into account correction factors) of the UAS crew's data displayed in the UCS. The correlation must cover the airspeed range over which the UA is to be operated, the range of altitude to which the UAS is limited, and 360° of heading. Correlation may be established on the ground as appropriate. (d) Each recorder container must: (1) Be either bright orange or bright yellow; (2) Have reflective tape affixed to its external surface to facilitate its location under water; and (3) Have an underwater locating device, when required by the operating rules, on or adjacent to the container which is secured in such a manner that they are not likely to be separated during crash impact. (e) Any novel or unique design or operational characteristics of the UAS must be evaluated to determine if any dedicated parameters must be recorded on flight recorders in addition to or in place of existing requirements. (f) The flight recorder is supplied with a method to record the last commanded actions received by the UA from the UCS.
1461	USAR.1461 Equipment containing high energy rotors (a) Equipment containing high energy rotors must meet sub-paragraphs (b), (c) or (d). (b) High energy rotors contained in equipment must be able to withstand damage caused by malfunctions, vibration, abnormal speeds and abnormal temperatures. In addition: (1) Auxiliary rotor cases must be able to contain damage caused by the failure of high energy rotor blades; and (2) Equipment control devices, systems and instrumentation must reasonably ensure that no operating limitations affecting the integrity of high energy rotors will be exceeded in service. (c) It must be shown by test that equipment containing high energy rotors can contain any failure of a high energy rotor that occurs at the highest speed obtainable with the normal speed control devices inoperative. (d) Equipment containing high energy rotors must be located where rotor failure will not adversely affect continued safe flight.
U1481	USAR.U1481 Payloads See AMC.U1481 (a) A payload is a device or equipment carried by the UA, which performs the mission assigned. The payload comprises all elements of the aircraft that are not necessary for flight but are carried for the purpose of fulfilling specific mission objectives. It is assumed that a UAS Type Certification Basis may be released for several payload configurations. (b) Where a UAS is designed to carry payloads, the integration and operation of those payloads must: (1) not adversely affect the safe flight and control of the UAS; (2) be shown as electromagnetically compatible (EMC) with systems on board of the UA;

	(3) meet safety objectives as provided in USAR.1309
U1485	USAR.U1485 Environmental Control System (ECS) See AMC.U1485 (a) If installed, the ECS must comply with the system safety requirements applicable to the UAS. (b) The ECS must meet all safety requirements when operating under installed conditions over the design envelope and maintain integration integrity to ensure the UAS safety-of-flight. (c) In the event that the primary ECS is non-operational the UAS design must comply with either (1) or (2) such that no single ECS subsystem failure shall result in loss of UA. (1) Incorporated secondary/emergency systems shall be capable of operating until either; the primary ECS is available or safe landing is achieved. (2) Allow the continued function of the systems required for safe flight and landing (e.g. flight controls, avionics and communications) (d) ECS normal and emergency procedures must be included in the UAS Flight Manual. (e) Adequate controls and displays for the ECS must be installed in the UCS or other appropriate locations to allow the ECS to function as intended. Sufficient cautions, warnings, and advisories must be provided to alert the UAS crew to problems in time for corrective action to be taken from a safety-of-flight perspective.

AUTOMATIC TAKE-OFF, TAXI, LANDING

U1490	USAR.U1490 Automatic take-off system, automatic landing system See AMC.U1490 (f)(2) When a UAS, designed for conventional take-off and landing on a runway is equipped with an automatic take-off system, an automatic landing system or any combination thereof, it must meet the following requirements: (a) Once the automatic take-off or landing mode has been engaged, the UAS crew monitors the whole process from the UCS, via the command and control data link, but is not required to perform any manual "piloting action", except manual abort, where required, as per provisions of USAR.U1492 (b) The automatic function will reside in the UAS airborne control laws algorithms and will utilize navigation and flight path tracking inputs in such a manner as to not degrade the overall redundancy or level of safety of the flight control system. When off-board sensors are utilized via data-links, the continued safe flight of the aircraft must be ensured in the event of a loss of that data-link. (c) The automatic system may cause no unsafe sustained oscillations or undue attitude changes or control activity as a result of configuration or power changes or any other disturbance to be expected in normal operation. (d) Automatic take-off system or automatic landing system data and status must be displayed in the UCS. All indications must be designed to minimise crew errors. (e) <i>Take-off</i> (1) Once the automatic take-off mode has been engaged, the brake release, the take-off run and the rotation are fully automatic: UA runway steering flightpath, speed, configuration, engine settings and UA flightpath after lift-off are controlled by the automatic take-off system.
-------	---

	(2) In case of failure that could adversely affect safe flight or exceedance from predefined limits occurring during the take-off run at every speed up to the rotation speed V_R or the proper refusal speed (if applicable), an automatic abort function shall be provided to stop the UA on the runway. (f) <i>Landing</i> (1) Once the automatic landing mode has been engaged, the approach, landing and ground roll are fully automatic until the UA reaches full stop or after a safe taxiing speed is reached and UAS crew changes to a manual taxi mode: UA flightpath, speed, configuration, engine settings, runway steering and braking after touch down are controlled by the automatic landing system. (2) In case of failure or exceedance from the predefined limits of a convergence window occurring during the approach, an automatic go around function shall be provided above a certain height called "Decision Point", at which such a go around may be safely performed (i.e. with no ground contact that may damage the UA).
U1492	USAR.U1492 Manual abort function Where a UAS is designed for conventional take-off and landing on a runway and is equipped with automatic systems for taxi, take-off, landing or any combination thereof, it must include the following functions: (a) The automatic system must incorporate a manual abort function. Its control shall be easily accessible to the UAS crew in order to: (1) stop the UA on the runway during the take-off run at every speed up to refusal speed or rotation speed V_R, whichever is less. (2) where it is safe to perform, initiate a go around during the landing phase at every height down to a Decision Point. (b) Specific go around procedure shall be provided in the UAS Flight Manual under USAR.1585 (j).
U1494	USAR.U1494 Automatic taxi system When a UAS, designed for conventional take-off and landing on a runway is equipped with an automatic taxi system, it must meet the following requirements: (a) Taxi (1) Before the automatic taxi mode can be engaged the system must provide to the UAS operator a clear and deterministic indication of the taxi path and planned speed that will be automatically followed. (2) Once the automatic taxi mode has been engaged, the UA steering, speed and braking are fully automatic until the UA reaches full stop or the UAS crew provides manual inputs into the automatic taxi settings, or changes to a manual taxi mode. UA taxi path, speed, configuration, engine settings, steering and braking during taxi are controlled by the automatic taxi system. (3) In case of failure that could adversely affect safe taxi, or exceeding of predefined limits occurring during the taxi, or a failure in the communication control link, an automatic abort function shall be provided to stop the UA. (b) The automatic system must incorporate a manual abort function. Its control shall be easily accessible to the UAS crew in order to stop the UA during all automated taxi operations on the taxiway or runway.

Subpart G – OPERATING LIMITATIONS AND INFORMATION

GENERAL	
1501	USAR.1501 General (a) Each operating limitation specified in USAR.1505 to USAR.1527 and other limitations and information necessary for safe operation must be established. (b) The operating limitations and other information necessary for safe operation must be made available to the UAS crew as prescribed in USAR.1541 to USAR.1589 (c) The requirements of this section may not be applicable to all UAS configurations.
1505	USAR.1505 Airspeed limitations (a) The never-exceed speed V_{NE} must be established so that it is: (1) Not less than 0.9 times the minimum value of V_D allowed under USAR.335; and (2) Not more than the lesser of: (i) 0.9 V_D established under USAR.335; or (ii) 0.9 times the maximum speed shown under USAR.251 (b) The maximum structural cruising speed V_{NO} must be established so that: (1) it is not more than the lesser of: (i) V_C established under USAR.335; or (ii) 0.89 V_{NE} established under sub-paragraph (a). (2) the maximum speed allowed by the flight envelope protection maintained by the flight control system is equal or less than V_{NO}. (c) Sub-paragraphs (a) and (b) do not apply to turbine UA or to UA for which a design diving speed V_D/M_D is established under USAR.335 (b) (2). For those UA, a maximum operating limit speed (V_{MO}/M_{MO} airspeed or Mach number, whichever is critical at a particular altitude) must be established as a speed that may not be deliberately exceeded in any regime of flight (climb, cruise, or descent) unless a higher speed is authorised for flight test or UAS crew training operations. V_{MO}/M_{MO} must be established so that it is not greater than the design cruising speed V_C/M_C and so that it is sufficiently below V_D/M_D and the maximum speed shown under USAR.251 to make it highly improbable that the latter speeds will be inadvertently exceeded in operations. The speed margin between V_{MO}/M_{MO} and V_D/M_D or the maximum speed shown under USAR.251 may not be less than the speed margin established between V_C/M_C and V_D/M_D under USAR.335(b), or the speed margin found necessary in the flight tests conducted under USAR.253. The maximum speed allowed by the flight envelope protection maintained by the flight control system must be equal or less than V_{MO}/M_{MO}.
1507	USAR.1507 Manoeuvring speed The maximum operating manoeuvring speed, V_O, must be established as an operating limitation. V_O is a selected speed that is not greater than $V_{Sn}^{1/2}$ established in USAR.335(c).

1511	USAR.1511 Flap extended speed (a) The flap extended speed V_{FE} must be established so that it is: (1) Not less than the minimum value of V_F allowed in USAR.345 (b); and (2) Not more than V_F established under USAR.345 (a), (c) and (d). (b) Additional combinations of flap setting, airspeed and engine power may be established if the structure has been proven for the corresponding design conditions.
1513	USAR.1513 Minimum control speed The minimum control speed(s) V_{MC}, determined under USAR.149 (b), must be established as an operating limitation(s).
1519	USAR.1519 Weight and centre of gravity The weight and centre of gravity ranges for each payload configuration, determined under USAR.23 must be established as operating limitations.
1521	USAR.1521 Powerplant limitations (a) General. The powerplant limitations prescribed in this section must be established so that they do not exceed the corresponding limits for which the engines or propellers are approved by the Certifying Authority. (b) Take-off operation. The powerplant take-off operation must be limited by: (1) The maximum rotational speed (rpm); (2) The maximum allowable manifold pressure where the UA is equipped with reciprocating engine with a controllable propeller or a turbocharger; (3) The maximum allowable gas temperature (for turbine engines); (4) The time limit for the use of the power or thrust corresponding to the limitations established in sub-paragraphs (1) to (3); and (5) The maximum allowable cylinder head (as applicable), liquid coolant and oil temperatures. (6) The minimum oil pressure, minimum gas temperature, minimum coolant temperature, and minimum oil temperature. (c) Continuous operation. The continuous operation must be limited by: (1) The maximum rotational speed; (2) The maximum allowable manifold pressure where the UA is equipped with reciprocating engine with a controllable propeller or a turbocharger; (3) The maximum allowable gas temperature (for turbine engines); and (4) The maximum allowable cylinder head, oil and liquid coolant temperatures. (5) The minimum oil pressure, minimum gas temperature, minimum coolant temperature, and minimum oil temperature. (d) Fuel grade / designation. The minimum fuel grade (for reciprocating engines), or fuel designation (for turbine engines), must be established so that it is not less than that required for the operation of the engines within the limitations in sub-paragraphs (b) and (c).

	(e) Ambient temperature. For all UAS, ambient temperature limitations (including limitations for winterisation installations if applicable) must be established as the maximum ambient atmospheric temperature at which compliance with the cooling provisions of USAR.1041 to USAR.1047 is shown.
1522	USAR.1522 Auxiliary power unit limitations If an auxiliary power unit is installed, the limitations established for the auxiliary power unit must be specified in the operating limitations for the UAS.
	1523 Minimum Flight Crew Not applicable in this subpart. (see USAR.U1704 Minimum UAS Crew)
	1524 Maximum passenger seating configuration Not applicable.
1525	USAR.1525 Kinds of operation The kinds of operation (such as flight rules, day or night) and the meteorological conditions (such as icing) to which the operation of the UAS is limited or from which it is prohibited, must be established appropriate to the installed equipment.
1527	USAR.1527 Maximum operating altitude The maximum altitude up to which operation is allowed, as limited by flight, structural, powerplant, functional, or equipment characteristics, must be established.
1529	USAR.1529 Instructions for continued airworthiness Instructions for continued airworthiness in accordance with Appendix G must be prepared.

INFORMATION, MARKINGS AND PLACARDS

1541	USAR.1541 General (a) Each element of the UAS (UA, UCS, launch and landing elements, ...) must contain: (1) The information, markings and placards specified in this subpart and in subpart I; and (2) Any additional information, markings and placards required for the safe operation if it has unusual design, operating, or handling characteristics. (b) Each marking and placard of the UAS prescribed in sub-paragraph (a) (1) Must be displayed in a conspicuous place; and (2) May not be easily erased, disfigured or obscured. (c) The information, placards and marking information must be furnished in the UAS Flight Manual. (d) The units of measurement used on placards must be the same as those furnished in the UAS Flight Manual or displayed to the UAS crew.
1543	USAR.1543 Instrument markings: General (see also USAR.U1733 Conventional controls and indicators) For each instrument installed on the UAS, (a) When markings are on the cover glass of the instrument, there must be means to maintain the correct alignment of the glass cover with the face of the dial; and (b) Each arc and line must be wide enough and located to be clearly visible to the ground staff. (c) All related instruments must be calibrated in compatible units.
	1545 Airspeed indicator

	Not applicable in this subpart. (see USAR.U1835 Airspeed data)
	1547 Magnetic direction indicator
	Not applicable in this subpart. (see USAR.U1837 Magnetic heading or track data)
	1549 Powerplant and auxiliary power unit instruments
	Not applicable in this subpart. (see USAR.U1839 Powerplant and auxiliary power unit data)
1551	USAR.1551 Oil quantity indicator (see also USAR.U1841 Oil quantity data) Where an oil quantity indicator is installed on the UA for use by the ground staff, it must be marked in sufficient increments to indicate readily and accurately the quantity of oil.
1553	USAR.1553 Fuel quantity indicator (see also USAR.U1843 Fuel quantity data) Where a fuel quantity indicator is installed on the UA for use by the ground staff, a red radial line must be marked on each indicator at the calibrated zero reading.
1555	USAR.1555 Control markings on the UAS (see also USAR.U1845 Control markings in the UCS) If installed on the UAS: (a) Each control (switch, button, selector, etc.) on the UAS must be plainly marked as to its function and method of operation. (b) Not applicable. (c) For powerplant fuel controls: (1) Each fuel tank selector control on the UAS must be marked to indicate the position corresponding to each tank and to each existing cross feed position. (2) Not applicable. (3) The conditions under which the full amount of usable fuel in any restricted usage fuel tank can safely be used must be stated on a placard adjacent to the selector valve for that tank. (4) Not applicable. (d) Not applicable. (e) For accessory, auxiliary and emergency controls: (1) Not applicable. (2) Each emergency control must be red and must be marked as to method of operation. No control other than an emergency control shall be this colour.
1557	USAR.1557 Miscellaneous information markings and placards (a) Payload compartments and ballast location. Each payload compartment, and each ballast location, must have a placard stating any limitations on contents, including weight, that are necessary under the loading requirements. (b) Not applicable.

	(c) Fuel filler openings must be marked at or near the filler cover with: (1) the type of the fluid, and (2) the permissible product designation, as indicated in the maintenance and operational handbook. (d) Not applicable. (e) The system voltage of each direct current installation must be clearly marked adjacent to its external power connection. (f) Emergency access placards. Each placard and operating control for each emergency access panel must be red. A placard must be near each emergency access panel control and must clearly indicate the location of that access panel control and its method of operation.
	1559 Operating limitations Not applicable in this subpart. (see USAR.U1849 Operating limitations indications)
	1561 Safety Equipment Not applicable.
	1563 Airspeed placards Not applicable in this subpart. (see USAR.U1835 (e))
	1567 Flight Manoeuvre placard Not applicable.

UAS FLIGHT MANUAL

1581	USAR.1581 General (a) A UAS Flight Manual must be submitted to the Certifying Authority and it must contain the following: (1) Information required by USAR.1583 to USAR.1589. (2) Other information that is necessary for safe operation because of design, operating or handling characteristics. In particular, any information associated with the UCS, the command and control data link, the communication system and any other ancillary elements necessary to enable flight such as launch and landing elements must be stated in specific subpart of the UAS Flight Manual with emphasis on deployment and operation limitations. (3) Further information necessary to comply with the relevant operating rules. (b) Approved information (1) Each part of the UAS Flight Manual containing information prescribed in USAR.1583 to USAR.1589 must be approved by the Certifying Authority, segregated, identified and clearly distinguished from each unapproved part of that UAS Flight Manual. (2) Not applicable. (c) The units used in the UAS Flight Manual must be the same as those marked on the appropriate indicators, data display and placards both on the UA and in the UCS. (d) All UAS Flight Manual operational airspeeds must, unless otherwise specified, be presented as indicated Airspeeds. (e) The UAS Flight Manual must be readily accessible to the UAS crew in the UCS.
------	---

	(f) Revisions and/or amendments. Each UAS Flight Manual must contain a means for recording the incorporation of revisions and/or amendments. (g) If the UCS is capable of operating more than one UA, the maximum number of UAs that can be safely controlled from the UCS must be stated in the UAS Flight Manual. In addition, the UAS Flight Manual must contain procedures for UA handover within a same UCS, where this capability exists. (h) Hazardous effects of radiation emitted from the UAS on fuel, ordinance, and personnel must be stated in the UAS Flight Manual, and appropriate stand-off distances provided if the UAS is emitting radiation. (i) For UAs with a battery, environmental limitations and potential hazards associated with the battery must be stated in the UAS Flight Manual.
1583	USAR.1583 Operating limitations The UAS Flight Manual must contain operating limitations determined under USAR, including the following: (a) Airspeed limitations (1) Information necessary for the marking of the airspeed limits on the indicator as required in USAR.U1835, and the significance of each of those limits and of the colour coding used on the indicator. (2) The speeds V_{MC}, V_O, V_{LE} and V_{LO} and their significance. (3) Not applicable. (b) Powerplant limitations (1) Limitations required by USAR.1521 (2) Explanation of the limitations, when appropriate. (3) Information necessary for marking the data displayed in the UCS and required by USAR.U1839 to USAR.U1843 (4) Fuel and oil designation and limitation. (5) For two-stroke engine, the fuel/oil ratio. (c) Weight (1) The maximum weight; and (2) The maximum landing weight, if the design landing weight selected by the Applicant is less than the maximum weight. (3) UAS performance operating limitations as follows: (i) The maximum take-off weight for each airfield altitude and ambient temperature within the range selected by the Applicant if the UA complies with the climb requirements of USAR.63 (c). (ii) The maximum landing weight for each airfield altitude and ambient temperature within the range selected by the Applicant if the UA complies with the landing requirements of USAR.63 (c).

	(4) The maximum take-off weight for each airfield altitude and ambient temperature within the ranges selected by the Applicant at which the critical field length determined under USAR.55 is equal to the available runway length plus the length of any stopway, if utilised. (5) Not applicable. (6) The maximum zero wing fuel weight where relevant as established in accordance with USAR.343 (d) Centre of gravity. The established centre of gravity limits. (e) Authorised manoeuvres. (f) Manoeuvre load factor. The positive and negative limit load factors in g's. (g) Minimum UAS crew and maximum ground station capacity. The number and functions of the minimum UAS crew determined under USAR.U1704 In addition, the maximum number of persons allowed to occupy the UCS must be stated in the UAS Flight Manual in accordance with USAR.U1702(a). (h) Kinds of operation. A list of the kinds of operation to which the UAS is limited or from which it is prohibited under USAR.1525, and also a list of installed equipment that affects any operating limitation and identification as to the equipment's required operational status for the kinds of operation for which approval has been granted. (i) Maximum operating altitude. The maximum altitude established under USAR.1527 (j) Not applicable. (k) Allowable lateral fuel loading. The maximum allowable lateral fuel loading differential, if less than the maximum possible. (l) Payload loading. The following information for each payload compartment or zone: (1) The maximum allowable load; and (2) The maximum intensity of loading. (m) Systems. Any limitations on the use of UAS Systems and equipment. (n) Ambient temperatures. Where appropriate maximum and minimum ambient air temperatures for operation. (o) Smoking. Any restrictions on smoking in the UCS and in the vicinity of any UAS elements. (p) Types of surface. A statement of the types of surface on which operation may be conducted (see USAR.45 (g) and USAR.1587 (a) (4), (c)(2) and (d)(4)). (q) Deployment limitations: All limitations induced by the deployment of the UCS, the command and control data link, the launch and landing elements and any ancillary systems must be stated in the UAS Flight Manual. (r) Communication system and command and control data link limitations: the limitations and performances of the communication system and of the command and control data link must be stated in the UAS Flight Manual and also the effect of link loss on performance limitation. The requested operating frequencies must be stated in the UAS Flight Manual. (s) Automatic take-off and landing system (if applicable). The UAS Flight Manual must state:
--	---

	(1) Limitations (wind, turbulence, ...) and performance for which the automatic take-off and landing system is certificated. (2) Permitted configurations (e.g. flap setting, number of engines operating, ...) (3) Normal and emergency procedures. (4) Minimum required equipment that must be serviceable before engaging the automatic take-off and landing system.
1585	USAR.1585 Operating procedures (a) Information concerning normal, degraded (if applicable) and emergency procedures and other pertinent information necessary for safe operation and the achievement of the scheduled performance must be furnished, including: (1) An explanation of significant or unusual flight or ground handling characteristics; (2) The maximum demonstrated values of crosswind for take-off and landing and procedures and information pertinent to operations in crosswinds; (3) A recommended speed for flight in rough air. This speed must be chosen to protect against the occurrence, as a result of gusts, of structural damage to the UA and loss of control; (4) If applicable, procedures for restarting any engine in flight, including the effects of altitude; (5) Procedures, speeds and configuration(s) for making a normal approach and landing in accordance with USAR.73 and USAR.75 and a transition to the balked landing condition. (b) In addition to sub-paragraph (a), for a single-engine UA (where emergency recovery includes a glide), the procedures, speeds and configuration(s) for a glide following engine failure in accordance with USAR.71 and the subsequent forced landing, must be furnished. (c) In addition to sub-paragraph (a), for all multi-engine UA, the following information must be furnished: (1) Procedures, speeds and configuration(s) for making an approach and landing with one or more engine(s) inoperative; (2) Procedures, speeds and configuration(s) for making a go-around with one or more engine(s) inoperative and the conditions under which a go-around can be performed safely, or a warning against attempting a go-around. (3) The V_{SSE} determined in USAR.149.(d) In addition to sub-paragraphs (a) and (b) or (c) as appropriate, the following information must be furnished: (i) Procedures, speeds and configuration(s) for making a normal take-off in accordance with USAR.51 (a) and (b) USAR.53 (a) and (b) and the subsequent climb in accordance with USAR.65 and USAR.69 (a); (ii) Procedures for abandoning a take-off due to engine failure or other cause. (d) In addition to sub-paragraphs (a), (c) and (d) for multi-engine UA, the information must include: (1) Procedures and speeds for continuing a take-off following engine failure and the conditions under which take-off can safely be continued, or a warning against attempting to continue the take-off; (2) Procedures, speeds and configurations for continuing a climb following engine failure, after take-off, in accordance with USAR.67, or en-route, in accordance with USAR.69 (b).

	(e) In addition to sub-paragraphs (a) and (c), the information must include: (1) Procedures, speeds and configuration(s) for making a normal take-off; (2) Procedures and speeds for carrying out an accelerate-stop in accordance with USAR.55; (3) Procedures and speeds for continuing a take-off following engine failure, (f) Fuel. (1) For multi-engine UA, information identifying each operating condition in which the fuel system independence prescribed in USAR.953 is necessary for safety must be furnished, together with instructions for placing the fuel system in a configuration used to show compliance with that section. (2) The UAS Flight Manual shall describe how maximum capacity refuelling is to be accomplished with the requested mission loads. (g) For each UAS showing compliance with USAR.1353 (g) (2) or (g) (3), the operating procedures for disconnecting the battery from its charging source must be furnished. (h) Information on the total quantity of usable fuel for each fuel tank and the effect on the usable fuel quantity as a result of a failure of any pump, must be furnished. (i) Procedures for the safe operation of the UAS's systems and equipment, both in normal use and in the event of malfunction, must be furnished. In particular, emergency recovery capability procedures and data link loss strategy must be stated in the UAS Flight Manual.
1587	USAR.1587 Performance information See AMC.1587 and AMC.1587 (f) Unless otherwise presented, performance information must be provided over the altitude and temperature ranges required by USAR.45 (b) and for intended UA configuration, engine configuration and weight ranges required by USAR.25. All performance shall be based on a minimum engine performance. If other engine performance is used, the flight manual shall state such. (a) The following information must be furnished: (1) The stalling speeds V_{SO}, and V_{S1} with the landing gear and wing flaps retracted, defined for the weight ranges required by USAR.25 and the effect on these stalling speeds at maximum bank angles allowed by the flight control system; (2) The steady rate and gradient of climb with all engines operating, determined under USAR.69 (a); (3) The landing distance, determined under USAR.75 for each airfield altitude and standard temperature and the type of surface for which it is valid; (4) The effect on landing distance of operation on other than smooth hard surfaces, when dry and when wet, determined under USAR.45 (g); and (5) The effect on landing distance of runway slope and 50% of the headwind component and 150% of the tailwind component. (6) The limitations on the flight performance imposed by the flight envelope protection as defined in USAR.U334

- (b) In addition to sub-paragraph (a), the steady angle of climb/descent determined under USAR.77 must be furnished.
- (c) In addition to sub-paragraph (a) and paragraph (b) the following information must be furnished:
- (1) The take-off distance, determined under USAR.53 and the type of surface for which it is valid;
 - (2) The effect on take-off distance of operation on other than smooth hard surfaces, when dry and when wet, determined under USAR.45 (g);
 - (3) The effect on take-off distance of runway slope and 50% of the headwind component and 150% of the tailwind component;
 - (4) For multi-engine powered UA, the one or more engine(s) inoperative take-off climb/descent gradient, determined under USAR.66;
 - (5) For multi-engine UA, the en-route rate and gradient of climb/descent with one or more engine(s) inoperative, determined under USAR.69 (b); and
 - (6) For single-engine UA, the glide performance determined under USAR.71
- (d) In addition to paragraph (a), the following information must be furnished:
- (1) The accelerate-stop distance determined under USAR.55;
 - (2) Not applicable.
 - (3) Not applicable.
 - (4) The effect on accelerate-stop distance, take-off distance and, if determined, take-off run, of operation on other than smooth hard surfaces, when dry and when wet, determined under USAR.45 (g);
 - (5) The effect on accelerate-stop distance, take-off distance and, if determined, take-off run, of runway slope and 50% of the headwind component and 150% of the tailwind component;
 - (6) Not applicable.
 - (7) The en-route gradient of climb/descent with one engine inoperative, determined under USAR.69 (b);
 - (8) The effect, on the en-route gradient of climb/descent with one or more engine(s) inoperative, of 50% of the headwind component and 150% of the tailwind component;
 - (9) Not applicable.
 - (10) The relationship between IAS and CAS determined in accordance with USAR.1323 (b) and (c); and
 - (11) The altimeter system calibration required by USAR.1325 (e).
- (e) Engine performance data shall be provided and maximum power performance data for take-off shall be specifically stated.
- (f) Flight Planning Performance data shall be provided.
- (g) For navigation systems that utilize an external reference source (such as GPS) as the primary means of ensuring navigation performance associated with USAR.171, the UAS Flight Manual must

	contain information on: (1) Navigation sensor accuracy (to include both normal and degraded modes), (2) Areas of navigator susceptibility that can result in the degraded mode (such as the navigation system's ability to compensate for constellation instabilities, clock timing errors, etc.), and (3) Any operational procedures that must be performed by the operator to compensate for the degraded navigation.
1589	USAR.1589 Loading information The following loading information must be furnished: (a) The weight and location of each item of equipment that can easily be removed, relocated, or replaced and that is installed when the UA was weighed under USAR.25 (b) Appropriate loading instructions for each possible loading condition between the maximum and minimum weights established under USAR.25, to facilitate the centre of gravity remaining within the limits established under USAR.23
U1591	USAR.U1591 Data link information The data link information furnished in the UAS Flight Manual must meet the requirements of USAR.U1611, USAR.U1613 (a) and USAR.U1615 (c).

Subpart H –COMMAND AND CONTROL DATA LINK

COMMAND AND CONTROL DATA LINK	
U1601	USAR.U1601 General See AMC.U1601 (a) The UAS communication system typically consists of the following subsystems: (1) the command and control data link subsystem, (2) the ATC communication subsystem, (3) the payload data link subsystem. The present subpart H only covers the command and control data link subsystem. ATC communication and payload data link are covered by Operating Regulations and Standards. (b) A UAS must include a command and control data link (a radio-frequency link, for example) for control of the UAS with the following functions: (1) Transmittal of UAS crew commands from the UCS to the UA (uplink), and (2) Transmittal of UAS status data from the UA to the UCS (downlink). This status data must include, to the appropriate extent, navigational information, response to UAS crew commands, and equipment operating parameters in accordance with subpart I requirements.
U1603	USAR.U1603 Command and control data link architecture See AMC.U1603 The command and control data link architecture must ensure that there is no single failure that could lead to a Hazardous or more serious event.
U1605	USAR.U1605 Electromagnetic interference and compatibility See AMC.U1605 and AMC.U1605 (b) (a) The command and control data link must be protected against electromagnetic interference (EMI). (b) Each command and control data link must be protected in such a way to prevent electromagnetic vulnerability (EMV). (c) The command and control data link electromagnetic compatibility (EMC) must meet USAR.1431 (d) The command and control data link, as system must comply with USAR.1309 requirement. (e) The command and control data link must be designed to be protected against electrostatic, lightning and EME hazards at a level agreed with the Certifying Authority.
U1607	USAR.U1607 Command and control data link performance and monitoring See AMC.U1607 (a) (a) The effective maximum range of each command and control data link must be stated in the UAS Flight Manual, for a range of altitudes up to the maximum operating altitude as defined in USAR.1527, and for a range of availability levels agreed with Certifying Authority for the uplink and for the downlink. (b) For each command and control data link, the effective maximum range which may include a safety margin to be agreed with the Certifying Authority must be displayed in the UCS for a specific availability

	level for both uplink and downlink on request of the UAS crew. The corresponding availability level must be displayable on UAS crew request at the appropriate position on the UCS display. (c) For each command and control data link, the integrity of the uplink and downlink must be continuously monitored at a refresh rate consistent with safe operation. (d) Maximum range cues must be provided in the UCS on UAS crew request or automatically in case of a likely breakdown of the command and control data link. (e) Intervisibility information must be displayed in the UCS and warning cues provided to the UAS crew in order to prevent a total loss of command and control data link.
U1611	USAR.U1611 Command and control data link latency See AMC.U1611 (a) (a) Time delays in the command and control data link (namely 'latency') shall be specified in the UAS Flight Manual as a function of all relevant conditions. (b) The command and control data link latency must not lead to an unsafe condition, considering all probable environmental conditions and envisaged type of operations and must be agreed with the Certifying Authority.
U1613	USAR.U1613 Command and control data link loss strategy See AMC.U1613 (a), AMC.U1613 (b) and AMC.U1613 (c) (a) A command and control data link loss strategy must be established, approved and presented in the UAS Flight Manual taking into account the emergency recovery capability as defined in USAR.U1412 (b) The command and control data link loss strategy shall include a reacquisition process in order to try to re-establish in a short/reasonable time the command and control data link. (c) There must be an alert for the UAS crew, via a clear and distinct aural and visual signal, for any total loss of the command and control data link.
U1615	USAR.U1615 Command and control data link antenna masking See AMC.U1615 (a) For all UA attitudes and orientations relative to the signal source within the design envelope, the UAS antenna margin must be consistent to maintain a sufficient link budget for safe operation. (b) The masking must be stated in the UAS Flight Manual. (c) Warning cues shall be provided to the UAS crew in case of approaching masking attitudes in order to prevent a total loss of command and control data link.
U1617	USAR.U1617 Command and control data link switchover function See AMC.U1617 (a), AMC.U1617 (b) and AMC.U1617 (c) (a) The operation that consists of performing the transfer of the UA command and control from one data link channel to another channel within the same UCS is called "switchover". (b) The switchover of a command and control data link shall not lead to an unsafe situation. (c) The UA must be under continuous positive control at all times during switchover and when changing data links from the same UCS or it shall be shown that no positive control will not lead to unsafe conditions.

Subpart I – UA CONTROL STATION

GENERAL	
U1701	USAR.U1701 General See AMC.U1701 and AMC.U1701 (e) (a) The UAS control station (UCS) of a UAS is the facility or device from which the UA is remotely controlled. A UAS may be comprised of more than one UCS. (b) The design of a UCS must facilitate the command and control of the UA by the UAS crew for safe operations as agreed with the Certifying Authority. (c) When the UCS is to be used on a moving platform, the Certifying Authority may require establishing a Special Condition. (d) It must be shown using the UAS System Safety Assessment (SSA) according to USAR.1309 that all identified hazards within the UCS have been reduced to a level consistent with safe operation of the system. (e) The UAS control station must be designed and its performance established by test for operation over the complete target environment as determined by an Environmental Life Cycle analysis. This analysis shall take into account the full range of operational and non-operational (storage, transportation, etc.) environments. (f) Some of the following sub-part 1 requirements related to data and markings to be displayed may not apply if the Applicant shows that specific design features (such as closed loop flight control with adequate stability characteristics and flight envelope protection) make the display or the marking of a specific parameter unnecessary as agreed with the Certifying Authority.
U1702	USAR.U1702 UCS infrastructure (a) The physical parameters (e.g. size, temperature, power supply, earth bonding, maximum capacity) deemed as essential for flight safety and that define the infrastructure suitable for the UCS must be stated in the UAS Flight Manual. (b) Where the UCS infrastructure is part of the UAS to be certified, its design must comply with the parameters prescribed in USAR.U1702 (a)
U1703	USAR.U1703 UAS crew work place (a) The UCS and its equipment must allow each UAS crew at work place to perform their duties without unreasonable concentration or fatigue. (b) The UAS crew work place conditions (temperature, humidity, vibration, noise, heat emissions, ...) must not hamper safe execution of the flights.
U1704	USAR.U1704 Minimum UAS crew See AMC.U1704 The minimum UAS crew must be established so that it is sufficient for safe operation considering: (a) The workload on individual UAS crew members taking into account at least the following tasks: (1) Operation and monitoring of all essential UAS elements, (2) Navigation, (3) Flight path control,

	(4) Communications, (5) Compliance with airspace, air traffic, and air traffic control requirements, (6) Command decisions including Crew resource management, (b) The accessibility and ease of operation of necessary controls.
U1705	USAR.U1705 UAS crew work place lights The UAS crew work place lights must: (a) Make each indicator, data display, information, markings, placard and control easily readable and discernible; (b) Be installed so that their direct rays, and rays reflected from any surface, are shielded from the UAS crew's eyes.
U1707	USAR.U1707 Communication system See AMC.U1707 (a) and AMC.U1707 (c) (a) For those UAS that require more than one UAS crew member in the UCS, or whose operation will require communication with more than one UAS crew member, the UCS must be evaluated to determine if the UAS crew, when at their work places, can converse without difficulty under the actual UCS environment. If the UCS design includes provisions for the use of communication headsets, the evaluation must also consider conditions where headsets are being used. If the evaluation shows conditions under which it will be difficult to converse, an intercommunication system must be provided. (b) If the communication equipment that is installed incorporates transmit switches, these switches must be such that when released, they return from the "transmit" to the "off" position. (c) If provisions for the use of communication headsets are provided, it must be demonstrated that the UAS crew will receive all aural warnings under the actual UCS noise conditions when any headset is being used.
U1709	USAR.U1709 Voice recorders See AMC.U1709 and AMC.U1709 (e) (a) If required by the Certifying Authority, the installation of voice recorder in the UCS must comply with the following: (1) Voice communications transmitted from or received in the UCS. (2) Voice communications in the UCS. (3) Voice communications of UAS crew using the UCS's interphone system. (4) Voice or audio signals introduced into a headset or speaker. (b) The recording requirements of sub-paragraph (a) (2) must be met by installing an area microphone, located in the best position for recording voice communications originating at UAS crew work place and voice communications of other ground staff when directed to those stations. The microphone must be so located and, if necessary, the preamplifiers and filters of the recorder must be so adjusted or supplemented, so that the intelligibility of the recorded communications is as high as practicable when recorded under UCS noise conditions and played back. Repeated aural or visual play-back of the record may be used in evaluating intelligibility. (c) Where more than three UAS crew members are involved in the ground station:

	(1) A channel must be dedicated to each boom, mask, headset or speaker used at each UAS crew work place. (2) A channel must be dedicated to each area microphone. (d) Each UCS voice recorder must be installed so that: (1) It receives its electric power from the control station bus that provides the maximum reliability for operation of the voice recorder. (2) There is an aural or visual means for pre-flight checking of the recorder for proper operation. (e) Universal time reference signal must be recorded on a special track of the voice recorder.
U1711	USAR.U1711 UCS data recorder See AMC.U1711 and AMC.U1711 (a) If required by the Certifying Authority, the installation of data recorder in the UCS must comply with the following: (a) The UCS must be equipped with a data recorder that continuously record all the data transmitted via the command and control data link and the status of the UCS. (b) The storage capacity of the data recorder should be compatible with recording the maximum duration of flight for which the certification is requested. (c) The time basis used by the data recorder must be synchronised with an uniform time basis, and (1) the downlink data must be labelled and recorded with the time basis at which they have been generated in the UA, (2) the uplink data must be labelled and recorded with the time basis at which they have been generated in the UCS, (3) A reference to Air Traffic Control time must be recorded. (d) Time basis of the required UAS recorders shall allow a post synchronisation of all recorded data or information with an accuracy better than half a second between any of the recorders. (e) The UCS should provide a function that is able to read the data recorder for post-flight operation.
U1712	USAR.U1712 UCS electrical systems (a) Each electrical system in the UCS must be: (1) free from hazards in itself, in its method of operation, and in its effects on other parts of the UCS. (2) so designed that the risk of electrical shock is reduced to a minimum. (3) designed to be protected against electrostatic, lightning and EME hazards. (b) The total electrical heat emission must be taken into account in the design of the UCS.
U1713	USAR.U1713 UCS power supply (a) Failure conditions of UCS power supply shall be assessed according to USAR.1309 (b) The minimum UCS power supply consistent with sub-paragraph (a) must be stated in the UAS Flight Manual.
U1714	USAR.U1714 Automated Mission Planning

	See AMC.U1714
	Automated mission planning calculations must not lead to unsafe conditions.

UCS FIRE AND HAZARD PROTECTION

U1715	USAR.U1715 Hand fire extinguishers There must be at least one hand fire extinguisher for use in the UCS that is located within easy access of UCS crew. For hand fire extinguishers the following apply: (a) The type and quantity of each extinguishing agent used must be appropriate to the kinds of fire likely to occur when that agent is used. (b) Each extinguisher must be designed to minimise the hazard of toxic gas concentrations.
U1716	USAR.U1716 UCS Fire Protection Sources of heat within the UCS that are capable of igniting the UCS contents must be shielded or insulated to prevent such ignition.
U1717	USAR.U1717 UCS Smoke or Fire Detector System The UCS must be equipped with a smoke detector or fire detection system to give warning to the UCS crew.
U1718	USAR.U1718 UCS Ventilation The UCS must be suitably ventilated.
U1719	USAR.U1719 Exits. (a) Each exit in the UCS must be internally marked and readily identifiable from inside the UCS. (b) Means must be provided to assist UCS crew in locating the exit(s) in conditions of dense smoke.
U1720	USAR.U1720 UCS Emergency Lighting An emergency lighting system must be provided

DATA DISPLAYED IN THE UAS CONTROL STATION

U1721	USAR.U1721 Arrangement and visibility See AMC.U1721 (a) Each flight, navigation, powerplant and UA status data must be clearly arranged and visible to UAS crew as required or UAS crew selectable. (b) For each multi-engine UA, identical powerplant data must be available and located so as to prevent confusion as to which engine the data relates. (c) Data required for safe operation of the system must be grouped appropriately and located within the normal scan pattern of the UAS crew. (d) If a visual indicator is provided to indicate malfunction of an instrument, it must be effective under all lighting conditions. (e) All displays, indications and warnings must be visible under all UCS lighting conditions.
U1722	USAR.U1722 Part-time data display See AMC.U1722

	Many UAS parameters or status indications are required in subpart H and I to be displayed, yet they may be only necessary or required in certain phases of flight. (a) When parameters are not displayed full-time, flight safety must not be impaired. (b) Part-time displays of UAS parameters or status indicators must be shown not to create any unsafe conditions.
U1723	USAR.U1723 Flight and navigation data (a) The following are the minimum required flight and navigational data that must be displayed at all times in the control station at an update rate consistent with safe operation: (1) indicated airspeed, (2) pressure altitude and related altimeter setting, (3) heading or track, (4) UA position: the UA position must be continuously displayed on a map at a scale selectable by the UAS crew at a level of detail to ensure safe flight. (5) where semi-automatic flight control modes as defined in USAR.U1330 are activated, the commanded flight or navigation parameters sent to the UA must be displayed in the UCS. (b) Considering USAR.U1722, the following are the minimum required flight and navigational data that shall be selectable or available when queried by the UAS crew for display in the control station at an update rate consistent with safe operation: (1) airspeed limitations identified under USAR.1505 to USAR.1513, (2) sideslip angle, (3) free air temperature, (4) A speed warning device for: (i) Turbine engine-powered UA; and (ii) Other UA for which V_{MO}/M_{MO} and V_D/M_D are established under USAR.335 (b) (1) and USAR.1505 (c) if V_{MO}/M_{MO} is greater than 0.8 V_D/M_D. The speed warning device must give effective aural warning (differing distinctively from aural warnings used for other purposes) to the UAS crew whenever the speed exceeds that agreed with the Certifying Authority. The upper limit of the production tolerance for the warning device may not exceed the prescribed warning speed. The lower limit must be set to minimise nuisance warnings. The need for or the exact setting of this speed warning device may nevertheless consider the existence of high speed protections maintained by the flight control system, when it may be shown that the UA is prevented from reaching such speeds. (5) UA position: (i) the UA position relative to the LOS data link transmitter/receiver must be also displayed in terms of range and bearing; (ii) the deviation between the planned ground track and the actual flightpath of the UA. (6) UA attitude in terms of roll and pitch, (7) vertical speed,

	(8) time (hours, minutes, seconds), (9) navigation systems status, (10) UA identification in accordance with USAR.U1883 (b), where multiple UAS are being operated. (11) Wind direction and speed at UA level, if only track data is displayed to the UAS crew.
U1725	USAR.U1725 Powerplant data (a) The following are the minimum required powerplant data that must be displayed at all times in the control station at an update rate consistent with safe operation: (1) remaining fuel quantity, (2) an indicating means to indicate the functioning of each engine. (b) For the following data, they are only required to be displayed full time if the system is not capable of providing a warning to the UAS crew if a safe range is exceeded. (1) For reciprocating engine-powered UA. In addition to the powerplant data required by sub-paragraph (a), the following powerplant data are required: (i) RPM for each engine. (ii) manifold pressure for each altitude engine and for each engine with a controllable propeller. (2) For turbine engine-powered UA. In addition to the powerplant data required by sub-paragraph (a), the following powerplant data are required: (i) gas temperature for each engine. (ii) speed of the rotors with established limiting speeds for each engine. (3) For turbojet/turbofan engine-powered UA. In addition to the powerplant data required by sub-paragraphs (a) and (b)(2), the following powerplant data are required: (i) For each engine, thrust or a parameter that can be related to thrust, including free air temperature if needed for this purpose. (ii) For each engine with a thrust reverser, an indication to the UAS crew of the thrust reverser position. (4) For turbopropeller-powered UA In addition to the powerplant data required by sub-paragraphs (a) and (b)(2), the following powerplant data is required: torque for each engine. (c) Considering USAR.U1722, the following are the minimum required powerplant data that shall be selectable or available when queried by the UAS crew for display in the control station at an update rate consistent with safe operation: (1) For all UA: (i) oil pressure for each engine, except for engines where the design does not include a separate lubrication device; (ii) oil temperature for each engine, except for engines where the design does not include a separate lubrication device;

	(iii) oil quantity for each oil tank which meets the requirements of USAR.1337(d), except for engines where the design does not include a separate lubrication device. (2) For reciprocating engine-powered UA. In addition to the powerplant data required by subparagraph (a), (b)(1) and (c)(1), the following powerplant data are required: (i) induction system air temperature for each engine equipped with a preheater and having induction air temperature limitations that can be exceeded with preheat. (ii) cylinder head temperature for each air-cooled engine with cowl flaps; (iii) fuel pressure for pump fed engines. (iv) For each turbocharger installation: - If limitations are established for either carburettor (or manifold) air inlet temperature or exhaust gas or turbocharger turbine inlet temperature, data must be furnished for each temperature for which the limitation is established unless it is shown that the limitation will not be exceeded in all intended operations. - If its oil system is separate from the engine oil system, oil pressure and oil temperature must be provided. (v) coolant temperature for each liquid-cooled engine. (3) For turbine engine-powered UA. In addition to the powerplant data required by subparagraph (a), (b)(2) and (c)(1), the following powerplant data are required: (i) fuel flow for each engine. (ii) fuel low pressure warning means for each engine. (iii) fuel low level warning means for any fuel tank that should not be depleted of fuel in normal operations. (iv) oil low pressure warning means for each engine. (v) An indicating means to indicate the functioning of the powerplant ice protection system for each engine. (vi) For each engine, a means to indicate the contamination of the fuel strainer or filter (required by USAR.997) before it reaches the capacity established in accordance with USAR.997(d). (vii) For each engine, a means to indicate the contamination of the oil strainer or filter (required by USAR.1019), if it has no bypass, before it reaches the established capacity. (viii) An indicating means to indicate the functioning of any heater used to prevent ice clogging of fuel system components. (4) For turbopropeller-powered UA In addition to the powerplant data required by subparagraphs (a), (b)(2), (b)(4), (c)(1) and (c)(3), the following powerplant data is required: a position indicating means to indicate to the UAS crew when the propeller blade angle is below the flight low pitch position, for each propeller, unless it can be shown that such occurrence is highly improbable.
U1726	USAR.U1726 Data display of equipment required by Operations regulation

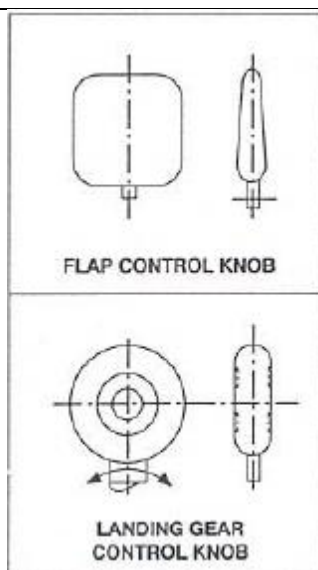
	The data and status of equipment required by Operation regulation must be capable of display in the UCS as agreed with the Certifying Authority.
U1727	USAR.U1727 Electronic data display See AMC.U1727 and AMC.U1727 (b) (a) Electronic data display systems must: (1) Meet the arrangement and visibility requirements of USAR.U1721; (2) Be easily legible under all the lighting conditions encountered by the workstation considering the expected electronic display brightness level at the end of an electronic display indicator's useful life. Specific limitations on display system useful life must be addressed in the instructions for continued airworthiness requirements of USAR.1529; (3) Incorporate sensory cues for the UAS crew that are easily comprehensible, and (4) Incorporate visual displays of indicators or data display markings, required by USAR.U1831 to USAR.U1843, or visual displays that alert the UAS crew to abnormal operational values or approaches to established limitation values, for each parameter required to be displayed in USAR. (b) The electronic display systems, including their subsystems and installations, and considering other UAs, must be designed so that one display of information essential for continued safe flight and landing will remain available to the UAS crew after any single failure or probable combination of failures. (c) Be designed such that failure of any single connection, processor, or display unit does not result in display of unsafe or misleading data without further indication of failure to the operator. Failure conditions of the electronic display shall be assessed according to USAR.1309
U1728	USAR.U1728 Data link data display, warnings and indicators Data link data display, warnings and indicators must meet the requirements of USAR.U1607
U1729	USAR.U1729 Fuel quantity and oil quantity data (a) Fuel quantity and consumption data. There must be means to indicate to the UAS crew the rate of fuel consumption and the quantity of usable fuel in each tank during flight. A calibrated scale in appropriate units and clearly marked to indicate those units, must be used. In addition: (1) Tanks with interconnected outlets and airspaces may be considered as one tank and need not display separate data; and (2) No fuel quantity data is required for an auxiliary tank that is used only to transfer fuel to other tanks if the relative size of the tank, the rate of fuel transfer and operating instructions are adequate to: (i) Prevent overflow; and (ii) Give to the UAS crew a prompt warning if a transfer malfunction occurs. (b) Oil quantity data. There must be a means to indicate in the UCS the quantity of oil in each tank in flight, if there is an oil transfer system or a reserve oil supply system.
U1730	USAR.U1730 Automatic take-off system or automatic landing system data For UAS equipped with an automatic take-off system or an automatic landing system or both, the following data must be continuously displayed to the UAS crew during the respective flight phases in compliance with USAR.U1490 (d) (a) the UA flightpath,

(b) the deviation between the UA flightpath and the planned flightpath.

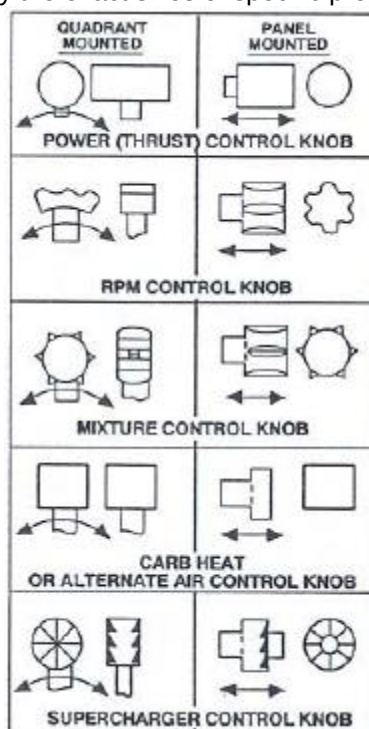
For landing the standard slope is as defined in USAR.75

CONTROLS

U1731	USAR.U1731 General See AMC.U1731 (a) Each control in the UCS must be located and (except where its function is obvious) identified to provide convenient operation and to prevent confusion and inadvertent operation. (b) The controls must be located and arranged so that the UAS crew, when at their workstation have full and unrestricted movement of each control without interference from either their clothing or the UCS structure. (c) The control system must be designed so that the controls needed for continued safe flight and landing remain available to the UAS crew in normal, degraded and emergency conditions.
U1732	USAR.U1732 Safety critical controls (a) The design, location and accessibility of safety critical controls (i.e. requiring immediate action of the UAS crew) must be compatible with a rapid and precise reaction of the UAS crew during emergency operation. (b) Where the interface with UAS crew is based on a “pull down menus” architecture: (1) the controls that necessitate a prompt reaction of the UAS crew must be accessible at the first level of the pull down menus, (2) otherwise, safety critical controls in the UCS must have dedicated knobs or levers. (c) Safety critical controls must be designed to prevent the possibility of confusion and subsequent inadvertent operation.
U1733	USAR.U1733 Conventional controls and indicators (a) Where conventional flying controls and indicators are employed, the form, the location and layout must ensure safe operation. (b) For each conventional indicators in the UCS: (1) When markings are on the cover glass of the indicator, there must be means to maintain the correct alignment of the glass cover with the face of the dial, (2) Each arc and line must be wide enough and located to be clearly visible to the UAS crew, (3) All related indicators must be calibrated in compatible units. (c) If conventional flap and landing gear control knobs are used by the UAS crew, they must conform to the general shapes (but not necessarily the exact sizes or specific proportions) in the following figure:



(d) If conventional powerplant control knobs are used by the UAS crew, they must conform to the general shapes (but not necessarily the exact sizes or specific proportions) in the following figures:



U1735 USAR.U1735 Motion and representation of controls

UCS controls must be designed so that they, or their representations, operate intuitively for the UAS crew. The representation of these controls must be similar to conventional flight controls that exist in manned aircraft.

U1741	USAR.U1741 UCS flight controls (a) The UCS flight controls are those used by the UAS crew to operate the UA with the semi-automatic mode of control as defined in USAR.U1330 (b) The design of UCS flight controls must allow the UAS crew to rapidly and easily change the following flight parameters of the UA: (1) heading or track, (2) altitude, and (3) airspeed.
U1742	USAR.U1742 Flight termination system control When the UAS is equipped with a flight termination system, (a) Its controls are safety critical controls as defined in USAR.U1732 (b) Its controls must be arranged and identified such that they are readily available and accessible. The use of these controls must be intuitive and minimise the possibility of confusion and subsequent inadvertent operation. (c) Two distinct actions of the UAS crew are required to initiate the flight termination command.
U1743	USAR.U1743 Fuel controls (a) There must be a means readily available to allow the UAS crew to rapidly shut off, in flight, the fuel to each engine individually. (b) In addition, there must be means to: (1) Prevent inadvertent operation of each shut-off valve; and (2) Allow appropriate UAS crew to reopen each valve rapidly after it has been closed. (c) Where fitted, fuel tank selector must: (1) Require a separate and distinct action to place the selector in the “OFF” position; and (2) Have the tank selector designed to operate so that it is impossible for the selector to pass through the “OFF” position when changing from one tank to another.
U1745	USAR.U1745 Fuel jettisoning control (a) Where existing, the fuel jettisoning valve must be designed to allow UAS crew to close the valve during any part of the jettisoning operation. (b) There must be an indication compliant with USAR.U1831, adjacent to the jettison control to warn the UAS crew against jettisoning fuel while any means (including flaps, slots and slats) of changing the airflow across or around the wings is in use, unless it is shown that using such means does not adversely affect fuel jettisoning. (c) Fuel jettisoning control must be designed to prevent inadvertent operation.
U1747	USAR.U1747 Air induction control Each automatic alternate air induction door must have an override means accessible to the UAS crew.
U1751	USAR.U1751 Engine and APU controls

	The controls necessary to perform all functions in normal, degraded and emergency modes must be provided to the UAS crew taking into account the level of automation substantiated by the flight control system.
U1753	USAR.U1753 Ignition switches See AMC.U1753 (b) (a) Ignition switches must control each ignition circuit on each engine. (b) There must be means readily available to the UAS crew to quickly shut off all ignition circuits on multi-engine UA. (c) Ignition switches must have safeguards to prevent inadvertent operation.
U1755	USAR.U1755 Mixture controls When mixture control is provided, there shall be a separate control for each engine. Each mixture control must be designed to prevent confusion and inadvertent operation. (a) The controls must be grouped and arranged to allow: (1) Separate control of each engine; and (2) Simultaneous control of all engines. (b) The control must require a separate and distinct operation to move the control towards lean or shut-off position.
U1757	USAR.U1757 Propeller speed and pitch controls (a) Where propeller speed or pitch controls exist, they must be grouped and arranged to allow: (1) Separate control of each propeller; and (2) Simultaneous control of all propellers. (b) The controls must allow ready synchronisation of all propellers on multi-engine UA.
U1759	USAR.U1759 Propeller feathering controls Where propeller feathering controls exist, it must be possible to feather each propeller separately. Each control must have means to prevent inadvertent operation.
U1761	USAR.U1761 Turbine engine reverse thrust and propeller pitch settings below the flight regime For turbine engine installations, each control for reverse thrust and for propeller pitch settings, where existing, below the flight regime must have means to prevent its inadvertent operation. The means must have a positive lock or stop at the flight idle position and must require a separate and distinct operation by the UAS crew to displace the control from the flight regime (forward thrust regime for turbojet powered UA).
U1763	USAR.U1763 Carburettor air temperature controls Where existing, there must be a separate carburettor air temperature control for each engine.
U1765	USAR.U1765 Shut-off controls (a) For each UAS function that can be shut-off from the UCS, there must be a means to prevent inadvertent operation of the shut-off control. In addition, there must be a means to restore the function after it has been shut-off. (b) For fuel shut-offs control, the requirement is addressed in USAR.U1743 (c) For ignition shut-off control, the requirement is addressed in USAR.U1753
U1769	USAR.U1769 "Abort" control for automatic take-off system or automatic landing system

	See AMC.U1769 Where a UAS is equipped with an automatic take-off system or an automatic landing system there must be a means readily available to the UAS crew to rapidly abort the take-off phase or the landing phase in compliance with USAR.U1492
--	---

INDICATORS AND WARNINGS

U1785	USAR.U1785 Warning, caution and advisory information colour code The warning, caution or advisory information installed in the UCS, must, unless otherwise approved by the Certifying Authority, be: (a) Red, for warning information (information indicating a hazard which may require immediate corrective action); (b) Amber, for caution information (information indicating the possible need for future corrective action); (c) Green, for safe operation information; and (d) Any other colour, including white, for information not described in sub-paragraphs (a) to (c), provided the colour differs sufficiently from the colours prescribed in sub-paragraphs (a) to (c) to avoid possible confusion. (e) Effective under all probable UCS lighting conditions.
U1787	USAR.U1787 UAS automatic diagnostic and monitoring See AMC.U1787 (a) (a) The UCS must include an automatic diagnostic and monitoring capability for the status of the UA and provide to the UAS crew appropriate warning indication. (b) Warnings, Cautions and advisories shall be provided in an organized, prioritized system, and the presentation of high priority information shall not be masked by the presence of older lower priority information. (c) Guidance for corrective actions shall be provided either automatically or in the UAS Flight Manual.
U1788	USAR.U1788 Degraded modes of operation warning The UCS must be configured to ensure that the UAS crew is informed of any degraded or emergency mode, including cases in which there is an automatic switching to an alternate mode of operation.
U1789	USAR.U1789 Low speed warning (a) There must be a clear and distinctive low speed warning in the UCS, with the flaps and landing gear in any normal position in straight and turning flight, in accordance with the following: (1) It should not be possible to command from the UCS speed values lower than the minimum steady flight speed (except take-off and landing) allowed by the flight envelope protection maintained by the flight control system. (2) Adequate low speed cues and warning should be provided in the UCS when approaching the stalling speed or $V_{\min \text{ DEMO}}$ if the stalling is not to be demonstrated. (3) During the tests required by USAR.201 the low speed warning must begin at a speed exceeding the stalling speed or $V_{\min \text{ DEMO}}$ (if the stalling is not to be demonstrated) by a margin of not less than 5 knots and must continue as long as the condition is true.

	(4) When following the procedures of USAR.1585, the low speed warning must not occur during a take-off with all engines operating, a take-off continued with one engine inoperative or during an approach to landing. (b) The low speed warning must be furnished by a device that will give clearly distinguishable indication. A visual low speed warning device that requires the attention of the UAS crew within the UCS is not acceptable by itself.
U1790	USAR.U1790 UAS mode of control indicator There must be a means in the UCS to indicate to the UAS crew the active mode of control of the flight control system. If semi-automatic mode is engaged, a specific indicator must be activated in clear view of the UAS crew.
U1791	USAR.U1791 Wing flaps position indicator Where a UA is equipped with wing flaps, there must be a wing flap position indicator in the UCS.
U1793	USAR.U1793 Landing gear position indicator and warning See AMC.U1793 (b) (a) Position indicator. If a retractable landing gear is used, there must be a landing gear position indicator in the UCS to inform the UAS crew that each gear is secured in the extended (or retracted) position. (b) Landing gear warning. If a retractable landing gear is used, an aural or equally effective landing gear warning devices must be provided to inform the crew of an imminent landing without the gear fully extended and locked. (c) Landing gear controlled by an automatic landing system (see USAR.U1490) must comply with (a) and (b).
U1795	USAR.U1795 Pressurised compartment indicator See AMC.U1795 Where a UA is equipped with a pressurised compartment, there must be a warning to indicate when the safe pressure differential is exceeded.
U1797	USAR.U1797 Fuel pumps warning Where a UA is equipped with fuel pumps, if both the main pump and emergency pump operate continuously, there must be a means to indicate to the UAS crew a malfunction of either pump.
U1799	USAR.U1799 Air induction indicator Where a UA is equipped with an air induction door, each alternate air induction door must have a means to indicate to the UAS crew when it is not closed.
U1801	USAR.U1801 Battery discharge warning See AMC.U1801 There must be means to warn the UAS crew if malfunctioning of any part of the electrical system is causing the continuous discharge of any battery which is relevant for safe flight.
U1803	USAR.U1803 Indicators for power-assisted valves in the powerplant See AMC.U1803 For power-assisted valves in the powerplant, there must be a means in the UCS to indicate to the UAS crew when the valve: (a) is in the fully open or fully closed position. (b) is moving between the fully open and fully closed position.

U1805	USAR.U1805 Shut off valves indicator Where a UAS is equipped with power operated shut off valves they must have means to indicate to the UAS crew when the valve has reached the selected position.
U1809	USAR.U1809 UAS electrical systems warning and indicator See AMC.U1809 (a) There must be a means to give immediate warning to the UAS crew of a failure of any UAS electrical power generating device. (b) A means must exist in the UCS to indicate to the UAS crew the electric power system quantities essential for safe operation. (c) A warning which is unambiguous and clearly distinguishable to the UAS crew shall be immediately provided for any UCS power supply failure which could result in an unsafe condition in any phase of UA flight, including landing and take-off.
U1811	USAR.U1811 De-icer boot system indicator If certification with ice protection provisions is desired and a de-icer boot system is installed there must be means to indicate to the UAS crew that the de-icer boot system is functioning normally.
U1813	USAR.U1813 Hydraulic systems indicator There must be a means to indicate to the UAS crew the pressure in each hydraulic system which supplies two or more primary functions.
U1817	USAR.U1817 Fire protection warning If action by the UAS crew is required to prevent or extinguish fire (e.g. equipment shut-down or actuation of a fire extinguisher) in the UA, quick acting means must be provided to immediately alert the UAS crew in the UCS.
U1819	USAR.U1819 Pitot heat indicator If a pitot heating system is installed to meet the requirements specified in USAR.1323 (b), an indication system must be provided to indicate to the UAS crew when that pitot heating system is not operating.
U1821	USAR.U1821 UCS Power distribution indicator See AMC.U1821 Each UCS power distribution circuit must have an indicator in the UCS to indicate when power is below safe minimum.
U1825	USAR.U1825 Flight control system lock warning If there is a device on the UAS to lock the flight controls as mentioned in USAR.679, the UAS crew must be warned when the device is engaged.
U1827	USAR.U1827 Flightpath deviation warning Where automatic flight control modes as defined in USAR.U1330 are activated, a warning must be displayed when excessive deviation from the pre-programmed flightpath occurs. The acceptable deviation shall be agreed with the Certifying Authority.
U1829	USAR.U1829 UAS safety status indication An indication must be provided in the UCS which shows the safety status of the UAS so approaching ground staff can be notified if the UA is in an unsafe state (e.g. radiation hazard present, laser energized, etc.)

INFORMATION, MARKINGS AND PLACARDS

U1831	USAR.U1831 General Each information, markings and placard displayed in the UCS prescribed in USAR.1541 (a) must be: (a) continuously displayed in a conspicuous place relative to the object, indicator or data it is assumed to be associated with; and (b) easily interpreted unambiguously by the UAS crew.
U1835	USAR.U1835 Airspeed data (a) If required to maintain safe flight, each airspeed data must be marked as specified in sub-paragraph (b), with the marks located at the corresponding indicated airspeeds. (b) The following markings must be made: (1) For the never-exceed speed V_{NE}, a red line. (2) For the caution range, a yellow band extending from the red line specified in sub-paragraph (1) to the upper limit of the green band specified in sub-paragraph (3). (3) For the normal operating range, a green band with the lower limit at V_{S1} with maximum weight and with landing gear and wing flaps retracted, and the upper limit at the maximum structural cruising speed V_{NO} established under USAR.1505 (b). (4) For the flap operating range, a white band with the lower limit at V_{S0} at the maximum weight and the upper limit at the flaps-extended speed V_{FE} established under USAR.1511 (5) For multi-engine powered UA, for the speed at which compliance has been shown with USAR.69 (b) relating to rate of climb, at maximum weight and at sea-level, a blue line. (6) For multi-engine powered UA, for the maximum value of minimum control speed (one or more engine inoperative) determined under USAR.149 (b), V_{MC}, a red line. (c) If V_{NE} or V_{NO} vary with altitude, there must be means to indicate to the UAS crew the appropriate limitations throughout the operating altitude range. (d) Sub-paragraphs (b) (1) to (b) (3) and sub-paragraph (c) do not apply to UA for which a maximum operating speed V_{MO}/M_{MO} is established under USAR.1505 (c). For those UA there must either be a maximum allowable airspeed indication showing the variation of V_{MO}/M_{MO} with altitude or compressibility limitations (as appropriate), or a red line marking for V_{MO}/M_{MO} must be made at lowest value of V_{MO}/M_{MO} established for any altitude up to the maximum operating altitude for the UA. (e) There must be an airspeed indication in clear view of the UAS crew and as close as practicable to the airspeed indicator. This indication must list: (1) The operating manoeuvring speed, V_o; (2) The maximum landing gear operating speed V_{LO}; and (3) For multi-engine-powered UA, the maximum value of the minimum control speed (one or more engine inoperative) determined under USAR.149 (b), V_{MC}.
U1837	USAR.U1837 Magnetic heading or track data See AMC.U1837 Where magnetic heading or track is displayed in the UCS, it must be automatically compensated for deviation.
U1839	USAR.U1839 Powerplant and auxiliary power unit data

	For each required powerplant and auxiliary power unit, data shall be available in the UCS, as appropriate to the type of powerplant. (a) Each maximum and if applicable, minimum safe operating limit must be marked with a red radial or a red line; (b) Each normal operating range must be marked with a green arc or green line not extending beyond the maximum and minimum safe limits; (c) Each take-off and precautionary range must be marked with a yellow arc or a yellow line; and (d) Each engine, auxiliary power unit or propeller range that is restricted because of excessive vibration stresses must be marked with red arcs or red lines.
U1841	USAR.U1841 Oil quantity data Each oil quantity data displayed in the UCS must be marked in sufficient increments to indicate readily and accurately the quantity of oil.
U1843	USAR.U1843 Fuel quantity data A red line must be marked on each data displayed in the UCS at the calibrated zero reading, as specified in USAR.1337 (b)(1).
U1845	USAR.U1845 Control markings See AMC.U1845 (e)(2) (a) Every control, switch, knob or lever in the UCS must be plainly marked as to its function and method of operation. (b) Each remote control, as defined in USAR.U1741, must be suitably marked. (c) For powerplant fuel controls: (1) Each fuel tank selector control must be marked to indicate the position corresponding to each tank and to each existing cross feed position; (2) If safe operation requires the use of any tanks in a specific sequence, that sequence must be marked on or near the selector for those tanks; (3) The conditions under which the full amount of usable fuel in any restricted usage fuel tank can safely be used must be stated adjacent to the selector valve for that tank; and (4) Each valve control for any engine of a multi-engine UA must be marked to indicate the position corresponding to each engine controlled. (5) For fuel jettisoning control, see USAR.U1745 (b). (d) Usable fuel capacity must be marked as follows: (1) For fuel systems having no selector controls, the usable fuel capacity of the system must be indicated near the fuel quantity data displayed in the UCS. (2) For fuel systems having selector controls, the usable fuel capacity available at each selector control position must be indicated near the selector control. (e) For accessory, auxiliary and emergency controls:

	(1) If retractable landing gear is used, the indicator required by USAR.U1793 must be marked so that the UAS crew can, at any time, ascertain that the wheels are secured in the extreme positions; and (2) Each emergency control must be red and must be marked as to method of operation. (3) No control other than an emergency control shall be this colour.
U1849	USAR.U1849 Operating limitations indications (a) There must be an indication in clear view of the UAS crew stating in the UCS that the UA must be operated in accordance with the UAS Flight Manual; (b) There must be an indication in clear view of the UAS crew that specifies the kind of operations to which the operation of the UA is limited or from which it is prohibited under USAR.1525

MISCELLANEOUS

U1881	USAR.U1881 UA handover between two UCS See AMC.U1881 (b), AMC.U1881 (c) and AMC.U1881 (d) Where the UAS is designed for UA handover between two UCS: (a) The in-control UCS must be clearly identified to all UAS crew members. (b) Positive control must be maintained during handover. (c) The command and control functions that are transferred during handover must be approved by the Certifying Authority and defined in the UAS Flight Manual. (d) Handover between two UCS must not lead to unsafe conditions. (e) The in-control UCS must have the required functionality to accommodate emergency situations.
U1883	USAR.U1883 Command and control of multiple UA Where a UCS is designed to command and control multiple UA, the following requirements apply: (a) The minimum UAS crew must be established so that it is sufficient for safe operation of each aircraft in compliance with USAR.U1704 and emergency conditions. (b) The UA data shall be displayed in the UCS in a manner that prevents confusion and inadvertent operation. (c) The UA controls must be available to the UAS crew for each UA of which it has command and control in a manner that prevents confusion and inadvertent operation. (d) All indicators and warnings must be available to the UAS crew for each UA in a manner that prevents confusion and inadvertent operation.
U1885	USAR.U1885 UA handover within the same UAS control station See AMC.U1885 (b) and AMC.U1885 (d) Where the UCS has more than one workstation designed for controlling the UA: (a) The in-control workstation must be clearly identified to all UAS crew members. (b) Positive control must be maintained during handover. (c) The command and control functions that are transferred during handover must be approved by the Certifying Authority and defined in the UAS Flight Manual.

	(d) Handover within the same UAS control station must not lead to unsafe conditions. (e) The in-control workstation must have the required functionality to accommodate emergency situations.
U1887	USAR.U1887 Multiple UA monitoring Where the UCS is designed to monitor multiple UA, there must be a means to clearly indicate to the UAS crew the UA over which it has command and control.

Appendices

Appendix C

Basic Landing Conditions

Condition		Tail Wheel Type		Nose Wheel Type		
		Level landing	Tail-down landing	Level landing with inclined reactions	Level landing with nose wheel just clear of the ground	Tail-down landing
Reference paragraph		USAR 479(a)(1)	USAR 481(a)(1)	USAR 479(a)(2)(i)	USAR 470(a)(2)(ii)	USAR 481(a)(2) and (b)
Vertical component at c.g.		nW	nW	nW	nW	nW
Fore and aft component at c.g.		KnW	0	KnW	KnW	0
Lateral component in either direction at c.g.		0	0	0	0	0
Shock absorber extension (hydraulic shock absorber)		Note (2)	Note (2)	Note (2)	Note (2)	Note (2)
Shock absorber extension (rubber or spring shock absorber)		100%	100%	100%	100%	100%
Tyre deflection		Static	Static	Static	Static	Static
Main wheel loads (both wheels)	Vr	$(n-L)W$	$(n-L)Wb/d$	$(n-L)Wa'/d'$	$(n-L)W$	$(n-L)W$
	Dr	KnW	0	$KnWa'/d'$	KnW	0
Tail (nose) Wheel loads	Vf	0	$(n-L)Wa/d$	$(n-L)Wb'/d'$	0	0
	Df	0	0	$KnWb'/d'$	0	0
Notes		(1), (3), and (4)	(4)	(1)	(1), (3), and (4)	(3), and (4)

NOTE (1)

K may be determined as follows: $K=0.25$ for $W=1361$ kg (3,000 pounds) or less; $K=0.33$ for $W=2722$ kg (6,000 pounds) or greater, with linear variation of K between these weights.

NOTE (2)

For the purpose of design, the maximum load factor is assumed to occur throughout the shock absorber stroke from 25% deflection to 100% deflection unless otherwise shown and the load factor must be used with whatever shock absorber extension is most critical for each element of the landing gear.

NOTE (3)

Unbalanced moments must be balanced by a rational conservative method.

NOTE (4)

L is defined in USAR.725(b).

NOTE (5)

n is the limit inertia load factor, at the c.g. of the UA, selected under USAR.473 (d), (f), and (g).

INTENTIONALLY BLANK

Appendix D

Wheel Spin-Up Loads

(a) The following method for determining wheel spin-up loads for landing-conditions is based on NACA T.N. 863. However, the drag component used for design may not be less than the drag load prescribed in USAR.479 (b).

$$FH_{max} = \frac{1}{re} \sqrt{\frac{2Iw(V_H - V_C)nFV_{max}}{t_z}}$$

$$FH_{max} = \frac{1}{re} \sqrt{\frac{2Iw(V_H - V_C)nFV_{max}}{t_z}}$$

where:

FH max = maximum rearward horizontal force acting on the wheel (in pounds);

re = effective rolling radius of wheel under impact based on recommended operating tyre pressure (which may be assumed to be equal to the rolling radius under a static load of njMe) in feet;

IW = rotation mass moment of inertia of rolling assembly (in slug ft²);

V_H = linear velocity of UA parallel to ground at instant of contact (assumed to be 1.2 V_{SO}, in feet per second);

V_C = peripheral speed of tyre, if pre-rotation is used (in feet per second) (there must be a positive means of pre-rotation before pre-rotation may be considered);

n = effective coefficient of friction (0.80 may be used);

FV max = maximum vertical force on wheel (pounds = njMe, where Me and nj) are defined in USAR.725 (note that the units for Me in USAR.725 are kg and must be converted to pounds);

t_z = time interval between ground contact and attainment of maximum vertical force on wheel (seconds). However, if the value of FH max, from the above equation exceeds 0.8 FV max, the latter value must be used for FH max.

(b) This equation assumes a linear variation of load factor with time until the peak load is reached and under this assumption, the equation determines the drag force at the time that the wheel peripheral velocity at radius re equals the UA velocity. Most shock absorbers do not exactly follow a linear variation of load factor with time. Therefore, rational or conservative allowances must be made to compensate for these variations. On most landing gears, the time for wheel spin-up will be less than the time required to develop maximum vertical load factor for the specified rate of descent and forward velocity. For exceptionally large wheels, a wheel peripheral velocity equal to the ground speed may not have been attained at the time of maximum vertical gear load. However, as stated above, the drag spin-up load need not exceed 0.8 of the maximum vertical loads.

(c) Dynamic spring-back of the landing gear and adjacent structure at the instant just after the wheels come up to speed may result in dynamic forward acting loads of considerable magnitude. This effect must be determined, in the level landing condition, by assuming that the wheel spin-up loads calculated by the methods of this appendix are reversed. Dynamic spring-back is likely to become critical for landing gear units having wheels of large mass or high landing speeds.

INTENTIONALLY BLANK

Appendix F

Test Procedure for Self-Extinguishing Materials in accordance with USAR.1359

(a) Conditioning. Specimens must be conditioned to $21^{\circ} \pm 3^{\circ}\text{C}$ ($70^{\circ} \pm 5^{\circ}\text{F}$), and at $50\% \pm 5\%$ relative humidity until moisture equilibrium is reached or for 24 hours. Only one specimen at a time may be removed from the conditioning environment immediately before subjecting it to the flame.

(b) Specimen configuration.

(1) Not applicable.

(2) Not applicable.

(3) When showing compliance with USAR.1359 (c) for materials used in electrical wire and cable insulation, the wire and cable specimens must be the same size as used in the UAS. In the case of fabrics, both the warp and fill direction of the weave must be tested to determine the most critical flammability condition. When performing the tests prescribed in sub-paragraphs (d) and (e) of this Appendix, the specimen must be mounted in a metal frame so that;

(i) in the vertical tests of sub-paragraph (d), the two long edges and the upper edge are held securely;

(ii) in the horizontal test of sub-paragraph (e), the two long edges and the edge away from the flame are held securely;

(iii) the exposed area of the specimen is at least 5 cm (2 in) wide and 30 cm (12 in) long, unless the actual size used in the UAS is smaller; and

(iv) the edge to which the burner flame is applied must not consist of the finished or protected edge of the specimen but must be representative of the actual cross-section of the material or part installed in the UAS. When performing the test prescribed in sub-paragraph (f) of this Appendix, the specimen must be mounted in a metal frame so that all four edges are held securely and the exposed area of the specimen is at least 20 cm by 20 cm (8 in by 8 in).

(c) Apparatus. Except as provided in sub-paragraph (e) of this Appendix, tests must be conducted in a draft-free cabinet in accordance with Federal Test Method Standard 191 Method 5903 (revised Method 5902) which is available from the General Services Administration, Business Service Centre, Region 3, Seventh and D Streets SW. Washington, D.C. 20407, or with some other approved equivalent method. Specimens which are too large for the cabinet must be tested in similar draft-free conditions.

(d) Vertical test. A minimum of three specimens must be tested and the results averaged. For fabrics, the direction of weave corresponding to the most critical flammability conditions must be parallel to the longest dimension. Each specimen must be supported vertically. The specimen must be exposed to a Bunsen or Tirrill burner with a nominal 9.5 mm (3/8 in) inner diameter tube adjusted to give a flame of 38 mm (1½ in) in height. The minimum flame temperature measured by a calibrated thermo-couple pyrometer in the centre of the flame must be 843°C (1550°F). The lower edge of the specimen must be 19 mm (¾ in) above the top edge of the burner. The flame must be applied to the centre line of the lower edge of the specimen. Flame time, burn length, and flaming time of drippings, if any, must be recorded. The burn length determined in accordance with sub-paragraph (h) of this Appendix must be measured to the nearest 2.5 mm (1.10 in).

(e) Horizontal test. A minimum of three specimens must be tested and the results averaged. Each specimen must be supported horizontally. The exposed surface when installed in the UAS must be face down for the test. The specimen must be exposed to a Bunsen burner or Tirrill burner with a nominal 9.5 mm (3/8 in) inner diameter tube adjusted to give a flame of 38 mm (1½ in) in height. The minimum flame temperature measured by a calibrated thermocouple pyrometer in the centre of the flame must be 843°C (1550°F). The specimen must be positioned so that the edge being tested is 19 mm (¾ in) above the top of, and on the centre line of, the burner. The flame must be applied for 15 seconds and then removed. A minimum of 25 cm (10 in) of the

specimen must be used for timing purposes, approximately 38 mm (1½ in) must burn before the burning front reaches the timing zone, and the average burn rate must be recorded.

(f) Forty-five degree test. Not applicable.

(g) Sixty-degree test. A minimum of three specimens of each wire specification (make and size) must be tested. The specimen of wire or cable (including insulation) must be placed at an angle of 60° with the horizontal in the cabinet specified in sub-paragraph (c) of this appendix with the cabinet door open during the test or placed within a chamber approximately 0.6 m (2 ft) high by 0.3 m by 0.3 m (1 ft by 1 ft), open at the top and at one vertical side (front), that allows sufficient flow of air for complete combustion but is free from drafts. The specimen must be parallel to and approximately 15 cm (6 in) from the front of the chamber. The lower end of the specimen must be held rigidly clamped. The upper end of the specimen must pass over a pulley or rod and must have an appropriate weight attached to it so that the specimen is held tautly throughout the flammability test. The test specimen span between lower clamp and upper pulley or rod must be 61 cm (24 in) and must be marked 20 cm (8 in) from the lower end to indicate the centre point for flame application. A flame from a Bunsen or Tirrill burner must be applied for 30 seconds at the test mark. The burner must be mounted underneath the test mark on the specimen, perpendicular to the specimen and at an angle of 30° to the vertical plane of the specimen. The burner must have a nominal bore of 9.5 mm (3/8 in), and must be adjusted to provide a 76 mm (3 in) high flame with an inner cone approximately one third of the flame height. The minimum temperature of the hottest portion of the flame, as measured with a calibrated thermocouple pyrometer may not be less than 954°C (1750°F). The burner must be positioned so that the hottest portion of the flame is applied to the test mark on the wire. Flame time, burn length, and flaming time of drippings, if any, must be recorded. The burn length determined in accordance with sub-paragraph (h) of this appendix must be measured to the nearest 2.5 mm (1/10 in). Breaking of the wire specimen is not considered a failure.

(h) Burn length. Burn length is the distance from the original edge to the farthest evidence of damage to the test specimen due to flame impingement, including areas of partial or complete consumption, charring, or embrittlement, but not including areas sooted, stained, warped, or discoloured, nor areas where material has shrunk or melted away from the heat source.

Appendix G

Instructions for continued airworthiness
G.1 General (a) This appendix specifies requirements for the preparation of instructions for continued airworthiness as required by USAR.1529 (b) The instructions for continued airworthiness for each UAS must include the instructions for continued airworthiness for each engine and propeller (hereinafter designated 'products'), for each appliance required by USAR, and any required information relating to the interface of those appliances and products with the UAS. If instructions for continued airworthiness are not supplied by the manufacturer of an appliance or product installed in the UAS, the instructions for continued airworthiness for the UAS must include the information essential to the continued airworthiness of the UAS. (c) The instructions for continued airworthiness for each UAS must include requirements associated with installation of only approved parts or appliances. The showing of compliance of parts and appliances installed in a UAS certificated under USAR must be made in accordance with officially recognized Standards and agreed with the Certifying Authority. G.2 Format (a) The instructions for continued airworthiness must be in the form of a manual or manuals as appropriate for the quantity of data to be provided. (b) The format of the manual or manuals must provide for a practical arrangement. G.3 Content The contents of the manual or manuals must be prepared in a language acceptable to the Certifying Authority. The instructions for continued airworthiness must contain the following manuals or sections, as appropriate and information: (a) UAS Maintenance Manual or section (1) Introduction information that includes an explanation of the UAS's features and data to the extent necessary for maintenance or preventive maintenance. (2) A description of the UAS and its systems and installations including its engines, propellers, and appliances. (3) Basic control and operation information describing how the UAS components and systems are controlled and how they operate, including any special procedures and limitations that apply. (4) Servicing information that covers details regarding servicing points, capacities of tanks, reservoirs, types of fluids to be used, pressures applicable to the various systems, location of access panels for inspection and servicing, locations of lubrication points, lubricants to be used, equipment required for servicing, tow instructions and limitations, mooring, jacking, and levelling information. (b) Ground segment and command and control data link (UCS – launch and landing elements) Maintenance Manual or section (1) Introduction information that includes an explanation of the ground segment's features and data to the extent necessary for maintenance or preventive maintenance. (2) A description of the ground segment and its systems and installations including UCS, launch and landing elements and command and control data link.

(3) Basic control and operation information describing how the ground segment components and systems are controlled and how they operate, including any special procedures and limitations that apply.

(4) Servicing information that covers details regarding servicing points, capacities of tanks, reservoirs, types of fluids to be used, pressures applicable to the various systems, location of access panels for inspection and servicing, locations of lubrication points, lubricants to be used, equipment required for servicing, tow instructions and limitations, mooring, jacking, and levelling information.

(c) Maintenance Instructions

(1) Scheduling information for each part of the UAS and its engines, auxiliary power units, propellers, accessories, instruments, and equipment that provides the recommended periods at which they should be cleaned, inspected, adjusted, tested, and lubricated, and the degree of inspection, the applicable wear tolerances, and work recommended at these periods. However, reference may be made to information from an accessory, instrument, or equipment manufacturer as the source of this information if it is shown that the item has an exceptionally high degree of complexity requiring specialised maintenance techniques, test equipment, or expertise. The recommended overhaul periods and necessary cross reference to the airworthiness limitations section of the manual must also be included. In addition, an inspection programme that includes the frequency and extent of the inspections necessary to provide for the continued airworthiness of the UAS must be included.

(2) Trouble-shooting information describing probable malfunctions, how to recognise those malfunctions, and the remedial action for those malfunctions.

(3) Information describing the order and method of removing and replacing products and parts with any necessary precautions to be taken.

(4) Other general procedural instructions including procedures for system testing during ground running, symmetry checks, weighing and determining the centre of gravity, lifting and shoring, and storage limitations.

(d) Diagrams of structural access plates and information needed to gain access for inspections when access plates are not provided.

(e) Details for the application of special inspection techniques including radiographic and ultrasonic testing where such processes are specified.

(f) Information needed to apply protective treatments to the structure after inspection.

(g) All data relative to structural fasteners such as identification, discard recommendations, and torque values.

(h) A list of special tools needed.

(i) Not applicable.

G.4 Airworthiness Limitations section

The instructions for continued airworthiness must contain a section titled airworthiness limitations that is segregated and clearly distinguishable from the rest of the document. This section must set forth each mandatory replacement time, structural inspection interval, and related structural inspection procedure required for type certification. This section must also set forth to review the approved Design Usage Spectrum (see USAR.U17) on a regular basis as specified by the Certifying Authority. If required by the Certifying Authority the Design Usage Spectrum should be updated. If the instructions for continued airworthiness consist of multiple documents, the section required by this paragraph must be included in the principal manual. This section must contain a legible statement in a prominent location that reads: The airworthiness limitations section is approved and variations must also be approved.

Appendix K

Other aspects of Transportation and Factors to consider

K.1 Introduction

This document provides guidance on the effects of transportation on Unmanned Aircraft Systems (UAS). In the context of this document, 'UAS' refers to the aircraft and all supporting systems that require transportation in order to support the aircraft. This document is designed to be generic so it can be applied to many different systems. However, aircrafts outside the weight range of 150kg to 20,000kg could potentially have unique transportation requirements so are outside the scope of this document.

Transportation by land, sea or air exposes components to potentially damaging levels of vibration. Examples include the vibrations in propeller aircraft, an uneven ride from a truck driving off-road or passage over rough seas. Vibration levels can be severe, so should be taken into account when calculating component life. The transportation environment has to be correctly defined if the risk of component failure is to be fully understood.

This document outlines one method of quantifying the transport environment. This is done by defining a usage profile (what transportation modes are used, how they are used and for how long) and attaching a vibration profile to each part of that usage profile. This allows a vibrations test to be defined which represents the entire transportation life of the UAS. The process described in this document is shown graphically in Figure 1 (in the red box). Since vibration can be a primary driver to the final design of the UAS or its packaging, the designer should be involved from the outset.

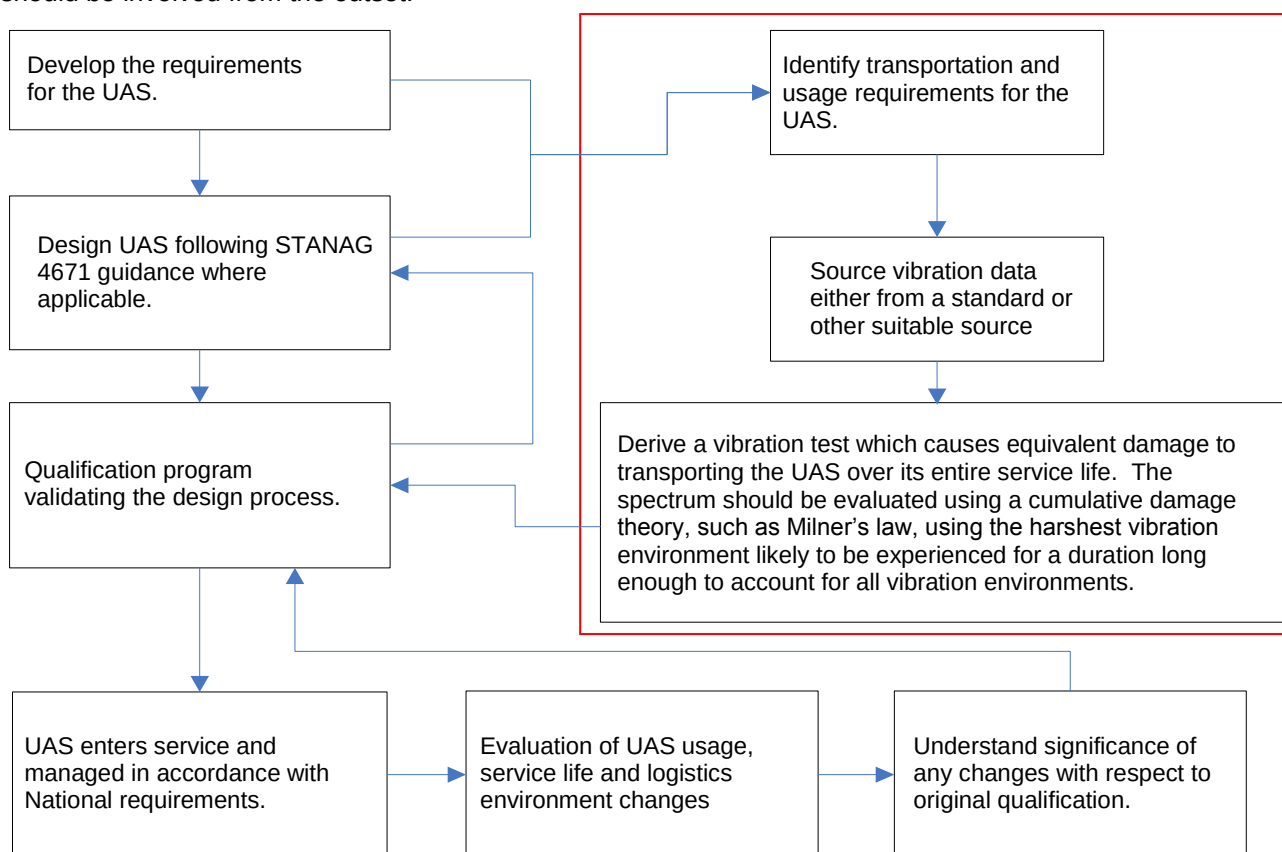


Figure 1 – Vibration qualification and management process

The purpose of this document is not to guide a particular design or suggest actual solutions but to aid the design process. This information should be used to drive the design of whatever system is to be

used to protect the UAS from vibration, be it a special container or inbuilt vibration isolation. This document is split into the following sections:

- K.2 The effects of vibration upon different materials and structures,
- K.3 Differences between the different transportation environments,
- K.4 Deriving transportation usage profiles,
- K.5 Deriving vibration profiles,
- K.6 Deriving test specifications from usage and vibration profiles,
- K.7 Performing a vibration test,
- K.8 Other transportation considerations which should be considered.

It is important to note that transportation exposes equipment to a wide range of potentially damaging conditions. This document only focuses on the vibration environment. Other transportation factors which should be considered are discussed in section 8.

K.2 The effects of vibration upon different materials and structures

When subjected to vibration, part of a structure or internal components will typically deflect or move in some way. Such motion will generate a certain level of strain and over time such strain can be detrimental to structural integrity. Vibration can also cause damage or affect the functionality of electronics and mechanical systems either by causing component failure or disruption to sensors or systems. Each part of the UAS design can therefore be affected so vibration should be considered from the outset:

Metals

Metallic components or structures which are exposed to cyclic loading (e.g. vibration) can be susceptible to fatigue, leading to a reduction in strength or ultimately, failure. If vibration causes stresses due to flexing or differential loading then this could lead to crack initiation and propagation. The designer should be aware of how vibration energy will manifest itself in the structure and internal components. Section 5 outlines a method for quantifying damage to metallic structure through vibration.

The effect of stresses on metallic structures (including electronic equipment with metal elements) is well understood so the comparative effect of vibration can be quantified and summed using classical fatigue theory. Miner's law can be used to arrive at a vibration test with a level and duration with an equivalent level of damage to the entire transportation life.

Composites

Composites behave differently to metallic materials. Various modes of failure such as matrix cracking, delamination, fibre breakage, fibre kinking, wedge effect and inter-laminate shear failure could develop and lead to either initial or final failure. These modes are created as a result of application of multi-axial loads. The interaction between the above modes of failure is material and loading dependent. Matrix cracking in an isolated ply leads to failure while matrix cracking in a multi-directional laminate usually takes place at small load amplitudes and does not immediately lead to final failure. In the absence of experimental and test data, matrix damage introduced by vibration could be modelled using an interactive failure criterion such as Tsai-Hill. Multiple matrix cracking could develop into delamination and this could lead to deteriorating effects on the integrity of the composite structure.

Miner's law may be used for composites in a manner similar to that of metals in order to assess the vibrational effects on the integrity of the structures. The value of exponent used for Miner's law calculations depends upon a number of parameters such as the composite lay-up, loading direction and environmental factors as well as the type of material used. An appropriate exponent should be selected by a composites specialist.

In addition, composites are particularly sensitive to temperature change and moisture up-take as well as to damage from impacts. Care should be exercised to avoid detrimental effects such as these in the presence of vibration if they could increase the risk of matrix damage.

Systems

Any equipment which could suffer reduced reliability through movement, accelerations or vibration is at risk due through transportation. Any components (such as circuit boards or other electronic components) constructed with metal can be damaged by fatigue if they are allowed to move. Most components of this type can be assessed using the same theory as any metal component, such as the technique specified within section 5.

Other systems, such as mechanical linkages, hydraulics and sensors which make use of moving parts can also be affected. Qualifying such components depends on their usage and any prior knowledge about how susceptible they are to vibration. Any final qualification tests should prove that there is no long term reliability issue.

K.3 Differences between the various transportation environments

Each type of transportation has its own vibration characteristics and therefore contributes differently to the total vibration spectrum; a summary of each is presented below. A greater description of these environments is found in Def Stan 00-35 Part 5 Section 2.

Propeller aircraft (for example C-130). Vibrations originate from the engines, mechanical components and pulses of air pressure (caused by the rotating propeller blades) acting against the fuselage and, possibly, from flow interaction with the empennage. The pressure waves occur at specific frequencies (e.g. blade passing frequency) and so can excite objects with appropriate resonant frequencies. Airflow past the fuselage can also produce vibration. The exposure to vibration depends on the position within the aircraft, for example, there is a significant increase in vibration when in line with the propellers or in close proximity to the fuselage wall.

Jet aircraft (for example C-17). Vibrations originate from rotating engine components and airflow past the fuselage. The magnitude of these and the risk of damage depend on the position within the aircraft but in general, compared to propeller aircraft, jet aircraft offer a benign transport environment.

Helicopters. Vibrations originate from the rotor blades and associated rotating mechanical components. Pressure waves caused by the rotating blades acting against the fuselage are also a significant factor. The magnitude of these depends heavily on the position within the aircraft. Vibration levels can be high in comparison with other forms of air transportation.

Underslung. Loads slung beneath a helicopter are generally isolated from vibrations generated by the aircraft. Aerodynamic forces such as turbulence can vibrate a slung load but it is likely to be benign.

Rail. Vibrations are generally limited to regular low level inputs due to movement along the tracks. Rail travel is generally benign when compared to other transport modes. The rail wagon size and type, rail quality and coupling method all affect the level of vibration experienced.

Sea. Ship vibrations are typically generated from the propulsion system and other mechanical systems. The level of vibration seen is therefore dependent on the distance between the payload and the mechanical system in question. Rough seas can create large motions of the ship and the action of waves against the hull can send vibrations throughout the structure. Ship motion (governed by the sea state) can excite objects with low natural frequencies. Given the potential length of sea journeys such long exposure could be damaging to susceptible structures.

Wheeled Vehicles. Vibrations are generated from mechanical systems and undulations in the terrain not filtered out by the vehicle's suspension. The quality of the ride is therefore dependent on a number of variables which need to be taken into account before deriving a vibration test schedule:

Vehicle type – there is a wide range of vehicles available for logistics efforts depending on the size and weight of the payload. Each vehicle type has different ride characteristics and proficiency at isolating the load area from undulating terrain.

The load state of the vehicle – logistics vehicle suspension is designed for a certain payload. If a vehicle is very lightly loaded the suspension will not be operating within an optimal range and the ride will suffer. This can significantly affect the ride quality.

Where the payload is positioned will affect the ride quality. Generally, ride quality is worst above axles.

The terrain type. A wide range of potential terrains exist which can produce very different vibration environments. Terrain is the main variable to consider, as some off-road terrains can be severe and unpredictable.

Tracked vehicles. The vibration environment associated with tracked vehicles is potentially the most severe of any transport option. The large powerplants generate a lot of noise and vibration while the stiff body and suspension allows easy transmission throughout the structure.

Another significant factor is track patter, a repetitive action caused by each section in the vehicle track impacting the ground. The repetitive nature of this load establishes itself as a vibration, the frequency and magnitude of which is dependent on the vehicle's speed, track design and condition and the type of terrain being traversed.

A lot of transport environments are difficult to quantify given the variables, but conservative vibration spectra are provided within Def Stan 00-35 Part 3. These are discussed in section 5 of this document, along with how to generate alternatives which could be more representative.

K.4 How to derive a transportation usage and transportation profile

The way in which a UAS will be transported operationally needs to be recognised. This can be detailed within a usage and transportation profile (simply referred to as a usage profile in this document) which should contain details of every type of journey likely to be carried out. Each journey should be sufficiently detailed so that a vibration spectrum can be defined and attached to it. Typically, the more remote the location that the UAS is launched/recovered from, the more complex the usage profile will be.

In order to define a usage profile, a number of assumptions need to be made regarding the operational use of the UAS and the logistics effort required to support it. There are different potential ways in which this can be defined, but a suggested process is detailed in Figure 2.

It will be necessary to repeat this process for each element of the UAS (i.e. any piece of equipment which will be transported). Some equipment may not travel directly with the UAS so will have different transportation requirements. Once the entire transportation usage profile of the UAS has been derived (i.e. until the out of service date) the total mileage for each transport medium/condition can be calculated.

It is important to use the correct vibration profile for each transportation type. Depending on the total payload, the vibration environment can change. The overall size and weight will determine what carriage options are available and what sort of vibration levels will be experienced by the UAS within its container (if applicable). The use of real trials data is always recommended. This is discussed in section 5.

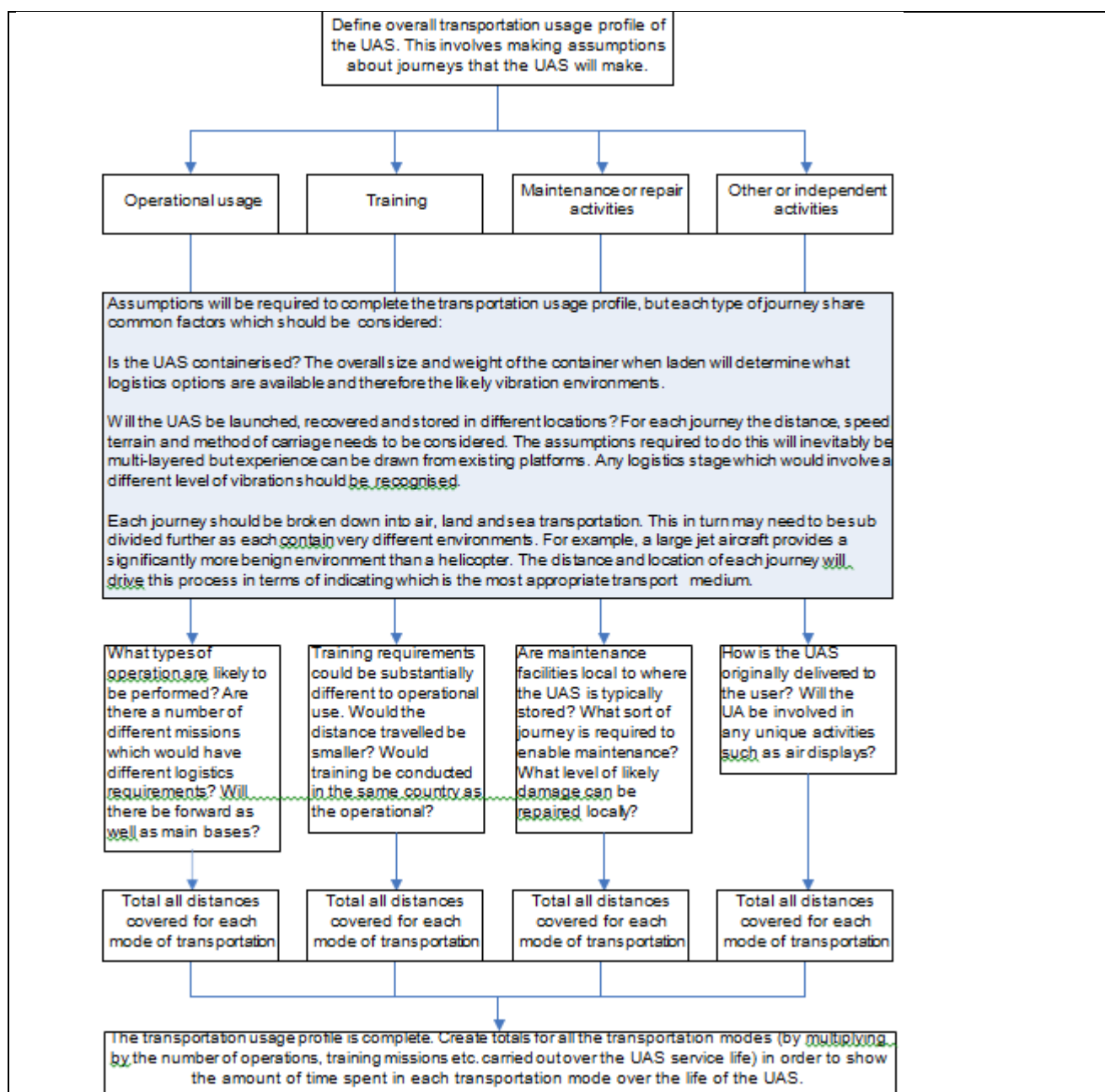


Figure 2 – Suggested process for the derivation of a transportation usage profile

K.5 How to derive vibration profiles

Defence Standard 00-35 Part 3 (Chapter 2 Annex A) provides default vibration profiles for most possible transportation modes. The Defence Standard advises that these should be used only if no specific vibration data is available. The use of real data is always recommended so the environment can be defined as accurately as possible.

Vibration data is collected during trials over a preset frequency range appropriate to the test – for example a typical range for a road vehicle would be 0-2000Hz. Data collected from the accelerometers is typically displayed in the form of power spectral density (PSD) plots which expresses how the power density (e.g. power per Hz) of an energy source is distributed across a given frequency range. This usually allows for easy identification of any resonance spikes or areas with particularly high vibration levels, provided that these have not been disguised by band averaging.

When sufficient vibration data has been gathered, it is possible to derive a vibration test which has an equivalent damage to the transportation life of the UAS. Once the transportation life has been established, and a vibration profile has been derived for each mode within that life, a single test can be derived.

The process is described within Defence Standard 00-35 (Part 5, chapter 6-01 Annex A) in greater detail but the concept is to use the worst vibration environment (such as off-road vehicle running) and proportionally scale the time spent testing at that environment so it reflects the same damage caused by all transportation modes, the majority of which are likely to be far more benign. This method allows for a significant drop in the test duration while at the same time avoiding exposing the UAS to an accelerated test regime which may create unrepresentative damage and ultimately an over-engineered design.

Miner's law for fatigue damage is used as the basis for the scaling of testing time with the equation below:

$$T_2 = T_1 \left[\frac{S_1}{S_2} \right]^N$$

Where:

S_1 = Vibration level attributed to a particular transport mode. This can be expressed as the grms (root mean square acceleration) figure derived from a particular PSD plot.

S_2 = Worst case vibration level, associated with the harshest likely mode of transportation. Depending on the final test method this level should have an additional factor added to it to allow for the limited range typically associated with vibration rig equipment. This is discussed in section 7.

T_1 = Length of time spent on a particular transportation type.

T_2 = Time required at S_2 level to produce equivalent damage to T_1 at the S_1 energy level.

N = Slope of the S-N curve for the predominant structural feature associated with the UAS. Once T_2 has been calculated for each transport condition, the total test duration can be calculated.

It is possible to adjust the resultant test duration for another/different vibration energy by using the same equation again, typically if the test duration is undesirably long. The duration can be reduced if a commensurate increase in test level can be tolerated. Sacrifices may be made in terms of test accuracy if this is taken too far so care should be taken in doing this and a vibration specialist consulted.

The tests suggested in this document are an appropriate way of representing exposure to levels of vibration seen during transportation. While Miner's law is used to represent vibration in terms of equivalent fatigue damage, it should be recognised that the method is only statistically reliable if variations in nominally identical structures are taken into account using a material scatter factor.

In addition to this, potential future changes to the usage profile should be allowed for. If it changes so that a greater proportion of a harsher transportation mode is used then the qualification test identified could be inadequate. This could be dealt with by applying a suitable safety factor, to cover this uncertainty.

The acceleration PSD plot can be used to derive the grms, found by calculating the area under the line and then taking the square root. The grms is an indication of the average severity of a particular vibration trace. It is this value which is used in section 6 of this document to derive an equivalent damage vibration test, representative of the entire transportation life.

This method should not be the only way of considering the vibration environment. As grms represents vibration power in an average form, details of peaks in vibration levels and the overall shape of the profile are lost. Peaks/spikes in the levels could represent resonances which could be significant to the design

of the UAS or its container. A certain amount of engineering judgement is required to identify how significant peaks and troughs in PSD plots are with respect to the data as a whole and whether the frequency at which the peak exists could affect the UA structurally. A vibration specialist should be consulted.

K.6 How to derive a test specification from usage and vibration profiles

Once the duration and vibration level has been derived, the shape of the test profile has to be established. This can be done by using the suggested profiles found in Defence Standard 00-35 (factored upwards/downwards accordingly to represent the required grms level S_2). This can then be tailored as required by adding any peaks or spikes from any trials data. The result of this will be a profile which includes all the worst features of the trials data and a grms level appropriate to the entire transportation life.

Different transportation modes (land, sea etc.) will have naturally different shape profiles. Engineering judgement should be used to determine how important it is to represent all transportation modes. For example, typically rail vibration profiles are typically more benign than that of road vehicles or propeller aircraft. In this case, depending on how much time is spent transporting the UAS on trains, the rail vibration profile may be ignored.

An example of a way to work out T_2 and S_2 is shown in Table 1. There are many possible iterations of this process, and judgement should be used to derive a method which fully reflects the variables. Many vibration profiles will be unique to each UAS, as the size and weight of a payload can affect where and how it is carried, and the quality of ride it receives

Type of Transportation	S_1	T_1	T_2
Each transportation mode broken down by speed, vehicle type, terrain etc.	grms level recorded for each given transportation type. ↓ S_2 = Highest S_1 value, with appropriate safety factors added (see sections 5 & 7). The final test will be carried out at this level.	Actual length of time spent in using that transportation mode according to the usage profile, per average transport hour.	Length of time required at S_2 levels to represent this part of the usage profile. ↓ Sum T_2 values to calculate the duration of test which has an equivalent damage to the transportation life of the UAS.

Table 1 – Example method of calculating equivalent damage

K.7 Performing a vibration test

Careful consideration should be made concerning how any resultant vibration test is conducted. The payload should be positioned, supported and secured in the same fashion as when in used in service so the transfer of vibration loads is as representative as possible. Any modelling carried out should be subject to the same intentions. There are three approaches which can provide confidence in a design:

Modelling

Computer modelling creates a complete picture of stresses in the UAS without the need for a complex expensive trial. However, in order to have confidence in the results, some practical testing will be required to substantiate the results. Reliability of the design and practicalities surrounding transportation cannot be evaluated with a purely analytical approach.

Vibration rig test

Trials on a vibration rig are repeatable and controllable, with few variables and data collection is less complex than on field trials. However, they can be expensive and time consuming to set up before a realistic environment is created. Vibration rigs typically have movement limitations (for example maximum acceleration or displacement limits) which do not exist on field trials. An extra factor may be required to account for this.

Representative field trial

A representative method is to instrument a UAS and transport it as it would be operationally and assess whether if this causes damage. This type of qualification can also be used to test reliability and practicality of operation. However, it can lack repeatability due to the amount of variables involved. Defence Standard 00-35 part 3 chapter 2-14 outlines a procedure for running a test track trial. Any trial should, as far as is practicable, reflect the transportation vehicle defined in the usage profile.

There are further drawbacks to a physical trial. It is difficult to represent the entire transportation life as it is harder to control the environment accurately. They are expensive and can be much more time consuming to carry out.

K.8 Other aspects of transportation and factors to consider

There are many aspects of transportation which can reduce the life or functionality of a system which fall outside the remit of this document. Such factors include:

Shock

Shock (transient) loads are associated with independent events such as an aircraft landing, a train shunting impact or a vehicle traversing a pot hole. Def Stan 00-35 part 5 chapter 2 discusses what shock loading is associated with transportation. Shocks can be damaging if they have sufficient acceleration and duration. They can be harder to quantify and predict if from an unexpected action (such as a handling error) but can be taken into account if only from a conservative stance.

Shock loading can be monitored in service by the installation of a g exceedance meter of some form. This can indicate if the airframe/container has been exposed to an undesirable pulse. Consideration will have to be given to the threshold level chosen, where it is measured and how/with what periodicity the data is collected and recorded.

Load restraint

This document assumes that an adequate restraint scheme is used at all times. Unrestrained items are potentially subjected to far higher levels of vibration than those firmly secured to the vehicle in question, due to moving around and coming into contact with other objects. Def-Stan 00-3 details methods of restraining cargo when transported on land, rail and sea or in the air. There should be negligible movement between the vehicle and the load (except movement due to anti-vibration mounts) as this could amplify any vibration. The load restraint scheme should be capable of firmly retaining all elements of the load under all reasonably likely carriage manoeuvres.

Logistics control

The mechanics and practicalities of the logistics effort should be considered. If UAS are to be mated with a specific container for any reason (such as shock monitoring) consideration should be given to how maintenance and repair tasks are to be carried out – how will parts requiring maintenance or repair be moved around? If the normal container is used will this prevent the rest of the UAS being used for operations and will this be efficient for smaller parts? How will the remainder be stored in the interim?

Environmental

Def-Stan 00-35 (part 4) discusses the environmental considerations. Temperature, moisture, ice, sand and dirt etc. can all have a detrimental effect on the integrity and performance of a UAS. The packaging/container used for transport should protect the UAS as far as is required.

Handling

Any transfer of equipment into/out of transportation containers, or container transfers between logistics vehicles will involve lifting or handling. The way that each component is supported during this process should be considered as well as the repercussions and risk of handling errors. Special care has to be taken to protect composite structures from impact during handling.

Storage

The UAS may be stored in containers for a long period of time during a ship transit or between deployments. There should be no detrimental effects of long term storage. Environmental considerations are particularly pertinent due to the length of time that the UAS/container could be stored. Storage conditions designed to minimise the environmental risk to the UAS should be provided.

Reliability

Moving a UAS presents other challenges which may affect the functionality of the UAS or its container over time. The reliability of components or mechanisms is related to the harshness of the environment in which they are operated and the frequency of their operation. Items such as hinges, door seals, foam padding, bolted joints etc. are susceptible to wear and degradation. It would be expected that the design should not suffer from early failure, which can be proven to a certain extent by trials similar in nature to those used to confirm resistance to vibration. Longer term durability should be dealt with by a comprehensive maintenance regime.

INTENTIONALLY BLANK

Appendix L

HIRF Environments and Equipment HIRF Test Levels

This appendix specifies the HIRF environments and equipment HIRF test levels for electrical and electronic systems under USAR.1308. The field strength values for the HIRF environments and laboratory equipment HIRF test levels are expressed in root-mean-square units measured during the peak of the modulation cycle.

(a) HIRF environment I is specified in the following table:

TABLE I.—HIRF ENVIRONMENT I

Frequency	Field strength (volts/meter)	
	Peak	Average
10 kHz-2 MHz	50	50
2 MHz-30 MHz	100	100
30 MHz-100 MHz	50	50
100 MHz-400 MHz	100	100
400 MHz-700 MHz	700	50
700 MHz-1 GHz	700	100
1 GHz-2 GHz	2,000	200
2 GHz-6 GHz	3,000	200
6 GHz-8 GHz	1,000	200
8 GHz-12 GHz	3,000	300
12 GHz-18 GHz	2,000	200
18 GHz-40 GHz	600	200

In this table, the higher field strength applies at the frequency band edges.

(b) HIRF environment II is specified in the following table:

TABLE II.—HIRF ENVIRONMENT II

Frequency	Field strength (volts/meter)	
	Peak	Average
10 kHz-500 kHz	20	20
500 kHz-2 MHz	30	30
2 MHz-30 MHz	100	100
30 MHz-100 MHz	10	10
100 MHz-200 MHz	30	10
200 MHz-400 MHz	10	10

400 MHz-1 GHz	700	40
1 GHz-2 GHz	1,300	160
2 GHz-4 GHz	3,000	120
4 GHz-6 GHz	3,000	160
6 GHz-8 GHz	400	170
8 GHz-12 GHz	1,230	230
12 GHz-18 GHz	730	190
18 GHz-40 GHz	600	150

In this table, the higher field strength applies at the frequency band edges.

(c) *Equipment HIRF Test Level 1.*

(1) From 10 kilohertz (kHz) to 400 megahertz (MHz), use conducted susceptibility tests with continuous wave (CW) and 1 kHz square wave modulation with 90% depth or greater. The conducted susceptibility current must start at a minimum of 0.6 milliamperes (mA) at 10 kHz, increasing 20 decibels (dB) per frequency decade to a minimum of 30 mA at 500 kHz.

(2) From 500 kHz to 40 MHz, the conducted susceptibility current must be at least 30 mA.

(3) From 40 MHz to 400 MHz, use conducted susceptibility tests, starting at a minimum of 30 mA at 40 MHz, decreasing 20 dB per frequency decade to a minimum of 3 mA at 400 MHz.

(4) From 100 MHz to 400 MHz, use radiated susceptibility tests at a minimum of 20 volts per meter (V/m) peak with CW and 1 kHz square wave modulation with 90% depth or greater.

(5) From 400 MHz to 8 gigahertz (GHz), use radiated susceptibility tests at a minimum of 150 V/m peak with pulse modulation of 4% duty cycle with a 1 kHz pulse repetition frequency. This signal must be switched on and off at a rate of 1 Hz with a duty cycle of 50%.

(d) *Equipment HIRF Test Level 2.* Equipment HIRF test level 2 is HIRF environment II in table II of this appendix reduced by acceptable aircraft transfer function and attenuation curves. Testing must cover the frequency band of 10 kHz to 8 GHz.

(e) *Equipment HIRF Test Level 3.*

(1) From 10 kHz to 400 MHz, use conducted susceptibility tests, starting at a minimum of 0.15 mA at 10 kHz, increasing 20 dB per frequency decade to a minimum of 7.5 mA at 500 kHz.

(2) From 500 kHz to 40 MHz, use conducted susceptibility tests at a minimum of 7.5 mA.

(3) From 40 MHz to 400 MHz, use conducted susceptibility tests, starting at a minimum of 7.5 mA at 40 MHz, decreasing 20 dB per frequency decade to a minimum of 0.75 mA at 400 MHz.

(4) From 100 MHz to 8 GHz, use radiated susceptibility tests at a minimum of 5 V/m.

BOOK 2 - ACCEPTABLE MEANS OF COMPLIANCE (AMC)

Subpart A – GENERAL

AMC.U17

Design usage spectrum

The Applicant uses this information as the basis for assumptions underpinning fatigue and damage tolerance and associated individual UAS tracking. The design usage spectrum is therefore necessary for producing and maintaining the Fatigue Type Record or equivalent document. The applicant also uses the design usage spectrum to identify any gross deviation between design assumptions and Service usage. The design usage spectrum is descriptive, rather than prescriptive. The design usage spectrum contains a breakdown of the typical Sortie Profiles Codes (SPCs) or any equivalent for the UAS type in each of its roles and at each typical operating location. SPCs or any equivalent are expressed in terms of height, time, speed, mass and configuration data, which are derived from recorded sortie information. The initial issue of the design usage spectrum should be produced as early as possible in the project life cycle and should be reviewed and updated throughout the life of type.

AMC.U19

Special military modes of operation

The methodology used to substantiate USAR.U19 should include consideration of potential system failures in both the inactive special modes and the normal system.

INTENTIONALLY BLANK

Subpart B – FLIGHT

AMC.21

Proof of compliance

During flight test, the variability in measurement of a desired performance parameter shall be less than 5%, accounting for flight test practice tolerances and instrumentation accuracies.

Critical items affected by weight should have a tolerance between +5% and -1% and C.G. tolerances of +/-7% total travel should be used.

In areas of critical handling or stability, notwithstanding the tolerance of 7% total travel, aft centre of gravity tests should be flown at a centre of gravity not more forward than the certificate aft centre of gravity limit. Tests which are critical on the forward centre of gravity limit should be flown at centres of gravity at least as forward as the certificate forward limit.

AMC.55

Accelerate-stop Distance or Critical Field Length

UAS crew and system response delays should be considered as part of the Accelerate-stop distance or Critical Field Length calculation. For example, if an arresting hook deployment requires activation from the UCS, latencies associated with this command may affect Accelerate-stop distance or Critical Field Length.

AMC.71

Glide

In addition to identifying the maximum horizontal distance travelled in still air per 305 m (1,000 ft) of altitude lost and the maximum range airspeed, where applicable, consideration should also be given to identify the best glide range angle of attack (AOA). Unless the flight controls are "G commanded" in the cruise configuration (gear up), best glide typically occurs at a constant AOA which is easily implemented by a pitch rate commanded flight control system. This constant AOA typically works through a large band of altitudes and is a function of weight and C.G. position.

AMC.141

General

The term "practical loading conditions" refers to the aircraft loadings as defined in recognized guidance such as Section 6.4.6 of JSSG-2001: "The loading of an aircraft is determined by what is in (internal loading) and attached to (external loading) the aircraft. The loading parameters that influence flying qualities are weight, c.g. position, and moments and products of inertia. In addition to these, external stores also affect aerodynamic coefficients.

Verification of the requirement for operation in environmental disturbances should be accomplished by analysis with a high fidelity simulation validated with flight test data. For example, Section 6.4.6 of JSSG-2001 provides definitions for discrete gust models, random turbulence spectra, and technical descriptions for common, uncommon, and extraordinary atmospheric disturbances. The loading conditions, types and magnitudes of environmental disturbances used in the simulation models, and flight test data needed to validate the simulation models should be negotiated with and approved by the Certifying Authority.

AMC.149 (b)

Minimum Control Speed

Because UAS emergency procedures associated with landing gear retraction, and the time required to affect them, may be different than those of manned aircraft, dynamic VMC values should be determined with the landing gear both extended and retracted.

AMC.171

Longitudinal and lateral stability and response

The following should be demonstrated in normal states by a combination of validated model-based analyses and flight test as agreed with the Certifying Authority. Acceptable means of compliance for degraded modes should be agreed with the Certifying Authority.

(a) Accuracy

The UA shall be capable of maintaining the desired flight parameters in smooth air with a sufficiently small static error, to be agreed with the Applicant and the Certifying Authority. This should be demonstrated by, for the following parameters, throughout the normal flight envelope:

- (1) Attitude: pitch and roll angles.
- (2) Airspeed, heading or track, turn rate, and altitude.

(b) Transient response

It shall be demonstrated for the entire flight envelope that:

- (1) Pitch and Roll response following an abrupt command input or gusts, are suitably damped so as not to cause exceedance of the:
 - (i) Limit load factor.
 - (ii) Maximum torque allowed by the control surface actuators.
- (2) Transition to a selected altitude, or engagement of an altitude hold function should not cause a deviation (overshoot) of the commanded value by a tolerance greater than 3 times the tolerance agreed with the authorities under paragraph (a).
- (3) Transition to a selected heading or engagement of a heading hold function should not cause transient deviation (overshoot) of the commanded value by a tolerance greater than 3 times the tolerance agreed with the authorities under paragraph (a).
- (4) Transition to a selected airspeed or selection of an airspeed hold function, within the permissible flight envelope protection, should not cause the air speed to:
 - (i) Fall below the minimum allowed air speed.
 - (ii) Exceed V_{NE} .

AMC.203

System flight characteristics and envelope protection test and evaluation

The guiding principle of 'envelope protection' in USAR is that the aircraft should not be able to depart controlled flight under any foreseeable operating conditions. (Noting that 'hands on throttle and stick' systems are not covered by USAR – see USAR.1). It is expected that designers will select system characteristics and equipment to sufficiently demonstrate that envelope protection is ensured throughout the clearance envelope to this requirement.

To meet this requirement, control and guidance systems, aerodynamic characteristics and all other relevant system components should be shown to possess suitably predictable operating and recovery characteristics for the intended operation. All system parameters should also be shown to have appropriate values such that the relevant envelope protection margins will be maintained at all phases of flight.

Where applicable, failure modes or degraded modes of operation should be shown to have the same level of envelope protection as that applicable to normal operation. It is expected that this may be accomplished by several methods. For instance, under certain conditions, the permissible values for rates of turn, climb or other flight parameter may be limited to maintain the relevant margin for envelope protection.

It is accepted that for many systems, it may be impractical to implement a flight test to determine (for instance) the stall speed under certain conditions, as the envelope protection may provide a large normal operating margin such that the system may not be designed to be capable of recovering from a real or incipient stall. Under many circumstances, disabling or manipulating the envelope protection system to achieve such a result may be impractical, or result in a large reduction in safety margin for other or all flight parameters.

Therefore, where the relevant parameter can be determined by other means acceptable to the authority such as calculation, simulation or alternative testing (e.g. wind tunnel), and the system envelope protection margins are also shown to be adequate, it should be sufficient for the flight test to show compliance with the envelope protection limits. This compliance test is to include operator input of demands outside the permitted envelope, and rapid dynamic changes close to the edges of the envelope to determine that an overshoot beyond permitted limits cannot occur.

AMC.235

Operations on unpaved surfaces

(a) Consideration should also be given to the relationship of the position of the engine intakes to the operating surface to establish whether there is any risk of damage from the ingestion of foreign objects, i.e. FOD.

(b) The following leaflets contained within the UK MoD Defence Standard 00-970 give additional guidance on the operation from surfaces other than smooth hard runways: Defence Standard 00-970 part 1 section 4 leaflets 48, 49, 50, 51, 52 and 53.

AMC.U240

Wet Runway Operations - Runway Spray

At a minimum, it should be shown by analysis that:

- (a) Runway spray will not damage the propellers or parts of the UA at any time during taxi, take-off, or landing, and
- (b) The location of the tyres and design of the UA minimizes the probability of throwing water, snow, or slush into the engine inlet, and
- (c) The location and design of pitot-static ports minimizes the probability of blockage from water, snow, slush, or ice, and
- (d) The design includes provisions such as seals and covers to prevent water intrusion into areas/cavities not designed for operation in that wet environment.

AMC.U283

Launch safety trace

Ballistic footprint, rocket misalignment errors, poor catapult launch performance, poor engine performance, flight control failures, etc., should be considered as part of the hazard analysis to determine the launch safety trace.

AMC.U293

Parachute landing safety trace

Factors such as, weight, altitude, wind condition, and temperature should be considered when determining the parachute landing safety trace.

Subpart C – STRUCTURE

AMC.301

Loads

(a) Static qualification evidence is usually in the form of a static type record (STR) or equivalent national document. The STR or equivalent document comprises a general arrangement and description of the UAS, a summary of static design assumptions and/or criteria, a summary of critical loading, shear force, bending moment, torque and mass distributions, and a summary of static reserve factors. The static evidence must normally list all relevant supporting stress and test reports and calculation files. For modern designs, evidence may have been obtained from aerodynamic and loads models (such as finite element models (FEM)) and so it may be beneficial to maintain and update these models through the life of the type, in the same manner as the STR document itself or equivalent document. The scope of the STR or equivalent document, in terms of the structural components to be covered, should match the list of items of structure whose failure would be *Catastrophic*.

(b) The following ultimate crash load factors should be used to protect people and structure on the ground in the event of a failure:

	Longitudinal		Vertical		Lateral	
	Fwd	Aft	Up	Down	Left	Right
(1) Internal fuel tanks (2/3 full)	9	1.5	2	4.5	1.5	1.5
(2) Installations of other items that could injure ground personnel	9	1.5	2	4.5	1.5	1.5

Notes: 1. The specified load factors will act separately or in combination, whichever produces the most critical loading.
2. The longitudinal, vertical, and lateral directions of load factor refer to the major aircraft axes (F.S., B.L., W.L.) and the loads will be applied in planes parallel to these axes.

(c) Load requirements may be matched to the FCS capability to prevent load excursions if it can be shown by the applicant that the FCS restrains the manoeuvres by control of rate or range.

AMC.U304

Structural Performance - Interaction of systems and structures

1 General

The following criteria should be used for showing compliance with USAR.U304 for UA equipped with flight control systems, stability augmentation systems, load alleviation systems, flutter control systems, and fuel management systems. If this AMC is used for other systems, it may be necessary to adapt the criteria to the specific system.

(a) The criteria defined herein only address the direct structural consequences of the system responses and performances and cannot be considered in isolation but should be included in the overall safety evaluation of the UA. These criteria may in some instances duplicate standards already established for this evaluation. These criteria are only applicable to structure whose failure could lead to a Hazardous or more severe effect. Specific criteria that define acceptable limits on handling characteristics or stability requirements when operating in the system degraded or inoperative mode are not provided in this AMC.

(b) Depending upon the specific characteristics of the UA, additional studies may be required that go beyond the criteria provided in this AMC in order to demonstrate the capability of the UA to meet other realistic conditions such as alternative gust or manoeuvre descriptions for an UA equipped with a load alleviation system.

(c) The following definitions are applicable to this AMC.

- **Structural performance:** Capability of the UA to meet the structural requirements of STANAG 4671.
- **Flight limitations:** Limitations that can be applied to the UA flight conditions following an in-flight occurrence and that are included in the flight manual (e.g. speed limitations, avoidance of severe weather conditions, etc.).
- **Operational limitations:** Limitations, including flight limitations that can be applied to the UA operating conditions before dispatch (e.g. fuel, payload and Master Minimum Equipment List limitations).
- **Probabilistic terms:** The probabilistic terms (probable, improbable, extremely improbable) used in this AMC are the same as those used in USAR.1309 and related AMC.1309 (b)
- **Failure condition:** The term failure condition is the same as that used in USAR.1309 and related AMC.1309 (b), however this AMC applies only to system failure conditions that affect the structural performance of the UA (e.g. system failure conditions that induce loads, change the response of the UA to inputs such as gusts or pilot actions or/and flight control system reactions, or lower flutter margins).

2 Effects of Systems on Structures.

(a) General. The following criteria will be used in determining the influence of a system and its failure conditions on the UA structure.

(b) System fully operative. With the system fully operative, the following apply:

(1) Limit loads must be derived in all normal operating configurations of the system from all the limit conditions specified in Subpart C, taking into account any special behaviour of such a system or associated functions or any effect on the structural performance of the UA that may occur up to the limit loads. In particular, any significant nonlinearity (rate of displacement of control surface, thresholds or any other system nonlinearities) must be accounted for in a realistic or conservative way when deriving limit loads from limit conditions.

(2) The UA must meet the strength requirements of STANAG 4671 (Static strength, residual strength), using the specified factors to derive ultimate loads from the limit loads defined above. The effect of nonlinearities must be investigated beyond limit conditions to ensure the behaviour of the system presents no anomaly compared to the behaviour below limit conditions. However, conditions beyond limit conditions need not be considered when it can be shown that the UA has design features that will not allow it to exceed those limit conditions.

(3) The UA must meet the aeroelastic stability requirements of USAR.629

(c) System in the failure condition. For any system failure condition not shown to be extremely improbable, the following apply:

(1) At the time of occurrence. Starting from representative mission flight conditions, a realistic scenario, including operator corrective actions and/or with flight control system reaction, should be established to determine the loads occurring at the time of failure and immediately after failure.

(i) For static strength substantiation, these loads multiplied by an appropriate factor of safety that is related to the probability of occurrence of the failure are ultimate loads to be considered for design. The relationship between the factor of safety and the probability of occurrence should take into consideration the probability criteria set forth in AMC.1309 (b) and representative load cases, subject to the Certifying Agency agreement.

(ii) For residual strength substantiation, the UA should be able to withstand two thirds of the ultimate loads defined in subparagraph (c)(1)(i).

(iii) Freedom from aeroelastic instability should be shown up to the speeds defined in USAR.629 For failure conditions that result in speed increases beyond VC/MC, freedom from aeroelastic instability should be shown to increased speeds, so that the margins intended by USAR.629 are maintained.

(iv) Failures of the system that result in forced structural vibrations (oscillatory failures) should not produce loads that could result in failure of primary structure.

(2) For the remainder of the flight in the system failed state when emergency recovery is initiated, it should be shown that the UA does not produce loads that lead to a failure of the primary structure considering UA limitations. If however the UA is to continue the flight, in the system failed state and considering any appropriate reconfiguration and flight limitations, the following apply:

(i) The loads derived from the following conditions at speeds up to VC / MC, or considering the limitations prescribed for the remainder of the flight should be determined:

(A) the limit symmetrical manoeuvring conditions specified in USAR.331 and in USAR.345

(B) the limit gust and turbulence conditions specified in USAR.341 and in USAR.345

(C) the limit rolling conditions specified in USAR.349 and the limit unsymmetrical conditions specified in USAR.367 and USAR.427(b) and (c)

(D) the limit yaw manoeuvring conditions specified in USAR.351

(E) the limit ground loading conditions specified in USAR.473

(ii) For static strength substantiation, each part of the structure should be able to withstand the loads in subparagraph (c)(2)(i) of this paragraph multiplied by a factor of safety depending on the probability of being in this failure state. The relationship between the factor of safety and the probability should take into consideration the probability criteria set forth in AMC.1309 (b) and representative load cases, subject to the Certifying Agency agreement.

(iii) For residual strength substantiation, the UA should be able to withstand two thirds of the ultimate loads defined in subparagraph (c)(2)(ii).

(iv) If the loads induced by the failure condition have a significant effect on fatigue or damage tolerance then their effects should be taken into account.

(v) Freedom from aeroelastic instability should be shown up to a speed determined as a function of the probability being in failure conditions, as per USAR.629

(d) Failure indications. For system failure detection and indication, the following apply:

(1) The system should be checked for failure conditions, not extremely improbable, that degrade the structural capability below the level required by STANAG 4671 or significantly reduce the reliability of the remaining system. As far as reasonably practicable, the operator should be made aware of these failures before flight. Certain elements of the control system, such as mechanical and hydraulic components, may use special periodic inspections, and electronic components may use daily checks, in lieu of detection and indication systems to achieve the objective of this requirement. These certification maintenance requirements should be limited to components that are not readily detectable by normal detection and indication systems and where service history shows that inspections will provide an adequate level of safety.

(2) The existence of any failure condition, not extremely improbable, during flight that could significantly affect the structural capability of the UA and for which the associated reduction in airworthiness can be minimized by suitable flight limitations, should be signalled to the operator.

(e) Dispatch with known failure conditions. If the UA is to be dispatched in a known system failure condition that affects structural performance, or affects the reliability of the remaining system to maintain structural performance, then the provisions of USAR.U304 must be met for the dispatched condition and for subsequent failures. Flight limitations and expected operational limitations may be taken into account in establishing the combined probability of being in the dispatched failure condition and the subsequent failure condition for the safety margins in (c)(2)(ii). These limitations should be such that the probability of being in this combined failure state and then subsequently encountering limit load conditions is extremely improbable. No reduction in these safety margins is allowed if the subsequent system failure rate is greater than 10^{-3} per hour.

AMC.307

Proof of Structure

In deciding the need for and the extent of testing including the load levels to be achieved the following factors will be considered by the Certifying Authority.

(a) The confidence which can be attached to the constructors' overall experience in respect to certain types of UAS in designing, building and testing UAS.

(b) Whether the UAS in question is a new type or a development of an existing type having the same basic structural design and having been previously tested, and how far static strength testing can be extrapolated to allow for development of the particular type of UAS.

(c) The importance and value of detail and/or component testing including representation of parts of structure not being tested, and

(d) The degree to which credit can be given for operating experience.

Analyses including finite element models used in place of tests must be demonstrated to be reliable for the structure under evaluation and the load levels that have to be covered. This would normally be provided by correlation with experimental results on the same structure or through comparison with other known and accepted methods and results or through a combination of both.

If the structure or parts thereof are outside the manufacturer's previous experience, the manufacturer should establish a strength test programme. In the case of a wing, wing carry through, fuselage and empennage this will usually involve ultimate load testing or structural analysis conducted in a rational and conservative manner.

When ultimate load static tests are conducted it is recommended that preliminary tests to limit load and back to zero are performed first, in order to demonstrate that no detrimental permanent deformation has taken place. During the ultimate test however, the limit load need not be removed provided that continuous readings of strains and deflections of the structure are measured at an adequate number of points, and also provided that a close examination of the structure is maintained throughout the tests with particular emphasis being placed upon close observation of the structure at limit load for any indications of local distress, yielding buckles, etc.

Static testing to ultimate load may be considered an adequate substitute for formal stress analysis where static loads are critical in the design of the component. In cases where a dynamic loading is critical, dynamic load tests may be considered equivalent to formal stress analysis. An example of components on which dynamic loading is usually critical is the landing gear and the landing gear structure of a UA. The same yield criteria apply to dynamic tests as to static tests.

Where proof of structure is being shown by an ultimate load test, the test article should conform to the same design specifications as the production article.

The manufacturer should ensure through his quality assurance organisation that the strength (e.g. material properties and dimensions) of the component tested conservatively represents the strength of the components used in production.

Test correction factors should be used to allow for process and material variability during production. This may be expected particularly when wood or composite-material is used. This factor may be varied according to the coefficient of variation that the manufacturer is able to show for his product (see Table 1).

TABLE 1
Test factor [Tf] vs. Coefficient of Variation [Cv%]

Cv%	5	6	7	8	9	10	12	14	15	20
Tf	1 00	1 03	1 06	1 10	1 12	1 15	1 22	1 30	1 33	1 55

Definition of Coefficient of Variation

For a population with mean M and standard deviation [sigma], the coefficient of variation expressed as a percentage, Cv%, is defined by:

$$Cv\% = 100 * [\sigma]/M$$

AMC.321 (c)

Flight Loads - General

For UA with an M_d less than 0.5 the effects of compressibility are unlikely to be significant.

AMC.341 (b)

Gust Loads Factors The gust alleviation factor K_g as specified in USAR.341(c) will not provide the conservatism required by /USAR341(b).

Using a gust alleviation factor of $K_g = 1.2$ in the calculation of the gust load of canard or tandem wing configuration may result in conservative net loads with respect to the gust criteria of USAR333(c).

An appropriate technique for selecting the gust load factor should be selected from those provided in ESDU data sheet 04024.

AMC.343 (b)

Design Fuel Loads

Fuel carried in the wing increases the inertia relief on the wing structure during manoeuvres and gusts which results in lower stresses and deflections. However, if the wing fuel tanks are empty the inertia load of the wing is reduced which, depending on the particular design, may lead to an increase of the bending stresses in the wing structure itself and in the wing attachments. In order not to over stress the UA's structure the maximum weight of the UA without any fuel in the wing tanks should therefore be established, taking into account the applicable manoeuvre and gust loadings.

AMC.345 (d)

High Lift Devices

The effect of propeller slipstream on the extended flaps may be limited to the flap area behind the propeller circle area.

AMC.371

Gyroscopic and Aerodynamic Loads

The aerodynamic loads specified in USAR.371 include asymmetric flow through the propeller disc. Experience has shown that the effects of this asymmetric flow on the engine mount and its supporting structure are relatively small and may be discounted, if propellers are installed having diameters of 2.74 m (9 ft) or less.

AMC.393 (a)

Loads parallel to hinge lines

On primary control surfaces and other movable surfaces, such as speedbrakes, flaps (in retracted position) and all-moving tailplanes the loads acting parallel to the hinge line should take into account the effect of wear and axial play between the surface and its supporting structure. Compliance may be shown by analysis or by test.

AMC.393 (b)

Loads parallel to hinge lines

For control surfaces of a wing or horizontal tail with a high dihedral angle and of a V-tail configuration the K-factor may be calculated as follows:

$$K = 12 \times \sqrt{4 - \left(\frac{3}{1 + \tan^2 v} \right)}$$

where:

v = dihedral angle measured to the horizontal plane

As a simplification the following K-factors may be assumed:

for dihedral angles up to $\pm 10^\circ$ $K = 12$

and for dihedral angles between 80° and 90° $K = 24$

AMC.441

Manoeuvring loads – Vertical surfaces

(a) If a manoeuvre analysis is used to predict the manoeuvring loads on the yaw control surfaces, the time for sudden deflection from neutral position to the stops or vice-versa should be measured.

(b) For UA where the horizontal tail is supported by the vertical tail, the tail surfaces and their supporting structure including the rear portion of the fuselage should be designed to withstand the prescribed loadings on the vertical tail and the rolling moment induced by the horizontal tail acting in the same direction.

(c) For T-tails, in the absence of a more rational analysis, the rolling moment induced by sideslip or deflection of the vertical rudder may be computed as follows:

$$M_r = 0.3 S_h \frac{\rho_0}{2} \beta V^2 b_h$$

where:

M_r = induced rolling moment at horizontal tail (Nm)

S_h = area of horizontal tail (m²)

b_h = span of horizontal tail (m)

β = effective sideslip angle of vertical tail (radians)

AMC.443

Gust Loads - Vertical Surfaces

For UA where the horizontal tail is supported by the vertical tail, the tail surfaces and their supporting structure including the rear portion of the fuselage should be designed to withstand the prescribed loading on the vertical tail and the rolling moment induced by the horizontal tail acting in the same direction.

For T-tails, in the absence of a more rational analysis, the rolling moment induced by gust load may be computed as follows:

$$M_r = 0.3 S_h \frac{\rho_0}{2} V U b_h K_{gt}$$

where:

M_r = induced rolling moment at horizontal tail (Nm)

S_h = area of horizontal tail (m²)

B_h = span of horizontal tail (m)

U = gust velocity (m/s) as specified in USAR.333(c)

K_{gt} = gust alleviation factor of vertical tail as specified in USAR.443(c)

In computing ' S_h ' and ' b_h ' the horizontal tail root has to be assumed on a vertical plane through the centreline of the UA fuselage.

AMC.471

General

Where a UA is designed to be re-used after a hard emergency parachute landing as defined in USAR.U290, consideration should be given to the requirements in USAR.473 through USAR.499

AMC.481

Tail-down landing conditions

Where i_y cannot be determined by more rational means, a value of: $i_y = 0.225 LR$ may be used.
 LR in this case to be taken as the overall length of the fuselage without rudder.
 In designing the tail skid, side loads should be accounted for in addition to the vertical load determined as above.

AMC.U570

General

(a) An example of fatigue evaluation guidance may be obtained from Defence Standard 00-970 Part 1 section 3 Leaflets 34, 35, 36, 37, 38, 40 and 41.

(b) Fatigue Qualification Evidence - Fatigue qualification evidence is usually in the form of a fatigue type record (FTR) or equivalent national document. The initial issue of the FTR or equivalent document generally consists of a single Part 1, which comprises a statement of the fatigue and/or damage tolerance principles used and a summary of fatigue and/or damage tolerance analyses for each item of structure whose failure would be *Catastrophic*. The FTR or equivalent document also lists all relevant reports and may be supported by computer models. The scope of the fatigue evidence should be determined by the list of structure whose failure would be *Catastrophic*. Furthermore, because both usage and fatigue evidence evolve significantly over the life of a UA type, the FTR or equivalent document should be developed in due course to incorporate additional sections, as follows:

Part 2 – Reassessment in light of Service usage and fatigue test results.

Part 3 – A reassessment of examination requirements shown to be necessary in Part 2.

Part 4 – A life extension document (as required).

AMC.572

Fatigue evaluation: metallic pressurised compartment structures, metallic fuselage, wing, empennage and associated structures

(a) In assessing the possibility of serious fatigue failures, the design should be examined to determine probable points of failure in service. In this examination, consideration should be given, as necessary, to the results of stress analysis, static tests, fatigue tests, strain gauge surveys, test of similar structural configurations, and service experience. Locations prone to accidental damage or to corrosion should also be considered.

Unless it is determined from the foregoing examination that the normal operating stresses in specific regions of the structure are of such a low order that serious damage growth is extremely improbable, repeated load analysis or tests should be conducted on structure representative of components or subcomponents of the wing (including canard and tandem wings, winglets and control surfaces), empennage, their carry-through and attaching structures, fuselage and pressurised cabin, landing gear, and their related primary attachments. Test specimens should include structure representative of attachment fittings, major joints, changes in section, cut-outs and discontinuities. Service experience has shown that special attention should be focused on the design details of important discontinuities, main attachment fittings, tension joints, splices, and cut-outs such as windows, doors, and other openings. Any method used in the analyses should be supported, as necessary, by tests or service experience. The nature and extent of tests on complete structures or on portions of the primary structure will depend upon evidence from applicable previous design and structural tests, and service experience with similar structures. The scope of the analyses and supporting test programmes should be agreed with the Certifying Authority.

(b) The use of the following stress levels should be taken as sufficient evidence, in conjunction with good design practices to eliminate points of stress concentration, that structural items have adequate safe-life design:

Material Used	Allowable normal stress level of maximum limit load
- Wood	According to ANC-18*
- Aluminium Alloy	Half of rupture tensile strength
- Steel Alloy	Half of rupture tensile strength

*ANC-18 is the ANC Bulletin 'Design of wood aircraft structures'; issued June 1944 by the Army-Navy-Civil Committee on Aircraft Design Criteria (USA).

Higher stress levels should need further fatigue investigation using one or a combination of the following methods:

- (1) Fatigue tests, based on a realistic operating spectrum.
- (2) Fatigue computation, using strength values which have been proved to be sufficient by fatigue tests of specimens or components.

For conventional metallic (2024, 6061, steel) structural item identified as flight critical structure, if the limit tensile stress levels does not exceed 50% of the ultimate allowable for the material then no further fatigue assessment is required. The limit tensile stress level must take into account any stress concentrations in the wing lower spar booms and attachment fittings, wing struts and lower wing-fuselage carry through structures.

AMC.573 (a) (1) & (3)

Damage tolerance and fatigue evaluation of structure – Composite airframe structure

In addition to the acceptable means of compliance and guidance material described in AMC 20-29 the following procedure may be adopted for residual strength tests of structure with built-in barely visible impact damage (BVID) and visible damage. Tests should be performed up to limit load level, then the visible damages may be repaired without substantially exceeding the original strength or characteristics of the type design and the test should be continued up to at least* ultimate load level in order to validate the BVID in the unrepaired structure.

* Experience has shown that continuation of testing to rupture should be considered in order to identify failure modes. Extrapolation by analysis of residual strength tests would not normally be acceptable for further development of the UA.

AMC.573 (b)

Damage tolerance and fatigue evaluation of structure – Metallic airframe structure

The damage-tolerance evaluation of structure is intended to ensure that, if serious fatigue, corrosion, or accidental damage occurs within the operational life of the UA, the remaining structure can withstand reasonable loads without failure or excessive structural deformation until the damage is detected. Design features which should be considered in attaining a damage-tolerant structure include the following:

- Multiple load path construction and the use of crack stoppers to control the rate of crack growth, and to provide adequate residual static strength;
- Materials and stress levels that, after initiation of cracks, provide a controlled slow rate of crack propagation combined with high residual strength. For single load path discrete items, such as control surface hinges, wing spar joints or stabiliser pivot fittings the failure of which could be Catastrophic, it should be clearly demonstrated that cracks starting from material flaws, manufacturing errors or accidental damage including corrosion have been properly accounted for in the crack propagation estimate and inspection method;
- Arrangements of design details to ensure a sufficiently high probability that a failure in any critical structural element will be detected before the strength has been reduced below the level necessary to withstand the loading conditions specified in USAR.573(b) so as to allow replacement or repair of the failed elements.

AMC.U590

Arresting system

In determining the limit loads on the UA, the Arresting System and the affected supporting structures, the following where applicable should be considered:

- (a) Attitude - 3 wheels on ground.
- (b) Speed at cable engagement (max consistent with FCS limit for max landing speed).
- (c) CG.
- (d) Design landing weight (Para USAR.473).
- (e) Descent velocity (assume zero).
- (f) Side load conditions (e.g. not hitting cable centre on).
- (g) Maximum swivel angle i.e. direction load acts.
- (h) Safety links or shear pins.
- (i) Arresting Cable being stuck in the arrestment device (e.g. the drum).
- (j) The hook load imparted by the arrester system.

AMC.U595 (a)

Limit load factor

The free drop height formula used to demonstrate USAR.595 (a) (2) should be the following:

$$h = \frac{Vzd^2}{2g}$$

where:

- Vzd is the maximum descent velocity of the UA under parachute,
- g is the acceleration due to gravity (m/s²).

However, the free drop height should not be less than 0.234m.

AMC.U598

Extracting devices

Any extracting device should be considered to be installed equipment.

INTENTIONALLY BLANK

Subpart D - DESIGN AND CONSTRUCTION

AMC.601

General

The following Military Standard/Handbook(s), FAA advisory circulars and policy documents may be applicable for UAS designed to USAR and may be cited by the Applicant in their certification evidence.

MIL-STD-810 Environmental Engineering Considerations and Laboratory Tests
 AC 20-107A Composite Aircraft Structure
 AC 21-26 Quality Control for the Manufacture of Composite Structures
 AC 20-53A Protection of Airplane Fuel Systems Fuel Vapor Ignition due to Lightning
 AC 20-66A Vibration and Fatigue Evaluation of Airplane Propellers
 AC 20-135 Powerplant Installation and Propulsion System Component Fire Protection Test Methods, Standards and Criteria
 AC 20-136 Protection of Aircraft Electrical/Electronic Systems Against the indirect Effects of Lightning
 AC 21-31 Quality Control for the Manufacture of non-metallic Compartment Interior Components
 AC 23-2 Flammability Test
 AC 23-20 Acceptance Guidance on Material Procurement and Process Specifications for Polymer Matrix Composite Systems
 AC 23.629-1A Means of Compliance with section 23.629 Flutter
 AC 145-6 Repair Stations for Composite and Bonded Aircraft Structures

>For Information

AC 29.2C MG8 Substantiation of Composite Rotorcraft Structure
 AC 35.37-1 Guidance Material for Fatigue Limit Tests and Composite Blade Fatigue Substantiation
 JSSG 2006, Aircraft Structures

>FAA Policy Documents Applicable to Part 23 Aircraft

PS-ACE100-2-18-1999 Temperature differential between Wet Glass Transition
 PS-ACE100-2005-10038 Bonded Joints and Structures
 PS-ACE100-2004-10030 Substantiation of Secondary Composite Structures
 PS-ACE100-2001-006 Static Strength of Composite Airplane Structure

AMC.603 (a)

Material and Workmanship for metallic UAS Structure

An acceptable means of compliance, but not the only means, of showing compliance with the provisions of USAR regarding airworthiness type requirements for metallic UAS structures can be found in FAA advisory circular AC23-13A.

AMC.603 (b)

Material and Workmanship Composite UAS Structure

EASA AMC 20-29 and FAA AC 20-107 are acceptable means of compliance for Composite Structures.

Due to the variation in location of some of the requirements within this USAR the following table is to be read in conjunction with Appendix I.

TABLE I
AMC paragraphs and related USAR texts

	AMC paragraph	USAR paragraph
1	Purpose	None
5	General	None

6	Materials and Fabrication	603, 605, 613, 619
6a	Quality Control	
6a	Production Specifications	603, 605
6e	Protection of Structure	609
7	Proof of Structure Static	305, 307
8	Proof of Structure – Fatigue/Damage Tolerance	573
9	Proof of Structure - Flutter	629
11	Additional Considerations	
11a	Impact Dynamics	601, 783, 787, 963
11b	Flammability	U850, 863, 865, 903, 967, 1121, 1181, 1183, 1189, 1191, 1193
11c	Lightning Protection	867, 863, 954
Appendix 2	Definitions	None

AMC.607 (b)

Fasteners

Locking devices of fasteners installed in engine compartments or other compartments affected by temperature and/or vibration should be of a type and material which is not influenced by such temperatures encountered under normal operating conditions.

AMC.609

Protection of structure

Adequate corrosion prevention and control practices should be in place for uniform surface corrosion, pitting, galvanic, crevice, filiform, exfoliation, inter-granular, fretting, high temperature oxidation (hot corrosion), corrosion fatigue, and stress corrosion cracking.

Corrosion prevention systems should remain effective during the service life, including the mitigation of environmentally assisted cracking. Specific corrosion prevention and control measures, procedures, and processes should be identified and established commensurate with the operational and maintenance capability. The finish system should provide adequate corrosion protection for specific parts, surfaces of similar and dissimilar materials, and attaching parts and fasteners.

At the request of the Certifying Authority, a Corrosion Prevention and Control Plan (CPCP) should be developed and submitted.

AMC.611

Accessibility provisions

(a) Non-destructive inspection aids may be used to inspect structural elements where it is impracticable to provide means for direct visual inspection if it is shown that the inspection is effective and the inspection procedures are specified in the Maintenance Manual required by USAR.1529

(b) For inspections repeated at short intervals (such as pre-flight or daily inspections) the means of inspection should be simple (e.g. visual with the aid of easily removable or hinged access panels). However, for inspections required only a few times, for example once or twice in the lifetime of the UA some disassembly of structure (e.g. deriveting a small skin panel may be acceptable).

AMC.613

Metallic strength properties and design values

Material properties should be those contained in documents accepted either specifically by the Certifying Authority or by having been prepared by an organisation or a person which the Certifying Authority recognises has the necessary capabilities.

Examples of such documents are:

- 1 MMPDS 'Metallic Materials Properties Development and Standardisation' issued by the FAA
- 2 CMH-17 'The Composite Materials Handbook'
- 3 Mil-HDBK-23 'Composite Construction for Flight Vehicles'
- 4 ANC-18 'Design of Wood Aircraft Structures'

In defining design specifications (e.g. allowable values) the material properties should be factored as necessary by the Designer to take into account the manufacturing practices (e.g. methods of construction, forming, machining and subsequent heat treatment). AMC 20-29 contains guidance information relevant to composite structure.

AMC.629

Flutter

Compliance with the rigidity and mass balance criteria (pages 4-12), in Airframe and Equipment Engineering Report No. 45 (as corrected) "Simplified Flutter Prevention Criteria" (published by the Federal Aviation Administration) may be accomplished to show that the UA is free from flutter, control reversal, or divergence if:

- (a) V_D/M_D for the UA is less than 482 km/h (260 knots) (EAS) and less than Mach 0.5;
- (b) The wing and aileron flutter prevention criteria, as represented by the wing torsional stiffness and aileron balance criteria, are limited to use to UA without large mass concentrations (such as engines, floats, or fuel tanks in outer wing panels) along the wing span; and
- (c) The UA:
 - (1) Does not have a T-tail or other unconventional tail configurations;
 - (2) Does not have unusual mass distributions or other unconventional design features that affect the applicability of the criteria; and
 - (3) Has fixed-fin and fixed stabiliser surfaces.

AMC.629 (j)

Flutter

The total backlash should be no greater than 0.2 degree for controls on the outer 50% of a flying surface, and 1 degree for inboard controls. Larger backlash is allowable if it is shown by rational analysis that control surface backlash will not lead to flutter.

AMC.U631

Bird strike

Consideration should be given in early stages in the design to the installation of items in essential services, such as control system components, and items which, if damaged, could cause a hazard, such as electrical equipment. As far as practicable, such items should not be installed immediately behind areas liable to be struck by birds.

Compliance with this section by provision of redundant structure and protected location of control system elements or protective devices such as splitter plates or energy absorbing material is acceptable.

AMC.671

Control systems – General

In designing and manufacturing primary control systems attention should be given to minimise friction in the systems and to avoid jamming and interference with other parts in operation, due to vibration and accelerations.

Each flight control system should have a fatigue evaluation determined in accordance with USAR.U570 at least equal to the specified life of the UA, unless otherwise agreed with the Certifying authority. This should be demonstrated by a fatigue test.

AMC.683

Operation tests

One method, but not the only one, for showing compliance with the requirements of USAR.683 is as follows:

Conduct the control system operation tests by operating the controls with the entire system loaded so as to correspond to the limit control forces established by the regulations for the control system being tested. The following conditions should be met:

(a) Under limit load, check each control surface for travel and detail parts for deflection. This may be accomplished as follows:

- (1) Support the control surface being tested while positioned at the neutral position.
- (2) Load the surface using loads corresponding to the limit control forces established in the regulations.
- (3) Load the controls until the control surface is just off the support.
- (4) Determine the available travel which is the amount of movement of the surface from neutral when the control is moved to the system stop.
- (5) Remove the surface load and command the surface position back to neutral. The resulting mechanical set (offset) from initial neutral position should be no greater than 1% of the total control system travel or the maximum value considered safe by modelling system response.
- (6) The minimum control surface travel from the neutral position in each direction being measured should be 10% of the control surface travel measured with no load on the surface. Regardless of the amount of travel of the surface when under limit load, the UA should have adequate flight characteristics as specified in paragraph USAR.141. Any derivative UA of a previous Type Certificated UAS need not exceed the control surface travel of the original UA; however, the flight characteristics should be flight tested to ensure compliance.

(b) Under limit load, no signs of jamming or of any permanent set of any connection, bracket, attachment, etc., may be present.

(c) Friction should be minimised so that the limit control forces and torques specified by the regulations may be met.

AMC.693

Joints

The specification for elastomeric bearings should be those contained in documents accepted specifically by the Certifying Authority. Such a specification is, for example "MIL-B-85598 General Specification for Elastomerics".

AMC.701

Flap Interconnection

Acceptable means of compliance include a mechanical interconnection between the movable flap surfaces independent of the flap drive system and/or an electromechanical implementation using mechanical interconnection and flap position sensors.

AMC.729 (g)

Equipment located in the Landing Gear Bay

In showing compliance with this requirement, consideration should include the effects that likely damage from hazards arising from other items of equipment such as high brake temperature and external sources such as slush, water and tyre burst/loose tyre tread will have on equipment/systems located on the landing gear or in the landing gear bay that are essential to continued safe flight and landing.

AMC.735 (c)
Brakes
As specified in the requirement, the pressure on the wheel brake must not exceed the pressure that is specified by the brake manufacturer. The requirement does not specify how the force that is applied to the braking command is transmitted to the brakes. This means may be mechanical, hydraulic or some other system, such as an electronic control system. By clarifying the applicability of the requirements to the force applied to the wheel brake assembly, it can be applied to any braking system that is included in the UA design.
AMC.U850
Fire Protection - General
Compliance with USAR Fire Protection requirements should ensure that:
(a) Electrical installation and propulsion systems (including related materials) are adequately designed (see USAR.1359, USAR.1181 and appendix F), and
(b) Necessary protection is assured to structural items and systems (such as the flight control system) in accordance with the risk reference system in AMC.1309, and
(c) The flammability, toxicity, smoke effects and thermal decomposition of the materials are considered in design.
AMC.863
Flammable fluid fire protection
The Designer should consider a means to ventilate and drain leaked flammable fluids and vapours.
AMC.865
Fire protection of flight controls, engine mounts and other flight structure
Engine mounts or portions of the engine mounts that are not constructed of fire proof material should be shielded to provide an equivalent level of safety to that provided by the use of fireproof materials. Care should be taken that any shielding does not invalidate the type certification of the engine.
AMC.867 (a)
Electrical bonding and protection against lightning and static electricity
The latest edition of the following documents should provide guidance as to acceptable means of compliance for USAR.867:
(a) SAE International standard ARP5412, "Aircraft Lightning Environment and Related Test Waveforms"
- This document describes the lightning environment, applicable to both direct (physical) and indirect (electromagnetically coupled) lightning effects.
(b) MIL-STD-464, "Electromagnetic Environmental Effects Requirements for Systems"
- This document provides requirements and guidance with regard to lightning, bonding, grounding, and static electricity.
(c) NATO STANAG 3614, "Electromagnetic Environmental Effects (E3) Requirements for Aircraft Systems and Equipment"
- This document is similar in scope to MIL-STD-464.
(d) RTCA, Inc. standard DO-160, "Environmental Conditions and Test Procedures for Airborne Equipment"

- This document describes equipment-level testing against direct lightning effects, indirect lightning effects and static-electric discharge from human contact.

- The direct lightning-effects test is limited in purview to externally-mounted airborne equipment (such as antennas, exterior lights, and exposed fuel-quantity probes). This test does not encompass aircraft structure, mechanical devices (such as fuel filler caps), or any item protected by a covering which is part of the aircraft itself.

- The indirect lightning-effects test applies principally to aircraft with metallic skin.

(e) AEP-29 (NATO STANAG 3856), "Protection of Aircraft, Crew, and Subsystems in Flight Against Electrostatic Charges"

- This document provides guidance with regard to precipitation static effects upon aircraft in flight.

(f) UK Ministry of Defence, Defence Standard 59-113 - Lightning Strike Protection Requirements for Service Aircraft.

AMC.U881 (a)

Parachute design

Where tests are carried out to demonstrate the compliance with USAR.U881 (a), the following requirements must be met. There shall be no failure to meet any of the requirements during the tests of this AMC. In case of failure, the cause must be found, corrected and all affected tests repeated.

(a) Environmental tests

Three drops must be made at the minimum speed of the UA flight envelope protection as defined in USAR.U334 under the following parachute assembly preconditioning:

- (1) Preconditioning for 16h at not less than +93.3°C, stabilise to ambient and test drop,
- (2) Preconditioning for 16h at not greater than -40°C, stabilise to ambient and test drop,
- (3) Preconditioning for not less than continuous 400h with a 889.6 N (200 lbf) or greater load applied to compress the pack in manner similar to that most likely to be encountered in actual use. Test drop must be made within 1h after removing the load.

(b) Strength tests

Tests may be conducted for either a complete parachute assembly or separate components. There should be no evidence of material, stitch, or functional failure that will affect airworthiness. The same canopy, harness, component and riser(s) must be used for all strength tests. Three drops must be carried out with the following characteristics:

- (1) Test weight is the UA maximum weight.
- (2) Test speed is 1.2 times the Maximum Pack Open Speed (MPOS) or maximum speed of the UA flight envelope protection as defined in USAR.U334
- (3) Where easily detachable hardware (such as snap and ring) is used to attach the canopy or riser(s) to the harness, a cross connector must be used and one of the drops should be with only one attachment engaged to test the cross connector and hardware.

The canopy must be functionally open. Functionally open shall mean a parachute sufficiently deployed and inflated to provide the designed rate of descent of not more than 24 ft/s (7.2m/s).

(c) Functional tests

There shall be a minimum of 12 drops with the following characteristics:

- (1) Test weight is the maximum UA take-off weight.
- (2) For 6 drops the test speed is the Maximum Pack Open Speed (MPOS) or the maximum speed of the UA flight envelope protection as defined in USAR.U334
- (3) For 6 drops the test speed is the minimum speed of the UA flight envelope protection as defined in USAR.U334

The canopy must be functionally open. Functionally open shall mean a parachute sufficiently deployed and inflated to provide the designed rate of descent of not more than 24 ft/s (7.2 m/s).

Subpart E – POWERPLANT

AMC.901
Installation
It is assumed that UA are designed with an automatic engine control (i.e. engine is controlled via FCS) with a direct UAS crew involvement as a Special Condition only. In case of semi-automatic or manually controlled engine(s), Special Conditions apply for: (a) data to be transmitted and displayed to the UAS crew. (b) set up of the UCS (i.e. levers, controls, etc.) which should be agreed with the Certifying Authority. The level of integration of engine control functions into the FCS should be agreed with the Certifying Authority.
AMC.903 (a)
Engines and auxiliary power units
Ingestion of hail, rain and foreign objects risk qualification and bird strike and ingestion risk qualification may be based on EASA CS-E790 and EASA CS E-800 or any equivalent qualification agreed with the Certifying Authority.
AMC.903 (f)
Engines and auxiliary power units
APU qualification may be based on EASA CS-APU or any equivalent qualification agreed with the Certifying Authority.
AMC.905 (d)
Propellers
Compliance with 905(d) may be based on EASA CP-P-210 requirement or equivalent document agreed with the Certifying Authority.
AMC.905 (e)
Propellers
Ice shed from the forward fuselage and the wings may cause significant damage to pusher propellers that are very close to the fuselage and well back from the UA nose. Similarly, ice shed from the wing may cause significant damage to wing mounted pusher propellers. Account should be taken of these possibilities. The term 'during any operating condition' may require tests also for intentional, or temporary unintentional entry into icing conditions. This may also be shown by analysis or a combination of both.
AMC.905 (g)
Propeller
In most pusher propeller installations, the engine exhaust gases pass through the propeller disc. Many factors affect the temperature of these gases when they contact the propellers and propeller tolerance to these gases varies with propeller design and materials.
AMC.907 (a)
Propeller Vibration
The definition of a conventional fixed pitch wooden propeller should be taken to include a propeller with a wooden core and a simple cover of composite material, but not a propeller where the load carrying structure is composite and the wood simply provides the form.

AMC.909 (a)
Turbo charger systems
Compliance with 909(a)(1) may be based on EASA CP-E-440 requirement or equivalent document agreed with the Certifying Authority.
AMC.909 (d) (1)
Turbo charger systems
Intercooler mounting provisions should have sufficient strength to withstand the flight and ground loads for the UA as a whole in combination with the local loads arising from the operation of the engine.
AMC.934
Turbojet and turbofan engine thrust reverser system tests
Compliance with USAR.934 may be based on EASA CS-E 650 and EASA CS-E 890 or equivalent document approved with the Certifying Authority.
AMC.939 (c)
Powerplant operating characteristics
Inlet Airflow Distortion Limits. The engine shall not surge, stall, flameout, or incur any damage with the steady state or time variant inlet distortion (pressure, temperature, or any combination of both) shown herein. The aerodynamic interface plane shall be identified on the engine configuration and envelope figure. Where exhaust nozzle backpressure effects on the engine affect tolerance of the engine to inlet air pressure variation, the effect shall be specified in the engine system specification. Engine stability and performance assessments shall use the methodology and inlet distortion descriptors defined in ARP 1420 and AIR 1419 (or other acceptable standards) for total pressure and temperature distortion. Temperature distortion methodology and concurrent distortion shall be specified herein.
AMC.959 (a)
Unusable fuel supply
The term 'most adverse fuel feed condition' is not intended to include radical or extreme manoeuvres not likely to be encountered in operation. Judgement should be used in determining what manoeuvres are appropriate to the type of UA being tested. A tank that is not needed to feed the engine under all flight conditions should be tested only for the flight regime for which it was designed (e.g. cruise conditions). Tests for this kind of tank should include slips and skids to simulate turbulence. Suitable instructions on the conditions under which the tank may be used should be provided in a placard or in the UAS Flight Manual. Analyse the fuel system and tank geometry to determine the critical manoeuvres for the specific tanks being considered (e.g. main, auxiliary, or cruise tanks and conduct only those tests considered applicable to the UA being tested). Particular attention should be directed towards the tank or cell geometry and orientation with respect to the longitudinal axis of the UA and location of supply ports. Care should be taken in planning how the critical altitude manoeuvres are tested so that the test procedure does not result in unconservative unusable fuel. The test manoeuvres should be selected using good judgement with regard to the kind of manoeuvres the UA under test will be subjected to in operation. Ground tests using equipment which accurately simulate the UAS fuel system and in-flight inertial effects may be considered acceptable. The quantity of fuel to be used for the tests should be sufficient for determination of unusable fuel by allowing the manoeuvres described herein to be performed. The manoeuvres are to be repeated until first evidence of engine malfunction. Repeated manoeuvres may result in fuel refilling some bays or tanks; therefore, minimum fuel should be used. For the tests, a malfunction will be considered when engine roughness, partial or total loss of power, fuel pressure loss of below minimum, or fuel flow fluctuations are experienced. To assure the most conservative unusable fuel supply value for each tank, another tank should be selected at the first indication of fuel interruption. The fuel remaining in the test tank at the time of malfunction should be drained, measured and recorded as unusable fuel. If header tanks (small tanks that accumulate fuel from one or more fuel tanks and supply the engine directly) are utilised, the fuel remaining in the header tank should be added to the unusable fuel but would not be shown on the fuel gauge marking. All tests should be conducted at a minimum practical weight or weight determined to be critical for the UA being tested. The flight testing of a single-engine UA with a one-tank system requires a separate temporary fuel system to supply the engine after fuel starvation occurs.

AMC.961**Fuel system hot weather operation**

Any fuel system that uses aviation gasoline is considered conducive to vapour formation. However a fuel system having a fuel pump with suction lift, is more critical with respect to vapour formation. Critical operating conditions which need to be considered during evaluation of hot weather tests should include at least the maximum fuel flow, high angles of attack, maximum fuel temperature, etc. The weight of the UA should be the weight with critical fuel level and the ballast necessary to maintain the centre of gravity within allowable limits. The critical fuel level in most cases would be low fuel; however, in some cases, full fuel may be critical. A flight test is normally necessary to complete the hot weather operation tests, however, if a ground test is performed, it should closely simulate flight conditions. Several methods of heating the fuel are available, such as circulating hot water or steam through a heat exchanger placed in the fuel tank to increase the fuel temperature, placing black plastic or other material on the fuel tanks in bright sunlight, or blowing hot air over the fuel tank. The fuel should not be agitated or handled excessively during the heating operation. The heating process should be completed in the shortest time period possible without causing excessive local temperature conditions at the heat exchanger. Raise the temperature of the fuel to the critical value as follows:

- For aviation gasoline, 43 °C – 0 to + 3 °C (110° F – 0 to + 5° F)
- For turbine fuel, 43 °C – 0 to + 3 °C (110° F – 0 to + 5° F)
- For automobile gasoline, 43 °C – 3 to + 0 °C (110° F – 5 to + 0° F)

Testing should commence immediately after the fuel temperature reaches its required value. The desirable outside air temperature measured 1.2 to 1.8 m (4 to 6 ft) above the runway surface should be at least 29 °C (85° F). If tests are performed in weather cold enough to interfere with the test results, steps should be taken to minimise the effects of cold temperature. This may be accomplished by insulating fuel tank surfaces, as appropriate, fuel lines, and other fuel system components from the cold air to simulate hot-day conditions. The take-off and climb should be made as soon as possible after the fuel in the tank reaches the required test temperature, and the engine oil temperature should be at least the minimum recommended for take-off. The airspeed in the climb should be the same as that used in demonstrating the requirements of USAR.65, except the UA should be at minimum weight with a critical quantity of fuel in the tanks. Power settings should be maintained at the maximum approved levels for take-off and climb to provide for the maximum fuel flow. The climb should be continued to the maximum operating altitude approved for the UAS. If a lower altitude is substantiated, appropriate limitations should be noted in the UAS Flight Manual. The following data should be recorded:

- Fuel temperature in the tank
- Fuel pressure at the start of the test and continuously during climb noting any pressure failure, fluctuation, or variations
- Main and emergency fuel pump operation, as applicable
- Pressure altitude
- Ambient air temperature, total or static as applicable
- Airspeed
- Engine power, i.e. engine pressure ratio, gas generator speed, torque, rpm, turbine inlet temperature, exhaust gas temperature, manifold pressure, and fuel flow, as appropriate
- Comments on engine operation
- Fuel quantities in the fuel tank(s) during take-off
- Fuel vapour pressure (for automobile gasoline only), determined prior to test
- Fuel grade or designation, determined prior to test

A fuel pressure failure is considered to occur when the fuel pressure decreases below the minimum prescribed by the engine manufacturer or the engine does not operate satisfactorily. The emergency fuel pump(s) should be inoperative if being considered for use as backup pump(s). This test may be used to establish the maximum pressure altitude for operation with the pump(s) off. If significant fuel pressure fluctuation occurs during testing

of the critical flight condition but pressure failure does not occur, additional testing should be considered to determine that pressure failure may not occur during any expected operating mode. Also, the fuel system should be evaluated for vapour formation during cruise flight at maximum approved altitude in smooth air at low to moderate power setting and low fuel flow and idling approach to landing. The hot weather tests may have to be repeated if the critical tank cannot be positively identified. Any limitations on the outside air temperature as a result of hot weather tests should be included in the UAS Flight Manual.

AMC.967

Fuel tank installation

There should be no external fuel leakage from the fuel system during normal operating conditions including ground refuelling.

AMC.973

Fuel tank filler connection

- (a) The UA, including external tanks, should be capable of being refuelled to any intermediate quantity or to the UA capacity.
- (b) The ground refueling adapter should be isolated from fuel pressure during all flight phases.
- (c) The engine and engine feed-lines should not be subjected to the pressures imposed on the refueling system during refueling.
- (d) The fuel system components should be electrically bonded to eliminate static charge accumulation, provide controlled current return paths, and provide lightning protection. See AIR5128A Electrical Bonding of Aircraft Fuel Systems.
- (e) A bonding receptacle should be provided adjacent to each refueling receptacle/port for bonding the refueling nozzle to the UA. The bonding receptacle should be located no more than 1.04 m (41 inches) from the gravity filler opening and no closer than 30 cm (12 inches) from fuel tank vent openings or filler cap openings.

AMC.975

Fuel tank vents and carburettor vapour vents

Pressure relief for fuel thermal expansion should be provided for all closed plumbing segments.

AMC.993

Fuel system lines and fittings

- (a) Clearance between all fuel system hoses and tubing should be provided to prevent contact with surrounding components, hardware, and structure during all fuel transfer and refueling operations on the ground and in-flight. Fuel lines should be supported so that the lines will not deflect out of position as a result of internal pressure or aircraft maneuvers. Fuel lines which may be subjected to damage due to maintenance, cargo handling, personnel traffic, or normal aircraft use should be protected.
- (b) The use of magnesium, cadmium, and copper on parts that come in contact with fuel should be avoided.

AMC.995

Fuel valves and controls

Controls and switches readily accessible to the ground staff necessary to meet the safety levels of AMC.1309 should be considered (e.g. emergency fuel cut-off switch).

AMC.997

Fuel strainer or filter

Depending on operating weather conditions likely to be encountered and unless means are provided in the fuel system to prevent the accumulation of ice on the filter, a means must be provided automatically to maintain the fuel flow if ice clogging of the filter occurs.

AMC.999

Fuel system drains

In the event of a wheels-up or collapsed skid landing, it should be possible to defuel each tank through the normal fuel servicing adapters or by suction through accessible openings in each tank.

AMC.1001

Fuel jettisoning system

Effect of fuel dump onto extended landing gear wheels and brakes should be considered.

AMC.1011 (b)

Oil System – General

The minimum allowable usable oil capacity can be determined from the endurance and the maximum allowable oil consumption. For either wet or dry sump engines, the maximum allowable fuel/oil supply ratio is equal to the minimum obtainable fuel/oil consumption ratio. This is expressed mathematically as follows:

Capacity Oil Usable Allowable Minimum Consumption Oil Specific Allowable Maximum

----- ≈ -----
Capacity Fuel Usable Allowable Maximum Consumption Fuel Specific Obtainable Minimum

Therefore, for both wet and dry sump engines, fuel/oil supply ratio equal to or less than the minimum obtainable fuel/oil consumption ratios are considered acceptable. For multi engine installations, unless an adequate oil reserve is provided, the endurance of a multi-engine UA employing a fuel crossfeed system or common fuel tank should be established on the basis that 50% of the specific total initial fuel capacity provided for a shutdown engine will be available to the other engine. The engine power levels to be considered for a multi engine UA having a crossfeed system are those that will allow maximum published endurance with both engines operating and adjusted as necessary (including mixture setting) to complete safely the flight with one engine inoperative after 50% of the fuel supply is consumed.

AMC.1045

Cooling test procedures for turbine engine-powered UA

Analysis and ground tests data may be considered by the Certifying Authority to assess the extent of flight testing to be carried out.

AMC.1045 (b)

Cooling test procedures for turbine engine-powered UA

For the cooling tests, a temperature is 'stabilised' when its rate of change is less than 1°C (2° F) per minute.

AMC.1047

Cooling test procedures for reciprocating engine-powered UA

(a) Analysis and ground tests data may be considered by the Certifying Authority to assess the extent of flight testing to be carried out.

(b) For the cooling tests, a temperature is 'stabilised' when its rate of change is less than 1°C (2° F) per minute.

AMC.1091

Air induction system

The maximum allowable shear, axial, and overhung moment loads at the engine inlet flange should be specified in the engine specification for static (1g) conditions. The allowable loads at the engine inlet flange should also be specified for the maximum allowable manoeuvre loads as defined in Figure 1 (combined linear and angular effects) of USAR.333

AMC.1093

Induction system icing protection

Pre-heaters should not be the only solution for carb icing prevention as small resistive heating elements placed on carb butterfly valves (surface on which ice is most likely to form) has been shown to be effective in minimizing ice formation; additionally, de-icing fluid has also been shown to, when added to fuel, greatly minimize carb butterfly ice formation.

AMC.1093 (b)

Induction system icing protection

Icing conditions may be defined using the EASA CS-definitions document.

AMC.1095 (a)

Carburettor deicing fluid flow rate

A rate of fluid flow for independent carburettor de-icing fluid, expressed in kg per hour, should not be less than 1.13 times the square root of the maximum continuous power of the engine.

AMC.1103 (g)

Induction system ducts

The best practice for manned systems is that the air inlet system should withstand proof pressure of 1.5 times the maximum over-pressure without degradation in performance or permanent deformation.

AMC.1141 (e)

UAS Powerplant controls: General

The effects of combination of failures should be assessed.

AMC.1143

Engine controls

(a) USAR.1143 (g) states that the UAS should be capable to continue safe flight or the maximum severity of the resulting failure is Major in case of engine control separation at the metering device.

(b) When linkage separation occurs, the fuel control should go to a setting that will allow to maintain level flight in the cruise configuration.

AMC.1147

Mixture controls

When mixture linkage separation occurs, the mixture control should go to a full rich setting.

AMC.1189 (a) (5)

Shut-off means

The *Hazardous* amount of flammable fluid for this requirement should be established as 1 L(one quart).

AMC.1191 (e)

Firewalls

(a) Compliance with the criteria for fireproof materials or components must be shown as follows:

(1) The flame will have the following characteristics:

(i) Temperature $1100^{\circ}\text{C} \pm 80^{\circ}\text{C}$ ($2012^{\circ}\text{F} \pm 176^{\circ}\text{F}$)

(ii) Heat Flux Density $116 \text{ KW/m}^2 \pm 10 \text{ KW/m}^2$ (10 BTU/sec/ft² heat flux)

(2) Sheet materials approximately 25 cm (10 in) square must be subjected to the flame from a suitable burner.

(3) The flame must be large enough to maintain the required test temperature over an area approximately 12.7 cm (5 in) square.

(4) Withstand the application of heat by a flame, for a period of 15 minutes without any failure that would create a hazard to the UA.

(b) The following materials with the minimum thickness as shown may be used in firewalls or shrouds without being tested as required by this paragraph:

(1) Stainless steel sheet, 0.38 mm (0.015 in) thick.

(2) Mild steel sheet (coated with aluminium or otherwise protected against corrosion) 0.45 mm (0.018 in) thick.

(3) Terne plate, 0.45 mm (0.018 in) thick.

(4) Monel metal, 0.45 mm (0.018 in) thick.

(5) Titanium sheet, 0.41 mm (0.016 in) thick.

(c) Firewall fittings may be constructed of steel or copper base alloy without being tested.

AMC.1193

Cowling and nacelle

Components in an engine nacelle or in the fuselage should be protected from damage which could result in spillage of enough fuel to constitute a fire hazard as a result of a wheels-up landing on a paved runway.

Subpart F – EQUIPMENT

AMC.U1307 Environmental control system (ECS) Thermal analyses and or tests should be conducted to demonstrate the thermal environment for critical avionics or electronics is maintained at or below the design limitations for ground and flight operations.
AMC.1308 High-intensity Radiated Fields (HIRF) protection See AMC 20-158 (a) The levels in Appendix L are the minimum levels. (b) Where the operational environment may be more severe, STANAG 3614 or MIL STD 464 may be used. (c) If the UAS does not comply with HIRF requirements, the Applicant shall identify and justify to which maximum HIRF environment threat levels and related frequency range the system does comply. As a result of this identification, or in case the system does not comply with known more severe HIRF actual environment levels, subsequent operational limitations shall be defined in order to avoid any unsafe condition.
AMC.1309 (b) and (c) System Design and Analysis (1) Introduction (a) This AMC describes an acceptable means for showing compliance with the requirements of USAR.1309 (b) and 1309(c) and sets a severity reference system and a probability reference system for UAS failure conditions in order to identify the risks deemed to be acceptable. While USAR.1309 does not apply specifically to UA powerplant, the most feared conditions, severity reference system, and safety objectives defined in paragraphs (4) through (6) below apply to all aspects of the UAS, including the engine integration to the UA propulsion system. While structural aspects are covered by subpart C and D, this AMC considers probable structural failures which have an effect on system safety and UAS system failures which affect UA flight structure integrity. (See also USAR.U304 and its AMC). (b) The requirements of USAR.1309 are generally applicable to the installation of engine, propeller, and propulsion system..., USAR.1309 (g) excludes UA powerplant systems provided as part of the certificated engine or propeller from compliance with USAR.1309 requirements. This does not exclude the necessity of the applicant to analyze powerplant systems related functions on UAS level in the Functional Hazard Assessment (FHA). (c) A failure condition is defined as a condition having an effect on either the UAS, people (including UAS crew, ground staff and third parties) either directly or consequentially, which is caused or contributed to by one or more failures, considering flight phase and relevant adverse operational or environmental conditions or external events. Failure conditions are classified in accordance with subparagraph (5) (c) hereunder and are identified during the FHA process at UAS level and then at UAS subsystem level. (except for UA powerplant systems provided as part of the certificated engine and except for UA flight structure, as stated in USAR.1309 (g)). (d) The extent to which the more structured methods and guidelines contained in this AMC should be applied is a function of system complexity and systems failure consequence. In general, the extent and structure of the analyses required to show compliance with USAR.1309 will be greater when the system is more complex and the effects of the failure conditions are more severe. (e) Terms such as “must” are used in this AMC only in the sense of ensuring applicability of this particular method of compliance when the acceptable means of compliance described herein is used. The word “must” is also used in this AMC when referring to a specific regulation or guidance that is essential when the applicant uses this AMC for the means of compliance. In this case there is no deviation. The word “should” is used to express a recommendation. Deviation from the specified recommendation may require justification.

(2) Definitions:

Analysis: An evaluation based on decomposition into simple elements. The terms "analysis" and "assessment" are used throughout. Each has a broad definition and the two terms are to some extent interchangeable. However, the term analysis generally implies a more specific, more detailed evaluation, while the term assessment may be a more general or broader evaluation but may include one or more types of analysis. In practice, the meaning comes from the specific application (e.g. fault tree analysis, Markov analysis, Preliminary System Safety Assessment (PSSA), etc.).

Assessment: An evaluation based on engineering judgment. See the definition of analysis above.

Average probability per flight hour: A representation of the number of times the subject failure condition is predicted to occur during the entire operating life of all UAs of a type, divided by the anticipated total operating hours of all UAs of that type.

Note: The average probability per flight hour is normally calculated as the probability of a failure condition occurring during a typical flight of mean duration divided by that mean duration. See FAA AC 23.1309-1E Appendix 3.

Complex hardware item: All items that are not simple are considered to be complex. See definition of simple hardware item. Source: RTCA/DO-254, Appendix C and Order 8110.105.

Complex system: A system is Complex when its operation, failure modes, or failure effects are difficult to comprehend without the aid of analytical methods. FMEA and FTA are examples of such structured assessment methods. Increased system complexity is often caused by such items as sophisticated components and multiple interrelationships. For example, for these types of systems, a portion of the compliance may be shown by the use of DALs such as by processes in RTCA/DO-178C or RTCA/DO-254 or equivalent. See the definitions for "conventional" and "simple" for more information.

Conventional system: A system is considered "conventional" if its function, the technological means to implement its function, technological implementation and its intended usage are all the same as, or closely similar to, that of previously approved systems that are commonly used. The systems that have established an adequate service history and the means of compliance for approval are generally accepted as "conventional." See the definitions for complex and simple systems for more information.

Design Assurance Level. All those planned and systematic actions used to substantiate, at an adequate level of confidence, that design errors have been identified and corrected such that the items (hardware, software) satisfy the applicable certification basis. This term may be used in some SAE and RTCA documents, but in this AMC it is intended that design assurance levels will correlate to the same levels as the DALs for the safety assessment process.

Development Assurance Level (DAL): All those planned and systematic actions used to substantiate, to an adequate level of confidence, that errors in requirements, design, and implementation have been identified and corrected such that the system satisfies the applicable certification basis.

- Function Development Assurance Level ("FDAL"): The level of rigor of development assurance tasks assigned to Functions.
- Item Development Assurance Level ("IDAL"): The level of rigor of development assurance tasks assigned to Item(s).

Note: FDALs in Figure 1 and IDAL throughout this AMC are also intended to correlate to software levels in RTCA/DO-178(C) and complex hardware design assurance levels in RTCA/DO-254 for the system or item. See paragraph (11) for more information.

Error: An omission or incorrect action by the UAS crew or ground staff, or a mistake in requirements, design, or implementation.

Essential Function: A function whose loss would reduce the capability of the UAS or the ability of the crew to cope with adverse operating conditions.

Event: An occurrence which has its origin distinct from the UAS, such as atmospheric conditions (e.g. gusts, temperature variations, icing and lightning strikes), runway conditions, conditions of communication, navigation, and surveillance services, bird-strike. The term is not intended to cover sabotage.

Failure: An occurrence, which affects the operation of a component, part, or element such that it can no longer function as intended, (this includes both loss of function and malfunction).

Note: Errors may cause failures, but are not considered to be failures.

Failure Conditions: A condition having an effect on the UAS, UAS crew, ground staff and third parties, either direct or consequential, which is caused or contributed to by one or more failures or errors, considering flight phase and relevant adverse operational or environmental conditions, or external events.

Function: Intended behavior of the UAS based on a defined set of requirements regardless of implementation. It may be further broken down to the lowest defined level of a specific action of a system, equipment, and UAS crew that, by itself, provides a completely recognizable operational capability (e.g. an airplane heading is a function). One or more systems may contain a specific function or one system may contain multiple functions.

Latent Failure: A failure is latent until it is made known to UAS Crew or ground staff.

Malfunction: Failure of a system, subsystem, unit, or part to operate in the normal or usual manner. The occurrence of a condition whereby the operation is outside specified limits.

Qualitative: Those analytical processes that assess system and UAS safety in an objective, non-numerical manner.

Quantitative: Those analytical processes that apply mathematical methods to assess system and UAS safety.

Redundancy: The presence of more than one independent means for accomplishing a given function or flight operation. Each means of accomplishing the function need not be identical.

Simple hardware item: An item with a comprehensive combination of deterministic tests and analyses appropriate to the design assurance level that ensures correct functional performance under all foreseeable operating conditions, with no anomalous behavior.--Source: RTCA/DO-254, paragraph 1.6 and Order 8110.105.

Simple system: Usually a system that can be evaluated by only qualitative analysis and it is not complex. Functional performance is determined by combination of tests and analyses. See the definitions for "conventional" and "complex" systems for more information.

(3) Highly Integrated Systems

(a) Design and analysis techniques traditionally applied to deterministic risks or to conventional, non-complex systems may not provide adequate safety coverage for more complex systems. Thus, other assurance techniques, such as development assurance utilizing a combination of process assurance and verification coverage criteria, or structured analysis or assessment techniques applied at the UAS level, if necessary, or at least across integrated or interacting systems, have been applied to these more complex systems. Their systematic use increases confidence that errors in requirements or design, and integration or interaction effects have been adequately identified and corrected.

(b) Considering the above developments, this AMC includes new approaches, both qualitative and quantitative, which may be used to assist in determining safety requirements and establishing compliance with these requirements, considering the UAS and its systems. It also provides guidance for determining when, or if, particular analyses or development assurance actions should be conducted in the frame of the development and safety assessment processes. Numerical values are assigned to the probabilistic terms included in the requirements for use in those cases where the impact of system failures is examined by quantitative methods of analysis. The analytical tools used in determining numerical values are intended to supplement, but not replace, qualitative methods based on engineering and operational judgment.

(4) Failure Condition Classifications and Probability Terms

Most feared conditions

The following conditions have been considered in developing the severity reference system specific to UAS:

(a) Uncontrolled flight and/or uncontrolled crash.

An uncontrolled flight (including flight outside of pre-planned or contingency flight profiles/areas) and/or uncontrolled crash is defined as a condition resulting from one or a combination of failure conditions that result in loss of UA control and or manoeuvrability.

Note: The consequences of failure conditions identified leading to flight outside of pre-planned or contingency flight profiles/areas may also be mitigated by additional regulations and requirements approved by the Certifying Authority. This statement does not imply that such failure conditions need not be assessed during the FHA process.

(b) Controlled-trajectory termination or forced landing potentially leading to the loss of the UA where it can be reasonably expected that a fatality will not occur.

A controlled-trajectory termination or forced landing is defined as a condition resulting from one or a combination of failure conditions that prevents the UA from landing on its planned main landing site although UAS control and/or manoeuvrability is maintained.

(c) Landing of the UA on a predefined site where it can be reasonably expected that a fatality will not occur.

A forced landing on a predefined site where it can be reasonably expected that a fatality will not occur is defined as a condition resulting from one or a combination of failure conditions that prevents the UA from landing on its planned main landing site although the UAS is controllable and manoeuvrable and can also able perform a forced landing on a predefined area that as far as is reasonably practicable has been cleared of people.

(5) Severity Reference System

(a) In international airworthiness codes applicable to manned aircraft, a crash is considered as *Catastrophic*, whether or not it occurs above a populated area since it is likely to result in numerous fatalities.

(b) Adaptation of these codes to UAS context must take into account the fact that a UA carries no passenger or crew. The consequences of an event in terms of casualties are therefore not to be considered with respect to the aircraft occupants, but with respect to people (including UAS crew, ground staff and third parties).

(c) On the basis of this analysis, the following definitions are adopted by USAR to define the severity of a failure condition. These severity definitions are listed in order of likely impact to UAS crew, ground staff and third parties with the highest impact being *Catastrophic* and the lowest being *Minor*. When quantifying the severity of the failure, the criteria that closest fits the situation should be selected.

(i) *Catastrophic*: Failure conditions that are expected to result in at least uncontrolled flight (including flight outside of pre-planned or contingency flight profiles/areas) and/or uncontrolled crash.

Or

Failure conditions which may result in a fatality to UAS crew, ground staff, or third parties.

(ii) *Hazardous*: Failure conditions that either by themselves or in conjunction with increased crew workload, are expected to result in a controlled-trajectory termination or forced landing potentially leading to the loss of the UA where it can be reasonably expected that a fatality will not occur.

Or

Failure conditions for which it can be reasonably expected that a fatality to UAS crew, ground staff or third parties will not occur.

(iii) *Major*: Failure conditions that either by themselves or in conjunction with increased crew workload, are expected to result in an emergency landing of the UA on a predefined site where it can be reasonably expected that a serious injury will not occur.

Or

Failure conditions which may result in an injury to UAS crew, ground staff, or third parties.

(iv) *Minor*: Failure conditions that do not significantly reduce UAS safety and involve UAS crew actions that are well within their capabilities. These conditions may include a slight reduction in safety margins or functional capabilities, and a slight increase in UAS crew workload.

(v) *No safety effect*: Failure conditions that have no effect on safety.

NOTE: The phrase “are expected to result in” is not intended to require 100% certainty that the effects will *always* be, for example, *Catastrophic*. Conversely, just because the effects of a given failure, or combination of failures, could *conceivably* be *Catastrophic* in *extreme* circumstances, it is not intended to imply that the failure condition will necessarily be considered *Catastrophic*.

(6) Safety Objectives

(a) The safety objectives should ensure that systems and equipment design will allow the UAS to achieve an acceptable safety level. A rational and acceptable inverse relationship should exist between the average probability per flight hour and the severity of failure condition effects. The relationship between probability and severity of failure condition effects is as follows.

- (i) Failure conditions with no safety effect have no probability requirement.
- (ii) Minor failure conditions must be no more frequent than probable.
- (iii) Major failure conditions must be no more frequent than remote.
- (iv) Hazardous failure conditions must be no more frequent than extremely remote.
- (v) Catastrophic failure conditions must be extremely improbable.

(b) The two-certification classes of UAS for STANAG 4671 are shown in Figure 1. They are:

- (i) Class I with Maximum Take-Off Weight-less than or equal to 5670 kg.
- (ii) Class II with Maximum Take-Off Weight greater than 5670 kg-

(c) The Applicant should consider the overall risk to crew, ground staff and third parties by showing that the Cumulative Probability of all Catastrophic Failure Conditions (Pcumcat) is extremely remote (for UA not heavier than 5670kg MTOW $P_{cumcat} \leq 10^{-5}$ per flt hour; for UA more than 5670kg MTOW $P_{cumcat} \leq 10^{-6}$ per flt hour) for the UA Class for which certification is requested. See Figure 1.

(d) Individual Failure Condition Allowable Quantitative Probabilities applicable to the USAR are provided in Figure 1. Also, the related Functional DALs for the various failure conditions are part of the matrix in Figure 1.

- (i) The allowable Individual Failure Condition Quantitative Probabilities in Figure 1 should be used as a guideline for each Individual Failure Condition. When there are more than 10 assumed Individual Catastrophic Failure Conditions at the UAS level, the allowable Individual Failure Condition Quantitative Probabilities in Figure 1 can be adapted as agreed with the certifying authority provided that Pcumcat for the UA Class for which certification is requested is met.
- (ii) For systems and equipment used only in certain phases of flight (e.g. landing gear), an average probability per flight hour" may be used as agreed with the Certifying Authority (see FAA AC 23.1309-1E).

(e) Exceptionally, for paragraph (6)(d) of this AMC, if it is not technologically or economically practicable to meet the numerical criteria for an Individual Catastrophic Failure Condition, the safety objective may be met by accomplishing all of the following.

- (i) Utilizing well proven methods for the design and construction of the system;
- (ii) Determining the Average Probability per Flight Hour of each Failure Condition using structured methods, such as Fault Tree Analysis, Markov Analysis, or Dependency Diagrams (see ARP 4761 for Markov Analysis and Dependency Diagrams), and
- (iii) Demonstrating compliance with Pcumcat for the UA Class for which certification is requested.

(f) Where the technology and architecture used do not permit the attainment of the objectives stated in subparagraphs (6)(c) and (6)(d), the UAS type certification should be dealt on a case by case basis, subject to the Certifying Authority agreement, either through operational restrictions or through a rationale justifying lesser value (e.g. considering UA weight or/and kinetic energy at impact) based upon the risk to UAS crew, ground staff, and third parties.

(7) Safety Assessments

(a) The applicant is responsible for identifying and classifying each failure condition and choosing the methods for safety assessment. The applicant should then obtain early concurrence from the cognizant certifying authority on the identification of failure conditions, their classifications, and the choice of an acceptable means of compliance.

(b) Compliance with USAR.1309 should be demonstrated by the approach as defined in "SAE ARP 4761 - Guidelines and Methods for Conducting the Safety Assessment Process on Civil Airborne Systems and Equipment" based on iterative approach through FHA, System FHA, PSSA, System Safety Assessment (SSA) and Common Cause Analyses (CCA) at UAS level, then at UAS subsystem level. Alternatively, a similar approach may be used employing the processes, requirements, task and products in MIL-STD-882 to ensure

hazards are identified and design requirements and management controls are implemented to prevent safety hazards.

(c) The application of ARP 4761 process, whether using ARP 4761 products and/or MIL-STD-882 tasks and/products, should be tailored by the Applicant according to the novelty, complexity and severity, and should be submitted for approval to the Certifying Authority.

(8) Identification of failure conditions and considerations when assessing their effects

(a) Failure Conditions should be identified by considering the potential effects of failures on the UAS, ground staff and third parties (see Figure 1). These should be considered from two perspectives:

(i) By considering failures of UAS level functions - Failure conditions identified at this level are not dependent on the way the functions are implemented and the systems' architecture.

(ii) By considering failures of functions at the system level - these failure conditions are identified through examination of the way that functions are implemented and the systems' architectures.

Note: It should be noted that a failure condition might result from a combination of lower level failure conditions. This requires that the analysis of complex, highly integrated systems, in particular, should be conducted in a highly methodical and structured manner to ensure that all significant failure conditions, which arise from multiple failures and combinations of lower level failure conditions, are properly identified and accounted for. The relevant combinations of failures and failure conditions should be determined by the whole safety assessment process that encompasses the UAS and system level FHA and CCA. The overall effect on the UAS of a combination of individual system failure conditions occurring as a result of a common or cascade failure may be more severe than the individual system effect. For example, failure conditions classified as Minor or Major by themselves may have Hazardous effects at the UAS level, when considered in combination.

Figure 1 - Relationship between Failure Probabilities, Severity of Failure Conditions, and FDAL

Classification of Failure	No Safety Effect	Minor	Major	Hazardous	Catastrophic
Severity Definition (See also para 5(c))	Frequent Failure conditions that have no effect on safety	Probable Failure conditions that do not significantly reduce UAS safety and involve UAS crew actions that are well within their capabilities. These conditions may include a slight reduction in safety margins or functional capabilities and a slight increase in UAS crew workload.	Remote Failure conditions that either by themselves or in conjunction with increased crew workload are expected to result in an emergency landing of the UA on predefined site where it can be reasonably expected that a serious injury will not occur. Or, Failure conditions which could potentially result in injury to UAS crew, ground staff, or third parties.	Extremely Remote Failure conditions that either by themselves or in conjunction with increased workload are expected to result in a controlled trajectory termination or forced landing potentially leading to loss of the UA where it can be reasonably expected that a fatality will not occur. Or, Failure conditions for which it can be reasonably expected that a fatality to UAS crew, ground staff, or third parties will not occur.	Extremely Improbable Failure conditions that are expected to result in at least uncontrolled flight (including flight outside of pre-planned or contingency flight profiles/areas), and/or uncontrolled crash, Or Failure conditions which may result in a fatality to UAS crew, ground staff, or third parties.
UA Less Than or Equal to 5670 kg MTOW					
Probability of Cumulative Catastrophic Failure	$\leq 10^{-5}$ per flt hour				

Individual Failure Condition Allowable Quantitative Probabilities	No Probability Requirements	$\leq 10^{-3}$ per flt hour	$\leq 10^{-4}$ per flt hour	$\leq 10^{-5}$ per flt hour	$\leq 10^{-6}$ per flt hour
Minimum Functional Development Assurance Level (FDAL)	FDAL E	FDAL D	FDAL C	FDAL C	FDAL B
UA Greater Than 5670 kg MTOW					
Probability of Cumulative Catastrophic Failure	$\leq 10^{-6}$ per flt hour				
Individual Failure Condition Allowable Quantitative Probabilities	No Probability Requirement	$\leq 10^{-3}$	$\leq 10^{-4}$	$\leq 10^{-6}$	$\leq 10^{-7}$
Minimum Functional Development Assurance Level (FDAL)	FDAL E	FDAL D	FDAL C	FDAL B	FDAL A
Note 1: Reserved.					
Note 2: At UAS level, no single failure will result in a Catastrophic Failure Condition. (See AMC 1309 para 8.(b)).					
Note 3: Probability of Cumulative Catastrophic Failure (Pcumcat) is the cumulative probability resulting from the probabilities per flight hour of all Individual Catastrophic Failure Conditions caused by all systems.					

(b) Functional Hazard Analysis (FHA)

(i) The FHA consists of:

- (A) identifying all the functions at the level under study and its interfaces (e. g. UA, UCS, data link, payloads, etc.),
- (B) identifying and describing the Failure Conditions associated with these functions, and,
- (C) determining the effects and the severity of these Failure Conditions.

(ii) Before an applicant proceeds with a detailed safety assessment, an FHA of the UAS and system functions to determine the need for and the scope of subsequent analysis should be prepared. This assessment may be conducted using service experience, engineering and operational judgment, or service experience and a top-down deductive qualitative examination of each function. An FHA is a systematic, comprehensive examination of UAS and system functions to identify potential No Safety Effect, Minor, Major, Hazardous, and Catastrophic Failure Conditions that may arise, not only as a result of malfunctions or failure to function but also as a result of normal responses to unusual or abnormal external factors. The FHA concerns the operational vulnerabilities of systems rather than a detailed analysis of the actual implementation.

(iii) Each system function should be examined regarding the other functions performed by the system because the loss or malfunction of all functions performed by the system may result in a more severe failure condition than the loss of a single function. In addition, each system function should be examined regarding functions performed by other UAS systems because the loss or malfunction of different but related functions, provided by separate systems, may affect the severity of failure conditions postulated for a particular system.

(iv) The FHA is an engineering tool that should be performed early in the design and updated as necessary. It is used to define the high-level UAS or system safety objectives that should be considered in the proposed system architectures. Also, it should be used to assist in determining the

FDALs for the systems. An FHA requires experienced engineering judgment and early coordination between the applicant and the certification authority.

(v) Depending on the extent of functions to be examined and the relationship between functions and systems, different approaches to FHA may be taken. Where there is a clear correlation between functions and systems, and where system and function interrelationships are relatively simple, it may be feasible to conduct separate FHA's for each system, provided that any interface aspects are properly considered and easily understood. However, a top-down approach from a UAS level perspective should be taken in planning and conducting FHAs where system and function interrelationships are more complex.

(vi) After each failure condition is classified, refer to Figure 1 to identify the Failure Condition Probability and FDAL objectives.

(vii) The classification of failure conditions does not depend on whether a system or function is required by any specific regulation. Some systems required by specific regulations, such as transponders, or position lights may have the potential for only Minor Failure Conditions. Conversely, other systems not required by any specific regulation, such as flight management systems and automatic landing systems, may have the potential for Major, Hazardous, or Catastrophic Failure Conditions.

(viii) The classification of failure conditions should consider all relevant factors. Examples of factors include the nature of the failure modes, which includes common mode failures, system degradation resulting from failures, UAS crew actions, UAS crew workload, performance degradation, reduced operational capability, effects on airframe, etc. It is particularly important to consider factors that would alleviate or intensify the severity of a failure condition. An example of an alleviating factor would be the continued performance of identical or operationally similar functions by other systems not affected by a failure condition. Examples of intensifying factors would include unrelated conditions that would reduce the ability of the UAS crew to cope with a failure condition, such as weather or other adverse operational or environmental conditions. The ability of a system to inform the UA operator of potential or real failure conditions so that timely corrective action can be taken to reduce the effects of the combination of events is desirable. This approach may reduce the severity of the failure condition.

(ix) Because of the large number of combinations of failures, various mitigating factors, UAS characteristic effects, and similar factors, a specific FHA and the related safety assessments may be significantly different for each evaluated UAS type and configuration. These factors preclude providing a concrete example of a FHA that applies across the board to every installation. However, general examples may be provided that illustrate the concepts involved in an FHA. It is critical to understand that significant engineering judgment and common sense are necessary to provide a practical and acceptable evaluation of the UAS and its systems.

Based on Failure Conditions identified in UAS level FHA and/or related system FHA, PSSA and SSA should be conducted to show that the implemented systems meet their safety requirements.

The PSSA and SSA should address all significant failure conditions identified in the FHA and aims at justifying their compliance with the quantitative safety objectives in Figure 1 and paragraph (9) hereunder and qualitative Fail-Safe criteria – at UAS function level, no single failure will result in a Catastrophic Failure Condition.

However, in certain cases, such as engine failure during limited risk time or some mechanical failures, where it can be reasonably shown that these failures are Extremely Improbable, account may be taken of their probability of occurrence and/or of experienced engineering judgment based upon the application of sound design techniques.

(9) Assessment of failure condition probabilities and analysis considerations

(a) An assessment of the probability of a failure condition may be either qualitative or quantitative. An analysis may range from a simple report that interprets test results or compares two similar systems to a detailed analysis that may or may not include estimated numerical probabilities. The depth and scope of an analysis depends on the type of functions performed by the system, the severity of failure conditions, and whether the system is complex. A quantitative analysis is intended to supplement, but not replace, qualitative methods based on engineering and operational judgment. A quantitative analysis is often used for Catastrophic or Hazardous

Failure Conditions of systems that are complex and Major Failure Conditions that are complex without redundancy, for the cases where there is insufficient service experience to help substantiate their safety, or that have attributes that differ significantly from those of conventional systems. The evaluation of each failure condition should take account of:

- (i) the failure of equipment or other functions performed by the subsystem.
- (ii) by 'interfacing subsystems'
- (iii) factors external to the system.
- (iv) operating phase and flight phase of the UAS.
- (v) exposure time.
- (vi) human Factors.
- (vii) potential for dormant faults, latent, or hidden failures.
- (viii) common cause failures (systemic failure).
- (ix) The probability of multiple failures and the probability of undetected faults'
- (x) The resulting effects on the UA, UCS, UAS, UAS crew, UAS ground staff and third parties, considering the stage of flight and operating conditions; and
- (xi) The UAS crew warning cues, corrective action required, and the UAS crew's capability of determining faults.

(b) A probability analysis may be either an FMEA or an FTA, which also includes numerical probability information. Numerical values are assigned to the probabilistic terms included in the requirements for use in those cases where the impact of system failures is examined by quantitative analysis methods.

(c) The probabilities of primary failures can be determined from failure rate data and exposure times using failure rates derived from either service experience on identical or similar items, manufacturer's accelerating testing data, or from acceptable industry standards. Conventional mathematics of probability can then be used to calculate the estimated probability of each failure condition as a function of the estimated probabilities of the various identified contributory failures or other events. See Appendix D of ARP 4761 for more information.

(d) When calculating the estimated probability of each failure condition, a margin may be necessary to account for uncertainty. A margin is not normally required for an analysis that is based on proven data or from operational experience and tests. Where data has limited background for substantiation, a margin may be required depending on the available justification.

(e) The applicant should obtain early certification authority concurrence for an acceptable classification of the failure conditions and probability for each minor, major, hazardous, and catastrophic failure condition. Early concurrence on the classification of the failure conditions may reduce the applicant's efforts in determining the probabilities resulting from changes.

(f) The details on how to calculate the "average probability per flight hour" for a failure condition is not provided in this AMC. The details on how to calculate Average Probability Per Flight Hour for a Failure Condition are given in Appendix 3 of AC 23.1309-1E or Appendix 3 of AMC 25.1309.

(g) The "average probability per flight hour" is the probability of occurrence, normalized by the flight time of a failure condition during a single flight. If the probability of a subject failure condition occurring during a typical flight of mean duration for the UA type, divided by the flight's mean duration in hours, is likely to be significantly different from the predicted average rate of occurrence of that failure condition during the entire operational life of all UAs of that type, then a risk model that better reflects the failure condition should be used. The single flight is analyzed to be representative of an average over all possible flights of the fleet of UAs to be certified. The calculation of the "average probability per flight hour" for a failure condition should consider the following:

- (i) The average flight duration and the average flight profile for the UA type to be certified.

Note: The assumption related to the average flight duration depends on UAS mission and should be agreed by Certification Authority;

- (ii) All combinations of failures and events that contribute to the failure condition;
- (iii) The conditional probability if a sequence of events is necessary to produce the failure condition;

- (iv) The relevant "at risk" time if an event is only relevant during certain flight phases; and,
- (v) The average exposure time if the failure can persist for multiple flights.

(10) Assessment Methods

(a) Assessment methods: Methods for qualitatively and quantitatively assessing the causes, severity, and likelihood of potential failure conditions are available to support experienced engineering and operational judgment. Some of these methods are structured. The various types of analyses are based on either inductive or deductive approaches. The applicant should select analyses to validate the safety of a particular design based on factors such as the system architecture, complexity, criticality of the function, etc. ARP 4761 has more details of the various methods.

(b) Failure Modes and Effects Analysis (FMEA): A structured, inductive, and bottom up analysis that is used to evaluate the effects on the system and the UAS of each possible element or component failure. When properly formatted, it should aid in identifying latent failures and the possible causes of each failure mode. ARP 4761 provides methodology and detailed guidelines that may be used to perform this type of analysis. An FMEA could be a piece-part FMEA or a functional FMEA. For modern microcircuit-based line replaceable units and systems, an exhaustive piece-part FMEA is not practically feasible with the present state of the art. In that context, an FMEA may be more functional than piece-part oriented. A functional-oriented FMEA can lead to uncertainties in the qualitative and quantitative aspects, which can be compensated for by more conservative assessments, such as the following: Assuming all failure modes result in failure conditions of interest, carefully choosing system architecture, and using lessons learned from similar technology.

(c) Fault Tree Analysis (FTA): A structured deductive, top-down analysis that is used to identify the conditions, failures, and events that would cause each defined Failure Condition. They are graphical methods of identifying the logical relationship between each particular Failure Condition and the primary element or component failures, other events, or combinations thereof that can cause it. The fault tree should be developed to the lowest level for which failure rates can be substantiated. Rates derived from applicable service experience, acceptable industry wide sources, manufacturer's accelerated testing data, or from an FMEA may be used as inputs to the lowest level events.

(d) Common Cause Analysis: The acceptance of adequate probability of failure conditions is often derived from the assessment of multiple systems based on the assumption that failures are independent. Therefore, it is necessary to recognize that such independence may not exist in the practical sense, and specific studies are necessary to ensure that independence can either be assured or deemed acceptable. The "common cause analysis" is divided into three areas of study:

(i) Zonal safety analysis. This analysis has the objective of ensuring that the equipment installations within each zone of the UAS are at an adequate safety standard regarding design and installation standards, interference between systems, and maintenance errors.

(ii) Particular risk analysis. Particular risks are defined as those events or influences outside the systems concerned (e.g. fire, leaking fluids, bird strike, tire burst, HIRF exposure, lightning, uncontained failure of high energy rotating machines, etc.). Each risk should be the subject of a specific study to examine and document the simultaneous or cascading effects, or influences, which may violate independence.

(iii) Common mode analysis. This analysis is performed to confirm the assumed independence of the events that were considered in combination for a given failure condition. The effects of specification, design, implementation, installation, maintenance errors, manufacturing errors, environmental factors other than those already considered in the particular risk analysis, and failures of system components should be considered.

Note: The need for Common Causes Analysis process as described within the SAE ARP 4761 (Common Mode Analysis, Zonal Safety Analysis and Particular risks analysis) should be tailored relatively to the specificity of the UAS and should be agreed by Certification Authority.

(11) Software and Complex Hardware DALs

(a) Acceptable application of software and complex hardware DALs

- (i) It is necessary to consider the possibility of requirement, design, and implementation errors in order to comply with the requirements of §USAR.1309. Errors made during the design and development of systems have traditionally been detected and corrected by exhaustive tests

conducted on the system and its components. These tests used direct inspection and other direct verification methods capable of completely characterizing system performance. These direct techniques may still be appropriate for simple systems, which perform a limited number of functions and which are not highly integrated with other UAS systems.

(ii) For more complex or integrated systems, exhaustive testing may either be impossible because all of the systems states cannot be determined or it may be impracticable due to the number of tests that must be accomplished. For these types of systems, compliance may be shown by the use of FDAL and item development assurance levels (IDAL). The FDALs should be determined by the severity of potential effects on the UAS in case of system malfunctions or loss of functions in accordance with Figure 1.

(b) System architecture for determination of the appropriate DALs

(i) SAE S-18, Airplane Safety Assessment Committee, revised ARP 4754 to ARP 4754A and is revising ARP 4761. The committee developed new concepts for DAL and Design Assurance Levels in ARP 4754A. These SAE documents are guidelines for assigning the DALs that start from the aircraft/system level (FDAL) and end at the IDAL. ARP 4754A addresses the development phases in two phases: Function Development Phase and Item Development Phase, with two different types of development processes.

(ii) These assignments depend on the failure condition classification, the number of independent failure paths, and their associated independence attributes. The independence attributes are the functional independence, design independence, and physical independence. In essence, functional independence ensures that the functional requirements that are implemented in the design are different, whereas design independence ensures that the hardware or software design, in which the functions are implemented, is different.

(iii) IDAL assignment considering system architecture may use the assignment process contained in ARP 4754A, however the FDALs in Figure 1 must be met. IDAL assignment considering system architecture must be agreed with the certifying authority.

(c) Software aspects

(i) DO-178C/ED-12C and its supplements should be used to demonstrate that the safety aspects of software hosted on airborne systems and equipment comply with airworthiness requirements.

(ii) The applicant may elect to use an alternative means of compliance. However, those alternative means of compliance must meet the relevant requirements, ensure an equivalent level of safety for software aspects and be approved by the Certifying Authority.

(iii) If alternative means of compliance are provided, a Plan for Software Aspects of Airworthiness (PSAA) should be provided and agreed with the Certifying Authority. Equivalence matrix between the objective tables contained in Annex A of DO-178C and provided means of compliance should be provided and agreed with the Certifying Authority.

(d) Complex Hardware Item aspects

(i) In case of PLD (Programmable Logic Devices) or ASIC (Application Specific Integrated Circuit) development and/or use of COTS digital components, development assurance process as per RTCA DO-254 should be applied.

(ii) DO-254/ED-80 should be used to demonstrate that the safety aspects of Complex Hardware Items hosted on airborne systems and equipment comply with airworthiness requirements.

(iii) The applicant may elect to use an alternative means of compliance. However, those alternative means of compliance must meet the relevant requirements, ensure an equivalent level of safety for Complex Hardware Items and be approved by the Certifying Authority.

(iv) If alternative means of compliance are provided, a Plan for Hardware Aspects of Airworthiness (PHAA) should be provided and agreed with the Certifying Authority. Equivalence matrix between the objective tables contained RTCA DO-254 and provided means of compliance should be provided and agreed with the Certifying Authority.

(e) UCS Software and Complex Hardware Items

(i) For UCS Software and Complex Hardware Items, AMC 1309 (12) b, c and d apply.

(ii) RTCA DO-278A/ ED-109A <<Guidelines for communication, navigation, surveillance, and air traffic management (CNS/ATM) systems software integrity assurance >> may be considered for COTS software of uncertain pedigree used in the UCS whose effects on safety are neither Catastrophic nor Hazardous.

AMC.1309 (d)

Power Load Shedding

In determining compliance with sub-paragraphs (c) (2) and (3), the power loads should be managed to allow for operational degradation commensurate with the remaining power available and the critical systems required for safe flight. The critical systems should be identified and priority given to maintaining their integrity. Consideration should be given to selective load shedding for operational contingencies.

AMC.1325(e)

Static pressure measuring device

The system error, in indicated pressure altitude, at sea-level, with a standard atmosphere, excluding measuring device calibration error, should not exceed ± 9 m (± 30 ft) per 185 km/h (100 knot) speed for the appropriate configuration in the speed range between $1.3 V_{SO}$ with flaps extended and $1.8 V_{S1}$ with flaps retracted. However, the error need not be less than ± 9 m (± 30 ft).

AMC.U1330(e)

Flight control performance

The Applicant should consider failure condition loads on the UA and failure conditions which cause unacceptable deviations from in the intended flight path.

AMC.U1330(i)

Flight control performance

Consideration should be given to FAA Advisory Circular 25.672-1, "Active Flight Controls" dated 15 Nov 1983, or equivalent means as approved by the Certifying Authority.

AMC.U1330(j)

Flight control performance

Provisions should be made for determining the status of the continuous built-in-test function and alerting the UAS crew of degradation of the system as appropriate.

AMC.1351 (a) (2)

Electrical Systems and Equipment, General

If compliance is shown by electrical measurements, the procedures should include sufficient testing to show that the electrical systems meet the requirements of USAR.1351. When laboratory tests of the electrical system are conducted:

- (a) The tests may be performed on a mock-up using the same generating equipment used in the UAS;
- (b) The equipment should simulate the electrical characteristics of the distribution wiring and connected loads to the extent necessary for rated test results; and
- (c) Laboratory generator drives should simulate the actual prime movers on the UAS with respect to their reaction to generator loading, including loading due to faults.

AMC.1357 (a)

Circuit protective devices

Circuits essential for continued safe operation should have a redundant circuit per USAR.1309

AMC.1361
Master switch arrangement
Controls and switches readily accessible to the ground staff necessary to meet the safety levels of AMC.1309 should be considered (e.g. an emergency stop switch).
AMC.U1412 (a)(2)
Emergency recovery capability
This emergency recovery capability may include, for instance:
(a) An automatic pre-programmed course of action to reach a predefined site where it can be reasonably expected that fatality will not occur, or
(b) An automatic flight profile to regain link (return home course), or
(c) An automatic orbit pattern around a pre-defined waypoint.
AMC.U1412 (e)
Emergency recovery capability
Where explosives are used as part of the flight termination system:
(a) A Master Armament Safety Switch should be provided for the UAS crew which can isolate all armament of explosive device circuits from all A.C. and D.C. electrical supplies.
(b) Only explosive devices approved by the Certifying Authority should be used.
(c) Explosive devices should be installed as to be easily accessible and should not require excessive handling during installation.
(d) Particular attention should be paid to protecting the insulation of circuits connected to explosives.
AMC.1419
See FAA AC 23.1419-2
AMC.1431
Electronic equipment
Before installation on the aircraft, compliance with subparagraphs (a), (b), (f), and (g) should be shown by equipment level EMC testing and analysis of test results per MIL-STD 461E, RS 105, "Radiated Susceptibility, Transient Electromagnetic Field". After installation on the aircraft, compliance with subparagraphs (a), (b), (f), and (g) should be shown by aircraft system level EMI test and analysis of the test results per MIL-STD-464A, para 5.5 and 5.6
AMC.1459
UA onboard flight recorders
On board flight recorders should comply with EUROCAE ED-112 or ETSO-C124a or any equivalent standard.
AMC.U1481 (b)
Payloads
Compliance with sub-paragraphs (b)(1) and (b)(2) may be shown by equipment level radiated emissions testing and by aircraft system level EMC test and analysis of the test results.

AMC.U1485
Environmental Control System (ECS) (a) Due consideration should be given to the cooling of avionic equipment, pressurisation requirements, bleed air and ducting systems and thermal management in the design of the ECS. (b) The ECS design (including emergency equipment and/or auxiliary methods) should provide an acceptable pressure environment for equipment affecting safety of flight. (c) Normal and emergency pressurization requirements and status should be indicated in the UCS. (d) Bleed air or other compressed air duct system should be monitored for leaks and structural integrity. Hot air leaking from damaged ducting should not cause ignition of any flammable fluids or other materials or cause damage to safety-critical equipment. Shutdown capability, with a UCS advisory or warning, should be provided when a potentially damaging or fire-producing leak occurs. The sensors for the leak detection system should recover their required leak detection function following exposure to a leak. (e) The UA thermal management system should be stable for all flight conditions and environments. The mass flow and delivery temperature of cooling medium should be sufficient for the aircraft heat loads and provide the necessary thermal stability to ensure safety-of-flight.
AMC.U1490 (f)(2)
General The size of the convergence window and associated tolerances should be defined with the Certifying Authority based on the tailoring of manned aircraft reference documents such as EASA CS-AWO.

Subpart G – OPERATING LIMITATIONS AND INFORMATION

AMC.1587

Performance information

(a) Performance information should be provided for each aircraft and engine configuration. If appropriate, a Drag Index System or an equivalent system approved by the Certifying Authority should be used. Performance data should be qualitatively assessed for each unique stores configuration. Performance data should be assessed for each unique stores configuration in conjunction with the use of a drag index approach for external stores. The drag index technique may not be valid for aircraft with large flight envelopes, especially for long-endurance, low-drag aircraft carrying comparatively high-drag external stores. The drag index technique collapses the store drag effects to one flight condition (reference) and then re-expands the data with a delta table for all off-reference conditions.

For estimated performance data, fuel flows should be increased by 5%. For flight tested performance data, this factor should not be used if approved by the Certifying Authority. If it is included, it should not exceed 5%.

(b) Dry/wet runway

(1) Runway conditions should be referenced to STANAG 3634 – Runway Friction and Braking Conditions.

(2) For example, “dry runway” means a runway with braking and directional control that are not reduced – whether the runway is actually “dry” or not.

AMC.1587 (f)

Performance information

The performance parameters should include, but not limited to: Climb performance (time, distance and fuel to climb, best climb speeds), Cruise performance (Specific Range, Specific Endurance, Best Cruise speed and altitude, Best Endurance Speed and Altitude), One Engine Inoperative Climb and Cruise performance, Divert Performance, Descent performance, etc.

Subpart H – COMMAND AND CONTROL DATA LINK

AMC.U1601 General (a) Command and control information consists of commands and parameters transmitted to the UA(s) from the UCS(s), and transmission of all relevant operating parameters required for operation of the system from the UA(s) to the UCS(s). In this context, commands to the UA include any parameter that is necessary to enable the operation of the system. (b) Command and control information transmitted to the UA from the UCS (transmitted via the ‘uplink’) should be shown to enable positive control of the UA during all normal operations, and transmitted at a rate consistent with safe operation. (c) Command and control information transmitted from the UA (transmitted via the ‘downlink’), should be shown to enable positive control of the UA during all normal operations, and transmitted at a rate consistent with safe operation. (d) Bandwidth and the latency of the overall communications system are to be considered when determining transmission rates consistent with safe operation. It should be noted that the terms ‘uplink’ and ‘downlink’ do not imply only a line of sight radio frequency channel, but include any configuration of any type(s) of communication device(s) capable of transmitting the required information. (e) It should be shown that a remote communications system is sufficient and robust enough to allow the safe operation of the system. (f) Reference should be made to STANAG 4586 for the design and assessment of the UAS communications system, including matters such as command formats and confirmation. (g) Any ATC communications system should be shown as compliant with USAR.1301 and USAR.1309 as installed equipment. (h) Any payload data link system used for command and control purposes (See paragraph (a)) should be shown to be compliant with USAR part H and USAR.1309 (i) The functionality of payload data links with no safety of flight impact need not to be assessed under USAR.U1601 (j) Where data is to be transmitted to a location other than the UCS (such as a remote viewing terminal), it should be shown that the communications paths used will not interfere with safe operation of the system.
AMC.U1603 Command and control link architecture Verification should be shown by analyses and tests necessary to show no single point of failure will cause a <i>Hazardous</i> or more serious event.
AMC.U1605 Electromagnetic interference and compatibility It should be shown that there is no interference (i.e. EMI/EMC, EME etc.) with the command and control link by the ATC communication link, any data link, or other UA systems, for all possible combinations of paths for both uplinks and downlinks. RTCA DO-160 and other relevant standards should be considered in design and compliance assessment. Verification should be shown by analyses and tests at the component level and as integrated at the system level.
AMC.U1605 (b)

Electromagnetic interference and compatibility
(a) Where multiple radiofrequency links are used for redundancy or for specific function such as local control for launch and landing elements, operating frequency should be chosen such as to rule out the possibility of mutual interference.
(b) Spread spectrum techniques may be effective in reducing mutual interference. Spread-spectrum communications are those in which the frequency of both the transmitted RF signal and the intended RF receiver are rapidly and identically varied. This is done to minimize disruption from other transmitters, as well as to prevent detection by or disruption to non-system RF receivers.

AMC.U1607 (a)
Command and control data link performance and monitoring
The maximum data link range depends on the data link frequency, the availability, the BER (bit error rate) for digital link, the relevant climate area parameters, the UA altitude, etc. Verification of data link performance should be accomplished through a combination of link analyses, component testing, and system level testing.

AMC.U1611 (a)
Command and control data link latency
Data link latency should be defined, predicted and verified for the expected relevant conditions.

AMC.U1613 (a)
Command and control data link loss strategy
The pre-programmed course of action should be stored in the UA to provide resilience to communication/data link anomalies.

AMC.U1613 (b)
Command and control data link loss strategy
The reacquisition process should be able to try to re-establish a command and control data link to the original or any other available UCS.

AMC.U1613 (c)
Command and control data link loss strategy
(a) Total loss of the command and control data link means that the data link has become degraded to a point that an emergency condition exists under USAR.U1412 and USAR.U1607 (c).
(b) The UCS should predict and display the geographic position of the UA flying without data link communications, based on the UAS Mission Plan, progress before losing the link, and the input link loss strategy.

AMC.U1615
Command and control data link antenna masking
Data link antenna masking should be verified by analyses and tests sufficient to verify safe operation of the UAS can be maintained.

AMC.U1617 (a)
Command and control data link switchover function
This operation is also called "changeover"

AMC.U1617 (b)
Command and control data link switchover function

A UAS may have multiple data links, some capable of UAS command and control, some capable of transferring payload information, and some capable of both. The protocols for data link switchover should be examined for every possible command and control switchover scenario, including scenarios where multiple UAs are being controlled from a single UAS control station. Specific emphasis should be given to determination and indication of which data link has actual control of the UA.

AMC.U1617 (c)

Command and control data link switchover function

Positive control is the practice of the command and control data link to mediate a transfer of control to the new controlling command and control data link for before severing the control connection to the UA (i.e. the entity wishing to transfer control cannot terminate the connection to the UA until the entity wishing to gain control has acknowledged the connection and required level of control).

Subpart I – UA CONTROL STATION

AMC.U1701

General

(a) It is assumed that regulatory requirements associated with occupational safety and health, fire safety, electrical safety, radiation safety etc., any effect on the UCS and/or the UAS crew are under national law. As one or more of these factors have a direct impact on airworthiness and may lead to an emergency situation, these factors should be included in the airworthiness assessment of the system. Any failure condition leading to UAS crew or ground staff injury or death is included under the AMC.1309 definitions of severity levels.

The design of the UCS should therefore take into account all relevant hazards and include mitigation techniques such as fire detection/suppression systems, emergency lighting, emergency egress, etc. that may have an impact on flight safety as deemed appropriate and agreed with the Certifying Authority.

(b) When considering the design of the UAS control station, guidance can be obtained from Defence Standard 00-251 Human Factors Integration for Defence Systems.

AMC.U1701 (e)

General

Reference for these environments are provided by MIL-STD-810 or any equivalent standard approved by the Certifying Authority.

AMC.U1704

Minimum UAS Crew

In determining the minimum UAS crew, the usability of the UCS interface by the UAS crew and associated workload should be measured under a full spectrum of conditions using a methodology such as that found in FAA Advisory Circular 23-1523. The crew workload associated with actions necessary for collision avoidance should be considered.

AMC.U1707 (a)

Communication system

Particular consideration should be given to communication intensive operations such as UA handover (USAR.U1881) and other operations where UAS crew may not be co-located in the same UCS.

AMC.U1707 (c)

Communication system

For those installations where all warnings are not provided through the radio/audio equipment, consideration should be given to the UAS crew ability to hear and recognise warnings when headsets are used, including noise cancelling headsets.

AMC.U1709

Voice recorders

(a) The storage capacity of the voice recorder should be compatible with recording of the maximum duration of flight for which certification is requested.

(b) The voice recorder should be designed to prevent unauthorised or unintentional editing.

AMC.U1709 (e)

Voice recorders

For instance, the GPS time signal can be used for this purpose.

AMC.U1711

UCS data recorder

- (a) The data recorder should be compatible with the UAS's classification guide and security procedures.
- (b) The data recorder should be designed to prevent unauthorised or unintentional editing.

AMC.U1711 (a)

UCS data recorder

- (a) Primary mission data should include but not be limited to;
 - Mission plans and updates
 - Aircraft commands
- (b) System data should include but not be limited to;
 - Data link up link commands
 - Data link down link aircraft data, to include;
 - Airspeed
 - Altitude
 - Vertical acceleration
 - Magnetic heading
 - Pitch attitude
 - Roll attitude
 - Engine performance data
 - Control surface position
 - Sense and avoid data
 - Other internal parameters necessary for monitoring safe operation of systems.
 - System health data
- (c) Consideration should be given to any unique design or operational features of the UAS to determine whether any relevant dedicated parameters should be recorded.
- (d) Guidance on data recorders can be obtained from:
 - (1) EUROCAE MOPS ED-112 (Minimum Operational Performance Specification for Crash Protected Airborne Recording Systems), or
 - (2) Air Force Instruction (AFI) 63-1401, Aircraft Information Program and Air Force Handbook (AFH) 63-1402, Aircraft Information Program.

AMC.U1714

Automated Mission Planning

The intent of USAR.U1714 is to ensure that safety critical calculations in mission planning wherever they are performed, are properly assessed by hazard analysis and where applicable software design assurance level application.

- (a) All of the UAS mission planning sub-systems that influence the safe operation of the UA (in particular those that provide aerodynamic performance predictions, UA manoeuvre commands or flight profile waypoints) should be verified, such that none of the UAS limits or performance capabilities are exceeded.
- (b) The verification process should range and type check UAS crew data entries against the UAS limits and performance capabilities.
- (c) The automated mission planning sub-systems should be compliant with USAR.1309

AMC.U1721

Arrangement and visibility

(a) The intention of USAR is to give system designers maximum flexibility to innovate safely. The selection of system data for display, and the nature of those displays, is a key safety factor. This AMC gives guidance on the categorization and assessment of data for display.

(b) 'Data' in this context means any system information required for the operation of the system, and includes, but is not limited to, information concerning navigation, flight management, system performance, cautions and warnings.

(c) In this context, USAR has three broad categories of system data:

(1) Data that is essential for the safe operation of the system and which must be displayed to the UAS crew at all times and at a refresh rate consistent with safe operation. Full-time display is mandatory for this category of data.

(2) Data where continuous display is not essential to safe operation but to which UAS crew access on demand is essential. Full-time display is not mandatory for this category of data (i.e. part-time display is an option).

(3) Data to which UAS crew display access on demand is not essential to safe operation but is nevertheless possible. Full-time display is not mandatory for this category of data (i.e. part-time display is an option).

(d) Data automatically monitored by the system that will result in a caution or warning when specified conditions are met or parameters exceeded is subject to the requirements of USAR.U1787. The display of a caution or warning itself is not subject to USAR.U1722 [U1723, U1725....] but the data used to generate the warning (for instance airspeed) may be also subject to full or part-time display.

(e) Given the diverse nature of UAS, it is difficult to prescribe a definitive list of data for which full-time display would be mandatory, and for which part-time or no display would be appropriate. Hence, each UAS should be analysed to determine which data shall be displayed full-time, and which data shall be displayed-part-time. This analysis and its results should be approved by the Certifying Authority.

AMC.U1722

Part-time display

If it is desired to inhibit some parameters from full-time display, an equivalent level of safety to full-time display should be demonstrated. Criteria to be considered include the following:

(a) Continuous display of the parameter is not required for safety of flight in all normal flight phases.

(b) The parameter is automatically displayed in flight phases where it is required.

(c) The inhibited parameter is automatically displayed when its value indicates an abnormal condition, or when the parameter reaches an abnormal value.

(d) Display of the inhibited parameter can be manually selected by the UAS crew without interfering with the display of other required information.

(e) If the parameter fails to be displayed when required, the failure effect and compounding effects must meet the requirements of USAR.1309. The analysis is to clearly demonstrate that the display(s) of data is consistent with safe operation under all probable operating conditions.

(f) The automatic, or requested, display of the inhibited parameter should not create unacceptable clutter on the display; simultaneous multiple "pop-ups" must be considered.

(g) If the presence of the new parameter is not sufficiently self-evident, suitable alerting must accompany the automatic presentation.

AMC.U1727

Electronic data display system

Colour Standardisation

(a) Although colour standardisation is desirable, during the initial certification of electronic displays colour standards for symbology were not imposed. At that time the expertise did not exist within industry, nor did sufficient service experience exist, to rationally establish a suitable colour standard.

(b) In spite of the permissive CRT colour atmosphere that existed at the time of initial EFIS certification programmes, an analysis of the major certifications to date reveals many areas of common colour design philosophy; however, if left unrestricted, in several years there will be few remaining common areas of colour selection. If that is the case, information transfer problems may begin to occur that have significant safety implications. To preclude this, the following colours are being recommended based on current-day common usage.

(c) The following depicts acceptable display colours related to their functional meaning recommended for electronic display systems.

(1) Display features should be colour coded as follows:

Warnings	Red
Flight envelope and system limits	Red
Cautions, abnormal sources	Amber/Yellow
Earth	Tan/Brown
Engaged modes	Green
Sky	Cyan/Blue
ATOL deviation pointer (See USAR.U1730(b))	Magenta
Flight director bar	Magenta/Green

(2) Specified display features should be allocated colours from one of the following colour sets:

	Colour Set 1	Colour Set 2
Fixed reference symbols	White	Yellow*
Current data, values	White	Green
Armed modes	White	Cyan
Selected data, values	Green	Cyan
Selected heading	Magenta**	Cyan
Active route/flight plan	Magenta	White

* The extensive use of the colour yellow for other than caution/abnormal information is discouraged.

** In colour Set 1, magenta is intended to be associated with those analogue parameters that constitute 'fly to' or 'keep centred' type information.

(3) Precipitation and turbulence areas should be coded as follows:

Precipitation	0-1	mm/hr	Black
	1-4	"	Green
	4-12	"	Amber/Yellow
	12-50	"	Red
	Above 50	"	Magenta
Turbulence		"	White or Magenta

(4) Background colour: Background colour may be used to enhance display presentation. (Grey or other shade)

(d) When deviating from any of the above symbol colour assignments, the manufacturer should ensure that the chosen colour set is not susceptible to confusion or colour meaning transference problems due to dissimilarities with this standard. The Certifying Authority should be familiar with other systems in use and evaluate the system specifically for confusion in colour meanings. In addition, compatibility with electro-mechanical instruments should be considered.

(e) The Certifying Authority does not intend to limit electronic displays to the above colours, although they have been shown to work well. The colours available from a symbol generator/display unit combination should be carefully selected on the basis of their chrominance separation. Research studies indicate that regions of relatively high colour confusion exist between red and magenta, magenta and purple, cyan and green, and yellow and orange (amber). Colours should track with brightness so that chrominance and relative chrominance separation are maintained as much as possible. Requiring the UAS crew to discriminate between shades of the same colour for symbol meaning in one display is not recommended.

AMC.U1727 (b)

Electronic data display

It should be shown that the transition of displayed information from one display screen to another may not lead to an unsafe situation.

AMC.U1731

General

The compliance with USAR.U1732 to USAR.U1769 may also be demonstrated by the use of modern synthetic interface based on the representation of controls on electronic display systems related to convenient man-machine interface.

AMC.U1753 (b)

Ignition switches

For example by grouping the switches or by a single master ignition control.

AMC.U1769

“Abort” control for automatic take-off system or automatic landing system

The Automatic take-off and landing “abort” control is a safety critical control as defined in USAR.U1732

AMC.U1787 (a)

Automatic diagnostic and monitoring

Compliance with USAR.U1787 (a) should address instances where the actual UAS parameters are substantially different from the commanded value such as airspeed, geospatial location, altitude, etc.

AMC.U1793 (b)

Landing gear position and warning

As a minimum, the following aural or equally effective landing gear warning devices should be provided:

(a) A device that functions continuously when one or more engine settings are beyond the power or thrust settings normally used for landing approach without the landing gear fully extended and locked. If there is a manual shut-off for the warning device prescribed in this paragraph, the warning system must be designed so that, when the warning has been suspended, subsequent retardation of any engine power or thrust to or beyond the setting for normal landing approach will re-activate the warning device.

(b) A device that functions continuously when the wing flaps are extended beyond the maximum approach flap position, using a normal landing procedure, without the landing gear fully extended and locked. There may not

be a manual shut-off for this warning device. The system for this device may use any part of the system (including the aural warning device) for the device required in sub-paragraph (a).

AMC.U1795

Pressurised compartment indicator

There should be an indicator in the UCS to display:

- (a) the pressure differential,
- (b) the compartment pressure altitude, and
- (c) the rate of change of compartment pressure altitude.

AMC.U1801

Battery discharge warning

Special attention should be given to batteries used for engine restart in flight.

AMC.U1803

Powerplant power assisted valves

The required means to indicate the valve position may be:

- a system which senses directly that the valve has attained the position selected, or
- other indications in the UCS which give the UAS crew a clear indication, that the valve has moved to the selected position.

Although a continuous display indicator would enable compliance with these requirements the alternative use of lights showing the fully open and fully closed position or transit of the valves are also acceptable means of compliance.

AMC.U1809

UAS electrical systems warning and indicator

There should be an indicator to inform the UAS crew about:

- (a) the available electric power (including power stored in batteries)
- (b) the actual power consumption, and
- (c) the electrical power required for safe flight.

AMC.U1821

Power distribution indicator

Safe minimums are a function of the minimum power needed for safe flight versus the distributions of this power over the distinguished circuits. Note that one sole circuit may be inadequate to provide all power needed for safe flight.

AMC.U1837

Magnetic heading or track data

The primary intent of the USAR.U1837 is to deal with deviation caused by the UAS. However, the Certifying Authority should consider Special Conditions during the taxi, launch and landing phase of UA.

AMC.U1845 (e) (2)
Control markings
(a) Reciprocating engine mixture control and turbine engine condition levers incorporating fuel stopcocks, or fuel stopcocks themselves, are considered to be emergency controls since they provide an immediate means to stop engine combustion. (b) Where emergency control interface is based upon electronic data display system, deviation from red colour assignment should be allowed if the Applicant is able to ensure that the chosen colour is not susceptible to confusion. The Certifying Authority should be familiar with other systems in use and evaluate the system specifically for confusion in colour meanings. In addition, compatibility with electro-mechanical instruments should be considered.
AMC.U1881 (b)
UA handover between two UCS
(a) Positive control is the practice of the UCS to mediate a transfer of control to the new controlling UCS before severing the control connection to the UA. I.e. The entity wishing to transfer control cannot terminate the connection to the UA until the entity wishing to gain control has acknowledged the connection and required level of control. (b) The transition time during which the UA is not under the control of any UCS must be assessed in order to demonstrate that UAS hand over procedure does not lead to any unsafe situation.
AMC.U1881 (c)
UA handover between two UCS
A specific description of the synchronisation procedure in each UCS should be presented in the UAS Flight Manual.
AMC.U1881 (d)
UA hand over between two UCS
Particular attention should be given to UCS settings during control handover to ensure operating parameters are identical before and after handover.
AMC.U1885 (b)
UA handover within the same UAS control station
Positive control is the practice of the UCS to mediate a transfer of control to the new controlling workstation before severing the control connection to the UA. I.e. The entity wishing to transfer control cannot terminate the connection to the UA until the entity wishing to gain control has acknowledged the connection and require level of control.
AMC.U1885 (d)
UA handover within the same UAS control station
Particular attention should be given to UCS settings during control handover to ensure operating parameters are identical before and after handover.

AEF-4671(B)(1)